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WILL AI EAT SOFTWARE?



For over a decade, software has come to “eat” the world. But the emergence of new agentic AI tools for software development in recent months has fueled concerns that AI could in turn “eat” software, triggering a sharp re-rating of the software sector as well as other sectors vulnerable to AI disruption. So are these fears about the future of software (and other industries) warranted, or overdone? GS’ Gabriela Borges, Sherlund Partners’ Rick Sherlund, and Cohesity’s Sanjay Poonen generally agree that AI won’t eat software but diverge on how radical a transformation legacy software firms must undergo to remain competitive and how durable incumbent moats are. With all this uncertainty, we then explore what it will take to stabilize share prices (earnings stability), whether stress in software-

exposed credit can catalyze a turn in the credit default cycle (unlikely), and how investors should be positioned, with Sherlund and GS analysts and strategists agreeing: investors should be selective, in software and beyond.

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Legacy software companies aren’t standing still. They are innovating as fast followers... That, combined with their moats, could ultimately leave incumbents in a better place from a soup-to-nuts platform perspective.

- Gabriela Borges

Moats buy incumbents time to adapt, as seen in prior software disruption cycles... But these moats aren’t insurmountable—AI will very likely erode them over time.

- Rick Sherlund

Agentic AI has already proven to be an incredibly powerful tool with the potential to be enormously disruptive... However, just like any technology wave, you must surf this tsunami, or it will demolish you.

- Sanjay Poonen

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The Goldman Sachs Group, Inc.

Macro news and views

We provide a brief snapshot on the most important economies for the global markets

US

Latest GS proprietary datapoints/major changes in views

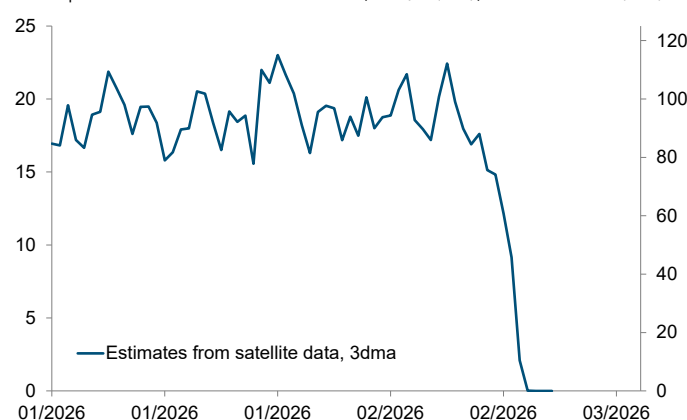
- No major changes in views.

Datapoints/trends we're focused on

- Oil disruptions related to the US-Iran conflict, which pose downside risks to growth and upside risks to inflation, depending on the longevity and severity of the conflict.
- Tariffs; we expect the global tariff that has replaced the IEEPA tariffs to lower the increase in the effective tariff rate since the start of 2025 to 9pp from 10pp.
- Labor market; the Feb jobs report suggests a higher risk that labor demand remains too weak to support the labor market stabilization we had seen in previous months.
- Concerns about nonbank lending, which have risen.

Strait of Hormuz oil transport grinds to a near halt

Oil exports via the Strait of Hormuz, mb/d (lhs), % of normal (rhs)



Source: Kpler, Goldman Sachs GIR.

Japan

Latest GS proprietary datapoints/major changes in views

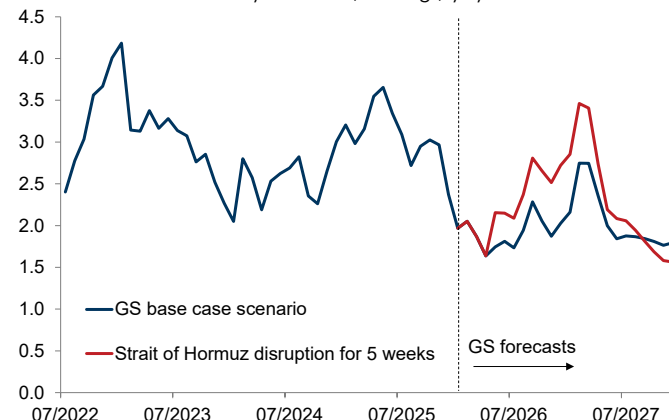
- We recently raised our FY26 average core CPI inflation forecast to 2.1% (from 1.6%) to reflect our higher energy price and USD/JPY forecasts.

Datapoints/trends we're focused on

- LDP's landslide victory in Japan's Lower House elections, which we think will make it easier for the Takaichi Administration to implement its proactive fiscal agenda, posing increased risk to Japan's fiscal position.
- BoJ policy; we continue to expect the next BoJ rate hike in July, and think two recent board member nominations have lowered the probability of an earlier rate hike.
- Shunto negotiations; we expect a base pay rise of 3.2-3.5%.

Rising Japanese inflation

Core CPI inflation rate by scenario, % chg., yoy



Source: Goldman Sachs GIR.

Europe

Latest GS proprietary datapoints/major changes in views

- We lowered our 2026 EA growth forecast and raised our 2026 EA headline inflation forecast to reflect higher energy prices amid the US-Iran conflict, with growth risks skewed further to the downside and inflation risks to the upside.
- We lowered our 2026 UK growth forecast to 1.0% yoy (from 1.1%) and raised our headline inflation forecast to 2.4% yoy (from 2.3%) to reflect higher energy prices; as such, we now expect the next BoE cut in April (vs. March before).

Datapoints/trends we're focused on

- European structural reforms; recent political momentum behind further reform steps to counter Europe's waning international competitiveness is promising, in our view.

Europe: still losing competitiveness

Real exchange rate with China (lhs, index, 2019=100) vs. survey-based assessment of int'l competitiveness (rhs, z-score)



Source: Haver Analytics, Goldman Sachs GIR.

Emerging Markets (EM)

Latest GS proprietary datapoints/major changes in views

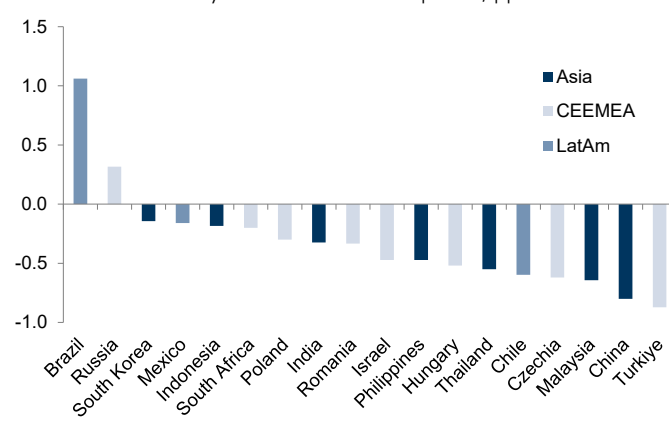
- We lowered our 2026 GDP growth forecasts for Saudi Arabia, Oman, Qatar, Bahrain, Kuwait, and UAE to reflect Strait of Hormuz disruptions and a fall in non-oil economic activity amid the US-Iran conflict, with risks skewed to the downside.

Datapoints/trends we're focused on

- US-Iran conflict impacts beyond the ME; we find that a 10% rise in oil prices would be especially negative for Türkiye and oil importers across Asia, while oil exporters Brazil and Russia are likely to be the biggest beneficiaries.
- China growth; we expect real GDP growth to fall to 4.8% yoy this year (from 5% in 2025), in line with the recently-announced official growth target of 4.5-5% this year.

Higher oil prices: negative for most EMs

Real GDP sensitivity to a 10% rise in oil prices, pp



Source: Goldman Sachs GIR.

Will AI eat software?

For over a decade, software has come to “eat” the world, dominating an ever-wider range of industries. But the emergence of new agentic AI tools for software development in recent months (see pg. 11) has fueled concerns that AI could in turn “eat” software, triggering a sharp re-rating of the software sector (see pg. 9) as well as other sectors vulnerable to disruption from these increasingly powerful and accessible AI tools. So are these fears about the future of software (and other industries) warranted, or overdone?

We ask three long-time software industry watchers: Gabriela Borges, GS US Software analyst, Rick Sherlund, Founder and CEO of Sherlund Partners, and Sanjay Poonen, CEO and President of data security and management firm Cohesity.

Borges explains that “AI is software” because it is code designed to perform tasks. So, rather than eat software, she expects AI to *expand* the software market. But she also expects competition to intensify as AI lowers the cost of code. The main risk for legacy software firms, she says, is that AI-native entrants capture the new AI opportunity while the incumbents are left with the old opportunity—namely, being the “system of record”—that shrinks as a share of the software stack. But Borges notes that legacy firms aren’t standing still but rather “innovating as fast followers.” This, together with moats such as domain experience that new entrants can’t quickly replicate, could ultimately lead incumbents to deliver better outcomes for customers, in her view. But she stresses that they must prove they can leverage their moats to do so, which hasn’t happened yet.

Sherlund shares Borges’ conviction that AI won’t eat software, viewing the current moment akin to past transformational technology shifts that were highly disruptive but ultimately propelled the software industry onto a new growth curve. In his view, software “is being reborn around AI”, necessitating a fundamental re-architecture of legacy systems around large language models (LLMs) and AI agents. While Sherlund acknowledges that incumbent moats can slow disruption, he is skeptical of their long-term durability, arguing that they will buy legacy software firms time rather than guarantee their survival.

And Poonen, for his part, agrees that new AI tools for software development mark a transformative moment for the industry, warning that “just like any technology wave, you must surf this tsunami, or it will demolish you.” In his view, how legacy software firms fare in the AI era will depend crucially on whether they can sustain their current price per user. Firms unable to defend pricing in their core offerings will need to expand and improve their products. But, like Sherlund, Poonen is explicit that just bolting AI agents onto legacy systems may not be sufficient—incumbents must ultimately be willing to rebuild, in some cases from scratch, to remain competitive.

So, with this uncertainty, what will it take to stabilize the sector and the firms most exposed to it? Ben Snider, GS Chief US Equity Strategist, argues that stabilization is unlikely to occur anytime soon, with history suggesting that share prices in industries facing structural disruption (e.g. newspapers) and an associated deep rethink of firms’ terminal value tend to stabilize only when earnings stabilize, underscoring Borges’ point that “the numbers need to contradict the narrative.”

Shamshad Ali, GS senior credit strategist, also expects pressure to persist in the broadly syndicated loan (BSL) market—which software comprises a meaningful share of—as impairment in software-exposed loans will likely be difficult to avoid. But he doesn’t see this as sufficient to catalyze a turn in the credit default cycle, assuming the benign macro backdrop and lower funding cost environment continue.

GS US Financials analysts Alexander Blostein and Michael Vinci are somewhat more positive on alternative asset managers, who have been hit hard given the meaningful software exposure in their private equity and direct lending portfolios. Blostein and Vinci argue that despite this exposure, firmwide risk remains manageable as software lending represents a relatively small share of management fees, software loans are largely senior secured and short duration, and managers have been highly selective in their software investments. But they caution that sentiment and headline risk will likely persist, with slower retail fundraising and accelerating redemption requests potentially weighing on growth despite the durability of underlying earnings streams.

So, how should investors be positioned? Sherlund and Borges agree: this is not an environment for binary bets on software’s survival or collapse, but one that demands selectivity. Sherlund argues that “investors must carefully assess which companies are most exposed to disruption and which possess complex workflows that are more difficult to encroach upon” and views the most promising companies as “those doing the heavy lifting to re-architect to put LLMs at their core and build robust agentic frameworks.” Borges, for her part, is focused on fast followers and firms with defensible moats, seeing select opportunities in applications, infrastructure, and cybersecurity. And GS US Software analysts Adam Hotchkiss and Matthew Martino see value in Vertical Software, Data Infrastructure, and Physical-to-Digital firms.

GS Head of Asset Allocation Research Christian Mueller-Glissmann then digs into how to manage AI risk in portfolios more broadly given the outsized role the technology sector plays in driving US equity valuations and returns. To achieve more balance between innovation and inflation risks, he believes investors may need to continue rotating from capital-light sectors, like technology, to capital-heavy sectors that AI is less likely to disrupt (think telecoms, industrials, and utilities), though he favors being more selective given the aggressive rotations that have already occurred.

Finally, with the “AI scare trade” having moved beyond just software to many other sectors—from consumer internet and information services to insurance, CRE services, financials, and transportation—GS equity analysts assess how new AI tools could impact these industries (see pgs. 21-23). Their key message: while AI could disrupt some firms, many may also benefit, reinforcing the idea that investors should be selective as AI technology continues to evolve.

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Interview with Gabriela Borges

Gabriela Borges is Senior Equity Research Analyst at Goldman Sachs covering US Software. Below, she argues that AI won't eat software and some of the recent price action has gone too far, but legacy software firms will need to prove themselves through better fundamentals.



Jenny Grimberg: How does the current software correction compare to past corrections?

Gabriela Borges: The YTD correction is squarely anchored to terminal value debates: instead of analyzing a change in the near-term demand trendline or the mathematical impact of higher interest rates—as was the case in the 2022 correction—the market is

questioning software moats and business models. While this debate is centered primarily on the application software firms, investors are also questioning whether the infrastructure /security stack could be impacted, and the ROI tied to hyperscaler capex. This is sparking a return to first principles questions about the value the industry provides and the nature of competition. And this is all happening against a backdrop of elevated uncertainty given the rapidly changing technology. So, the end state is difficult, if not impossible, to predict.

Jenny Grimberg: Generative AI technology has been around for several years. So, what's changed recently?

Gabriela Borges: The technology is now beginning to impact enterprises. Most AI chatbot applications that have emerged since ChatGPT's debut have primarily impacted the consumer-facing world. Concerns about how data is maintained, cleaned, and validated have been a huge limiting factor for enterprise AI adoption, and regulatory, compliance, and security frameworks are in the process of adapting to make enterprises comfortable adopting these mission-critical yet invasive technologies. At the same time that progress has been made on these fronts, tools more relevant for enterprise use cases have emerged.

For example, Anthropic's recently-released Claude Cowork product makes its Claude Code product accessible to all knowledge workers, not just coders. And its new large language model (LLM) Claude Opus 4.6 has several domain-specific applications, including in cybersecurity, legal, and financial services. OpenAI debuted Frontier, an enterprise platform for building, deploying, and managing AI agents and is strategically leaning into enterprise monetization. And personal AI assistant OpenClaw launched with capabilities much more powerful than a typical AI chatbot's; usage today is primarily limited to technically-savvy consumers, but this may expand as security and usability challenges are solved. So, AI use cases and the power to execute them has progressed, leading the technology to intersect more with the traditional software world and creating a competitive bake-off dynamic between new entrants and today's software leaders at enterprises now better equipped to utilize the technology.

Jenny Grimberg: So, what's your response to those who argue that AI will "eat" software?

Gabriela Borges: The key question is, what do you have to believe to be bullish on software? You have to believe that the

total addressable market (TAM) will grow more than enough to offset the rise in competition. We feel comfortable that the TAM will grow because, despite the current nomenclature, AI *is* software—it is code designed to perform a set of tasks, which is in many ways simply an extension of the software category. But the competition part of the question is currently the bigger unknown as the rapid pace of technological change is attracting new entrants and a wave of VC money. At the same time, the cost of code is falling, which also catalyzes more competition. In our view, the biggest risk for today's software leaders is that the new opportunity AI creates accrues to these new competitors, and the incumbents are left with outsized exposure to the old opportunity that compresses as a percentage of the total software stack.

One way to illustrate this is with the concept of a "system of record." Many of today's application software leaders are systems of record. These systems, such as Enterprise Resource Planning, Customer Relationship Management, and Human Resources, are the digital backbone of the business, storing the foundational data upon which it operates. AI may find it hard to disrupt these for two reasons. First, the data and historical context sitting in the system of record cannot be extracted holistically and in a way that captures ongoing data updates owing to nuances with meta context and data labeling as well as with stream processing that are proprietary to the system of record and break when extracted, even when using sophisticated APIs. Second, using generative AI technology to replace deterministic logic is expensive and ineffective. Put simply, it makes more strategic and economic sense for new AI entrants to leverage existing systems of record for high quality data and context than to replace them. The question then becomes whether the incumbents can also capture the new opportunity because of this domain experience and context.

Jenny Grimberg: So, is it possible to be bullish on agentic AI and *not* bearish on legacy software?

Gabriela Borges: Yes, because some legacy software firms will likely capture the upside opportunity of AI. We expect some incumbents will be able to prove to customers that their domain experience drives better AI outcomes. It's worth highlighting that incumbents are not static—many have world-class engineering teams that benefit from the same AI productivity gains as start-ups, founder involvement, and strong cash balances to respond to new competitive threats.

But the playbook to capture AI upside is three-fold, and companies must succeed on all three fronts, which won't be easy but is possible. First, companies must minimize any technical debt in their tech stack. That means ensuring all acquisitions integrate nicely into the company's core technology backbone and data flows across different modules and is centralized and uniformly cleaned. Second, companies must understand the holes in their roadmaps. One misconception is that a software leader must be a leader in technology. It can instead be a fast follower, letting the VC

ecosystem fund next-gen technologies, seeing which have durability and the best product market fit, and then acquiring them or using internal R&D to capture that product market fit, which is arguably a better use of capital than spending heavily to develop every new technology in-house.

Third, companies must figure out how to monetize AI, which may require a change in pricing model. For example, instead of charging per customer service representative, a software company could instead charge per successfully completed customer service call. Another type of pricing model is enterprise license agreements (ELAs), which allow customers to access many different products for one price. ELAs have existed in software for some time. Salesforce recently shifted to agentic ELAs (AELAs), giving customers access to their AI and other products. This is important because it shifts the software pricing model more holistically to include agentic value and allows customers to have fluidity between seats and outcomes. Importantly, we don't think customers can reduce seat count and maintain/enhance productivity unless software tools are picking up the slack and delivering more value.

Jenny Grimberg: What will allow the sector to stabilize?

Gabriela Borges: The numbers need to contradict the narrative. The software sector has been in a four-year-long depression; for 14 consecutive quarters, revenue growth has decelerated and the cost of acquiring new customers has risen. So, the numbers need to stabilize. It would be hard to argue that software companies are being disintermediated if they put up better fundamentals, because the only way they can do so in the AI era is by having exposure to the new opportunities and budgets that AI creates. Evidence of this would be clean beat-and-raised quarters and improved unit economics, either on an industry or, more likely, company-specific basis.

Jenny Grimberg: But couldn't that just push out the bear case for software disruption rather than prove it wrong?

Gabriela Borges: This is where moats become important. A company must prove its fundamentals are holding up because it has a defensible moat. One example is data architecture and collection moats, which are a feature of some security software companies and are outside the realm of AI's current and likely future capabilities. Cloudflare, for example, has a physical edge compute network co-located in telecom data centers, meaning that it owns physical computing infrastructure deployed close to end-users. Palo Alto Networks and CrowdStrike directly collect data, either through physical hardware deployed at the network edge or, in CrowdStrike's case, through an agent installed on an end-user device. These are classic examples of network effects that have compounded over 10+ years.

Another defensible moat is domain experience. Software firms with domain experience understand how knowledge workers like salespeople, accountants, and financial analysts work and use their tools owing to years of being embedded in an enterprise. That usage also allows these firms to collect proprietary data, some of which is very nuanced and difficult to abstract. Beneath the surface of Microsoft Copilot, for example, is a highly nuanced graph showing how knowledge workers in a specific enterprise collaborate, which Microsoft has the intelligence to create because it has observed how these workers interact with its software for many years. That is a powerful type of domain experience. Microsoft is also vertically integrated, operating across all three layers of software—the platform, infrastructure, and application layers—which allows it to observe and optimize how enterprise work gets done.

Jenny Grimberg: But couldn't AI native companies replicate those moats over time?

Gabriela Borges: Yes, but time matters, as does whether it makes strategic sense to do so. If everything were frozen in place right now and AI-native firms were given two years to figure out how to replicate domain experience, for example, they probably would. But, again, legacy software companies aren't standing still. They are innovating as fast followers. The incumbents have access to the same AI technology as the AI firms themselves—the LLMs, the coding productivity tools, etc.—and so are benefitting from the same productivity improvements. That, combined with their moats, could ultimately leave incumbents in a better place from a soup-to-nuts platform perspective. They already have the system of record, domain experience, and user interface. Layering AI on top of that gives incumbents everything they need to build a successful agentic solution more powerful than any one new competitor can offer. But ultimately, incumbents will have to demonstrate they can leverage their moats to deliver such an outcome, and so far, that hasn't happened in a broad enough way. So, the pressure is on for these companies.

Jenny Grimberg: Given everything we've discussed, has the recent sharp selloff in software gone too far?

Gabriela Borges: The pace of competition and the level of uncertainty is undoubtedly high. However, two areas of discovery value allow us to be bullish on many software assets. First is the dynamic we've discussed on incumbents being able to leverage domain experience to drive better AI outcomes. Second is the potential for more companies to manage to GAAP earnings, which provide a level of valuation support for the stocks. The level of competitive noise is likely near a local maximum, and the perception vs. reality gap should narrow ahead for the majority of software firms. Looking at it from that lens, some of the recent price action has likely gone too far.

Interview with Rick Sherlund

Rick Sherlund is Founder and CEO of Sherlund Partners LLC, an investment advisory firm focused on the rapidly evolving AI software sector and traditional enterprise SaaS space and is a Senior Advisor at Wedbush Securities. Below, he argues that the software industry is being reborn around AI, necessitating a complete re-architecture of existing software systems.

The views stated herein are those of the interviewee and do not necessarily reflect those of Goldman Sachs.



Allison Nathan: Does the emergence of new agentic AI tools for software development mark a transformative moment for the traditional software industry?

Rick Sherlund: Very much so. The software industry undergoes a major platform shift roughly every 10-20 years as new technologies emerge.

While these shifts can be highly disruptive, they also serve as powerful catalysts for growth, propelling the entire software industry onto new growth curves. We are at such a juncture today, with over 10,000 venture-backed AI-native companies poised to encroach on and disrupt the ~\$400bn global Software as a Service (SaaS) market.

Allison Nathan: How does this compare to past periods of transformation?

Rick Sherlund: Some technology shifts, such as the move to mobile, allowed companies to adapt their existing architecture without fundamentally altering business processes. However, other shifts created true discontinuities that forced companies to completely rebuild their systems from the ground up. The move to relational databases in the 1980s/1990s is a prime example. Companies could not simply tweak their existing frameworks. Instead, they had to reconstruct their architectures entirely—a task akin to demolishing and rebuilding a skyscraper from its foundation. Once-dominant players that were unable to reinvent their platforms to compete with Oracle, which was a leader in promoting relational database architecture, were ultimately left behind. Similar upheavals occurred during the transitions to client-server models and later to cloud computing.

The rise of AI marks a similarly transformative shift, enabling processes that legacy system architectures cannot readily accommodate. Faced with falling valuations, incumbent SaaS providers must overhaul their systems. Remaining competitive in the AI era requires software companies to do much more than add incremental AI capabilities to automate narrow legacy human tasks. While such bolt-on functions may be quicker and look good in the marketing literature, they are superficial and risk missing the more critical and transformative opportunity to rearchitect systems around large language model (LLM) intelligence with reasoning at the core to orchestrate evolving autonomous agents, which can power a new generation of business processes. This shift is not only disruptive for the software industry, but for businesses and sectors globally, representing a fundamental change in technology platforms.

Allison Nathan: What new processes will such AI-enabled platforms be able to achieve?

Rick Sherlund: Traditional SaaS platforms rely on hard-coded workflows and human-centric, menu-driven interfaces. By contrast, AI-based systems feature more conversational interfaces, embedding LLM intelligence at their core to orchestrate agents capable of reasoning, decision-making, and executing tasks previously handled by humans. This enables the integration of entirely new processes. For example, legacy customer support consists of menu-based interactive voice responses (IVR) and irritating cascading menus. Users may frequently scream “representative” in frustration to exit the menu system to connect with a human to address their issue, but even that representative may not be able to do so. And customer support remains costly.

So, companies are turning to AI-powered virtual agents that are available 24/7, multilingual, and trained on comprehensive support materials. These agents run on hyperscaler systems that avoid problems of availability and scalability in periods of peak demand. Agents may also identify that a customer has, say, an older version of a product that may no longer be supported. The agent can then pivot from a cost center support function to a revenue function by recommending a new product. So, AI agents can seamlessly shift from support to sales and billing, orchestrating complex, multistep workflows across disparate systems with minimal human intervention.

As infrastructure improves to provide governance and guardrails, AI agents will increasingly handle autonomous, agent-to-agent processes, with humans overseeing compliance and exceptions. This shift will unlock even more processes and expand automation as systems become more intelligent and capable of advanced reasoning.

Allison Nathan: So, are concerns that AI could “eat” software, which have sparked a major selloff in legacy software companies, warranted, or overblown?

Rick Sherlund: At the end of the day, AI is expressed in software, so declaring software “dead” is misguided. Software will endure, even if the industry looks different than it does today. Certain segments of the software market—particularly applications—will experience considerable disruption, but the broader industry will be reincarnated, adapting to new processes and reaping substantial benefits from the widespread integration of AI capabilities across all business functions. This will lead to a step function increase in productivity that was not possible with traditional SaaS architectures.

Allison Nathan: Won't the rise of "vibe coding" eventually make software replaceable, as some argue?

Rick Sherlund: Concerns that "vibe coding" will make it so simple to write software that software itself becomes trivial are largely unfounded. Coding accounts for only around 15–20% of writing enterprise systems. So, while automating code generation is a major efficiency gain, it is just one aspect of a far more complex process. Designing effective workflows, defining data structures, and establishing secure access controls are all critical and complex tasks. Developers must decide how databases are organized, what data is stored, and how sensitive information, like payroll data, is protected. These challenges extend well beyond basic coding. So, easier coding won't sound the industry's death knell. On the contrary, the industry will likely become more vibrant than ever as firms leverage the new technology to remain competitive. So, the software market is not dead; it is being reborn around AI.

Allison Nathan: How defensible are legacy software moats amid the AI transition?

Rick Sherlund: Complex workflows are a powerful moat, demanding deep domain expertise to design effective interactions with vendors and suppliers in the supply chain. This expertise extends not only to vertical markets but also across horizontal complexities. Data access is another powerful moat, as software companies fiercely protect their customers' data and resist third-party access, recognizing its tremendous value. Scale and distribution moats are powerful as well—building and maintaining an enterprise sales force is extremely costly, so established distribution networks further protect incumbents.

Together, these moats buy incumbents time to adapt, as seen in prior software disruption cycles. For example, larger companies with complex workflows like SAP and Oracle were slower to adopt cloud architecture but eventually did so, leveraging their moats to buy time. So, firms with these protective moats, like the large SaaS incumbents, have more time to adjust amid the AI transition. But these moats aren't insurmountable—AI will very likely erode them over time.

Allison Nathan: How are incumbent software companies responding?

Rick Sherlund: Incumbent software companies are working hard to fundamentally re-architect their systems. Large firms like Salesforce, Workday, and ServiceNow—who were disruptors in the shift from client-server to the cloud—are now integrating LLMs into the core of their systems and developing agentic frameworks to support intelligent, autonomous workflows. Having been disruptors themselves, these vendors recognize that major technological shifts require comprehensive transformation, not just superficial bolt-ons.

These companies are also making significant efforts to address broader issues such as governance, security, and the need for advanced data structures like knowledge and context graphs. Such measures are vital for understanding data and workflow intent, ensuring that systems can operate autonomously while maintaining compliance and security. While these efforts do not guarantee that the incumbent SaaS companies will "win" against emerging AI-native competitors, they represent significant steps toward adapting and maintaining relevance in

an evolving market. Whether they can move fast enough and reposition to leverage their market presence, distribution, and scale advantages to be net beneficiaries of the AI era remains to be seen.

Allison Nathan: By contrast, how can AI-native firms compete and win in the space?

Rick Sherlund: Disruptors have historically succeeded by entering at the lower end of the market, where there is less invested in incumbent systems and companies are more open to adopting innovative solutions. This was the case with Salesforce and Workday, who began by serving small- and medium-sized businesses rather than directly competing with SAP.

Over time, as new entrants enhance their offerings and scale their distribution, they can move gradually upmarket, competing with larger incumbents—adopting a "land and expand" strategy. Salesforce, for example, began with sales automation before expanding into marketing and customer service, while Workday started in Human Resources before adding financial capabilities. Ultimately, the path to market disruption starts at the bottom where the protective moats are fewer and fresh ideas can gain traction, and scales upward.

Allison Nathan: Given all that, how do you foresee the competitive landscape for software evolving?

Rick Sherlund: As more AI-native companies enter the lower end of the market, the software industry will likely become increasingly fragmented. We saw this with the move from client-server systems to the cloud/SaaS model, where specialized "best-of-breed" software solutions were embraced over traditional monolithic suites from providers like SAP and Oracle. Now, with the rise of agentic AI systems, we will likely see a further disaggregation as agents enable a fragmentation of functions and an unbundling of existing SaaS functions. This is already evident in the customer support segment, with AI agents transforming Contact Center as a Service (CCaaS) and IVR systems, as well as in the emergence of dedicated sales and marketing agents.

Allison Nathan: How can investors distinguish between software companies that will likely remain resilient in the face of AI-driven disruption, and those that may be at risk of becoming obsolete?

Rick Sherlund: Investing in software has now become riskier, making it important to distinguish between layers of the software stack—applications are probably more at risk than infrastructure software, while areas such as governance, security, and data management stand to benefit most from AI advances. Investors must carefully assess which companies are most exposed to disruption and which possess complex workflows that are more difficult to encroach upon, even if these firms will eventually need to undergo significant transformation to remain competitive.

Ultimately, the most promising companies are those doing the heavy lifting to re-architect to put LLMs at their core and build robust agentic frameworks. At the end of the day, only those with the right architecture will be able to adapt and capitalize as AI capabilities rapidly advance.

The software ecosystem, at a glance

The software industry can be thought of in three layers: Applications, Platforms, and Infrastructure. Companies operating within each layer are considered software companies because their core offering is the creation, deployment, or sale of software. A company can appear in multiple layers of the software stack depending on its product offerings, in a strategy known as “vertical integration”.

Applications

The application layer represents the software products that end-users directly interact with to perform specific tasks.

Applications include an array of software such as enterprise resource planning (ERP) systems, customer relationship management (CRM) tools, productivity suites, and industry-specific applications.



Platforms

The platform layer provides the software environment, tools, and services that enable developers to build, deploy, and manage applications. Platforms include operating systems, databases, development frameworks, cloud Platform-as-a-Service (PaaS) offerings, and API platforms.



Infrastructure

The infrastructure layer comprises the core computing resources, networking capabilities, and security and storage solutions that underpin all software.

Infrastructure includes physical and virtual servers, data centers, cloud

Infrastructure-as-a-Service (IaaS)

offerings, databases, and cybersecurity essentials.



Note: List of companies in each layer is not exhaustive.
Source: Goldman Sachs GIR.

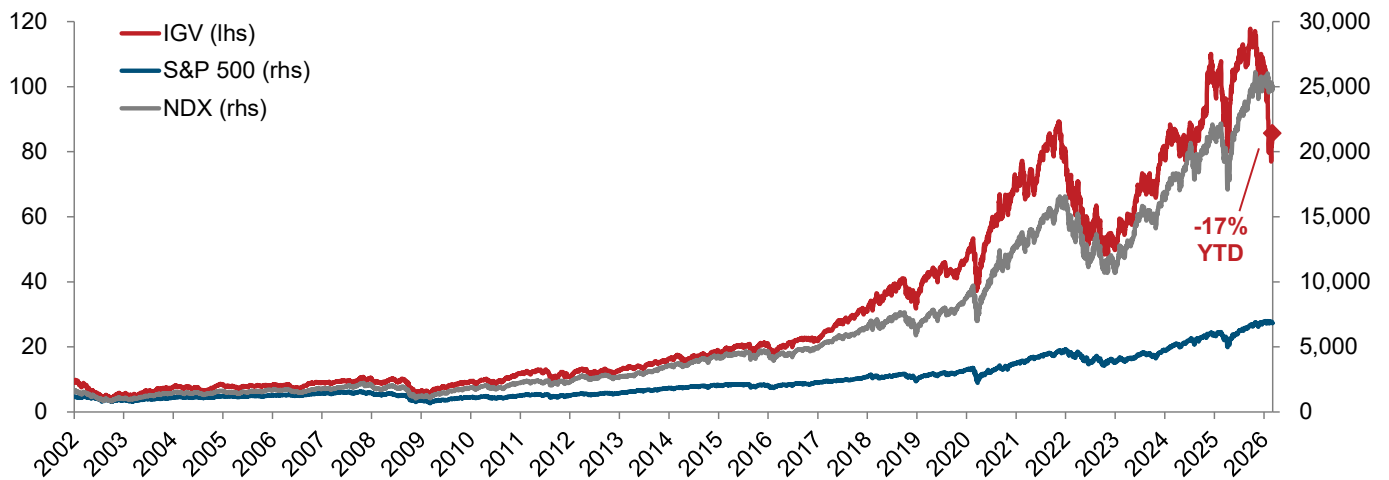
Special thanks to GS US Software Analyst Matthew Martino for data and guidance.

Illustration created using random AI generated code.

The software selloff, visualized

The iShares Expanded Tech-Software Sector ETF (IGV), which tracks the performance of around 120 North American stocks in the software industry, has declined by around 17% since the beginning of the year and 26% from its fall 2025 highs

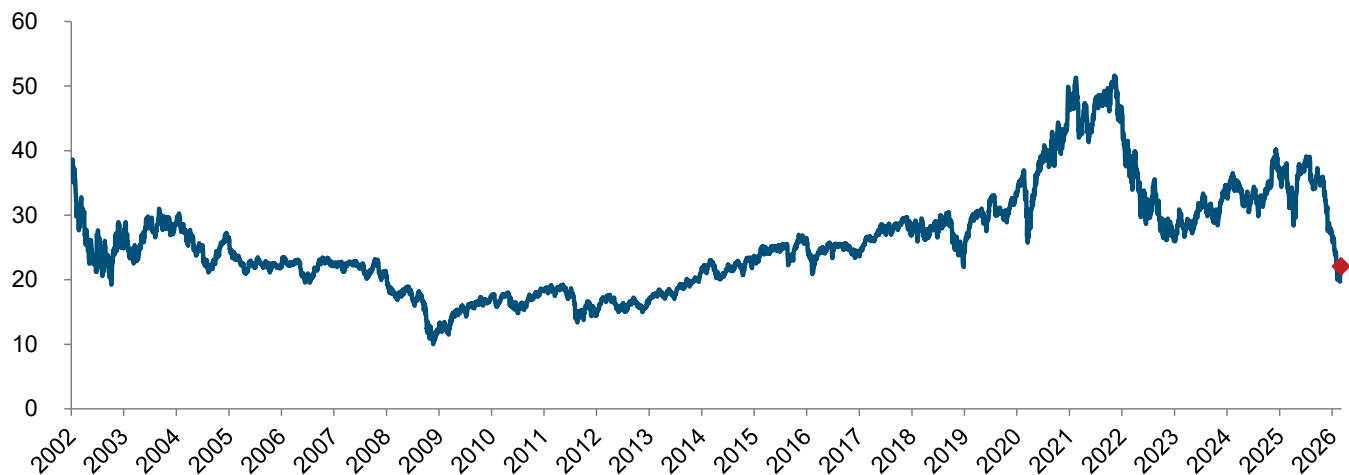
Price



See [here](#) for more details about the IGV ETF.

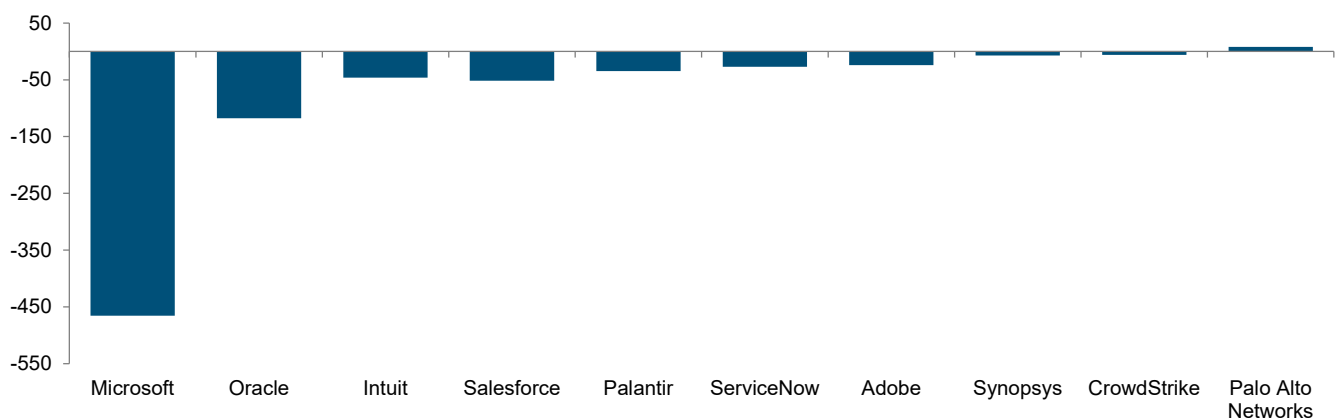
Source: FactSet, Goldman Sachs GIR.

Valuations have also fallen sharply, with the companies in the ETF trading at roughly 22x, well below their Covid-era peak of 51x IGV NTM P/E



Source: FactSet, Goldman Sachs GIR.

The 10 largest stocks in the IGV ETF have in total lost nearly \$800 billion in market capitalization since the beginning of the year
Market cap change for each of the 10 largest stocks in IGV since January 2, 2026, \$bn



Source: Bloomberg, Goldman Sachs GIR.

Special thanks to GS US Portfolio Strategist Christophe Sung for data.

AI disruption: software subsector impacts

Q: Vertical Software, Data Infrastructure, and Physical-to-Digital are key subsectors of the software space. To what extent are legacy firms in these subsectors vulnerable to AI disruption?

Vertical Software

A: Vertical software refers to products built for specific industries rather than general-purpose systems like email or accounting tools. These systems store critical data, reflect industry-specific workflows, and are deeply embedded in companies' daily operations. Vertical vendors include firms like Guidewire Software (insurance), Tyler Technologies (public sector), and Veeva Systems (life sciences). As AI technology continues to rapidly evolve, the key question is whether new agentic technologies will make vertical systems less valuable or even replace them. Our view is nuanced but ultimately constructive. First, we believe vertical software systems will remain the core "systems of record" (see pg. 4-5) for their industries. The data and context embedded in these platforms are both unique to complex, industry-specific networks and essential for AI to generate meaningful, reliable outputs. Second, AI gives vertical software vendors an opportunity to enhance connectivity between complicated industry workflows and get customers more value out of their data. Over time, this will likely drive synergies between products and result in higher revenue, even if competition grows.

The important nuance here is execution. Management teams that are fast followers—those that actively integrate new AI capabilities and maximize the value of available LLM technology—are better positioned to limit value abstraction. Conversely, less innovative vendors, rather than be valued as an immersive software ecosystem, could see their value erode to that of a database, which would likely weigh on pricing power and growth.

Competition in this layer will likely intensify. We are already seeing early signs of this with businesses that deploy engineers directly within a customer's environment to implement AI (often called forward deployed engineering), as well as early-stage tools launched by LLM companies themselves (i.e., Claude plugins). That said, incumbents that already own the underlying data and industry context layer start from a position of strength. Customers are also likely to be relatively patient with these vendors given the strength of their relationships and deep embeddedness of the software.

These dynamics could ultimately drive more M&A. Incumbents seeking to remain competitive may increasingly turn to M&A to keep pace with AI-native innovation, while private equity firms may find vertical businesses with clear data and context moats increasingly attractive given the compression of public market valuations.



*Adam Hotchkiss,
US Emerging Software Analyst*

Data Infrastructure and Physical-to-Digital

A: Data infrastructure refers to platforms that store, govern, and secure data so it can be accessed and used across an organization, such as Snowflake, MongoDB, and Rubrik. We view AI as a force multiplier in this space. As organizations deploy AI copilots and agents, they will create and query more data, demand faster response times, and tolerate less downtime. This raises the value of this foundational layer, because it determines whether AI can run reliably and securely in production.

The primary risk this subsector faces is that, over time, workloads could shift to newer or cheaper platforms—a familiar pattern in technology transitions—and more integrated AI stacks could reshape enterprises' spending priorities. However, we see these as competitive dynamics to monitor rather than immediate existential threats.

The disruption debate is more pointed for physical-to-digital companies. These are software firms that sell digital products to physical asset-intensive industries like construction and transportation. On the surface, these products can look like everyday applications. In reality, they sit inside mission-critical workflows with governance and compliance baked in, which raises the bar for AI-native challengers.

The long-term risk is not that AI replaces these platforms outright, but that a general-purpose agent layer could sit above them and reduce their standalone importance. But our view is that AI adoption will gravitate toward systems that control proprietary data and link directly to real-world outcomes. For such systems, AI will likely become a source of reinforcement rather than disruption.



*Matthew Martino,
US Software Analyst*

AI tools for enterprise

Tool	Summary
Core enterprise AI platforms	
Anthropic Claude Cowork	An AI productivity tool that operates as an AI agent that can perform tasks directly on a user's computer, including automating repetitive tasks such as organizing files, extracting data from PDFs into spreadsheets, and generating presentations.
OpenAI Frontier	An enterprise-grade platform designed to build, deploy, and manage autonomous AI agents that act as "digital co-workers." AI agents can perform complex, multi-step tasks across business systems like CRM and data warehouses.
Microsoft Copilot	An AI digital assistant integrated across Microsoft 365 apps (Word, Excel, Teams), Windows, and web browsers, designed to boost productivity and streamline workflows.
Google Gemini for Workspace	An AI digital assistant integrated directly into Gmail, Docs, Drive, Slides, Sheets, and Meet that helps users boost productivity through drafting, organizing, summarizing, and brainstorming.
OpenClaw	An open-source autonomous AI agent that acts as a personal digital assistant capable of managing files, running terminal commands, and automating tasks across platforms like Discord, Slack, and WhatsApp.
Adept AI	Uses autonomous AI agents that can execute manual, repetitive, end-to-end workflow tasks across existing software applications.
AI for customer service and sales	
Sierra AI	An AI-powered customer service agent that can handle complex conversations across digital channels including around tracking orders or resolving issues.
Ada CX	An AI customer service platform designed for enterprises to automate support across chat, email, and voice channels.
Gong AI	A revenue intelligence platform using AI to analyze customer interactions to improve sales performance and identify risks.
Clari AI	Provides AI-driven revenue forecasting and deal intelligence.
AI for HR	
HR Plugin for Claude Cowork	Allows HR professionals to deploy specialized agents that operate across local files and enterprise apps. It can draft offer letters, build onboarding plans, and run compensation analyses by pulling data from multiple spreadsheets.
Paradox	A tool for high-volume recruitment that handles candidate screening, answers FAQs, and automatically schedules interviews based on recruiter availability.
AI for financial services	
AlphaSense	An AI-powered market intelligence platform that aggregates millions of data points from equity research, filings, and news to provide real-time insights for analysts.
Alteryx	Automates repetitive data workflows and analytical modeling, allowing finance teams to run various risk scenarios and consolidate disparate data sources without manual effort.
Prophix One	A finance-native platform embedding AI agents directly into budgeting, forecasting, and "continuous close" workflows to reduce human error in reporting.
Trullion AI	An AI accounting platform that automates lease accounting and revenue recognition.
Anaplan (PlanIQ)	An AI platform that detects anomalies and generates predictive forecasts across finance, sales, and supply chain operations.
AI for manufacturing and supply chains	
Siemens & NVIDIA (Industrial AI OS)	A joint operating system that transforms "passive" digital twins into active intelligence, allowing real-time simulation and optimization of factory floors.
Blue Yonder	An AI-driven supply chain platform that provides cognitive analytics to predict disruptions and automatically adjust production and logistics schedules accordingly.
Cognex Vision AI	Leverages deep learning for high-speed machine vision quality control, detecting subtle defects that traditional rule-based systems miss.
Interos	An AI risk management platform that monitors global supply chains for geopolitical, financial, and ESG-related disruptions.
Phaidra	Provides autonomous AI control systems for industrial manufacturing and mission-critical facilities, such as data centers.
AI for legal and professional services	
Harvey AI	A domain-specific AI platform used by major law firms for regulatory research, document analysis, and drafting documents.
Luminance	An AI platform specialized in M&A due diligence and autonomous NDA negotiation, capable of identifying anomalies across large document sets.
Ironclad	A contract lifecycle management tool that uses AI to automate the drafting, negotiation, and tracking of legal obligations at scale.
Casetext (CoCounsel)	An AI legal assistant that performs case law research, summarizes depositions, and reviews documents for relevance using advanced reasoning.
AI for retail and e-commerce	
Salesforce Einstein	An integrated AI layer for CRM that generates personalized product descriptions, predicts customer churn, and provides merchandising insights.
Shopify Magic	Suite of AI tools that automates inventory management, generates marketing copy, and personalizes product recommendations.
ConversionBox	An AI shopping assistant that uses conversational AI to guide customers through product discovery, acting as a digital salesperson on e-commerce sites.

Note: Table does not constitute an exhaustive list of all AI tools for enterprise.

Source: Company data, various news sources, compiled by Goldman Sachs GIR.

Interview with Sanjay Poonen

Sanjay Poonen is CEO and President of Cohesity, an AI-powered data security and management firm, and serves on the boards of Philips and Snyk. Previously, Poonen was COO at VMware and President of SAP. Below, he argues that agentic AI has the potential to be enormously disruptive for the software industry, and legacy firms must adapt by leveraging the technology to improve their product offerings or risk being demolished by it.

The views stated herein are those of the interviewee and do not necessarily reflect those of Goldman Sachs.



Allison Nathan: Based on your vast experience in the software industry, does the emergence of new agentic AI tools for software development mark a transformative moment for the industry?

Sanjay Poonen: Agentic AI has already proven to be an incredibly powerful tool with the potential to be enormously disruptive. As an engineer who began my career writing code, this is a truly exciting and transformative moment—back in the 1990s, I often stayed up late trying to debug code, hoping to find another engineer also working late whom I could consult. Today, that “person” is an AI assistant.

However, just like any technology wave, you must surf this tsunami, or it will demolish you. If software engineers aren’t leveraging AI technology to write code and improve productivity today, it’s a crime. The best software engineers I know, both at Cohesity and beyond, say that the technology is revolutionizing the way they code. So, we’ve thoroughly embraced it at Cohesity. But every business should be thinking about how these tools can support and improve not only their software engineers, sales reps, and human resources personnel, but also their products and services. I see limitless potential to leverage generative AI in almost every industry, from medtech to aeronautics and even space.

“Agentic AI has already proven to be an incredibly powerful tool with the potential to be enormously disruptive... However, just like any technology wave, you must surf this tsunami, or it will demolish you.”

Allison Nathan: You paint a very positive picture of the benefits of AI for software and productivity. But the whole software sector has re-rated substantially lower because of concerns that legacy software companies won’t be able to compete even if they are embracing AI technology themselves. So, are these concerns warranted?

Sanjay Poonen: With such transformative technology, every software company must assess whether they can sustain their current price per user or will instead be threatened by AI-native or other competitors leveraging the technology. At the core of every Software as a Service (SaaS) application is a CRUD—Create, Read, Update, Delete—database that maintains data, application logic, and workflows, which AI tools have the potential to radically disrupt. If a company can sustain its price

per user amid this disruption, it will likely be fine. But some companies may need to lower their price per user to remain competitive. That doesn’t necessarily mean the company will go bankrupt or disappear, but it does have important implications for revenue growth and terminal value. So, every CEO, analyst, and investor is rightly assessing this risk.

“With such transformative technology, every software company must assess whether they can sustain their current price per user or will instead be threatened by AI native or other competitors leveraging the technology.”

Allison Nathan: Is it realistic to think that these companies will be able to effectively respond to this pricing pressure and stay competitive?

Sanjay Poonen: Yes, but it requires innovation. Firms whose core products are under pressure may need to add a second or third module with enhanced capabilities that garners a higher price per user to compensate for the lost revenue. For example, a company could offer a small, medium, and large packaging of its product, with the price per user of the small, base product coming down, but the price per user of the larger product offerings higher.

Alternatively, firms can expand into new areas to grow their total addressable market (TAM) and sustain revenue growth even as price per user declines. When I was at SAP, we recognized that a price per user ceiling existed for our ERP software and responded by adding a system of engagement on top of the system of record by acquiring analytics companies and building new business objects. This allowed us to raise the effective price per user for the combined product offering. Every company facing disruption of their core product will need to make these types of adjustments.

SaaS companies can also attempt to shift from a price per user to a price per outcome model, where fees are based on the measurable value they provide to the customer. That pricing model is harder to sustain because tracking outcomes can be challenging and an IT buyer typically has a fixed budget and is often unable or unwilling to enter into a vendor contract with fees that depend on the broader company’s performance. Because of that, most companies price per user. The only company I am aware of that uses a price per outcome model is Palantir, which often sells its product at the CEO level, where the power to make these decisions resides.

Allison Nathan: So, is the indiscriminate selloff across the software space overdone?

Sanjay Poonen: The answer to that will ultimately be determined by the revenue growth of these companies. As both a student of software and the CEO of a software company, I will be closely watching whether growth is holding up or slowing down. That said, in the security world Cohesity lives in, the large market reaction to Anthropic's announcement that it had built a code scanning tool for security was akin to Anthropic sneezing and everyone coming down with pneumonia. But the market seemed to quickly realize that this was an overreaction and that this tool won't enable users to build a firewall or a data or endpoint security product overnight.

Allison Nathan: Even if that's true today, are you concerned that it won't be in the future given the rapid rate at which generative/agent AI technology is advancing?

Sanjay Poonen: Former Intel CEO Andy Grove said it best: "only the paranoid survive." Nobody in this space can afford to be cavalier about the potential for disruption. As a CEO, I have to assume that someone could use AI tools to build a competitor. In fact, I recently asked an AI assistant to build a Cohesity competitor to see what it would do. Unsurprisingly, its first response was that this was a "tall order." But it was able to think about the challenge in logical steps, which revealed an understanding of how to build a competitor. So, it's only safe to assume that, given time, these tools will eventually be able to do just that. For that reason, it is incumbent upon myself and every other CEO in a similar position to take advantage of these same tools to expand and improve our own product offerings. Again, we must ride the tsunami to our benefit rather than let it demolish us.

“Former Intel CEO Andy Grove said it best: “only the paranoid survive.” Nobody in this space can afford to be cavalier about the potential for disruption.”

Allison Nathan: But is just bolting AI agents onto legacy layers ever going to be as successful and powerful as native AI product offerings, or will legacy companies eventually have to start from scratch?

Sanjay Poonen: Software is typically built in architectural layers, with the top layers easiest to refresh and the core, bottom layers much harder to. But even timeless software evolves every decade or two, so engineers must be comfortable with the idea of completely rewriting part of or even the whole architectural stack. The good news is that using new tools and approaches to rewrite code from scratch will probably take a small fraction of the time it took to write the original code. Even then, disrupting the whole stack is undoubtedly hard, even if necessary. In such instances, companies might decide to shelter in place and keep the

business and cashflow running while simultaneously building or buying the next generation of the company. I see this as akin to living in your old house while building a new one next door and only tearing down the old house once the new one is ready.

We dealt with this at every organization I've worked at. At SAP, as new companies that were building a SaaS version of ERP in the cloud threatened to disrupt our on-prem product, we had to consider whether we could build a cloud-native product fast enough without disrupting our core platform. The best companies find a way to constantly evolve their stack to remain competitive.

Allison Nathan: While large players like a Microsoft or Salesforce may be able rebuild, is there an inherent disadvantage for the smaller companies that have less resources to do so?

Sanjay Poonen: It's wrong to assume that reinvention is easier at a Microsoft or Salesforce, and in fact the opposite may be true. It took new leadership for Microsoft to transform from a Windows company to a cloud company and now an AI company. In my experience, the bigger the company, the greater the molasses, dinosaur mindsets, and resistance to change. Smaller companies may not have as many resources, but they have the advantage of speed. It's a bit like a speedboat vs. an aircraft carrier—small firms can usually make decisions faster and are less likely to have a large legacy install base that they must worry about transforming. Microsoft pivoting away from Windows to Azure was a big and highly risky shift. A company starting completely from scratch has no legacy to deal with. I'm reminded of the joke that God created the world in seven days because he didn't have an install base. The bigger the install base, the tougher it is to pivot and change.

All that said, whether small, medium, or large, as long as a company maintains a mindset and culture that is open to innovation, none of these challenges are too daunting to achieve. Companies with such a mindset can start things from scratch and build on them without destroying the house they live in, which allows them to create S-curve upon S-curve of continued innovation.

Allison Nathan: So, what could the software industry look like in five or 10 years?

Sanjay Poonen: I expect more vertical industry software capabilities in industries like space, healthcare, material sciences, and robotics. Consider SpaceX's vision of data centers in space, where cooling isn't a problem. That endeavor will undoubtedly require software. The same could be said for biotech, medtech, and robotics, which will almost certainly increasingly attract top software talent that previously went to the Googles and Salesforces of the world, driving innovation that could help people live healthier and longer lives. So, my hope is that the software industry spreads its skillset into other vertical industries with positive implications for society and the world. And AI will undoubtedly be a part of every one of those industries.

Software stocks: earnings stability needed

Ben Snider argues that a rebound in the stocks most exposed to AI disruption risk likely won't come until earnings stabilize

The AI narrative has shifted. After three years of equity investor enthusiasm regarding the tailwinds from AI, recent equity market rotations reflect a growing focus on the risk of AI disruption. Software stocks have been at the center of this shift, falling by 18% from their YTD highs and now sitting roughly 26% below late 2025 levels. Stocks in a range of data-intensive industries beyond software, including insurance, media, and business services, have also declined sharply in recent weeks. While the moves have been swift and severe, we think a rebound in the stocks most exposed to AI disruption risk will take time, as disproving the disruption narrative won't be easy or quick.

Re-rating: a decomposition

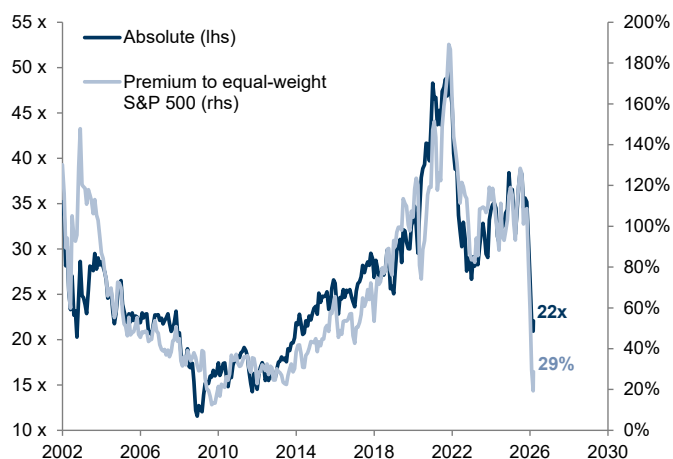
After entering 2026 at extremely elevated valuations, growth expectations, and weights in investor portfolios, the software industry has sharply re-rated to reflect declining expectations for future earnings growth as AI disruption risk grows.

The industry's forward P/E multiple has fallen from roughly 35x in late 2025—consistent with 2028 consensus revenue growth in the 15-20% range, over twice the roughly 6% growth expected for the median S&P 500 stock and the highest expected growth in at least two decades—to around 22x currently, the low end of the multiple's recent range and the lowest absolute level since 2014, consistent with 5-10% earnings growth.

Given the high starting point of valuations, software continues to trade at a valuation premium to the equal-weight S&P 500 despite the recent selloff, and its 9% S&P 500 weight is roughly double its weight a decade ago.

Software continues to trade at a valuation premium to the equal-weight S&P 500 despite the recent selloff...

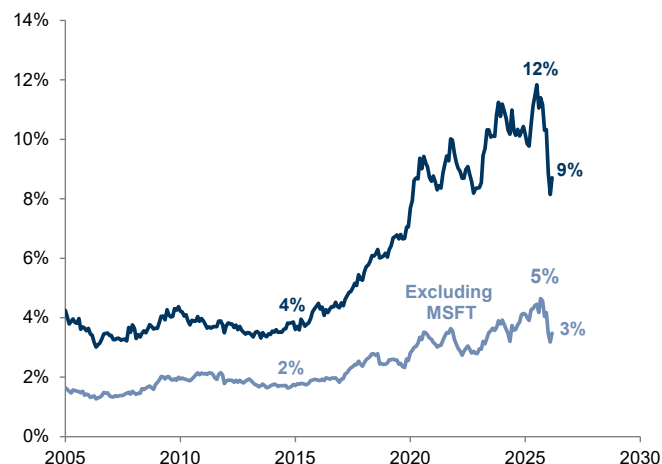
Software forward P/E, X (lhs), % (rhs)



Source: FactSet, Goldman Sachs GIR.

...with its weight in the S&P 500 roughly double that of a decade ago

Software weight in the S&P 500, %



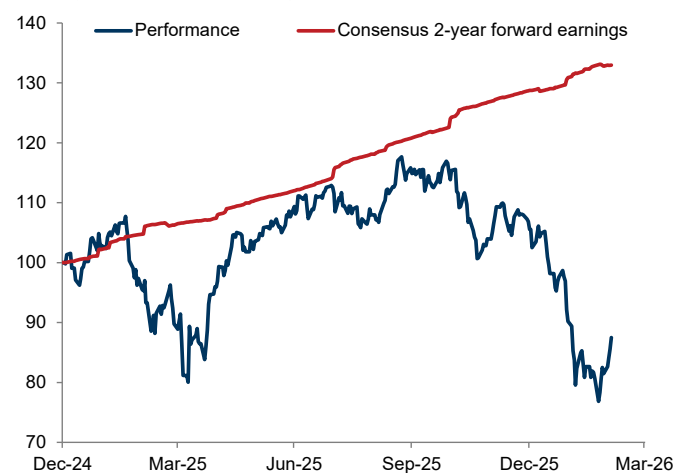
Source: FactSet, Goldman Sachs GIR.

In addition to lower growth expectations, the recent re-rating reflects rising uncertainty around the long-term trajectory of the industry and the “terminal value” of stocks previously viewed as long-duration growth assets. In other words, share prices now reflect not just slower expected growth, but also doubts about the durability of business models as AI disruption raises questions about pricing power and competitive barriers.

The impact of this long-term uncertainty can be seen in the juxtaposition of recent share price volatility and 4Q25 earnings results. In contrast with falling share prices, recent earnings results have reflected strong profitability—including double-digit earnings growth last quarter, with profit margins for the industry sitting at record highs and roughly twice those of the rest of the market—and have driven upward revisions to near-term analyst earnings estimates.

The performance and earnings of software stocks have diverged

Indexed performance and earnings of software stocks



Source: FactSet, Goldman Sachs GIR.

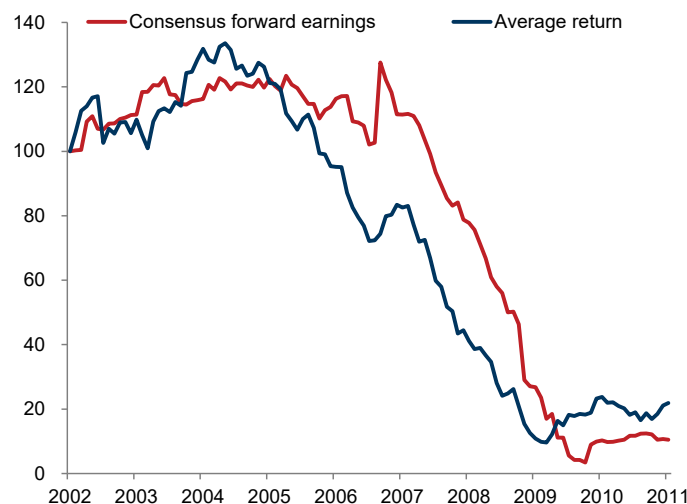
Lessons from history: don't expect a V-shaped rebound

Historical episodes of disruption risk suggest that share prices in affected industries will likely stabilize only when earnings stabilize, and that these periods of uncertainty can last for years. In the early 2000s, US newspaper stocks fell by an average of around 95% over the course of five years, and the group bottomed out only slightly ahead of the stabilization in

forward earnings estimates. Tobacco stocks in the late 1990s experienced disruption risk driven by litigation, rather than technology, but the pattern was similar. In that episode, share prices declined by over 50% in 1997-1998 and troughed only as the litigation settlement reduced uncertainty and the near-term earnings outlook stabilized. Given the rapid pace of AI advancement today, disproving the narrative of disruption risk is difficult when near-term earnings outlooks in the vulnerable industry have barely begun to weaken, much less stabilized.

US newspaper stock prices stabilized only when earnings stabilized

Indexed avg return and consensus forward earnings of US newspaper stocks



Source: FactSet, Goldman Sachs GIR.

For investors, the lesson is not to expect a V-shaped rebound in the parts of the equity market most affected by recent disruption fears. For these industries, near-term earnings can be an important signal of business resilience, but in many cases will be insufficient to disprove long-term downside risk. For the group to re-rate higher, investors will likely require more clarity on AI's eventual business impact, which will take time, or a substantially lower starting point for valuations and ownership. This may be an easier lift for the parts of the software and business services industries with particularly resilient business moats, but those moats, too, will need to be proven.

Of course, in any broad-based selloff there will likely be some proverbial "babies thrown out with the bathwater" that represent value opportunities for stock pickers. In the current episode, the key is the speed with which companies can disprove disruption risk.

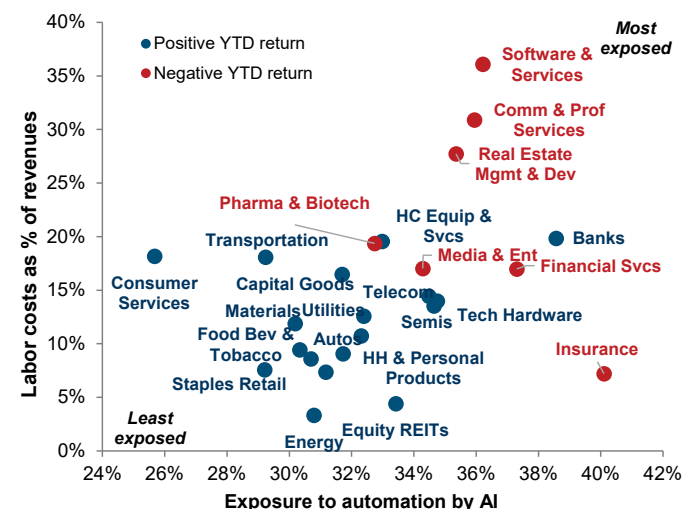
Avoiding AI disruption risk: look to "real world" assets

Another consequence of the recent focus on disruption risk is a rotation of investor portfolios toward "real world" assets with perceived insulation from AI disruption risk. Recent outperformers in this vein include a handful of cyclical industries like industrials and materials that have also enjoyed tailwinds from an improving economic growth backdrop. GS forecasts for above-trend economic growth have lent fundamental support to cyclicals' recent strength, and we entered 2026 recommending investors own many of these stocks. However, the tailwind from cyclical economic acceleration also highlights the broader equity market vulnerability if the growth backdrop proves less robust than

expected or industry-specific AI disruption fears spread to broader concerns about labor market disruption. This risk helps explain the recent outperformance of more defensive "AI-insulated" sectors, including consumer staples and biopharma, and we think the appropriate strategy is to balance cyclical exposures with some relatively defensive equity positions.

The recent focus on AI disruption risk has triggered a rotation toward "real world" assets

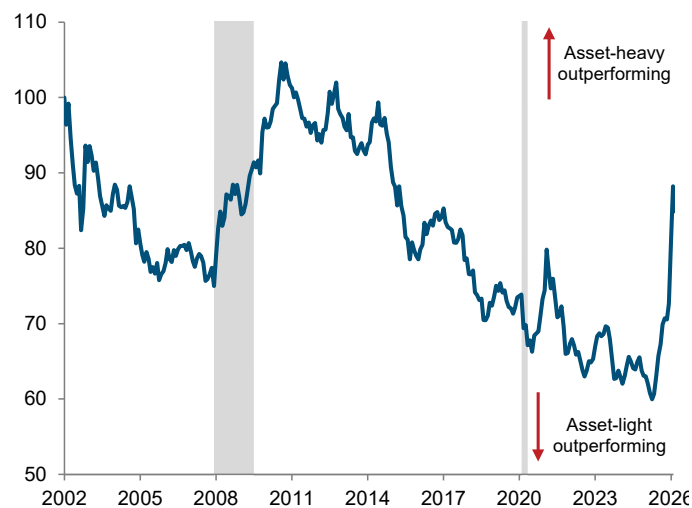
Russell 1000 industry median



Source: Revelio, company filings, Goldman Sachs GIR.

Increased investor desire for "AI insulation" has led asset-heavy stocks to sharply outperform

Asset-heavy vs. asset-light indexed performance



Note: Grey bars indicate recessions.

Source: Goldman Sachs GIR.

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Managing AI risk in portfolios

Christian Mueller-Glissmann discusses the why and how of managing AI risk in portfolios amid growing AI disruption fears

Investor concerns about AI risks for the global technology sector have grown amid ongoing worries about hyperscalers' massive AI capex spend and, most recently, rising fears that new AI tools will disrupt tech incumbents, especially legacy software firms. Such concerns matter because global portfolios are highly exposed to the technology sector, especially within US equities, where software stocks have a relatively large weight, with substantial exposures to technology and software specifically also a feature of private markets.

But the fate of the tech sector also matters more broadly given how closely tied structural cycles in US equity valuations and outperformance have been to the performance of tech stocks. To achieve more balance between innovation and inflation risks within portfolios, we think investors may need to continue rotating from capital-light to capital-heavy businesses that AI won't easily disrupt, though the aggressive rotations we've already seen warrant being more selective.

A tech-dependent equity market...

What happens to the tech sector has broader implications for the equity market for several reasons. The sector has historically been a key driver of structural cycles for US equities, with major waves of technological innovation often significantly impacting the broader market. During the 1960s PC boom, 1990s Internet boom, and the post-Global Financial Crisis (GFC) ascent of the Magnificent 7, tech's share of total market cap rose sharply, and significant market corrections often followed these periods. Today, the TMT sector once again commands the largest weight within US equities, comparable to Tech Bubble levels, but the sector's contribution to earnings is now substantially greater compared to then.

Tech's weight in the US equity market has risen sharply during major waves of technological innovation

Tech market cap and earnings weight within US equities, %

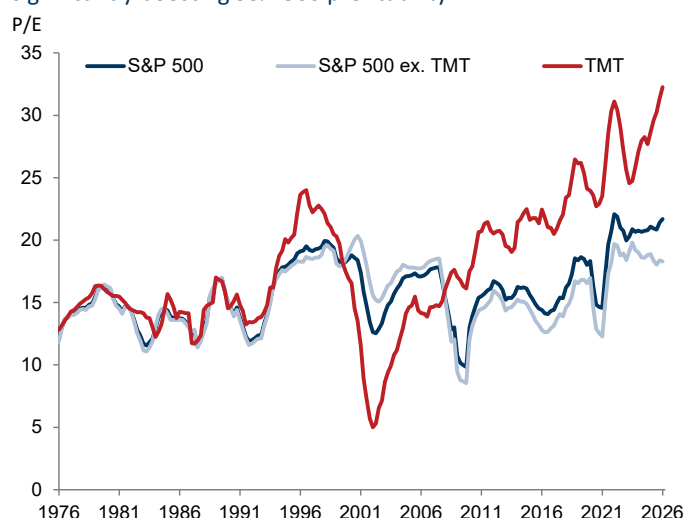


Source: Datastream, Goldman Sachs GIR.

Tech has also played a key role in driving structurally higher S&P 500 profitability and valuations. Since the GFC, the return on equity (ROE) across the TMT sector has nearly doubled,

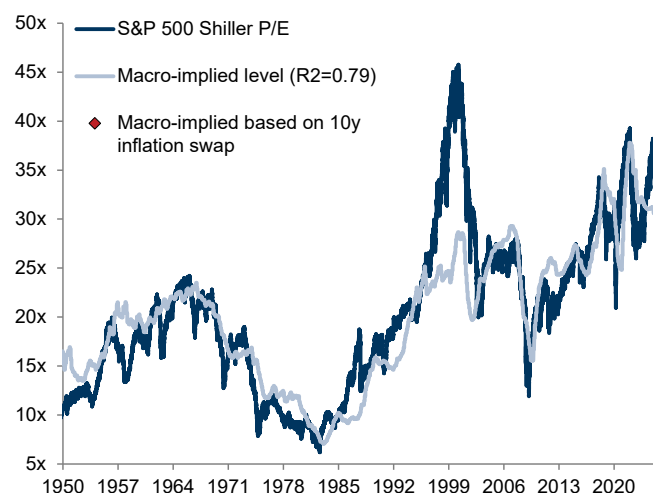
significantly lifting the S&P 500's overall profitability. This trend accelerated during the pandemic as technology adoption surged. Higher profitability has, in turn, supported higher equity valuations. Indeed, the rise in the S&P 500's Shiller P/E over the past two decades reflects two interconnected elements: improved corporate profitability—driven by the growing weight of the highly profitable, capital-light US tech sector—and persistently low and stable inflation. And the strong performance of large cap tech stocks has driven a large amount of the US equity outperformance since the GFC.

The ROE for the TMT sector has nearly doubled since the GFC, significantly boosting S&P 500 profitability



Source: Compustat, Goldman Sachs GIR.

High profitability, coupled with low inflation expectations, have fueled a rise in US equity valuations



Source: Haver Analytics, Kenneth French, Robert Shiller, Goldman Sachs GIR.

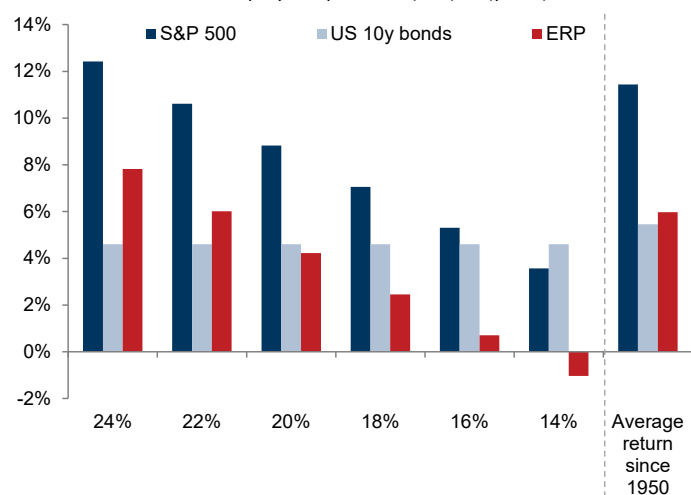
...meets AI disruption fears

Investors have extrapolated the TMT sector's high ROE, even as these firms pay out relatively little to shareholders, essentially betting that major tech conglomerates will maintain their profitability as they make substantial reinvestments. However, last year anxieties around AI capex began to impact some of the largest spenders in the sector, particularly the hyperscalers. Beyond concerns around potential future AI revenues, the big spenders are also becoming more capital-heavy, which might weigh on ROE in the medium term.

More recently, rising concerns about AI disruption have weighed on parts of the TMT sector, especially software stocks, with investors increasingly worried that the tech sector may end up disrupting itself. Advances in AI-powered task automation, for example, could undermine the pricing power of some of the most profitable large-cap software firms. While AI adoption may also boost other businesses' ROEs, a decline in the profitability of the market's largest and most lucrative segment could adversely affect the S&P 500's overall ROE and dampen long-term return expectations, which currently rely on the compounding of those ROEs. And markets have been pricing negative AI disruption faster than potential tailwinds from AI adoption.

Lower corporate profitability could portend lower long-term equity returns and less attractive equity risk premia

S&P 500 ROE in 10 years scenarios, % (x-axis) vs. annualized nominal returns for stocks and bonds and equity risk premium (ERP), % (y-axis)*



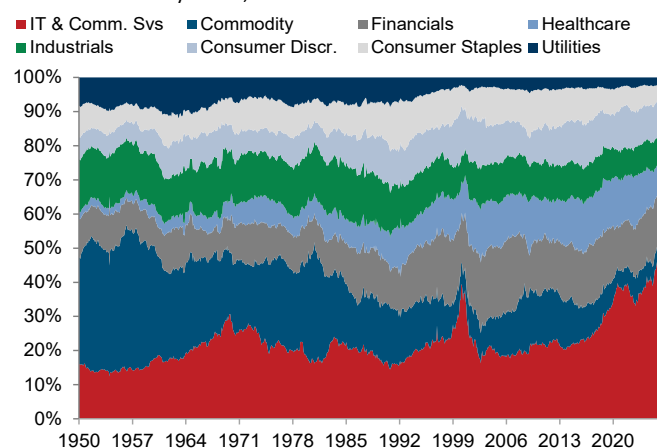
*See this [note](#) for more detail about the long-term return calculations.
Source: Haver Analytics, Goldman Sachs GIR.

Balancing innovation and inflation risks

We expect an AI-driven rise in productivity to eventually support corporate profitability. But concerns about AI disruption may continue to be priced faster than the benefits of AI adoption. On top of that, the AI application firms driving disruption are mostly private companies and, as such, not easily investable, which makes it difficult to diversify AI disruption risk. Investors may therefore demand more risk premium for at-risk businesses, which has already driven a rotation and value convergence between capital-light growth sectors like TMT and capital-heavy value sectors like telecoms, industrials, and utilities that AI is unlikely to disrupt.

Owing to a boost from innovation, the TMT sector has become a much larger part of the S&P 500 since the GFC, while capital-heavy sectors have underperformed, partly owing to a lack of inflation. As investors continue to worry about AI disruption, this may lead them to demand a higher equity risk premium, which could result in some de-rating of equities vs. bonds, with bonds potentially also benefitting from disinflation due to AI disruption.

The TMT sector holds a dominant position within the US equity market, with capital-heavy sectors holding a relatively small one



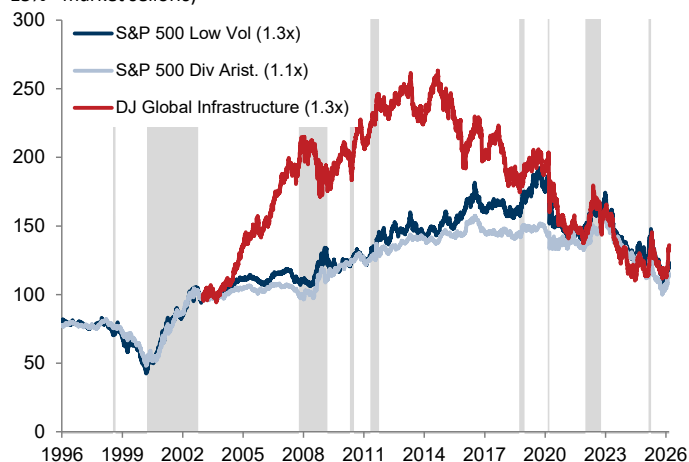
Source: Bloomberg, Datastream, Goldman Sachs GIR.

To manage innovation and inflation risks within portfolios, investors may need to continue rotating further from growth to value and capital-light to capital-heavy businesses, both in equity and credit. However, after the aggressive rotations since the beginning of the year, we favor being more selective. Within equities, we continue to see diversification benefits from defensive styles such as low volatility stocks, dividend aristocrats, and infrastructure, which materially outperformed on a risk-adjusted basis during the Tech bubble burst.

We also expect more dispersion as AI will likely continue to produce more winners and losers. This should support strategies that buy individual stock and sell index options and long/short hedge funds in both equity and credit. And we continue to view geographical diversification as an attractive strategy. TMT exposure is typically lower in non-US equity markets and tends to be more hardware-focused, with such markets also offering more capital-heavy sector opportunities.

Defensive styles have lagged in recent years but outperformed over the long run

Total return relative performance vs. S&P 500 (grey bars indicate periods of 15%+ market selloffs)*



Source: Datastream, Bloomberg, Goldman Sachs GIR.

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Goldman Sachs International

Software disruption & credit: no hard fall

Shamshad Ali argues that software loan losses owing to AI disruption fears are unlikely to catalyze a turn in the credit default cycle

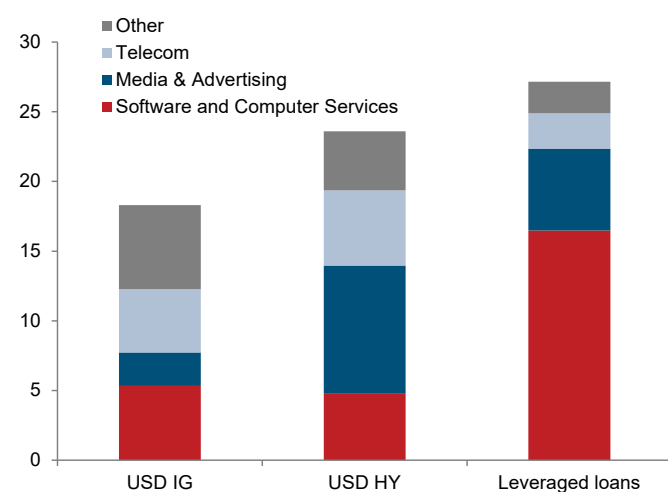
Recent headlines around AI-driven disruption—and the associated sharp drawdown in software stocks—have weighed on software-exposed credit, most notably in the broadly syndicated loan (BSL) market. As sectors like insurance, brokers, and other services exposed to AI disintermediation have also come under pressure as more new AI tools have emerged and investors worry about downstream implications, we believe the BSL market faces the greatest risk should disruption concerns translate into weaker earnings trajectories for traditional software companies, which would erode interest coverage ratios for these already levered borrowers. Given that software accounts for a meaningful share of the BSL market, some degree of index-level stress would be difficult to avoid if fundamentals continue to deteriorate. That said, the benign macro backdrop and lower funding costs should help limit the risk of a turn in the credit default cycle.

A more exposed BSL market

Software accounts for only a small slice of the USD IG and HY bond indices (roughly 5%) but a meaningfully larger share of the BSL market (16%). While many IG software issuers are large incumbents (including the AI hyperscalers), software-exposed issuers in the HY bond and leveraged loan markets have underperformed broader benchmarks as investors have repriced disruption risk. HY software has shifted from trading tighter than the average spread level of the overall HY index to roughly double index spreads, while secondary prices for software loans have fallen to the lowest levels since April 2025.

Software accounts for only a small slice of IG and HY but a meaningfully larger share of the BSL market

TMT subindex exposure in USD credit, %



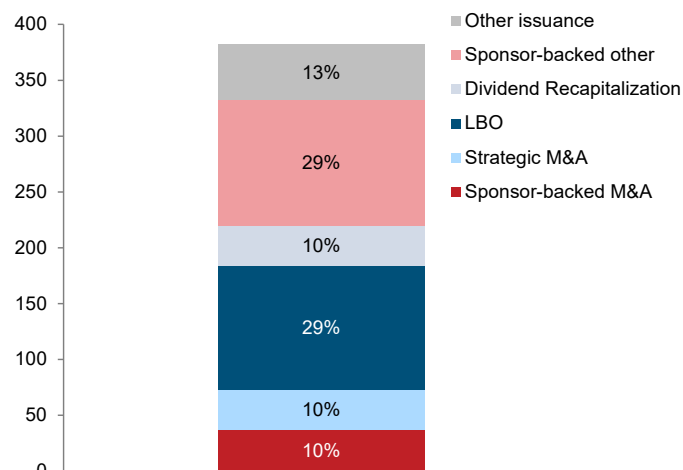
Source: Bloomberg, Pitchbook LCD, Goldman Sachs GIR.

A defining feature of software-related loans is the linkage to sponsor-backed (typically private equity) activity. Since 2021, nearly 40% of technology leveraged loan supply has been tied to sponsor-driven M&A and LBO financing. Another 10% of issuance stemmed from dividend recapitalization—firms taking on new debt to pay dividends—as sponsors sought liquidity amid a less favorable exit environment. The upshot is that

underwriting is no longer just about borrower fundamentals; investors also need to carefully consider sponsor incentives, time horizon, and sponsors' willingness to support a firm's capital structure through difficult periods as the industry navigates a changing paradigm.

80% of tech loan issuance since 2021 has been sponsor-backed

Share of total gross Computer and Electronics loan issuance since 2021 by category, \$bn



Source: Pitchbook LCD, Goldman Sachs GIR.

Some impairment likely, but not a turn in the default cycle

The potential persistence of AI disruption pressures would likely most clearly manifest via downgrades and lower secondary prices in the software-heavy portion of the loan index. Particularly important to watch would be the implications of such shifts for the CLO community, the main buyer of leveraged loans. CLOs have built-in rules or covenants designed to ensure that the overall diversification and quality of the underlying loans in their portfolios remain within acceptable limits (concentration, over-collateralization and collateral-quality tests). The downgrading of too many loans to a CCC rating (BSL CLOs typically have a limit of 7.5% for CCC exposure) could trigger various penalties. As a result, prices for these loans could move beyond fundamentals as the investor base potentially shifts from CCC-averse CLO buyers.

That said, bifurcation in loans isn't new. Even in periods when lower-quality loans cheapen, the broader index can remain resilient when higher-quality borrowers repeatedly reprice their loans at lower spreads and new loan supply is muted, as has been the case this year. Overall, while some impairment in software-exposed loans is hard to avoid, we don't think this impairment is sufficient to catalyze a credit default cycle, for two reasons. First, the macro backdrop remains benign, and history suggests that defaults typically coincide with rather than predict the broader cycle. Second, lower interest rates and improved funding conditions should continue to provide incremental relief to floating-rate borrowers.

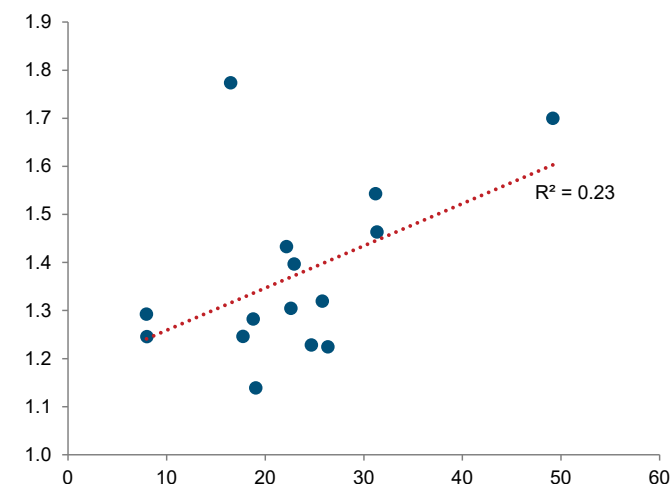
Pain in direct lending, but de-leveraging risk looks contained

Beyond the BSL market, direct lending—and by extension business development companies (BDCs) that invest in small and mid-sized, often private, firms—has been in the crosshairs given meaningful allocations to software. Indeed, our equity analysts estimate that software portfolio companies account for

22% of BDC loans (see pg. 20). Consistent with this, BDC equities have tended to trade at wider discounts to net asset value (NAV) while BDC credit spreads generally have widened in accordance with software exposure.

BDC credit spread widening correlates with software exposure

Software exposure as of 3Q25, % (x-axis) vs. on-the-run BDC senior unsecured bond spread change, ratio relative to Jan. 23 level (y-axis)



Source: Bloomberg, Goldman Sachs GIR.

We are closely monitoring signs of stress in BDC portfolios and have long argued that some writedown in asset valuations is likely. Recent liquidations of assets in BDC portfolios have also fueled speculation about asset health despite the loans selling at high, near-par, dollar prices. While full transparency isn't available on the sales, our equity analysts continue to think that credit trends in BDC portfolio data seem benign for now, but near-term portfolio data will remain under scrutiny.

While the rising rate of redemptions in BDCs is worth monitoring, we continue to think that the probability of a liquidity event or forced de-leveraging in direct lending is low owing to generally moderate leverage and substantial dry powder that can re-enter the complex and provide support, especially at more attractive valuations.

More dispersion from the next leg of the AI trade

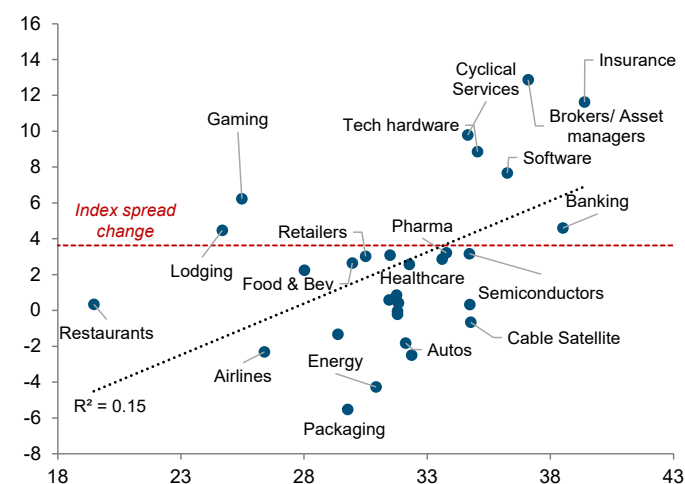
While AI disruption fears remain centered on software, they have now spread to several other sectors, including insurance, brokers, and broader financial services (see pgs. 21-23), as the foundational model providers continue to roll out new domain-specific products. Amid this, IG and HY sector performance has become correlated with exposure to AI automation, mirroring the broader risk asset rotation toward "real economy" sectors that are less exposed at the expense of white-collar services that could face more disruption risk.

Looking ahead, we think this higher dispersion dynamic will persist, for two reasons. First, continued improvements in AI capabilities will likely impact more industries. Second, dispersion within industries should rise as company outcomes diverge based on the execution of AI adoption strategies. This dispersion will make it more difficult for investors to calibrate fair value for affected sectors, and issuer selection will become increasingly important. But we still see limited scope for serious impairment, especially for higher quality firms. Beyond their incumbent status and low starting points for leverage, we

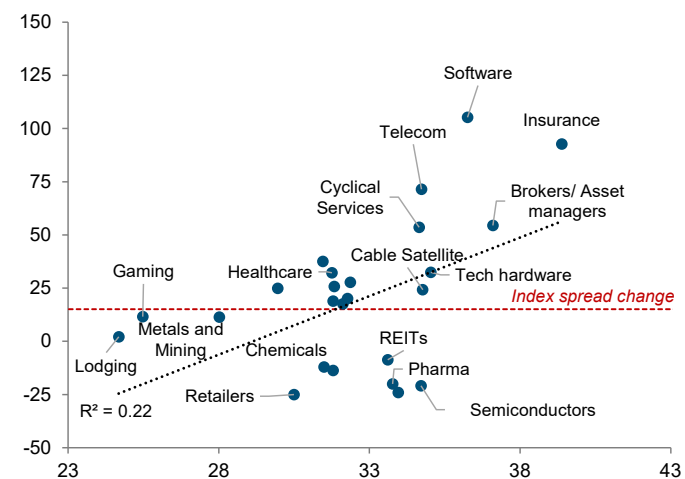
would note that the concerns around terminal values that have fueled much of the equity volatility are less salient for fixed duration credit assets, which are concerned with repayment of interest and principal.

Credit sector performance has become correlated with AI exposure

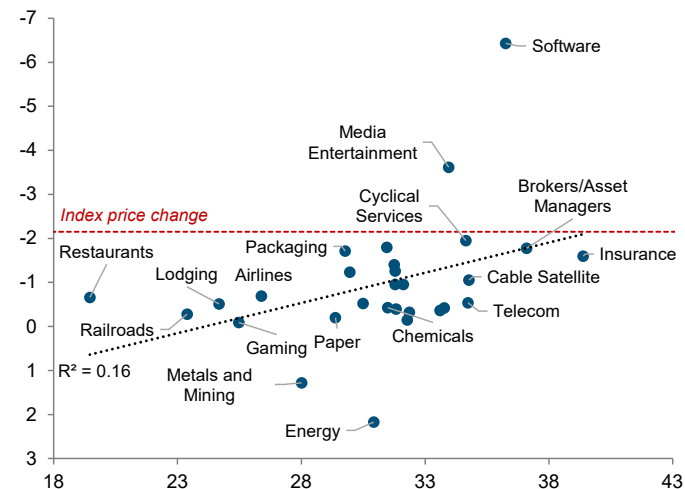
AI automation exposure, % (x-axis) vs. IG spread change YTD, bp (y-axis)



AI automation exposure, % (x-axis) vs. HY spread change YTD, bp (y-axis)



AI automation exposure, % (x-axis) vs. leveraged loan price change YTD, \$ (y-axis, inverted)



Source: Revelio, Bloomberg, Goldman GIR.

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Alts managers: limited AI risk so far

Alexander Blostein and Michael Vinci argue that Alternative Asset Managers have limited exposure to AI-related software disruptions

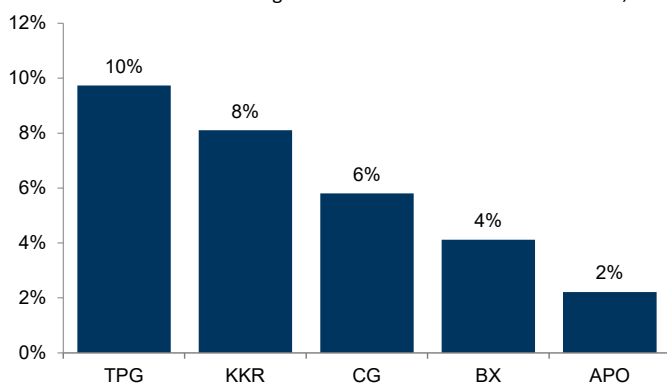
Concerns about AI-driven disruption risk have once again weighed on Alternative Asset Managers' stocks, which have declined ~27% YTD, bringing share prices and valuations below post-Liberation Day lows (NTM P/E net of stock-based compensation (SBC) at ~15X currently). This decline largely reflects concerns about software exposure across both private equity and private credit as well as fears that weaker investment performance could hamper growth. However, firmwide software exposure remains limited—averaging ~7% of total AUM—with software lending only modestly contributing to management fees, and managers have also been highly selective in their software investments, choosing those with more robust moats. While we think the recent selloff is overdone and continue to see significant upside across the sector, we remain cautious around names with outsized Retail Credit exposure given accelerating redemption requests.

Modest software exposure

Overall, we view Alternative Asset Managers' exposure to software investments as manageable. On the private equity (PE) front, while software has accounted for ~25% of industry-wide deployment volumes in recent years, exposure at Public Alternative Managers is more modest, with software activity among PE firms averaging only ~6% of total management fees. These estimates may even prove conservative as actual net exposure could be lower after managers monetized many investments in prior years.

PE software investments average ~6% of total management fees

2025 estimated firmwide management fees from Info Tech investments, %



Source: Preqin, Goldman Sachs GIR.

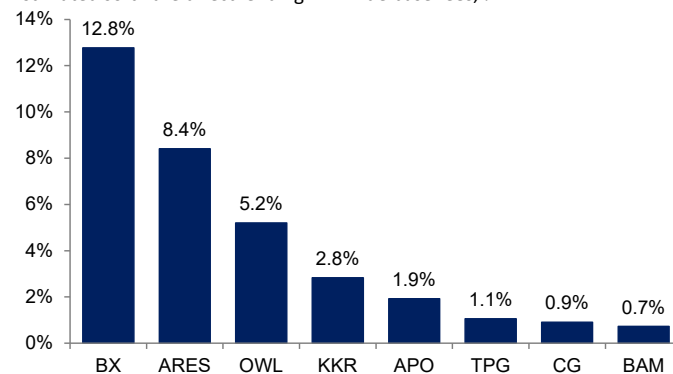
On the credit front, across BDCs (where public data is available), we estimate that software loans represent ~22% of gross loans at an industry level, with a similar share among Public Alternative Managers—~20% of Direct Lending FPAUM on average. This translates to around 25% of Direct Lending base management fees.

But software lending is a relatively small portion of firmwide management fees when accounting for the broader spectrum of Alt asset classes that GPs manage. We estimate that Direct Lending to software-oriented portfolio companies accounts for 4% of firmwide base management fees on average. Even in

the most adverse scenario of a significant markdown in software loans, our stress analysis (5%-15% mark-to-market) suggests the effect on firmwide base fees would remain in the low single-digits. And, importantly, these software loans are mostly senior secured and short duration, with the underlying companies generally showing strong EBITDA and cash flow.

Direct Lending to software-oriented portfolio companies accounts for 4% of firmwide base management fees on average

Estimated software direct lending firmwide base fees, %



Source: Company data, Goldman Sachs GIR.

Selective software investments

We also note that not all AI-related risk is created equal. Software holdings of most managers skew toward vertical, mission-critical businesses with proprietary data and platforms. These businesses are more likely to benefit from AI as opposed to the more disruption-prone horizontal software applications. AI-related risk has also become a central focus of investment processes, with one senior leader recently noting that AI is now among the first and most critical topics examined when bringing an investment before the investment committee. Evaluating AI-related risk has become a deeply embedded practice within GP investment committee deliberations and portfolio management frameworks, not just a reaction to recent market concerns.

We remain selective looking forward

While heightened market volatility and greater uncertainty related to AI disruption risks could dampen near-term PE exit activity, our conversations with management teams have given us more confidence around drawdown fund portfolio construction. That said, we continue to believe that sentiment/headlines will weigh on retail/wealth flows. While the group's earnings streams remain highly durable, we see a slower pace of retail fundraising/accelerating Credit redemption trends potentially weighing on growth. Despite the disruption, deployment outlooks are constructive, and for Opportunistic Credit managers, most see opportunity in the current backdrop. We continue to think the group's earnings base and management fee growth are considerably more durable than current valuations suggest and see significant upside across the space.

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AI disruption: equity sector impacts

What types of companies in your coverage universe could be disrupted by and, conversely, benefit from new AI tools, and how much of this is already priced into stocks?

Consumer Internet

Eric Sheridan and team, GS Equity Research

- AI tools are currently having a democratizing effect on product development and innovation within the consumer internet sector, presenting disintermediation risk while also driving efficiency gains. Within our TMT coverage universe, digital advertising and gaming are the most exposed subsectors to AI disintermediation risk and have also seen the most advanced adoption of AI tools—a trend that the recent declines in several gaming and advertising stocks reflects.
- Digital advertising, in particular, has advanced the furthest in AI-driven product development and end-user adoption (outside of cloud computing). Companies within the subsector are currently integrating AI into four key workflows: campaign planning and strategy, creative development, media buying and optimization, and CRM and personalization. GOOGL's Performance Max and META's Advantage+ have been exceptionally successful in integrating multiple AI use cases into a single product. These AI-powered tools have the potential to both "abstract away"/automate many advertising services as well as improve the overall performance and efficiency of ad spend. This should result in shifting profit dynamics including: 1) accelerating the shift of ad spend from traditional to digital channels (an estimated \$170bn addressable opportunity), 2) automating ad creative generation (~\$114bn), 3) consolidating ad tech "intermediaries" (~\$25bn), and 4) disrupting the traditional role of ad agencies (~\$161bn).
- Beyond online advertising, the specter of AI disruption has caused outsized volatility across many other subsectors—including media & advertising, online marketplaces, online dating, travel, online education, and online gaming & gambling—which we expect to persist. Incumbent consumer internet platforms are playing a mix of "offense" (launching AI-driven consumer applications that enhance their existing offerings and revenue) and "defense" (developing AI capabilities to counter competitive applications that could be disruptive to existing legacy applications) to counter AI disruption risk. Several companies are also partnering directly with leading AI platforms to drive direct traffic and monetization. Examples include ChatGPT in-app integrations, agentic commerce, in-app chatbot placement, and data/content licensing for model training.

US Business and Information Services

George Tong, GS Equity Research

- Concerns over AI-driven disintermediation have sparked a sharp de-rating of the Information Services sector since mid-2025, with the sector experiencing a particularly notable valuation step-down in early Feb 2026. The median NTM P/E multiple in our Info Services coverage has fallen from 32x in June 2025 to 20x currently as investors worry that free or cheaper AI tools could diminish or eliminate demand for existing Info Services offerings. However, we believe some companies in the sector are better positioned to withstand AI disruption than others.
- Companies with a sustainable moat against AI are those with proprietary data, entrenched workflows with customers, scale and brand advantages, and/or a regulatory moat, and have made meaningful AI investments. These firms are not only insulated from AI risks but also are positioned to benefit from AI integration via product upgrades, accelerated pricing increases, increased cross-selling, higher retention rates, and new business growth. From a stock perspective, the multiples of many of these firms have recently contracted, making them attractive buying opportunities. The market may be overestimating the AI-related hit to these firms' organic revenue growth (with some estimates in the mid-to-high single digits) and we believe earnings at these firms remain stable and are not in a negative revision cycle. Upcoming earnings results that demonstrate beat-and-raise patterns and sustained momentum in organic revenue growth should act as a catalyst for valuation upside, in our view.
- By contrast, firms most vulnerable to AI disruption are those that have limited proprietary content, weak integration with client workflows, limited scale or brand advantages, no regulatory moat, and/or haven't invested enough in AI themselves. They also face potential headwinds from lower demand for white collar staffing roles due to AI automation.

North American Insurance

Robert Cox and team, GS Equity Research

- Recent reports that OpenAI has incorporated insurance services directly into ChatGPT sparked a significant selloff in insurance providers. We believe Private and Commercial (P&C) insurance companies will need to adapt to keep pace with these emerging AI tools. Certain personal lines & micro business insurance brokers are the most vulnerable to disruption as competitors increasingly leverage AI to simplify the purchase of lower complexity products like auto/home insurance and disrupt various consulting services. Shifts in distribution could also impact auto/home insurers' customer acquisition models, with auto insurers also facing long-term TAM concerns from increasing autonomous vehicle (AV) penetration. Commercial (re)insurers are well positioned as trusted, efficient, regulated capital allocators, especially if they quickly move to embrace AI efficiencies. Larger insurance brokers are also well positioned due to the data and structural moats associated with complex consulting and brokerage, however certain small account broking and less complex consulting could be exposed to AI disruption.
- We think both insurers and brokers can achieve significant efficiency gains through AI automation. Larger market cap firms with big technology budgets will likely benefit first. Insurers are using AI to automate claims processing and speed up policy handling, while brokers aim to automate routine processes/services. First movers are likely to see near- to medium- term margin/growth benefits, although whether these benefits will ultimately lead to lower premiums for customers remains an open question.
- The market has primarily focused on the risk of revenue disruptions from AI rather than on potential efficiency gains. While valuations for P&C insurers are under pressure from concerns about a softer pricing cycle, making it more difficult to isolate AI impacts, the market is clearly penalizing insurance brokers for potential revenue risks—a move which we think is overdone. Auto insurers have faced multiple contraction from AVs and could see revenue disruption from AI-based distribution platforms, but we think these headwinds will ultimately be manageable for underwriters with historically strong direct underwriting platforms. By contrast, commercial (re)insurers are in some cases benefiting from reported AI efficiency gains—CB has reported a ~20% headcount reduction over 3-4 years—and are seen as relatively safe from revenue disruption.

CRE Services

Julien Blouin, GS Equity Research

- We believe AI disruption concerns across CRE services businesses are significantly overstated. The physical nature of the CRE ecosystem provides meaningful insulation versus digitally-native industries. Historical precedents—listings platforms, online property management, virtual tours, automated valuation models—show that CRE technology adoption unfolds over years and decades, not quarters, and most importantly these innovations improved transparency, efficiency, and productivity but did not structurally disintermediate CRE brokerage and service providers.
- Transactional brokerage businesses face the most theoretical AI exposure, particularly for standardized mid-market mandates where AI can drive underwriting automation, OM generation, lease abstraction, and buyer matching. Yet wholesale disintermediation remains unlikely given the relationship-driven, localized, negotiated, and liability-bearing nature of CRE transactions (particularly for larger deals). Facilities, property, project, and investment management businesses are even more insulated as they are operationally complex, and often require on-premises workers/technicians. We believe AI will enhance functions like scheduling, cost control, and predictive maintenance rather than eliminate demand.
- The valuation opportunity is compelling. CRE services companies are benefiting from a transaction recovery that remains mid-cycle, with volumes still 44% below 2021–2022 peaks and 20% below 2017–2019 levels. Yet sector multiples have contracted sharply on AI fears, reverting to 2023 discounts versus the S&P Equal Weight—levels last seen when transactions were under extreme pressure. We view this selloff as an attractive entry point into a sector with powerful structural and cyclical tailwinds, supporting double-digit earnings CAGRs through 2028.

European Financials

Oliver Carruthers and team, GS Equity Research

- Within our European Financials coverage universe, AI disruption concerns have centered on market structure firms such as LSEG and DB1, though their AI exposure varies. LSEG's stock has declined since mid-2025 amid concerns that AI-powered analysis could disrupt its data business.
- More recently, focus has also turned to private markets and their exposure to software-based service business models. A de-rating of public software comparables off the back of AI-disruption concerns could lead to negative mark-to-market adjustments for portfolio companies and limit near-term monetization. That said, many private equity firms are benefitting from rising AI demand, including through digital infrastructure investments. And traditional asset managers have largely been shielded from AI disruptions with managers actively leveraging AI to boost efficiency and alleviate margin pressure as well as enhance efficiency across client servicing, risk management, and investment intelligence.
- AI disruption poses greater risk for private credit managers as fund performance is more sensitive to dispersion and, unlike buyout funds, private credit managers cannot offset losses with investments into "winners". We've seen significant underperformance among Alternative Asset Managers, with private credit-focused firms most impacted. Wealth managers and investment platforms also experienced an AI-related selloff after Altruist launched an AI-powered tax planning tool, which raised concerns that automated advice tools could increase competition and compress fees. However, we note that these tools also have the potential to enhance adviser productivity and increase efficiency for firms able to strategically integrate AI into their offerings.

European transport, infrastructure, and construction

Patrick Creuset and team, GS Equity Research

- When it comes to AI risk, the main debates in the European transport, infrastructure, and construction sector center around 1) whether the incumbents will be disrupted by AI technological progress, and 2) the extent to which AI productivity gains will be deflated away.
- We view the AI disruption risk as relatively low for the sector as incumbents benefit from large physical networks (e.g., sizable blue-collar workforces as well as numerous physical gateways and warehouses), have higher value-add proposition (e.g., end-to-end supply chain management), and benefit from network effects—all of which will be difficult for new AI-native entrants to match. Incumbents also stand to unlock significant advantages from integrating AI technology into their existing operations, further supporting their resilience.
- We are also less concerned about the prospect of AI-related productivity gains being deflated away. While some productivity gains will likely be passed on to customers—as has historically been the case within the industry—we expect the magnitude of productivity gains to be meaningfully larger than the potential yield erosion.
- Asset-light logistics companies are particularly well-positioned to benefit from integrating AI tools into their ongoing digitization push. The sector's SG&A operations have historically involved many repetitive, documentation-intensive tasks, including quoting, booking, carrier payment tracking, and customs clearance. We believe that further digitization and AI deployment will eventually enable industry players to automate some of these workflows and improve efficiency.

Glossary of GS proprietary indices

Current Activity Indicator (CAI)

GS CAIs measure the growth signal in a broad range of weekly and monthly indicators, offering an alternative to Gross Domestic Product (GDP). GDP is an imperfect guide to current activity: In most countries, it is only available quarterly and is released with a substantial delay, and its initial estimates are often heavily revised. GDP also ignores important measures of real activity, such as employment and the purchasing managers' indexes (PMIs). All of these problems reduce the effectiveness of GDP for investment and policy decisions. Our CAIs aim to address GDP's shortcomings and provide a timelier read on the pace of growth.

For more, see our CAI page and Global Economics Comment: Technical Updates to Our Global CAIs.

Dynamic Equilibrium Exchange Rates (DEER)

The GSDEER framework establishes an equilibrium (or "fair") value of the real exchange rate based on relative productivity and terms-of-trade differentials.

For more, see our GSDEER page, Global Economics Paper No. 227: Finding Fair Value in EM FX, 26 January 2016, and Global Markets Analyst: A Look at Valuation Across G10 FX, 29 June 2017.

Financial Conditions Index (FCI)

GS FCIs gauge the "looseness" or "tightness" of financial conditions across the world's major economies, incorporating variables that directly affect spending on domestically produced goods and services. FCIs can provide valuable information about the economic growth outlook and the direct and indirect effects of monetary policy on real economic activity.

FCIs for the G10 economies are calculated as a weighted average of a policy rate, a long-term risk-free bond yield, a corporate credit spread, an equity price variable, and a trade-weighted exchange rate; the Euro area FCI also includes a sovereign credit spread. The weights mirror the effects of the financial variables on real GDP growth in our models over a one-year horizon. FCIs for emerging markets are calculated as a weighted average of a short-term interest rate, a long-term swap rate, a CDS spread, an equity price variable, a trade-weighted exchange rate, and—in economies with large foreign-currency-denominated debt stocks—a debt-weighted exchange rate index.

For more, see our FCI page, Global Economics Analyst: Our New G10 Financial Conditions Indices, 20 April 2017, and Global Economics Analyst: Tracking EM Financial Conditions – Our New FCIs, 6 October 2017.

Goldman Sachs Analyst Index (GSAI)

The US GSAI is based on a monthly survey of GS equity analysts to obtain their assessments of business conditions in the industries they follow. The results provide timely "bottom-up" information about US economic activity to supplement and cross-check our analysis of "top-down" data. Based on analysts' responses, we create a diffusion index for economic activity comparable to the ISM's indexes for activity in the manufacturing and nonmanufacturing sectors.

Macro-Data Assessment Platform (MAP)

GS MAP scores facilitate rapid interpretation of new data releases for economic indicators worldwide. MAP summarizes the importance of a specific data release (i.e., its historical correlation with GDP) and the degree of surprise relative to the consensus forecast. The sign on the degree of surprise characterizes underperformance with a negative number and outperformance with a positive number. Each of these two components is ranked on a scale from 0 to 5, with the MAP score being the product of the two, i.e., from -25 to +25. For example, a MAP score of +20 (5;+4) would indicate that the data has a very high correlation to GDP (5) and that it came out well above consensus expectations (+4), for a total MAP value of +20.

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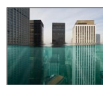
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December 15, 2022

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Disclosure Appendix

Reg AC

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TOP^{of} MIND

AMERICAS FIRST



The Trump Administration kicked off 2026 with a new, more forceful foreign policy, driven by the so-called “Donroe Doctrine”. So, what could “America First” evolving to “Americas First” mean for economies, assets, and geopolitical risk? Trump’s former Special Envoy for Latin America, Mauricio Claver-Carone, and Johns Hopkins SAIS’ Hal Brands discuss the motives behind the Donroe Doctrine and agree it portends more action in the Western Hemisphere. GS’ Alberto Ramos then explores the implications for regional economies, and GS’ Neil Mehta and Daan Struyven dig into the implications for the asset at the center of recent events: oil. But with perhaps the most profound implication being the rise in geopolitical risk, we ask the Centre for Liberal Strategies’ Ivan Krastev what’s ahead for US-Europe relations

(a rocky period, but not a divorce) and GS’ Christian Mueller-Glissmann how to protect portfolios (diversify, don’t time), with GS’ Kamakshya Trivedi also seeing value in a surprisingly resilient corner of the market: EM assets.



The prospect of a fundamental shift in US policy is more real now than at any point in the last 70 years.

- Hal Brands

President Trump has long believed that the US can’t be the preeminent global power if it’s not the preeminent regional power.

- Mauricio Claver-Carone

The [US-Europe] relationship looks to be headed for a rocky period, but not a divorce.

- Ivan Krastev



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INTERVIEWS WITH:

Hal Brands, Professor, Johns Hopkins School of Advanced International Studies, Senior Fellow, AEI

Mauricio Claver-Carone, Managing Partner, LARA Fund, Former US Special Envoy for Latin America and Senior Director for Western Hemisphere Affairs at the NSC in the Trump Administration

Ivan Krastev, Chair, Centre for Liberal Strategies in Sofia

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Kamakshya Trivedi, GS FX and EM Research

Q&A ON DEFENSE SECTOR GEOPOLITICAL RISK

Noah Poponak and Sam Burgess, GS Equity Research ...AND MORE

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We provide a brief snapshot on the most important economies for the global markets

US

Latest GS proprietary datapoints/major changes in views

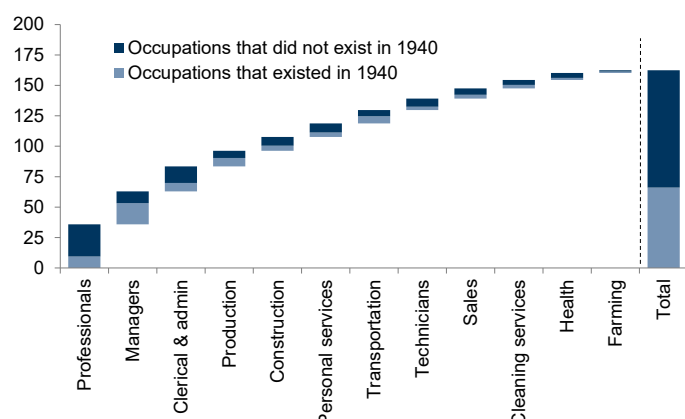
- We now expect two 25bp cuts in Jun and Sept (vs. Mar and Jun previously) to a terminal rate range of 3-3.25%.
- We recently lowered our 12m US recession probability to 20% (vs. 30%) partly owing to early signs of labor market stabilization.

Datapoints/trends we're focused on

- US growth; we expect somewhat stronger growth of 2.5% in 2026 (Q4/Q4) as the tariff drag gives way to a fiscal boost.
- US government shutdown, the odds of which have spiked.
- DoJ investigation into Fed Chair Powell, which we don't expect to impact Fed policy for now, but we find that a less independent Fed would put upward pressure on inflation.
- AI job apocalypse concerns, which we think are overblown.

Technological change: a key driver of job growth

Employment by new and pre-existing occupations, millions



Source: Goldman Sachs GIR.

Japan

Latest GS proprietary datapoints/major changes in views

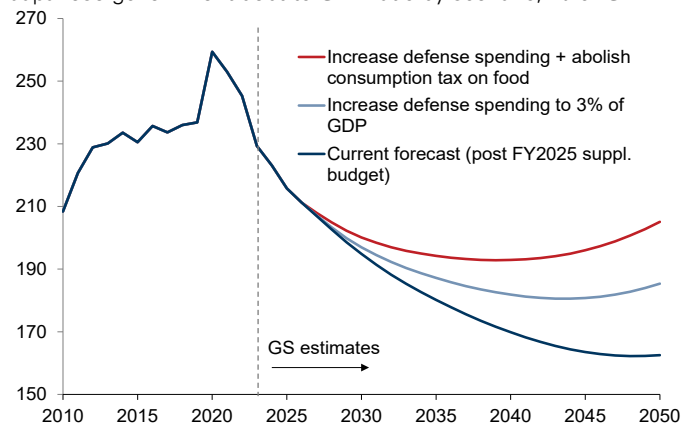
- No major changes in views.

Datapoints/trends we're focused on

- Japanese growth; we expect still-steady growth of 0.8% in 2026 (yoy) led by domestic demand, but see external demand slowing slightly amid higher Japan-China tensions.
- BoJ policy; we expect the BoJ to hike at a semiannual pace, with the next hike in July, to a policy rate of 1.5% by mid-2027, though the timing of the next hike is uncertain.
- Japanese fiscal fears, which have risen on the back of the looming Lower House election (February 8), with all political parties advocating for a consumption tax reduction.

Japan: a potential undermining of fiscal health ahead

Japanese government debt-to-GDP ratio by scenario, % of GDP



Source: Ministry of Finance, Cabinet Office, Goldman Sachs GIR.

Europe

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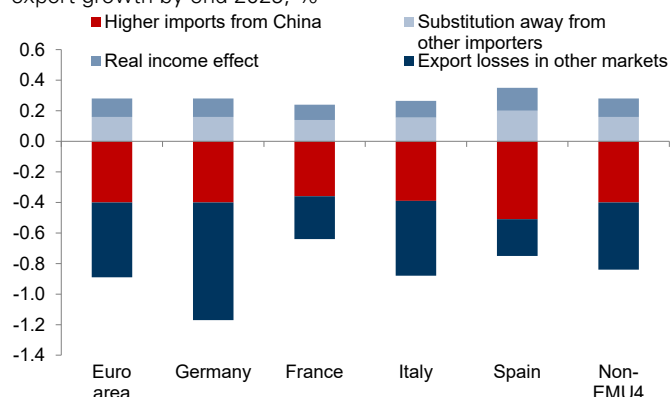
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Datapoints/trends we're focused on

- EA growth; we expect real GDP growth of 1.2% in 2026 (yoy) reflecting real income gains and a fading fiscal drag but growing headwinds from increased competition with China.
- UK economy, for which we expect another mixed year in 2026 characterized by trend-like growth and a further rise in unemployment but much lower inflation and more BoE cuts.
- Trade policy; we see only limited downside to European growth and limited upside to inflation if the US-EU trade deal implementation remains stalled.

Europe: a meaningful drag from China Shock 2.0

Estimated country-level real GDP impact of higher Chinese export growth by end-2029, %



Source: Haver Analytics, Goldman Sachs GIR.

Emerging Markets (EM)

Latest GS proprietary datapoints/major changes in views

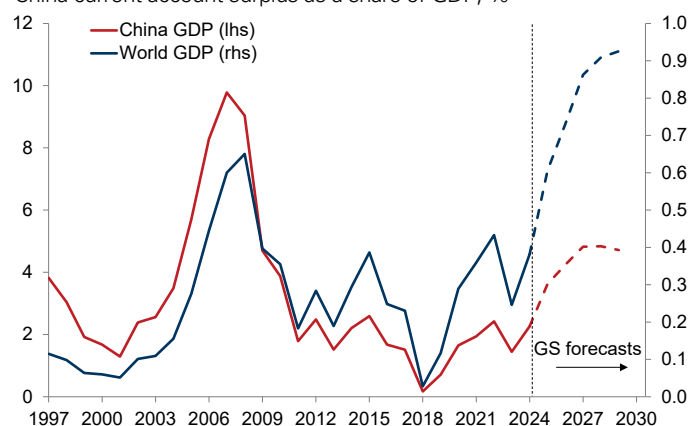
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Datapoints/trends we're focused on

- China's current account surplus, which we expect to rise further over the next several years to almost 1% of global GDP, the biggest of any country in recorded history, owing to China's manufacturing strength and weak domestic demand.
- China policy easing; we expect more easing this year, including two 10bp rate cuts, which should support our above-consensus growth and inflation forecasts.
- EM policy normalization, which we expect to continue amid further inflation normalization, including in Brazil and CEEMEA.

China current account surplus: rising to record levels

China current account surplus as a share of GDP, %



Source: IMF, Haver Analytics, Goldman Sachs GIR.

Americas First

The Trump Administration kicked off 2026 with a new, more forceful foreign policy, driven by the so-called “Donroe Doctrine”—a corollary to the Monroe Doctrine that aims to expand US influence and control in the Western Hemisphere (see pg. 8). What the evolution of “America First” to “Americas First”—and potentially beyond—means for economies, assets, and geopolitical risk more broadly is Top of Mind.

We first turn to Mauricio Claver-Carone, President Trump’s former Special Envoy for Latin America and Senior Director for Western Hemisphere Affairs at the NSC, and Johns Hopkins School of Advanced International Studies’ Hal Brands for more insight into the Donroe Doctrine and the Trump Administration’s potential next moves.

Claver-Carone explains that the Donroe Doctrine reflects Trump’s long-held belief that “the US can’t be the preeminent global power if it’s not the preeminent regional power” (see pg. 9). This belief is driving Trump to reassert US dominance in Latin America with the overarching goal of strengthening US security—namely, national security and, importantly, energy security, which Claver-Carone says are closely linked. He sees no contradiction between this and Trump’s “America First” policy, arguing that “Americas First” is its natural extension.

Brands, for his part, sees two motivating factors behind the Donroe Doctrine, which he characterizes as “a reversion to an earlier style of statecraft”: the US’ desire to consolidate its position in the Western Hemisphere amid a new era of great power rivalry, and Trump’s desire to increase America’s control of hemispheric resources (think oil and territory).

But Claver-Carone and Brands agree: the Donroe Doctrine portends more action in the Western Hemisphere ahead. Claver-Carone expects the Trump Administration to continue pursuing “practical partnerships” with Latin American countries. And Brands underscores the importance of the Greenland issue (see pg. 14), suggesting that it likely won’t be easily resolved, though he also argues that “we shouldn’t view the Donroe Doctrine as a constraint on action elsewhere” (e.g. Iran).

So, what could this all mean for regional economies? Alberto Ramos, GS Head of LatAm Economics Research, doesn’t expect major economic spillovers from the developments in Venezuela to the rest of Latin America given their minimal linkages owing to Venezuela’s strikingly prolonged and deep depression and hyperinflation. He sees similarly limited political implications as voters in several Latin American countries (Peru, Brazil, and Colombia) are set to head to the polls later this year.

And what about exposed assets? Ramos recounts the many uncertainties facing Venezuela’s defaulted bonds amid potential debt restructuring. Neil Mehta, GS Head of North American Natural Resources Equity Research, and Daan Struyven, Co-head of Global Commodities Research, then dig into the global

asset at the center of the recent developments—oil. Struyven expects the Administration’s plan to revitalize Venezuela’s beleaguered oil sector to result in a modest increase in Venezuelan oil production and a modestly negative impact on oil prices over the next several years, with more significant oil price impacts over the longer run as production ramps up further (see pg. 11 for a map of Venezuelan oil assets).

But the most profound and potentially lasting implication of the recent developments is the rise in geopolitical risk (see pg. 18). We speak with the Centre for Liberal Strategies’ Ivan Krastev about the future of the US-Europe relationship, which he describes as at a “turning point but not a breaking point” given Europe’s security, economic, and technological dependence on the US as well as American political realities that may constrain US policy. But he expects a rocky period in the relationship ahead given that trust has been broken, not just because of Greenland, but also because of US actions vis-à-vis Ukraine and the US’ broader treatment of Europe.

And Brands sees even broader geopolitical implications, arguing that US designs on Greenland may test the global territorial status quo that has prevailed post WWII. He warns that the US deciding to play a fundamentally different role in the world than it has in the past 70 years could lead to significant changes in everything from nuclear proliferation, to the security of seaborne transit and democracy, to Dollar dominance.

So, what does this elevated geopolitical risk environment mean for global investors? Christian Mueller-Glissmann, GS Head of Asset Allocation Research, offers guidance on how to protect portfolios against this risk. He argues that rather than trying to time geopolitical shocks, investors’ first line of defense should be diversification. He sees value in several diversification strategies: turning to bonds and other safe havens like the Swiss Franc, increasing allocations to assets that could benefit from geopolitical shocks (like gold and the defense sector, which GS defense analysts Noah Poponak and Sam Burgess believe has not yet fully priced in elevated geopolitical risk), limiting direct and indirect exposures, and hedging with options.

Kamakshya Trivedi, GS Chief FX & EM Strategist, also sees value in a surprisingly resilient corner of the market: EM assets, which have held up despite elevated geopolitical risk, which typically weighs on EM relative performance. He expects this resilience to continue this year, with EM assets’ role as diversifiers further increasing their value proposition. In this context, Trivedi believes frontier assets are also worth a look.

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Interview with Hal Brands

Hal Brands is Henry A. Kissinger Distinguished Professor of Global Affairs at the Johns Hopkins School of Advanced International Studies and Senior Fellow at the American Enterprise Institute. Below, he discusses the US' renewed assertiveness in the Western Hemisphere, which he argues could have major repercussions for the global order.

The views stated herein are those of the interviewee and do not necessarily reflect those of Goldman Sachs.



Allison Nathan: In 2024, you argued that a second Trump presidency would feature a “revitalized Monroe Doctrine”. What factors are motivating the US’ more forceful posture in the Western Hemisphere?

Hal Brands: I see two motivating factors—one structural, one personal. The structural factor is the US’ historic

pattern of attempting to consolidate its position in the Western Hemisphere when the rest of the world is falling apart. During both World Wars and the Cold War, the US worked hard to try to ensure its dominance in the Western Hemisphere. So, it's not surprising that the US is now doing the same as the world enters a new era of great power rivalry.

The personal factor revolves around President Trump's long-held interest in renewing the Monroe Doctrine to increase the US' control of hemispheric resources, like oil, and territory, specifically from Panama, Canada, and Greenland. The President is also intensely focused on tangible threats to American sovereignty and security rooted in hemispheric issues like drug trafficking and migration. Putting all this together, it's not surprising that this Administration has focused on reconsolidating US primacy in the Western Hemisphere.

What is a bit more surprising is how energetic the Administration has been in doing so. The Venezuela intervention has received substantial attention, but it's just the capstone of a much bigger and broader campaign that has featured diplomatic pressure on Colombia and Panama, economic support to allies like the Milei government in Argentina, partnerships with El Salvador's Bukele government and other ideological fellow travelers, lethal strikes on drug boats, and an array of other military, diplomatic, and economic initiatives. In many ways, this has been the centerpiece of Trump's foreign policy so far.

Allison Nathan: So, it's right to think that oil was a key motivation for the Venezuela actions?

Hal Brands: Resource control is undoubtedly at the forefront of the President's calculus. He has long associated it with national power, often remarking that the primary mistake in the American invasion of Iraq was not “taking the oil”. In describing US aims in Venezuela since the intervention, Trump has said very little about democracy and good governance, and even relatively little about drug trafficking. He talks *a lot* about the US exerting greater control over the disposition and sale of Venezuelan oil and the desire for greater openness of the Venezuelan oil industry to the US oil majors. Others in the Administration may value different things in this Trump Corollary to the Monroe Doctrine, but for Trump himself, the

primary issues are essentially about economic gain and a neo-mercantilist approach to international economics.

Allison Nathan: Is that what differentiates the “Donroe Doctrine” from the Monroe Doctrine?

Hal Brands: The Monroe Doctrine's meaning has evolved over time. It was initially about preventing the reimposition of colonial control by European powers. During the 20th century, it became about keeping fascist and communist regimes out of the hemisphere. In some ways, the “Donroe Doctrine” is a throwback—it is a reversion to an earlier style of statecraft that is much less apologetic about the desire to control the resources of weaker countries and much more willing to talk about outright territorial acquisition and other ideas that went out of style with the end of the age of imperialism. It's sometimes said that Trump is trying to revive a 19th century style of foreign policy. I agree that he would likely feel quite at home if he were presiding over some of the US' interventions in Latin America in the late 19th and early 20th centuries. Indeed, the tactics he has used in Venezuela—overthrowing a government and exerting control of the country's trade revenues—aren't that different from what the US did in places like the Dominican Republic well over a century ago.

Allison Nathan: With that context, why do you think the Administration chose to leave the rest of the Maduro regime in place?

Hal Brands: Trump is seeking maximum leverage and gain with minimum long-term investment of resources and political capital. He burst onto the domestic political scene 10 years ago by arguing that the US had made terrible mistakes by undertaking nation-building missions in Afghanistan and Iraq. And flying Maria Corina Machado or Edmundo Gonzalez into Caracas in the hours after Maduro's apprehension probably would have incited a lot of instability by putting them in conflict with the remnants of the Maduro regime. So, the mechanism Trump has settled on is a 21st century version of gunboat diplomacy, where the US will work through the existing authorities while threatening them with punishment if they don't do the Administration's bidding.

Allison Nathan: What do these developments mean for other countries in the region, many of which are slated to hold elections this year?

Hal Brands: They serve notice to any vulnerable anti-American regimes playing strategic footsie with China and Russia that they will likely face greater US pressure. Cuba might be the best example. That said, I'm skeptical that the US would undertake a large-scale military intervention in Cuba because a collapse of that regime might produce a flood of refugees headed for the US. More broadly, countries from Colombia to Mexico will likely feel pressure to act with greater urgency on

issues related to narcotrafficking, which we're already seeing evidence of in the discussions the US and Mexico appear to be having about a potential role for the US military in going after cartels operating in Mexico, as well as some of Trump's comments about Colombia.

The domestic political implications of these shifts will vary given the array of political cultures and levels of susceptibility to US interests throughout the region. In Brazil, the political culture to some degree has long been interwoven with resistance to US influence in Latin America, even in periods when conservative military governments in Brazil worked very closely with the US. So, I wouldn't be shocked if US efforts backfire to some degree in the next Brazilian election. But it could go the other way there and/or in other countries, where the US either has more influence or where a homegrown swing to the right or in the direction of neo-populist politics occurs.

Allison Nathan: What lessons do you think China and Russia are taking from the recent US actions, and how are they shifting their calculus in the region?

Hal Brands: Beijing and Moscow have run up hard against the reality of US hard power in the Western Hemisphere. Whether it's the US Navy and Coast Guard seizing shadow fleet tankers, the US forcibly deposing Maduro, or the Pentagon shooting up drug boats in the Caribbean, neither China nor Russia can do much to contest the application of American military power in the Western Hemisphere. That said, China's influence in the region in particular is multifaceted and deeply embedded in everything from trade relationships to the diffusion of technology to physical and digital infrastructure, all of which will persist to some degree.

It's important to note that China's influence is not simply imposed on Latin American countries, but a response to their needs. Big gaps in physical and digital infrastructure exist in the region, which China is well-placed to address. So, Beijing will play the long game and continue to attempt to win influence in the region through economic, political, and, in some cases, low profile security relationships that will focus more on internal security than traditional military functions. If anything, the recent developments likely mark the start of an even more intense phase of competition for influence in the region.

Allison Nathan: How does Greenland fit into Trump's plan, and what might that suggest about his next steps there?

Hal Brands: Greenland is both a hemispheric and global issue. It's a hemispheric issue in the sense that the Trump Administration views consolidating US control of Greenland as part of the larger effort to revive the Monroe Doctrine. And they can point to historic examples of the US increasing its influence in Greenland as part of a strategy of hemispheric defense, as President Roosevelt did in the run-up to US entry into WWII.

It's also a global issue, for two reasons. First, Greenland powerfully affects the fate of NATO and the transatlantic community; it's hard to overstate the degree of outright alarm that Canada and many European countries feel about US designs on the territory. Second, Greenland is the best test of whether the US will seek changes to the global territorial status quo. That's an explosive issue because China and Russia are attempting to do much the same thing in the South China

Sea/Himalayas and Ukraine, respectively. So, a world in which the US also aims to redraw borders through coercion or even force is a world in which the three most powerful countries are all violently or coercively disrupting the territorial status quo. That's something we haven't seen since the 1930s and could be deeply corrosive to the post-1945 global order.

Allison Nathan: So, recent US actions have helped further the ambitions of China and Russia in some respects?

Hal Brands: Both countries were undoubtedly unhappy to see Maduro go, especially at the hands of ruthlessly efficient US military power. But recent developments aren't entirely bad news for Beijing and Moscow. A world in which international law and norms matter less, and great powers are free to do as they like, especially in their own backyards, is a world in which Chinese and Russian leaders would feel comfortable. In fact, that's exactly the world they aim to bring about. And if an intensification of US military activity in the Western Hemisphere pulls resources away from other regions, as was the case when Trump redeployed an aircraft carrier covering the Middle East and Europe from the Mediterranean to the Caribbean, that's even better news for them.

Allison Nathan: How does Iran fit in here?

Hal Brands: Iran is a key reason why we should be skeptical of the idea that the world is rapidly transitioning into a spheres of influence arrangement. Trump wants an American sphere of influence in the Western Hemisphere, but he also likes to maintain unhindered freedom of action globally when he views it as advantageous to him. This was clear in the recently-released national security strategy. This Administration has also conducted two major military interventions in the Middle East and threatened others. We shouldn't view the Donroe Doctrine as a constraint on action elsewhere.

Allison Nathan: Amid all this, what geopolitical risks should companies and investors be most focused on?

Hal Brands: The mega risk is the potential for the US to decide to play a fundamentally different role in the world. We have lived for so long in the post-1945 international order in which US power was wielded to secure strong prohibitions on forcible conquest, freedom of navigation to support and propel the global economy, and protection of human rights and the spread of democracy, that it's difficult to conceive of a world without that leadership. Such a world would likely see more countries pursuing nuclear weapons, greater disorder on the high seas that would impact global trade and finance, a challenge to Dollar dominance, and a global retreat of democratic values.

30 years ago, I would have said there was little chance of such a shift. 15 years ago, I would have said signs of retrenchment-minded inclinations in the US foreign policy debate had grown. We now have a US president who often expresses a desire for fundamental change in US foreign policy and the way the US engages in the global economy, even if the policies don't always match the rhetoric. So, the prospect of a fundamental shift in US policy is more real now than at any point in the last 70 years. The stakes are high, and the choices the US makes in the coming years will not only determine its own role on the world stage, but also the trajectory of the international order.

Interview with Mauricio Claver-Carone

Mauricio Claver-Carone is Managing Partner of the Latin America Real Assets Opportunity Fund. He served as US Special Envoy for Latin America and Senior Director for Western Hemisphere Affairs at the National Security Council in the Trump Administration. Below, he argues that President Trump views US dominance in the Western Hemisphere as essential for global preeminence, and that “Americas First” is a natural extension of “America First”.

The views stated herein are those of the interviewee and do not necessarily reflect those of Goldman Sachs.



Jenny Grimberg: You played a key role in foreign policy in the Western Hemisphere during both Trump Administrations. What’s behind President Trump’s focus on the region, and do you sense a shift in that focus from his first term?

Mauricio Claver-Carone: President Trump has long believed that the US can’t be the preeminent global power if it’s not the preeminent regional power. That is the “Donroe Doctrine”. And contrary to widespread perception, this reflects the President’s own thinking, not his advisors’ influence. President Trump is a developer from New York—he thinks in terms of building neighborhoods. He has traveled throughout the region and knows it well. And he believes that no region in the world impacts Americans’ daily lives more than Latin America and the Caribbean from a security, economic, and social perspective.

I’ve seen firsthand many times how focused the President has been on this, and on the US’ waning influence and positioning in the region over the last two decades, especially vis-à-vis China. During periods of conflict in Afghanistan and Syria in his first term, President Trump would incredulously remark on how eagerly and easily the US sent troops to those countries yet was too scared to have any type of presence in its own hemisphere. And I remember at one point during the intensifying crisis that gripped Venezuela in 2019, the President being disturbed to learn that the US Navy had no ships in the vicinity of the Caribbean because they were all deployed to the Persian Gulf, South Asia, and the Mediterranean.

That said, given his team was distracted by the Middle East and Asia at the time, President Trump’s first term was largely about facilitating a pivot from China in the region. The Administration focused on setting frameworks to, mostly through soft power, execute this pivot, which led to the creation of the America Crece Initiative to catalyze private sector investment in energy and infrastructure projects across Latin America and the Caribbean.

The second term, by contrast, is about the implementation of President Trump’s vision for the region. He made that vision extraordinarily clear during his second inauguration as he harkened back to President McKinley and the golden era of American influence, projected not only through policy—the Monroe Doctrine and, later, the Roosevelt Corollary to it—but also through brick-and-mortar initiatives like the Panama Canal. And this time around, his team is aligned with that vision. Secretary of State and National Security Advisor Marco Rubio understands and cares about the region and President Trump’s vision for it. Deputy Chief of Staff for Policy and Homeland

Security Advisor Stephen Miller is also focused on the region owing to his holistic approach on immigration and his view that immigration would be less of a problem for the US if Latin American countries were thriving. And, in contrast to the secretary of defense in the first Administration, who didn’t believe that the Department of Defense should be involved in counternarcotics, Secretary of Defense Pete Hegseth is very involved.

Jenny Grimberg: What does the President hope to achieve by reasserting US influence in Latin America?

Mauricio Claver-Carone: The Trump Administration’s actions in Latin America—from the Venezuela operation to the relationship it created with Mexico to the historic support it lent Argentina during its economic crisis last year—are based first and foremost on US security. This runs the gamut from national security in terms of halting narcotic and migrant flows and reducing terrorism risk to energy security. The Achilles heel of Latin America and the Caribbean in particular is energy. Fidel Castro always understood that, which is why he helped install Hugo Chavez and viewed Venezuela as his golden crown. And Caribbean countries’ pressing need for energy led to the formation of PetroCaribe, an initiative launched by Chavez to supply Venezuelan oil to these countries on favorable financial terms, which allowed Venezuela to essentially control the Caribbean for many years.

Energy security is closely intertwined with national security, as high energy costs have been a key driver of migration throughout Central America. So, energy security is incredibly important, which is consistent with President Trump’s desire to increase oil production in the Western Hemisphere. The US is an energy powerhouse in its own right, and many Latin American countries could be too with proper management. US influence can play a crucial role here. Guyana currently produces more oil than Venezuela, largely because the Guyanese government has worked hand-in-hand with US companies to develop the right policies from the get-go and leave behind the nationalist ideological strains that were embedded into the psyche of the regional population during the 20th century revolutions.

Jenny Grimberg: Does expanding US influence in Latin America conflict with the “America First” pillar of President Trump’s Make America Great Again (MAGA) platform?

Mauricio Claver-Carone: Not at all. “Americas First” is the natural extension of America First as President Trump works to counter the notion of every DC think tank expert that the 21st century would be the Chinese century. The President was adamant during his inauguration that it would instead be

another American century, and potentially the greatest American century. And contrary to what many believe, the President's Latin America strategy is not about boots on the ground or nation-building. It is about practical partnerships between nations as the US shifts from a policy of non-interventionism—which served neither the local population nor the US well by leading to the rise of Chavez and China—to non-imperial expansionism, a term coined to explain the strategy of deepening US influence in the region through partnerships.

These are not political partnerships—the US is currently working with the right-leaning governments of Argentina, El Salvador, Costa Rica, and Chile as well as the left-leaning governments of Mexico and Venezuela—but practical, business-oriented relationships that benefit not only the US but also these countries, whose populations desire security and economic opportunities. The best example of that is Argentina. Many people, including myself, worried that the Trump Administration's financial support ahead of last year's midterm elections could spark a backlash against President Milei's Conservative party given that strains of Peronism still exist in Argentina. But the election results showed that most of the party's support came from the poor, who are fed up with ideologies that neither pay the bills nor put food on the table. So, the region has undergone a dynamic social shift.

Jenny Grimberg: How does the Trump Administration's decision to depose Maduro fit into that?

Mauricio Claver-Carone: Let me first clarify that the Administration didn't "depose" Maduro. This was a law enforcement operation with military support to capture an individual indicted for narcoterrorism in the US. Now, this was a particularly complicated operation because Maduro was de facto running a nation-state with a military, which showed the world that nobody is outside of US reach and sent a message of US influence, ability, and capability. But the operation also underscored the Trump Administration's preference for partnership. President Trump gave Maduro plenty of opportunities to work together; he expressly rejected every offer, which led the Administration to execute a carefully thought-out operation to capture him. That sends an important message: when President Trump offers you a deal or an opportunity, you take it.

The Trump Administration's subsequent actions in Venezuela are also a testament to its partnership preference. The US undoubtedly had the capabilities for a broader operation, but assessed that the other people in Venezuela controlling the institutions of power would want to work with the US. And so far, that assessment has proven right. In the days following the operation, the US reached a deal to purchase 30-50 million barrels of Venezuelan oil and several million barrels moved shortly after. Venezuela is also increasingly and consistently releasing political prisoners. And the US is working with the Venezuelan authorities to put the country on a path toward becoming a more stable and secure presence in the region with an open, participative process. Much remains uncertain, but I'm optimistic.

Jenny Grimberg: The million-dollar question is what President Trump will do next in Latin America and the broader Western Hemisphere. What's your view?

Mauricio Claver-Carone: Further efforts at forming and consolidating partnerships with countries in the region are likely. Upcoming elections in Brazil and Colombia will be important. Conservative candidates recently won Honduras' and Chile's presidential elections, and if right-wing candidates also prevail in Colombia and Brazil, it would result in a historic ideological consolidation for the region. But again, this isn't about ideology but whether the new leaders will choose the US first and foremost as their partner. That would give the US a historic opportunity to modernize its business relationships and commercial partnerships across Latin America.

That could start with Mexico and the upcoming USMCA renegotiation, which seems to be the path the President is heading down. But President Trump's intense focus on trade is reactive. He feels that the trade deals of the 1990s/early 2000s were all flawed, forcing him to focus on fixing them. As those issues resolve, the focus of US partnerships with regional allies could shift to making deals that ensure a US strategic presence in critical sectors like energy and infrastructure. The hope is that by the end of President Trump's term, nobody will ever again talk about China's expanding influence in the region, because it will be unquestionably clear that the US is the partner of choice for Latin America and the Caribbean.

Jenny Grimberg: As an investor in the region, what advice do you have for investors attempting to navigate Latin America's evolving geopolitical landscape?

Mauricio Claver-Carone: Investors looking to monetize the "Donroe Doctrine trade" by focusing more on Latin America should look beyond the big-ticket deals in Mexico and Brazil, which have often been walloped by currency issues, to smaller countries. The deals are smaller, but the returns are bigger as a percentage, and these countries are among the region's fastest-growing economies, offering great opportunities for investors. The secret sauce is local partners.

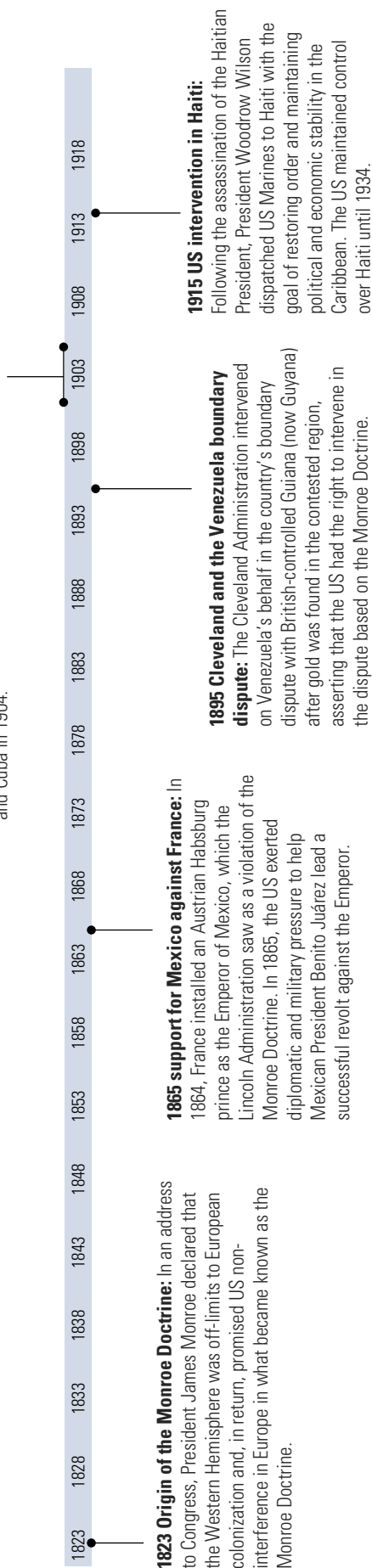
I would also advise against relying on multilateral development banks. As a former president of the Inter-American Development Bank, I've seen firsthand that these institutions aren't focused on finding or creating the best deals. Investors should instead focus on US domestic finance institutions like the International Development Finance Corporation (DFC), which partners with the private sector to mobilize capital for strategic investments around the world and recently secured a six-year reauthorization, to gain exposure to long-term deals in strategic sectors.

I also see significant short and medium-term investment opportunities in the energy and infrastructure space. The timing is right for these investments because the US government is focused on and geopolitically invested in Latin America and wants to encourage private investment in the region, and that focus isn't going away given President Trump's "Americas First" philosophy.

The Monroe Doctrine through the years

INVOKING THE MONROE DOCTRINE

The Monroe Doctrine, first articulated by President James Monroe in 1823, declared that European powers should not interfere in the affairs of the newly independent nations of the Americas, a stance motivated by concerns over potential European re-colonization. Since its inception, US presidents have frequently invoked the Doctrine to justify various foreign policy actions across the Western Hemisphere.



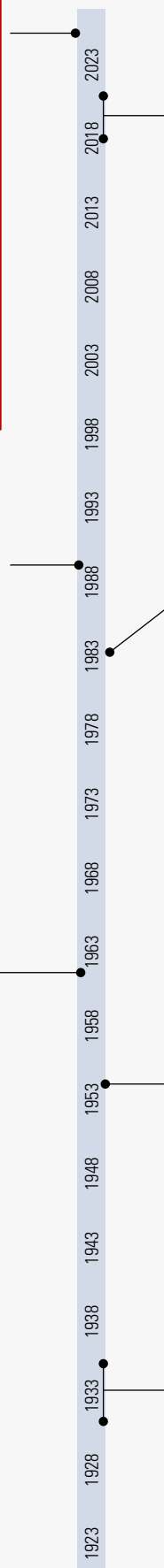
1962 JFK and the Cuban Missile Crisis: President Kennedy invoked the Monroe Doctrine during a televised address in October 1962, announcing the discovery that the Soviet Union was secretly installing offensive nuclear missile sites in Cuba. Kennedy ordered a naval blockade around the island, characterizing the Soviet action as a direct challenge to the long-standing principle of preventing hostile foreign powers from establishing military bases in the Western Hemisphere.

1989 capture of Manuel Noriega in Panama:

President George H.W. Bush deployed US troops to Panama to arrest and extradite Manuel Noriega on drug trafficking and money laundering charges. The US cited the Monroe Doctrine to justify the invasion, claiming that Noriega's regime threatened US citizens living in Panama as well as the security of the Panama Canal.

2026 Trump Corollary and the Donroe Doctrine:

President Trump laid out a Trump Corollary to the Monroe Doctrine that aims to restore US preeminence in the Western Hemisphere. Referring to it as the "Donroe Doctrine," Trump used it as justification for the arrest and extradition of Venezuelan leader Nicolás Maduro on drug trafficking charges.



*Not an exhaustive list of all policy actions related to the Monroe Doctrine.
Source: US Department of State, National Archives, Office of the Historian, compiled by Goldman Sachs GIR.*

April 1, 2020: Trump deploys US Navy destroyers, surveillance aircraft, and additional military personnel to the Caribbean near Venezuela, saying “As governments and nations focus on the coronavirus, there is a growing threat that cartels, criminals, terrorists, and other malign actors will try to exploit the situation for their own gain. We must not let that happen.”

September 25, 2019: "For all of the countries of the Western Hemisphere, our goal is to help people invest in the bright futures of their own nation... We want every nation in our region to flourish and its people to thrive in freedom and peace. In that mission, we are also committed to supporting those people in the Western Hemisphere who live under brutal oppression, such as those in Cuba, Nicaragua, and Venezuela... We are watching the Venezuela situation very closely. We await the day when democracy will be restored, when Venezuela will be free, and when liberty will prevail throughout this hemisphere."

January 14, 2026: "The United States needs Greenland for the purpose of National Security. It is vital for the Golden Dome that we are building. NATO should be leading the way for us to get it. IF WE DON'T RUSSIA OR CHINA WILL, AND THAT IS NOT GOING TO HAPPEN!"

January 3, 2026: “The Monroe Doctrine is a big deal, but we’ve superseded it by a lot, by a real lot. They now call it the **Donroe document**... [the] Monroe Doctrine, we sort of forgot about it. It was very important, but we forgot about it. We don’t forget about it anymore. Under our new national security strategy, American dominance in the Western Hemisphere will never be questioned again. It won’t happen... For decades other Administrations have neglected or even contributed to these growing security threats in the Western Hemisphere. Under the Trump Administration, we are reasserting American power in a very powerful way in our home region.”

January 14, 2026: "We are making tremendous progress, as we help Venezuela stabilize and recover... This partnership between the United States and Venezuela will be a spectacular one FOR ALL. Venezuela will soon be great and prosperous again, perhaps more so than ever before."

January 16, 2026: "I may put a tariff on countries if they don't go along with Greenland, because we need Greenland for national security."

January 21, 2026:
“We probably won’t get anything [on Greenland] unless I decide to use excessive strength and force where we would be, frankly, unstoppable. But I won’t do that.”

9

Venezuela: what's next?

Alberto Ramos explores what developments in Venezuela could mean for the country's economy and the broader LatAm region

Following the US operation in Venezuela, the situation on the ground remains very fluid and the outlook complex and uncertain. The Trump Administration is now working with the remnants of the Maduro regime to stabilize the Venezuelan economy, focusing first and foremost on the battered oil sector, and eventually hold open elections.

An economic collapse of historic proportions

Venezuela has been trapped in a seemingly never-ending economic depression and hyperinflationary spiral. Large macro and financial imbalances have spawned an economic collapse of historic proportions. In fact, Venezuela has experienced one of the worst peacetime economic downfalls in modern history, with real GDP contracting far more than in the US during the Great Depression (-28%), Spain during the Spanish Civil War (-28%), and the 2009 Greece crisis (-27%). Per the IMF, in Dollar terms, Venezuelan GDP has shrunk by ~80% since 2012, with seven consecutive years of negative real GDP growth over 2014-2020 for an astonishing cumulative decline of 74.3%.

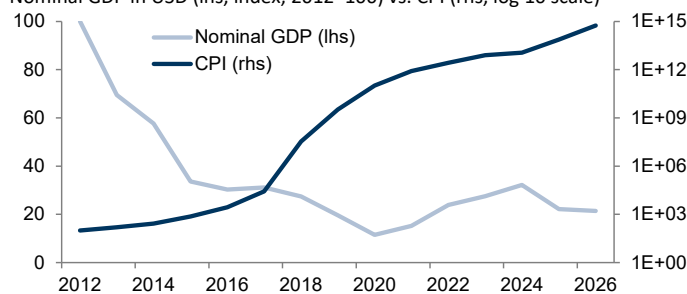
Quite telling and illustrative of the major economic collapse of the last 20 years is the dramatic shift within Venezuela's oil sector—the country's key source of foreign exchange. Oil production fell from 2.7mb/d in 2015 to 0.93mb/d in 2025, resulting in an oil-export hard-currency crunch that imposed a severe and debilitating constraint on imports. As a result, Venezuela's economy has contracted to ~\$80bn today, roughly the size of Uruguay and less than a quarter the size of Chile.

Hyperbolic inflation immiserizes

Annual inflation over the decade leading up to 2025 was an astronomical 14,544% on average—reflecting years of hyperinflation (inflation was ~130,000% in 2018). Over the same period, Venezuela's currency (the Bolivar) lost ~98% of its value. The economic hardship has helped generate a mass exodus of people over the last 16 years. According to the UNHCR, nearly 8mn people fled the country, close to a quarter of the population and one of the world's largest displacements, with most settling in nearby countries, particularly Colombia.

Venezuela has experienced hyperinflation and depression

Nominal GDP in USD (lhs, index, 2012=100) vs. CPI (rhs, log 10 scale)



Source: IMF, Goldman Sachs GIR.

Likely limited economic and financial spillovers...

It therefore comes as no surprise that the economic and financial linkages between Venezuela and the rest of LatAm have become minor. Accordingly, we do not anticipate major economic spillovers from the situation in Venezuela to the rest

of LatAm. However, if the situation on the ground were to deteriorate into widespread lawlessness and violence, we could see renewed migration flows, likely affecting neighboring countries first and foremost. Beyond migration, the situation in Venezuela could indirectly impact the region through their impact on key commodity prices, particularly oil and gold.

...and political implications

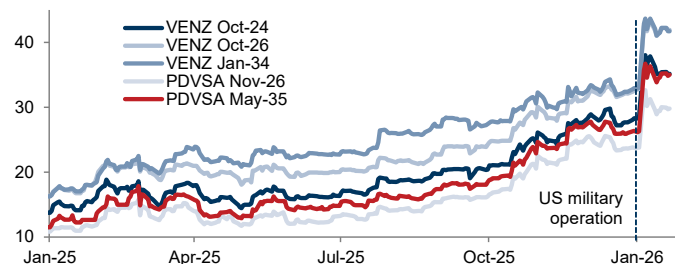
The political implications for the rest of LatAm also seem limited. The reactions of regional governments were mixed and clearly divided along political/ideological lines, but even those critical of the situation were cautious and contained in their responses. However, US actions in Venezuela could stress its relationship with Mexico, Colombia, and Brazil, especially if the US expands its focus to the regime in Cuba and/or conducts military strikes on drug cartels inside Mexico or Colombia. At this juncture, we do not expect the ongoing developments in Venezuela to impact upcoming elections in Brazil (general election in October), Colombia (congressional elections in March, presidential in May), or Peru (general elections in April).

Debt restructuring key to watch from here

Venezuela defaulted on international bonds in late 2017 due to the compounding effects of the deep economic crisis and sanctions. Defaulted debt is estimated at north of \$150bn, including around \$60bn (face value) in defaulted bonds (sovereign and state oil company PDVSA), other PDVSA obligations (e.g., suppliers' debt), bilateral loans (e.g., with China, Russia), and multilateral debt, with significant unpaid interest as well as legal claims from expropriation cases compounding the total since the 2017 default.

Venezuela bond prices rallied following the US military operation

Venezuela and PDVSA bond price, bp



Source: Bloomberg, Goldman Sachs GIR.

Venezuela bond prices rallied in 4Q25 on expectations that rising US pressure and military buildup would eventually lead to regime change. And the initial market reaction to the January 3 US airstrikes was positive. Indeed, defaulted Venezuelan bond prices rallied 8-9pts (including PDVSA) on January 5 on expectations of a brighter outlook for Venezuela's economy and oil sector as well as the potential for debt restructuring.

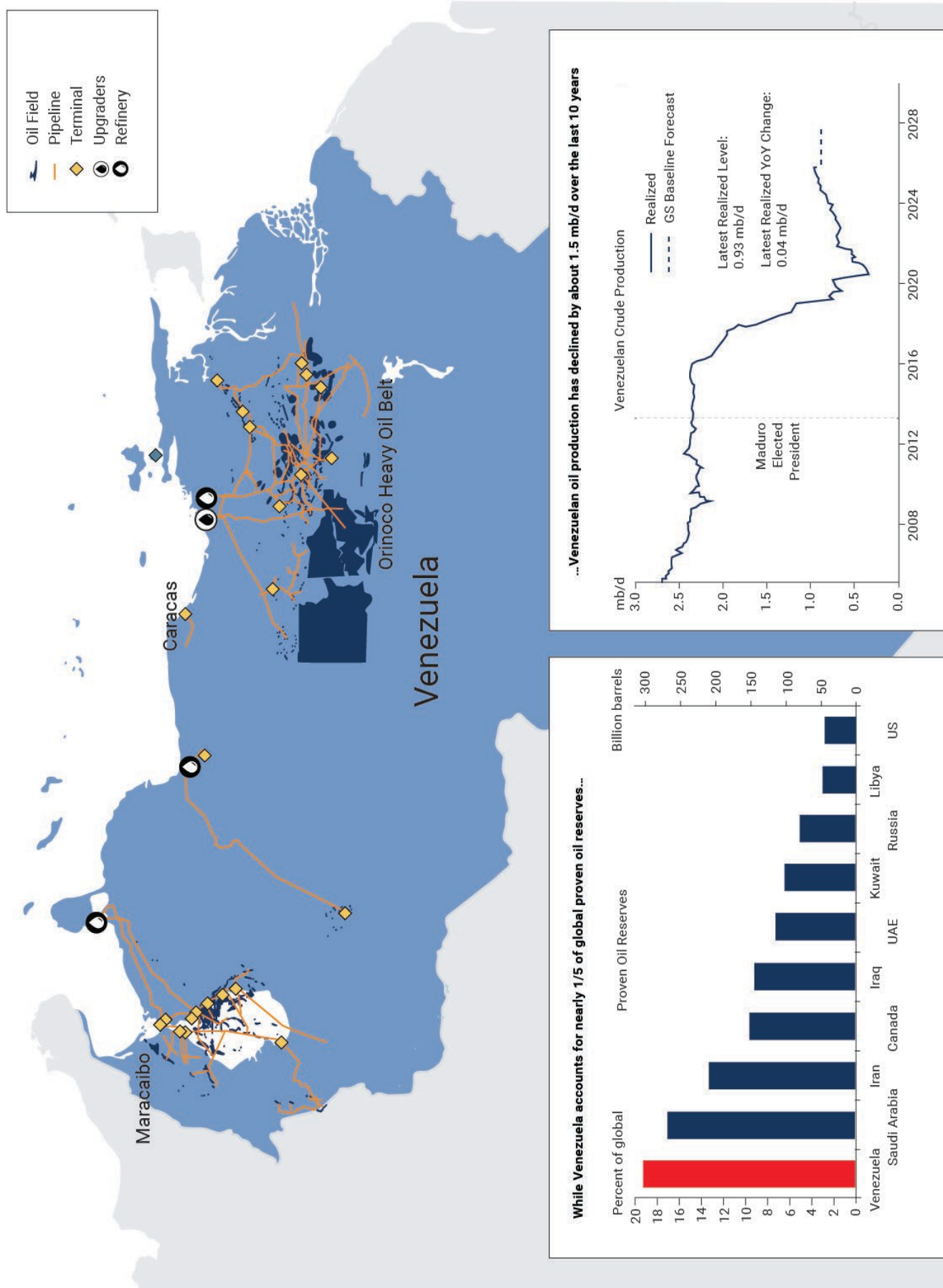
However, bond prices have since given back some of these gains due to the complex and still highly uncertain economic and political picture, the perception that debt restructuring may not be a near-term priority, President Trump's executive order protecting Venezuelan oil revenue held in US Treasury accounts, and bond valuations commensurate with lower-bound recovery levels. The market debate has now shifted to "what type of restructuring" and when and with whom creditors will eventually be negotiating.

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Venezuela oil infrastructure, mapped out



*We use OPEC data on proven crude oil reserves as of 2024 for all the countries above except Canada, where we use Energy Institute proven reserves estimates as of 2020 (latest available).
Sources: OPEC, Energy Institute, secondary sources, S&P Global Energy, Goldman Sachs GIR.
Special thanks to GS Commodities Research team for charts.

Q&A on implications for Venezuelan oil

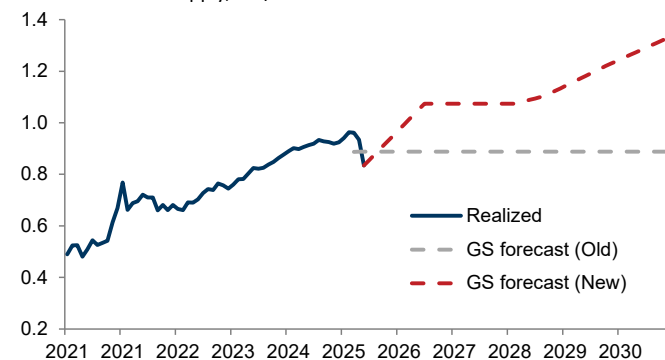
Neil Mehta and Daan Struyven answer key questions about the implications of US plans to revitalize Venezuela's oil sector¹

Q: What's the size of the opportunity for the oil industry of the US' plan to revitalize Venezuela's oil sector?

A: Venezuela accounts for roughly one-fifth of the world's proven oil reserves², yet its production is constrained at ~800-900kb/d, which is less than ~1% of global output. The Trump Administration has [called for](#) ~\$100bn of investment into the country over the next decade to rebuild infrastructure and bring production back to peak levels of ~3mb/d. We believe gains of a few hundred thousand barrels per day are achievable in the near term. Specifically, we expect Venezuelan crude production to rise by 0.2mb/d over the next 12m (from 0.8-0.9mb/d in Dec 2025) as an anticipated easing of sanctions and modest investment in existing assets help ease some operational bottlenecks. We expect production to rise by an additional 0.2mb/d by 2030 (to 1.3mb/d). But a complete restoration of the country's output to ~3mb/d would likely extend into the next decade and require significant capital investment. Some estimates suggest an initial ~\$5-10bn of capital investment over the next 2-3 years could yield an additional ~500kb/d in Venezuela, but the subsequent ~1mb/d would require an estimated ~\$70-80bn. Upstream production expansions would also require a reactivation of damaged upgraders that process heavy crude, extension of pipeline infrastructure, investments in the power grid, and continued political stability.

We expect a modest increase in Venezuelan crude supply

Venezuelan crude supply, mb/d



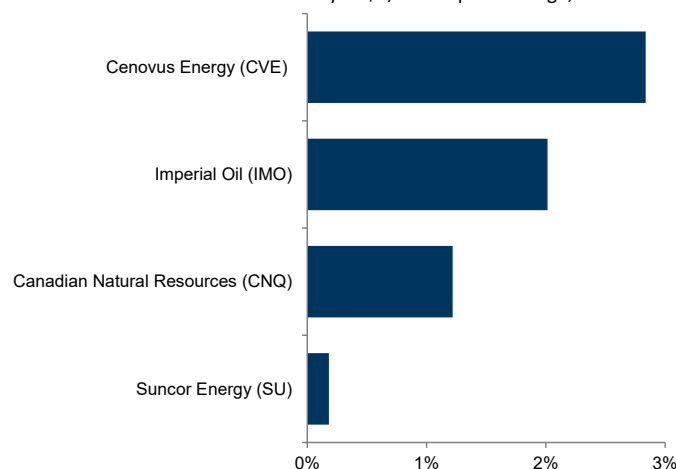
Source: Secondary sources, Goldman Sachs GIR.

Q: What types of companies could be harmed by the revitalization of the Venezuelan oil industry?

A: On a relative basis, Canadian oil companies could be negatively impacted by higher oil production from Venezuela. In particular, Venezuelan barrels could displace Canadian heavy crude on the Gulf Coast and result in wider WTI-WCS differentials, which would reduce realized prices for Canadian producers.

Venezuelan barrels could displace Canadian heavy crude on the Gulf Coast and result in wider WTI-WCS differentials

Estimated annual cash flow sensitivity to \$1/b WCS price change, %



Source: Goldman Sachs GIR.

Q: How have US oil companies responded to these potential opportunities/risks so far?

A: Among the US Majors, Chevron is doubling its liftings from the country immediately and raising its production by 50% (or ~100kb/d) over the next 18-24 months. By contrast, ConocoPhillips and Exxon remain focused on recovering previous debt following the nationalization of their assets. Exxon has also publicly discussed its view that Venezuela is difficult to invest in without significant changes to the legal and commercial environment. Among oilfield services companies, SLB has discussed its ability to scale quickly as needed in the country, as it currently has operations on the ground. Halliburton and Weatherford have also commented on their ability and interest in allocating resources to Venezuela if activity and production ramp up.

Q: How much of the upside/downside of these developments is already priced into stocks?

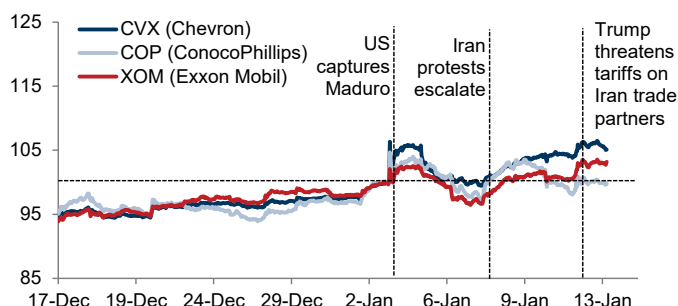
A: Energy equities reacted sharply on the back of the initial Venezuela headlines, including the stocks of US majors, US refiners, international service companies, and Canadian oil producers. While we believe the latest developments are now largely reflected in the equities, we continue to highlight strong fundamentals across the energy complex.

¹ The research views in this section are those of the analysts who cover the relevant areas. Equity research views are expressed by Neil Mehta, an Equity Research Analyst covering Americas Integrated Oil & Refiners. Commodities views are expressed by Daan Struyven, a Macro Research Analyst covering global commodities.

² According to OPEC data.

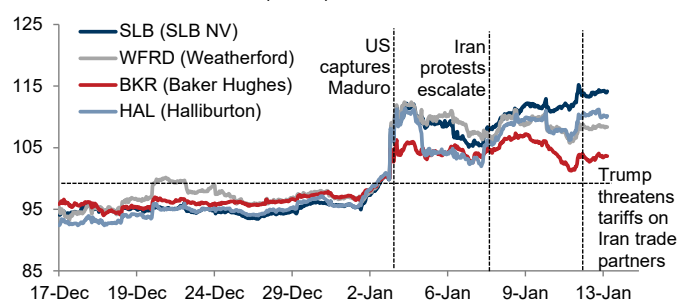
Energy equities reacted sharply on the back of the initial Venezuela headlines, including across US Majors...

Stocks of US Majors, index, Jan 2 close=100



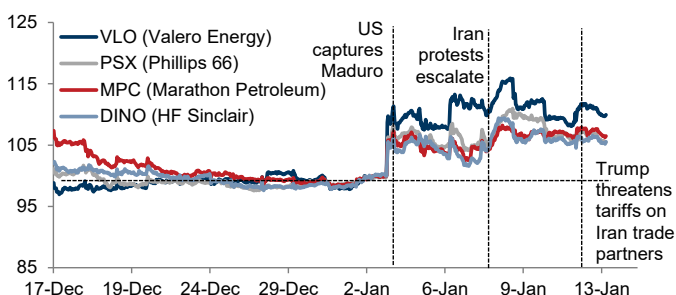
...international services...

Stocks of international services, index, Jan 2 close=100



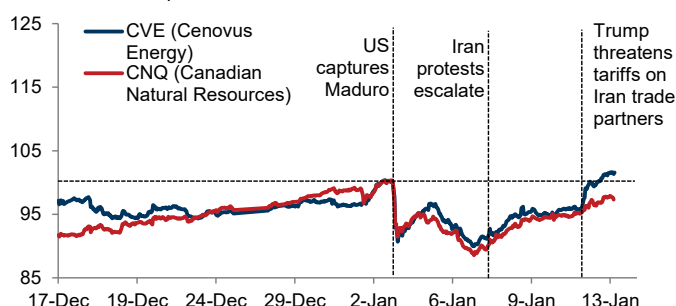
...US Gulf Coast refineries...

Stocks of US Gulf Coast refineries, index, Jan 2 close=100



...and Canadian producers

Stocks of Canadian producers, index, Jan 2 close=100



Source: Bloomberg, Goldman Sachs GIR.

Q: What, if anything, are investors underappreciating about the potential impacts of the Venezuela developments for US oil companies?

A: Given the state of Venezuela's energy infrastructure—which has deteriorated amid underinvestment—investors may be underestimating the challenges associated with sustainably ramping Venezuelan volumes. We also believe the scale of investment required to increase their Venezuelan presence as well as the political and legal risks of doing so could deter companies from making significant new capital commitments in the country, which is a major reason we expect an only modest increase in Venezuela's production by 2030.

Q: What are you watching most closely ahead to gauge how this could all play out for the US oil industry?

A: Despite the recent developments in Venezuela, the outlook for the US oil industry should remain largely unchanged from our base case scenario, with three primary forces continuing to drive the industry's trajectory. First, after rising from 5-6mb/d in early 2000s to 14mb/d of crude today, the industry is now entering a period of slowing growth and maturity. Accordingly, US producers will likely focus on harvesting cash flow domestically, investing in international exploration, and driving consolidation through M&A.

Second, the larger companies have structural advantages of scale and the ability to execute an international development plan. That leaves the US majors, as well as international services companies, well-positioned for growth. Third, we believe that refiners are structurally well-positioned on a multi-year basis as crack spreads are likely to remain above historical averages and global oil demand is expected to outpace new capacity additions between now and 2030.

Q: What could the Venezuela developments mean for oil prices?

A: We estimate that the 0.2mb/d increase in Venezuelan oil production we expect over the next 12m will have a \$1 impact on Brent/WTI prices this year (baseline forecast of \$56/\$52/bbl average 2026 prices vs. \$57/\$53/bbl in a scenario of flat production). But the impact on long-run oil prices will likely be more significant. We estimate that the cumulative 0.4mb/d increase in Venezuelan production we expect by 2030 will have a \$2 impact on long-term oil prices. This, together with higher expected US and Russian production, led us to lower our long-run Brent/WTI price forecasts to \$75/\$71/bbl earlier this month (2030-35, from \$80/\$76/bbl). We estimate that a larger recovery in Venezuelan production to 2mb/d by 2030 would pose \$3 of downside to these revised forecasts.

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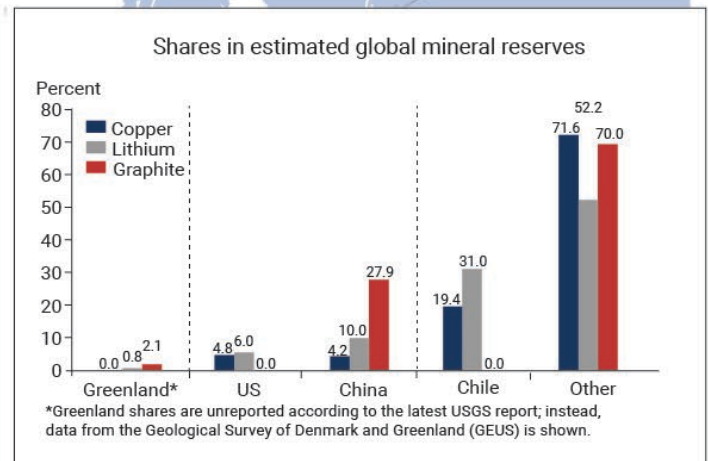
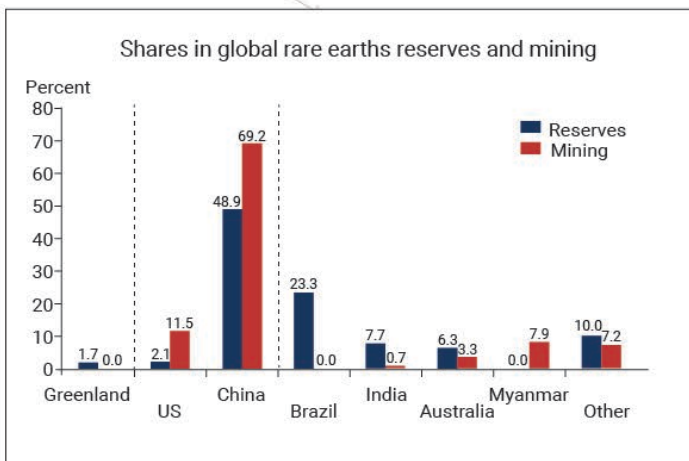
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Goldman Sachs & Co. LLC

Greenland's strategic value, mapped out



Source: Bloomberg, USGS, GEUS, Goldman Sachs GIR.
Special thanks to GS Commodities Research team for charts.

Interview with Ivan Krastev

Ivan Krastev is Chair of the Centre for Liberal Strategies in Sofia, Bulgaria. Below, he argues that US-Europe relations have reached a turning point, but not a breaking point.

The views stated herein are those of the interviewee and do not necessarily reflect those of Goldman Sachs.



Allison Nathan: Do the Greenland developments mark a turning point in US-Europe relations?

Ivan Krastev: They mark a turning point but not a breaking point because alliances don't exist on paper, but in the minds of the public and policymakers. And the Greenland story, combined with the Trump

Administration's recently-released national security strategy and Venezuela actions, have substantially shifted how Europe views transatlantic relations. I was part of a recent European Council on Foreign Relations group that conducted global polling revealing that only 16% of Europeans see the US as an ally, while 20% view it as a rival, with most others seeing it as only a necessary partner. This marks a big shift from the past.

The Greenland issue has been especially difficult for Europeans to digest because the Danes are ready to cooperate with the US on almost anything it wanted from a security and even economic perspective. And the Danish government is one of the most pro-Atlantic governments in the European Union. So, Europeans are shockingly no longer sure about three things: one, the extent to which the Trump Administration believes that its relationships with Russia and Europe would be more difficult in NATO's absence; two, that one of the objectives of US trade policies is not the European Union's destruction; and, three, that the EU would be able to count on the US in times of major crisis, which was a key part of the magic of NATO.

Allison Nathan: So, why isn't this a breaking point?

Ivan Krastev: First of all, it's one thing to understand that your marriage is not working in the same way as before, but it's completely different to imagine what divorced life would look like. Europe is too dependent on the US not only for its security, but also technologically and economically, to prefer the divorced life today. And the stakes are too high to find out what such a life would look like, with the very survival of the European project on the line. Remember that the European Union was largely an American project. In the moment of crisis, Americans played a critical role in securing European unity.

Second, some Europeans believe that Trump's actions don't reflect the consensus among American society or the elite, and point to the walking back of the potential use of military force in Greenland as evidence of the constraints Trump faces. So, Europeans believe that when they stand against Trump on this issue, they are not standing against America. Third, while Trump has strong allies within some European countries on issues like immigration, they have not sided with him on Greenland because no European leader can convince their voters that the US getting its way in Greenland is in Europe's best interest. It's noteworthy that the European Parliament stalled ratification of the US-EU trade deal, and all MPs, from the mainstream parties to the nationalistic right, applauded the decision to stand behind Denmark and Greenland.

Now, this is largely about territory for both sides. But while Trump views land from the lens of a real estate manager, European countries, especially small ones, view land as a critically important part of their national sovereignty, and the idea that small countries don't matter is not good news for anyone in Europe. So, European leaders agree that caving on Greenland would be disastrous for Europe and its credibility. In that light, this issue has strengthened European unity. And Europe's weakness paradoxically means that it cannot *not* retaliate when it comes to this red line, because its weakness would be exposed. My sense is that Trump initially miscalculated on this point but now seems to understand it.

Allison Nathan: How could US-Europe relations evolve?

Ivan Krastev: The relationship looks to be headed for a rocky period, but not a divorce. For Europe, while there is much talk about finding new allies, aligning with others won't be a quick or easy process. The European Parliament essentially blocking the EU-Mercosur trade deal shows that too many competing interests exist among EU countries. And I'm skeptical that Europe and China will suddenly have a smoother relationship just because the US-Europe marriage has hit hard times. Instead, Europe will be focused on showing the US that Europe matters, not only because it can be helpful to the US, but also because it can be harmful. One action Europe can take to show that it is strong and serious about consolidating its economic power is making major progress on capital market integration.

Europe will also look inward in terms of focusing on building up its defenses, but here too it will encounter difficulties. Europe is struggling against time, and the blame doesn't lie with Trump. 15 years ago, then US Secretary of Defense Robert Gates warned Europe that the US would not pay for Europe's defense if it wasn't willing to increase its defense budgets, which Europeans chose to ignore, and now time is not on its side. But Europe's biggest problem is that budgets don't fight wars—people do. And Europe culturally is not prepared for the more hostile world it now finds itself in. That said, some European countries are taking their defense more seriously, and I expect closer cooperation among those countries, especially those closest to the Ukrainian crisis, as well as Germany and France in its capacity as the only European nuclear power. The Greenland issue has also brought the UK closer to the EU on defense, which will be critically important in bolstering European defense capabilities.

For the US, midterm elections loom large, which I suspect will continue to serve as a constraint on the Administration's actions. So, both the US and Europe will likely look for ways to de-escalate tensions and compromise, as has already begun. But the trust between the US and Europe has been broken, not just because of Greenland but also because of Ukraine in terms of US policy related to the conflict, and the broader US treatment of Europe. So, I don't expect the relationship to go back to the way it was even if, as we've discussed, the current moment doesn't prove to be a breaking point.

When geopolitics meets portfolios

Christian Mueller-Glissmann discusses how to navigate and hedge against geopolitical risk

Geopolitical risk has returned to the forefront following the recent developments around Venezuela, Iran, and Greenland, and is likely to remain there for some time as the US continues to reshape its geopolitical and economic relationships with the world. This has raised questions about how investors can protect themselves from geopolitical shocks, which can have material impacts on global growth, inflation, and sentiment, especially as the linkages between economies and markets have become increasingly complex. Rather than try to time geopolitical shocks, which can be difficult, we think robust portfolio construction and diversification should be investors' first line of defense.

Geopolitical shocks: rhyming, but seldom repeating

While geopolitical shocks are a key risk for portfolios, they tend to differ every time and are particularly difficult to position for as the timing and potential economic and market impacts are hard to anticipate. Geopolitical tensions can also linger for a long time before they escalate. To further complicate matters, geopolitical shocks often start out as local shocks but can extend globally and have lasting knock-on effects.

The broader macro backdrop also affects how markets react to geopolitical shocks. We find an imperfect relationship between the news-based Geopolitical Risk Index and equity volatility, with an increase in geopolitical tensions often leading to higher—but not extreme—volatility. Geopolitical risks alone have seldom pushed the VIX above 40. But the combination of such risks and an already-weak macro backdrop has often sent volatility sharply higher. A more resilient US economy and a Fed able to buffer negative growth shocks may help markets digest geopolitical shocks better today, though low volatility, elevated equity valuations, and bullish investor sentiment to start the year increase the risk of a volatility spike.

The individual asset impacts of geopolitical shocks can also vary. Historically, large geopolitical shocks like the Middle East wars have driven oil prices sharply higher, often resulting in a worsening global growth/inflation mix. By contrast, oil prices have remained in a downtrend amid the recent geopolitical developments due to excess supply, with the Dollar also weakening as investors have become more concerned about US assets.

Finally, the inherent unpredictability of geopolitical events, including the possibility of a reversal of the shock, creates material market timing risk. Equities on average have recovered and risen after geopolitical shocks. In some cases, uncertainty can decline rapidly as more information becomes available or a policy pivot occurs, which can drive sharp equity recoveries, as happened around "Liberation Day" last April, which showcased the difficulty of timing equity moves.

Protecting portfolios: diversify, don't time

Given these challenges, investors' first line of defense against geopolitical shocks should be robust portfolio construction and diversification. In a balanced multi-asset portfolio, bonds can provide a buffer for risky assets against unanticipated growth

shocks. However, bonds become a less effective hedge if the geopolitical shock increases inflation risk. US policy uncertainty and global fiscal risks have also weighed on bonds, with recent concerns about potential fiscal expansion in Japan increasing the upward pressure on long-dated bond yields globally. Long-dated bonds may provide a smaller buffer for risky assets going forward if fiscal risks persist.

Investors can also turn to other global safe haven assets like the Swiss Franc, which has become a more effective "risk-off" hedge vs. the Euro in periods of rising geopolitical risk. But it's important to keep in mind that while safe haven assets tend to outperform during risk-off episodes, they usually have lower returns through the cycle. So, as with bonds, investors need to consider the tradeoffs between risk reduction and returns.

Increased allocations to assets that could benefit from an escalating shock can also buffer geopolitical risk. Commodities like oil and gold have often been useful diversifiers in periods of geopolitical uncertainty, though their hedge value can vary depending on the drivers (if there are oil supply disruptions, for example) and again they tend to have relatively low risk-adjusted returns over longer horizons. That said, gold and precious metals more broadly have recently helped buffer both geopolitical and fiscal risks, with higher central bank demand further boosting their risk-adjusted returns. Defense stocks are another potential beneficiary of geopolitical shocks (see pg. 20).

Investors can also manage geopolitical risk by avoiding or reducing direct and indirect exposures. This could include avoiding the country/region facing elevated geopolitical risk or large energy consumers (in the event of sharply higher oil prices) as well as reducing indirect exposure from trade linkages. With global investor portfolios currently dominated by US assets due to their large weights in equity and bond benchmarks, investors could also increase home bias in portfolios to reduce FX risk, though the potential impacts of large shifts out of US assets on portfolio returns and risk should be carefully considered.

Finally, the fact that geopolitical shocks can reverse quickly suggests that options can be a valuable hedge. Hedging tail risk with options can undoubtedly be expensive, making prolonged exposure to risk management strategies that protect portfolios systematically unfeasible. That said, realized and implied volatility has been very low coming into the year, increasing the value of tactical portfolio overlay hedges.

So, for investors looking for ways to protect their portfolios from geopolitical shocks, the message is: diversify, don't time. Specifically, we recommend more regional, sector, and style diversification relative to global benchmarks and like safe assets such as gold and selectively hedging Dollar risk. With the volatility reset, we also like shorter-dated rates receivers as a hedge for major growth shocks, credit payer spreads/VIX call spreads to hedge corrections, and longer-dated calls to maintain overweight equity allocations despite the potential for more geopolitical shocks.

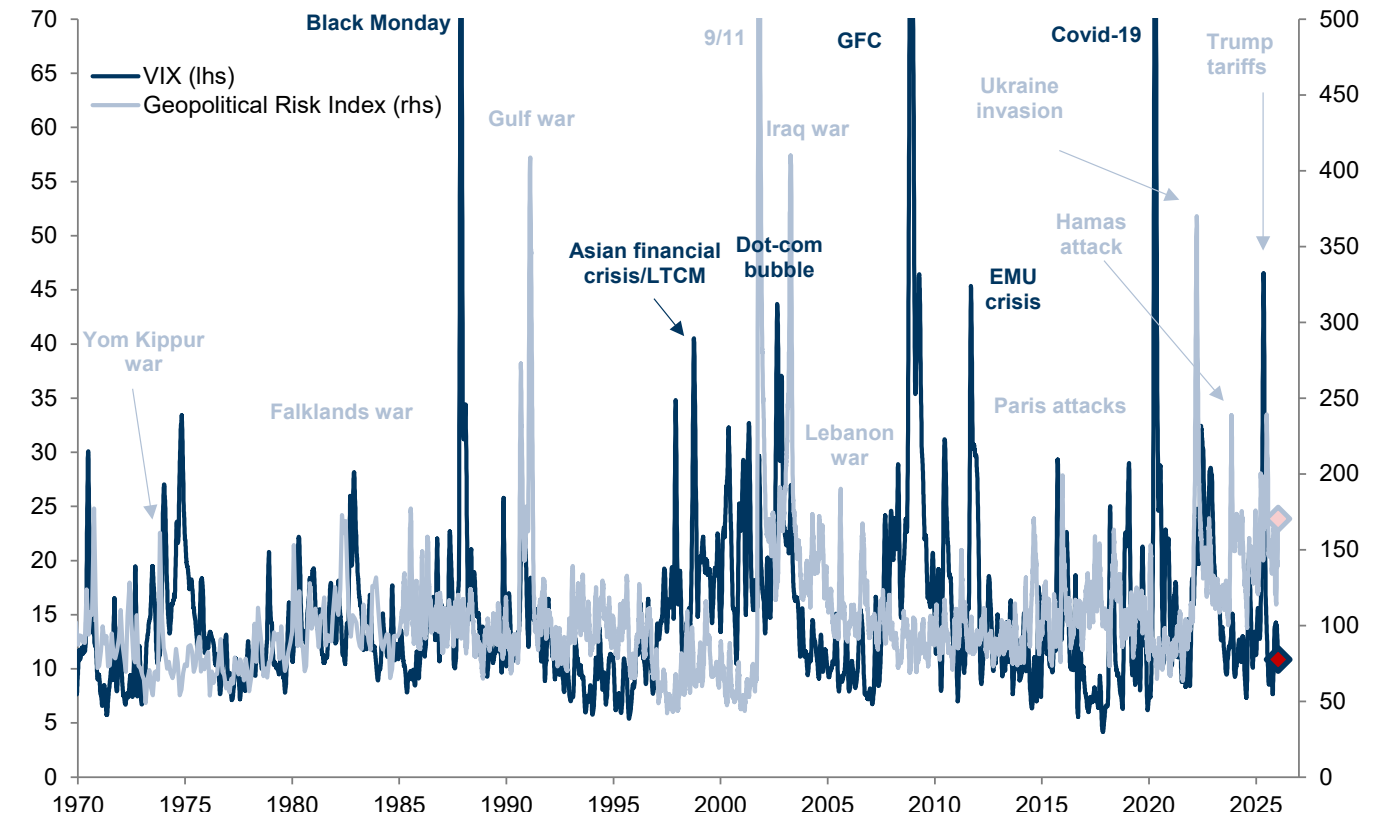
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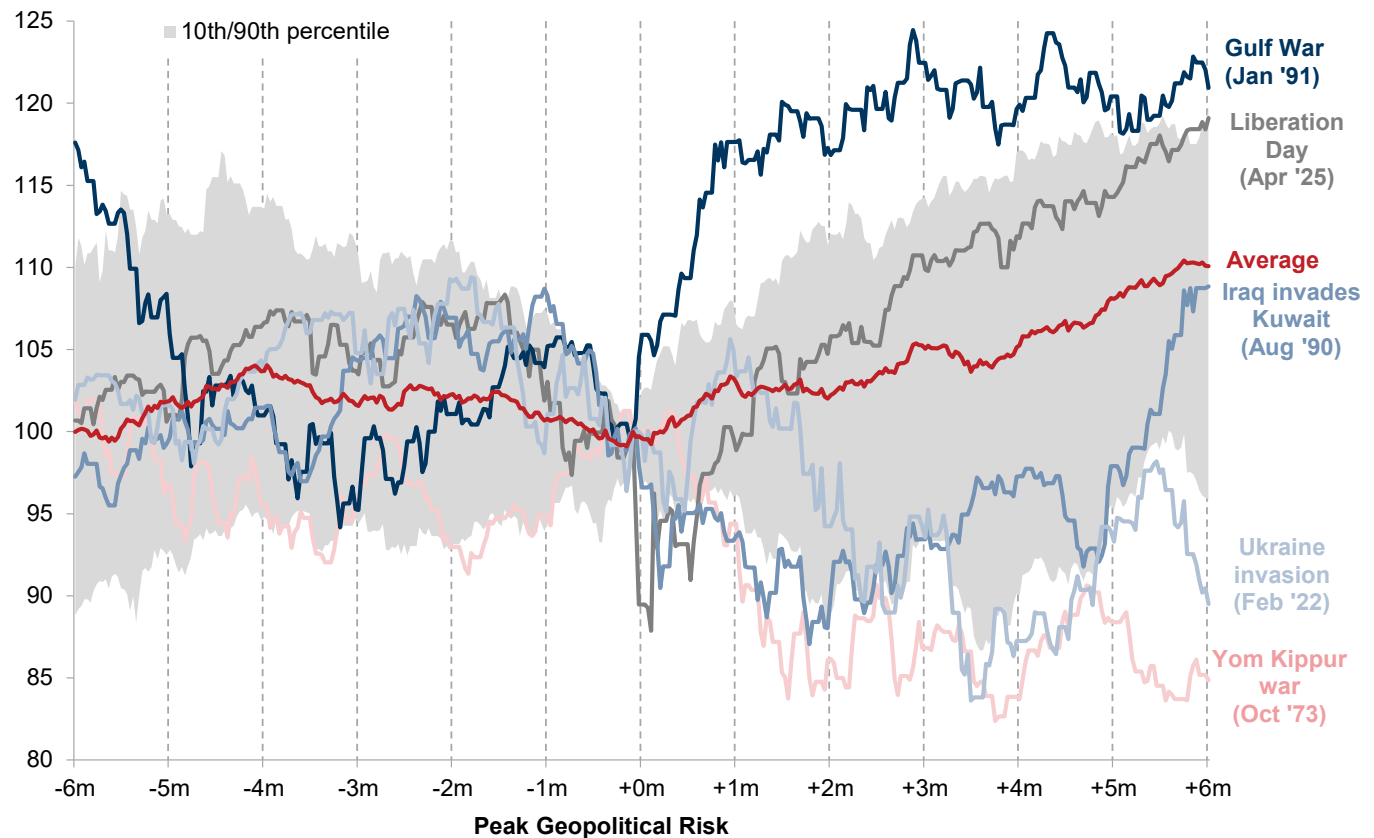
The relationship between geopolitical risk and equity volatility is mixed

VIX Index (lhs) vs. Geopolitical Risk Index (rhs), 1-month average



Timing equities around geopolitical shocks has been very difficult

Equity response to geopolitical shocks since 1970 (15 geopolitical shocks included)



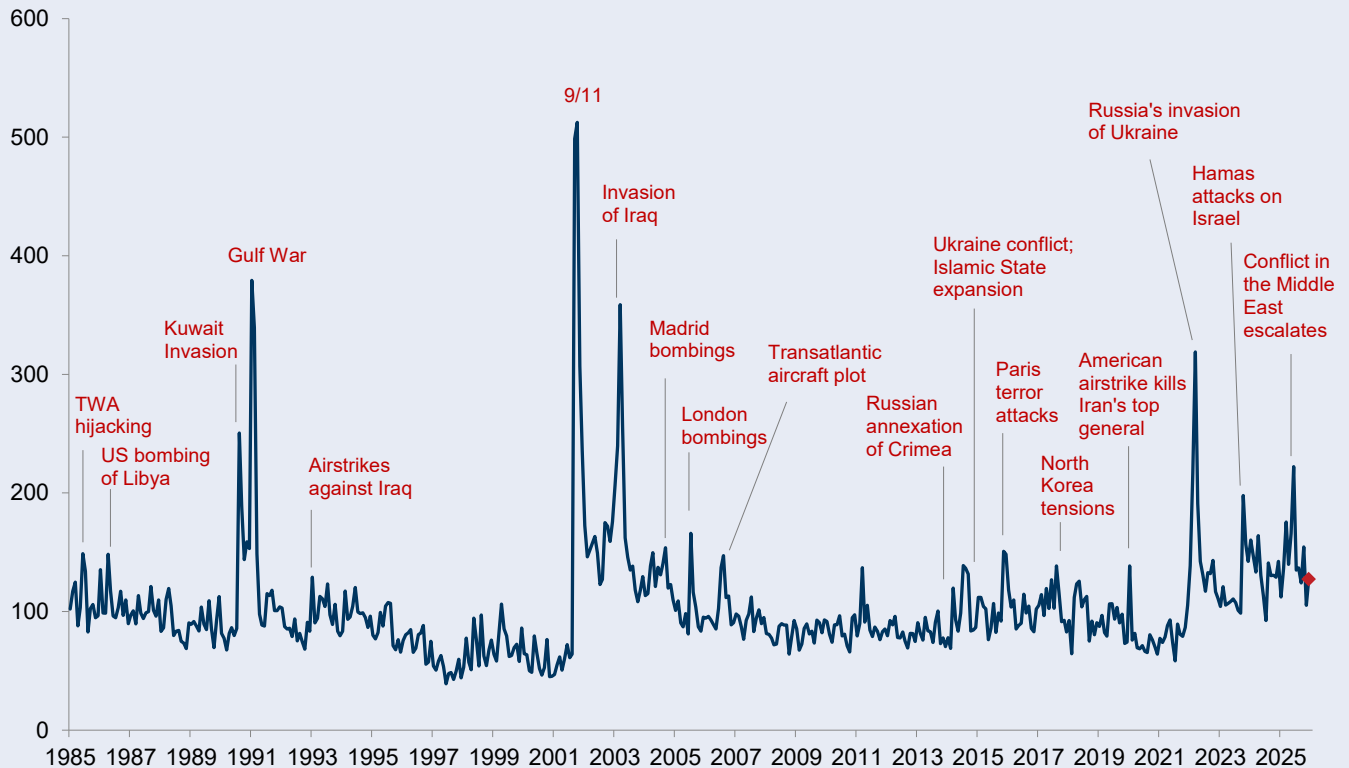
Source: Datastream, Bloomberg, Dario Caldara and Matteo Iacoviello, Goldman Sachs GIR.

Geopolitical risk through the years

Geopolitical tensions, which take many different forms, are difficult to measure. One proxy for assessing the geopolitical environment is the news-based Geopolitical Risk Index developed by economists from the Federal Reserve Board.

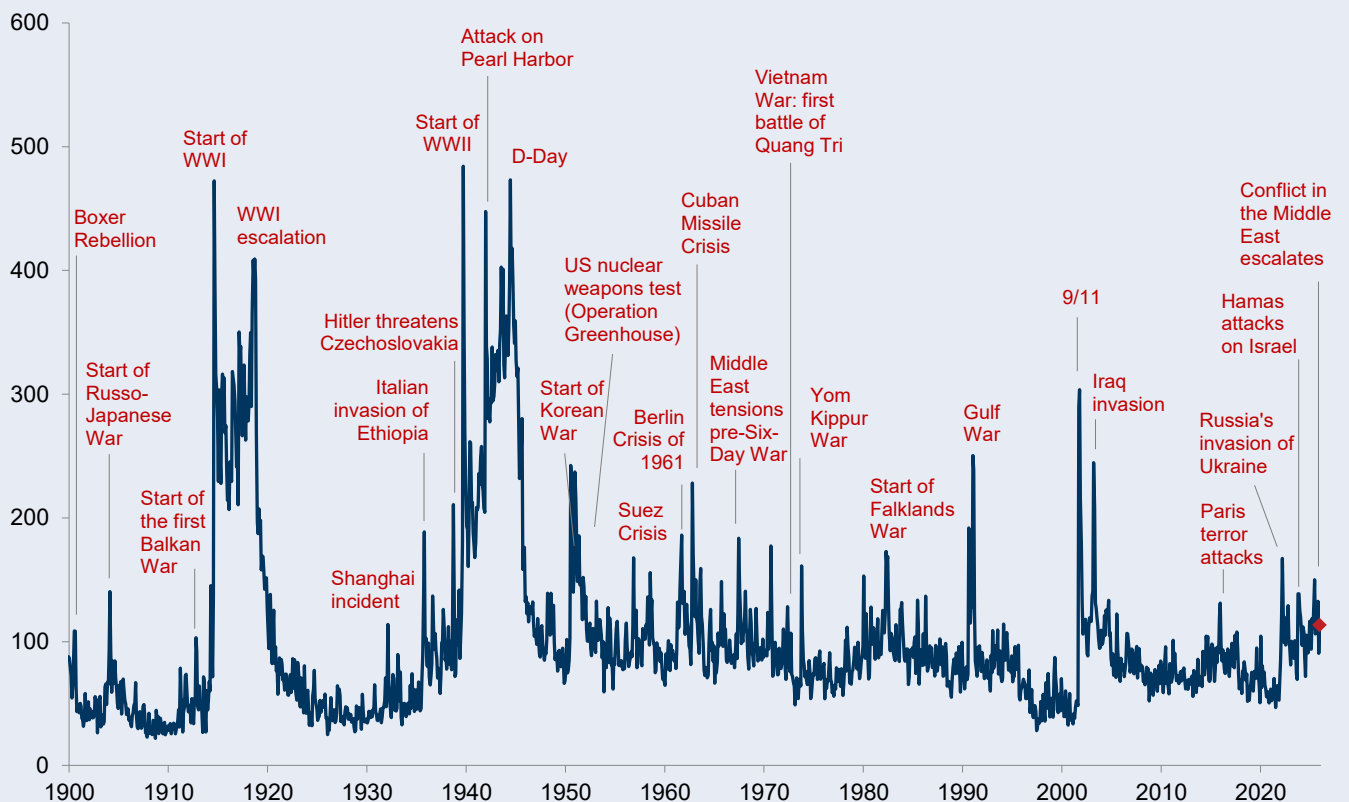
1985-Present

Geopolitical Risk Index, 1900-2019=100



1900-Present

Geopolitical Risk Index, 1985-2019=100



Note: The index from 1985 on counts the number of articles in 11 US, UK, and Canadian newspapers mentioning phrases related to geopolitical tensions. The index from 1900 on performs the same analysis using the archives of three newspapers: the New York Times, the Wall Street Journal, and the Financial Times. The choice of newspapers for both indices implies a measure of geopolitical risk as covered by the Anglo-Saxon press. See [here](#) for more information.

Source: [Dario Caldara and Matteo Iacoviello](#), Federal Reserve Board, Goldman Sachs GIR.

EM assets: continued resilience

Kamakshya Trivedi discusses the outlook for EM assets amid elevated geopolitical risk

Geopolitical risks have dominated headlines in 2026, from developments in Venezuela and Iran to Greenland. Despite that, EM equities have started the year strong and have comfortably outpaced DM equity markets, EM currencies are among the top performers year-to-date, and EM rates and spreads have held on. We think this is indicative of the resilient upside that we expect out of EM this year. Indeed, despite heightened geopolitical risks and several upcoming elections, our global macro outlook of sturdy growth, low inflation, and further Fed cuts is a friendly one for EM assets, even if less intensely so than in 2025. Better local EM fundamentals and still-low global investor allocations should also support returns in 2026.

Limited spillovers from Venezuela

Venezuelan developments—while highly pertinent from a geopolitical standpoint—have had limited spillovers in part because of limited exposure in EM investor portfolios. Political shifts can create significant volatility in local markets—as seen in Argentina last year and the political ructions in Türkiye before that—but their relatively small share of aggregate EM portfolios typically limits any impact. Equally, the improvement in EM fundamentals in recent years—solid growth, further inflation normalization, and scope for rate cuts as well as contained fiscal and current account positions—has helped insulate the broader EM complex from political and geopolitical shocks.

A positive returns story for EM assets

We continue to expect positive returns for EM assets this year amid a friendly global macro backdrop. In particular, we expect EM equities to outperform DM equities again, though we look for more moderate returns of ~15% (vs. last year's record 30% gains) driven primarily by solid earnings growth (19%/12% in 2026/2027) spread across the EM geographies. In EM FX, after the year of carry in 2025, we think that cyclical exposure—via both equity and commodity price linkages—will be as important as carry levels in 2026 for relative spot returns. This includes ZAR, CLP, and KRW, which have high betas to cyclical pricing and to the appreciation we expect in the CNY anchor and remain among the more undervalued currencies vs. USD. In EM fixed income, the scope for deeper cutting cycles in EM high-yielders (like Hungary and Brazil) should provide the best opportunities in local rates. We remain defensive on EM sovereign credit, with total returns likely to be largely a function of yield given our expectation of a modest widening in very tight spreads.

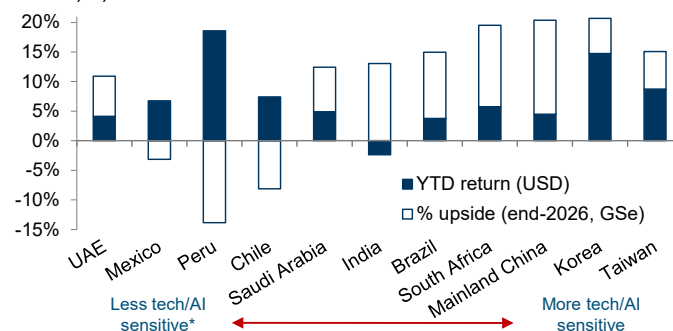
EM assets as diversifiers

EM assets can also provide an attractive opportunity for investors given their role as diversifiers. In particular, EM assets can help balance AI exposure amid growing investor concerns about valuations. For example, within EM, while we continue to favor the tech-sensitive equity markets of Korea, Taiwan (strong semiconductor earnings), and Mainland China (AI-driven growth and external market share gains), we prefer rotating some portfolio weight into the more domestic-oriented equity markets of South Africa (where a cyclical growth recovery and easing rates should help laggard domestic sectors catch up to the miners), India (with its mass-consumption revival), and

Brazil (where rate cuts should drive further gains).

EM has seen a broad-based rally this year

Returns, %, USD



*MSCI weight in tech/hardware, semis, internet, software, media/entertainment.
Source: FactSet, Goldman Sachs GIR.

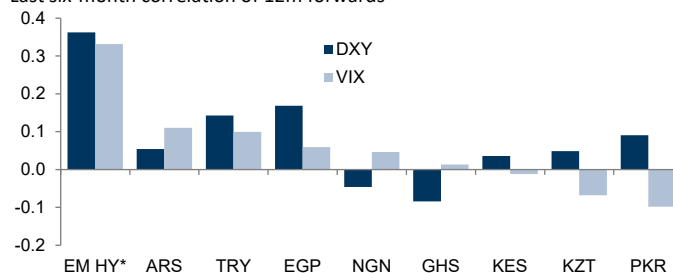
Uncorrelated risks and returns on the EM Frontier

Frontier markets also provide a source of returns that are relatively uncorrelated to the broader macro backdrop (although they come with their own risks, as Venezuela shows). Amid renewed geopolitical risks and the potential for higher volatility this year, we think that Frontier currencies continue to represent an attractive investment proposition in a more range-bound Dollar environment owing to elevated carry levels together with low correlation to Dollar strength and global macro volatility relative to other EM counterparts.

In particular, given still-healthy GSDEER model valuations and our economists' constructive external outlook, we favor carry longs in CEEMEA, including EGP, NGN, and KES, where risks are skewed toward appreciation given historically high FX reserves. Specifically, a still-cautious central bank in Egypt, which could keep real rates elevated for longer than we previously expected, led us to recently extend the target on our short USD/EGP trade recommendation (opened last July). And in Türkiye, we expect TRY monthly total returns to remain around 1-1.5% as a slight reduction in carry is offset by a decline in the pace of nominal depreciation amid slowing year-on-year inflation. The main risk to this constructive view is elevated positioning in still-relatively illiquid FX markets that could reverse sharply on a negative shock. Finally, we expect ARS to weaken along wider FX flotation bands given Argentine policymakers' objective to rapidly accumulate international reserves this year from critically low levels.

Frontier currencies are less correlated with USD and global vol

Last six-month correlation of 12m forwards



Note: We take log changes of all series before computing correlations.

*Median of BRL, COP, MXN, and ZAR.

Source: Goldman Sachs GIR.

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Q&A on defense sector geopolitical risk

We ask our equity analysts about the potential implications of elevated geopolitical risk for the defense sector.

US Defense

Noah Poponak, GS Equity Research

Q: Is the upside for defense stocks on the back of elevated geopolitical risk already fully priced in?

A: A lot of upside is priced in some stocks, but it is not fully priced across the sector. The US Defense sector has outperformed year-to-date on the back of geopolitical developments and statements from the White House regarding potentially much higher defense budgets. The Administration currently appears to want to further grow the US defense budget, but efforts are also underway to shift more financial and technology development risk onto the industrial base, creating a somewhat mixed picture for defense firms' financials. Product category prioritizations are also driving different growth profiles across verticals and individual companies in the sector. And with the US defense budget already at an all-time high and questions remaining about the appropriate level of US government spending, the annual level of defense spending will continue to be debated notwithstanding the clear focus on national security. Valuations at the sector level are on average above their historical midpoint but below their highs, with enough dispersion at the company level to still find value.

Q: Where does value remain in the sector?

A: We see value in stocks leveraged to product and vertical prioritizations that have growth and margin upside potential (rather than facing more financial risk as the Administration and Pentagon seek to shift more of the investment burden onto firms) and trade at relatively attractive valuations. We also see value in select Defense Tech companies.

Q: What key catalysts are you watching to realize the upside?

A: The White House recently discussed the possibility of a much higher US defense budget. The government typically makes its official budget request for the next fiscal year in the spring, which then goes through the congressional armed services committees through the summer, and should be finalized by the fall. Where that budget ultimately lands will go a long way toward informing future growth. We're also watching the moving pieces under the hood to continue to understand the verticals and products in higher demand. New Administration frameworks have been incrementally informing major changes in the Pentagon's defense procurement process, capital requirements, and risk sharing processes. And Defense budgets have been rising in other places, especially Europe, which will impact defense companies in those regions as well as US companies given that other countries procure military technologies from the US.

European Defense

Sam Burgess, GS Equity Research

Q: Is the upside for defense stocks on the back of elevated geopolitical risk already fully priced in?

A: No. While geopolitical tensions, alongside an increasingly fractious NATO alliance, have driven a sharp re-rating in European defense equities—with the GS EU Defense basket (GSSBDEFE) up over 20% year-to-date—we still see value in the sector. On growth-adjusted metrics, European defense valuations remain materially cheaper than US peers, with the discount currently at one of the highest points over the past three years. As a result, we would push back on the idea of a European defense bubble—the sector has clearly re-rated, but valuations are not uniformly stretched.

Q: What key catalysts are you watching to realize the upside?

A: As global geopolitical developments continue to emerge, we are watching for catalysts that reinforce defense spending as a necessary "insurance" outlay rather than a discretionary budget item for European policymakers. We are also keeping an eye on European defense budgets as well as disbursements from the ~€150bn European Security Action for Europe (SAFE) fund, a financial instrument providing loans to EU member states for defense procurement, which should drive meaningful orders for European primes. And we will be monitoring European primes' ability to expand capacity and meet demand in the new environment of increased military spending.

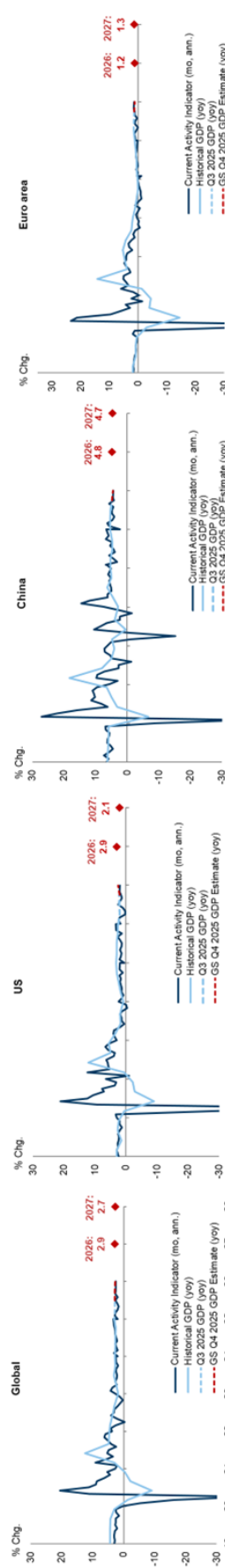
GS GIR: Macro at a glance

Watching

- **Globally**, we expect sturdy and above-consensus real GDP growth of 2.9% yoy in 2026 amid fading headwinds from US tariffs and rising real income growth. We expect global core inflation to decline to 2.1% by the end of 2026 as the tariff-driven boost fades and shelter and wage inflation fall further.
- **In the US**, we expect real GDP growth to rise to an above-consensus 2.5% on a Q4/Q4 basis in 2026 as tax cuts provide a tailwind to disposable incomes and financial conditions ease, but uncertainty around trade policy presents a potential risk. We expect core PCE inflation to decline to 2.1% yoy by the end of 2026 as tariff pass-through fades while underlying wage and shelter inflation trends improve. We expect the unemployment rate to end 2026 at 4.5%.
- **We expect the Fed** to deliver two 25bp cuts this year in June and September for a terminal rate range of 3-3.25%.
- **In the Euro area**, we expect real GDP growth of 1.2% yoy in 2026 reflecting real income gains and a fading fiscal policy drag but growing headwinds from more intense export competition with China. We expect core inflation to decline to 1.8% by end-2026 amid lower energy prices, a stronger Euro, and the potential for a negative inflation effect from Chinese trade rerouting.
- **We expect the ECB** to remain on hold for the foreseeable future given our and the ECB's forecast of better growth ahead and target-consistent inflation over the medium term.
- **In China**, we expect above-consensus real GDP growth of 4.8% yoy in 2026 given resilient export growth, continued government policy easing, and a decreasing drag from the ongoing property market downturn, although sluggish domestic demand should remain a headwind. On the inflation front, China's reflation process will likely remain gradual, but CPI/PP1 inflation should rise to 0.6%/-0.7% yoy this year owing to government policy efforts and a low base.
- **WATCH US POLICY, GEOPOLITICAL DEVELOPMENTS, AND FISCAL DYNAMICS.** Uncertainty about US policy remains elevated, especially around Greenland, tariffs, and a possible US government shutdown. Geopolitical developments more broadly pose risks worth tracking, especially as the situation in Venezuela develops, tension in the Middle East continues, a potential Russia-Ukraine peace deal remains uncertain, and US-China relations remain fragile. And fiscal pressures—including in the US, UK, France, and Japan—remain important to watch.

Goldman Sachs Global Investment Research.

Growth



Source: Haver Analytics, Goldman Sachs Global Investment Research.
Note: GS CAI is a measure of current growth. For more information on the methodology of the CAI please see "Technical Updates to Our Global CAIs," *Global Economics Comment*, Sep. 01, 2025.

Forecasts

Economics										Markets					Equities										
GDP growth (%)		2026		2027		Interest rates 10Yr (%)		Last	E2026	E2027	FX	Last	3m	12m	E2026		E2027		Returns (%)		YTD	E2026 P/E			
		GS (Q4/Q4)	Cons. (Q4/Q4)	GS (CY)	Cons. (CY)	GS (CY)	Cons. (CY)								GS	Cons.	GS	Cons.							
Policy rates (%)	Global	2.9	--	2.9	2.6	2.7	2.6	US	4.24	4.20	4.25	1.18	1.18	1.25	Price	7.600				S&P 500	9.9	1.0	22.6x		
	US	2.5	2.1	2.9	2.0	2.1	2.3	Germany	2.86	3.25	3.00	1.36	1.33	1.36	EPS	\$305	\$311	\$356	MXAPJ	12.0	5.7	15.2x			
	China	4.7	4.7	4.8	4.9	4.7	4.5	Japan	2.26	2.00	2.00	\$JPY	158	157	152	Growth	12%	15%	10%	Topix	7.4	6.5	18.1x		
	Euro area	1.4	1.4	1.2	1.4	1.3	1.2	UK	4.42	4.00	4.00	\$CNY	6.94	6.95	6.85					STOXX 600	2.7	2.8	15.4x		
Commodities		2026		2027		Commodities		Last	3m	12m	Credit (bp)	Last	2026	4Q26	2026		2027		Wage Tracker 2025 (%)						
		GS	Mkt.	GS	Mkt.	GS	Unemp. Rate (%)								CPI (%)	Unemp. Rate (%)	Q1	Q2	Q3	Q4					
Oil		3.13	3.23	3.13	3.33	3.13	3.33	Crude Oil, Brent (\$/bbl)	56	57	55	IG	72	85	90	US	2.4	4.5	1.9	4.5	4.1	3.9	3.7	3.4	
Gas		2.00	1.93	2.00	2.20	2.00	2.20	Nat Gas, NYMEX (\$/mmBtu)	5.28	3.75	4.00	USD	IG	72	85	90	US	2.4	4.5	1.9	4.5	4.1	3.9	3.7	3.4
Euro area		1.20	1.34	1.10	--	1.10	--	Nat Gas, TTF (EUR/MWh)	39.47	38	30	HY	256	295	315	Euro area	1.8	6.4	1.8	6.3	--	--	--	--	
China		1.00	1.22	1.50	1.60	1.50	1.60	Copper (\$/mt)	13,049	12,665	10,870	EUR	IG	85	95	99	China	0.6	--	--	--	--	--	--	--
Japan		1.00	1.22	1.50	1.60	1.50	1.60	Gold (\$/troy oz)	4,946	4,805	5,445	HY	257	288	315					--	--	--	--	--	--

Source: Bloomberg, Goldman Sachs Global Investment Research. For important disclosures, see the Disclosure Appendix or go to www.gs.com/research/hedoe.html.

Glossary of GS proprietary indices

Current Activity Indicator (CAI)

GS CAIs measure the growth signal in a broad range of weekly and monthly indicators, offering an alternative to Gross Domestic Product (GDP). GDP is an imperfect guide to current activity: In most countries, it is only available quarterly and is released with a substantial delay, and its initial estimates are often heavily revised. GDP also ignores important measures of real activity, such as employment and the purchasing managers' indexes (PMIs). All of these problems reduce the effectiveness of GDP for investment and policy decisions. Our CAIs aim to address GDP's shortcomings and provide a timelier read on the pace of growth.

For more, see our CAI page and Global Economics Comment: Technical Updates to Our Global CAIs.

Dynamic Equilibrium Exchange Rates (DEER)

The GSDEER framework establishes an equilibrium (or "fair") value of the real exchange rate based on relative productivity and terms-of-trade differentials.

For more, see our GSDEER page, Global Economics Paper No. 227: Finding Fair Value in EM FX, 26 January 2016, and Global Markets Analyst: A Look at Valuation Across G10 FX, 29 June 2017.

Financial Conditions Index (FCI)

GS FCIs gauge the "looseness" or "tightness" of financial conditions across the world's major economies, incorporating variables that directly affect spending on domestically produced goods and services. FCIs can provide valuable information about the economic growth outlook and the direct and indirect effects of monetary policy on real economic activity.

FCIs for the G10 economies are calculated as a weighted average of a policy rate, a long-term risk-free bond yield, a corporate credit spread, an equity price variable, and a trade-weighted exchange rate; the Euro area FCI also includes a sovereign credit spread. The weights mirror the effects of the financial variables on real GDP growth in our models over a one-year horizon. FCIs for emerging markets are calculated as a weighted average of a short-term interest rate, a long-term swap rate, a CDS spread, an equity price variable, a trade-weighted exchange rate, and—in economies with large foreign-currency-denominated debt stocks—a debt-weighted exchange rate index.

For more, see our FCI page, Global Economics Analyst: Our New G10 Financial Conditions Indices, 20 April 2017, and Global Economics Analyst: Tracking EM Financial Conditions – Our New FCIs, 6 October 2017.

Goldman Sachs Analyst Index (GSAI)

The US GSAI is based on a monthly survey of GS equity analysts to obtain their assessments of business conditions in the industries they follow. The results provide timely "bottom-up" information about US economic activity to supplement and cross-check our analysis of "top-down" data. Based on analysts' responses, we create a diffusion index for economic activity comparable to the ISM's indexes for activity in the manufacturing and nonmanufacturing sectors.

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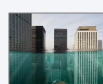
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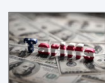
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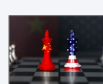
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Special Issue | December 15, 2025 | 3:00PM EST

GLOBAL MACRO RESEARCH

TOP^{of} MIND

2025: 4 THEMES IN CHARTS

Our special Top of Mind year-end edition
Features charts, quotes, and a crossword, per our tradition.
They tell the story of a year that was anything but boring,
In four themes that we found well worth exploring.

The year began with **tariff** talk galore,
As Donald Trump assumed the US presidency once more,
And upended decades of trade policy through significant action,
Which led to worries about a US economic contraction.

The other focus of the year remained **innovation**,
With AI still at the center, even at higher valuations.
These prompted fiercer bubble concerns than ever,
While stablecoins also had a moment, thanks to a key legislative endeavor.

Amid all that, **institutional credibility** became an obsession,
With fiscal sustainability, data reliability, and Fed independence all questioned.
These led to unease that higher bond yields may not fade,
And created a tailwind for the debasement trade.

Finally, important **geopolitical developments** continued to take shape,
As President Trump shook up the post-Cold War security landscape,
With significant implications for the Russia-Ukraine war and European defense,
While the race between the US and China for tech superiority grew even more intense.

We hope our favorite 2025 charts on these themes bring you cheer,
As well as our annual crossword—the answers to which this year's pieces will make clear.
We wish you a very happy holiday season, and thank you for this year's clicks,
We're looking forward to engaging with you again in 2026.

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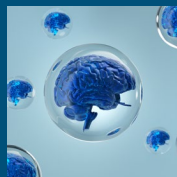


Which led to worries about a US economic contraction.

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2

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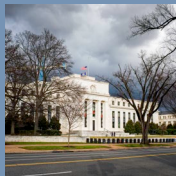
While stablecoins also had a moment, thanks to a key legislative endeavor.



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3

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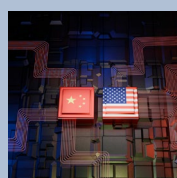
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4

Finally, important **geopolitical developments** continued to take shape,

As President Trump shook up the post-Cold War security landscape,

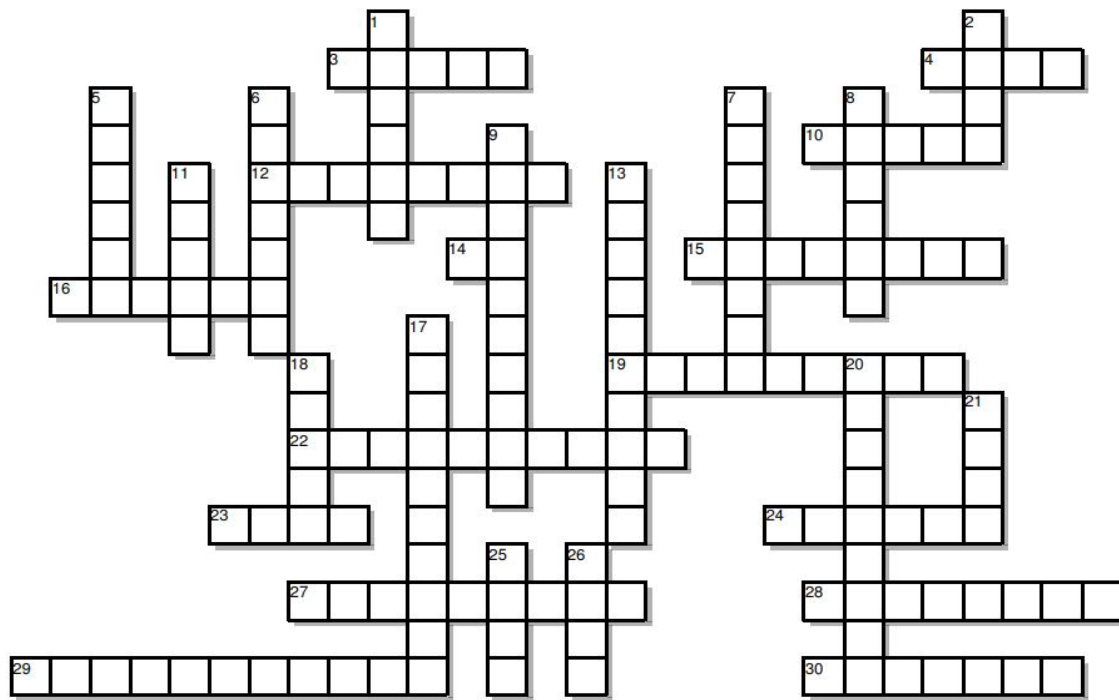
With significant implications for the Russia-Ukraine war and European defense,



While the race between the US and China for tech superiority grew even more intense.

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Revisiting 2025 themes, crossword-style



Across:

3. Former Chief of Britain's MI6 Sir Alex Younger argued that the world has transitioned from a rules- and institutions-based order to a strongman and _____-based order (Issue 137).
4. GS equity analyst Will Nance believes that the potential _____ savings from using stablecoins rather than traditional channels for remittances is overstated (Issue 141).
10. GS commodity strategist Daan Struyven explained the importance of diverging _____ capacity in the US-China tech race (Issue 144).
12. Erica Groshen, Former BLS Commissioner, explained that revisions are _____, not bugs, of the statistical system as producing statistics that don't get revised sacrifices either quality or timeliness (Issue 142).
14. Byron Deeter, Partner at Bessemer Venture Partners, argues that _____ has the power to reshape businesses in ways previously unimaginable (Issue 143).
15. GS equity analyst Eric Sheridan points out that unlike in the lead up to the Dot-Com bubble, the firms tied to the AI theme today have outsized levels of free cash flow and engage in stock _____ and pay dividends (Issue 143).
16. Former Acting US Comptroller of the Currency Brian Brooks says that the _____ Act will create a sense of safety in stablecoins, driving greater stablecoin adoption (Issue 141).
19. President Trump invoked the International _____ Economic Powers Act to impose tariffs on Canada and Mexico soon after his inauguration (Issue 136).
22. According to former Deputy US Trade Representative Jeff Gerrish, a key goal of President Trump's bold and broad use of tariffs is to address significant trade _____ (Issue 136).
23. Former Fed Vice Chair Richard Clarida clarified that while the Fed has "instrument" independence, it does not have "_____ " independence (Issue 139).
24. The US deficit and debt as a share of GDP are at their highest levels outside of _____ periods per the latest IMF data (Issue 140).
27. GS global economist Joseph Briggs finds global evidence that public _____ pressure on central banks results in higher inflation and long-end rates (Issue 139).
28. GS US economist Ronnie Walker's analysis of over thirty economic indicators suggests that US data quality has generally _____ over the last decade (Issue 142).
29. Nobel Laureate Paul Krugman was more concerned about the _____ around tariffs, rather than the scale of them, in assessing the risk of a potential tariff-induced recession (Issue 138).
30. According to MIT's Ricardo Caballero, one driver of lower equilibrium interest rates over time will be a rebuilding of the equity risk _____ (Issue 135).

Down:

1. Bridgewater's Ray Dalio argued that history has shown that unless currencies are effective stores of _____, they are eventually devalued and abandoned (Issue 140).
2. GS LatAm economist Alberto Ramos finds that despite decades of protectionist policies in Latin America, the performance of tradable sectors in these economies has been notoriously _____ (Issue 138).
5. GS equity strategist Peter Oppenheimer argues that we are not in an AI _____ ...yet (Issue 143).
6. GS European economist Filippo Taddei argued that the process to increase European _____ spending will be gradual and complicated (Issue 137).
7. In March 2025, President Trump attempted to fire two FTC commissioners without cause, which the dismissed commissioners argued violates the Humphrey's _____ precedent (Issue 139).
8. GS strategist Kamakshya Trivedi finds that the effect of fiscal expansion on the _____ is empirically mixed and typically depends on the state of the business cycle and monetary policy (Issue 140).
9. A key aspect of President Trump's trade policy has been the imposition of _____ tariffs (Issue 136).
11. Camille Grand, Former Assistant Secretary General for Defense Investment at NATO, argued that the primary driver of the potential sea change in European defense spending was not Trump, but rather this leader (Issue 137).
13. The University of California, Berkeley's Barry Eichengreen is concerned that the proliferation of stablecoins could violate the _____ of money, which is essential for economic stability (Issue 141).
17. PGIM's Greg Peters argued that bond market _____ will remain a key feature of the market (Issue 135).
18. DGA-Albright Stonebridge Group's Paul Triolo makes the case that if any country can achieve self-sufficiency in semiconductors, it would be _____ (Issue 144).
20. GS China economist Hui Shan finds that the economic impact of tariffs is _____ (Issue 138).
21. GS rates strategist Will Marshall argues that few corners of the market would be impacted as directly by a loss of confidence in official economic data than the _____ market (Issue 142).
25. GS US economists made the case that consensus around the level of the neutral rate looked too low in the last cycle but as of February it looked too _____ (Issue 135).
26. In the US-China tech race, China has leverage in its global dominance of _____ earths (Issue 144).

Puzzle made at www.puzzle-maker.com. Solutions on pg. 45.

In the words of our 2025 interviewees

"It's not the size of the [trade] policy shift, but the uncertainty around it that could cause a recession... [and] at this point, policy reversals may actually worsen the situation because they would enhance uncertainty."

– Paul Krugman, Nobel Laureate
(Issue 138, April 17)

"Two mechanisms will likely exert significant downward pressure on the equilibrium rate over time. The first is a rebuilding of the equity risk premium... The second mechanism is a global phenomenon known as 'indebted demand'."

– Ricardo Caballero, Professor, MIT
(Issue 135, February 5)

"The reality is that the primary driver of this sea change in European defense is not the behavior of Donald Trump, but the behavior of Vladimir Putin, who remains unthwarted and might even become further emboldened if the conflict in Ukraine ends in his favor."

– Camille Grand, ECFR, Former NATO Assistant Secretary General for Defense Investment
(Issue 137, March 13)

"This isn't a 'hope-and-hype' cycle like the Dot-Com era."

– Byron Deeter, Partner, Bessemer Venture Partners
(Issue 143, October 22)

"I don't disagree that the uncertainty effect could inflict ongoing damage to the economy. But I do think a pullback on tariff policy would nevertheless help stabilize conditions in the near term."

– Jan Hatzius, Head of Global Investment Research and Chief Economist, Goldman Sachs
(Issue 138, April 17)

"We should be very worried [about a 'heart attack' US debt crisis] as all the previously mentioned conditions and my indicators that reflect them point toward an impending crisis."

– Ray Dalio, Founder, Bridgewater Associates
(Issue 140, June 12)

"It is entirely possible that neither the US nor China emerge as the outright victor in the tech race."

– Mark Kennedy, Founding Director, Wahba Initiative for Strategic Competition, New York University's Development Research Institute
(Issue 144, December 4)

"It's important to understand that revisions are features, not bugs, of the statistical system. Producing statistics that never get revised is easy, but it requires sacrificing either quality or timeliness."

– Erica Groshen, Former US BLS Commissioner
(Issue 142, September 11)

"The Fed has what's known as 'instrument independence', which means that the Fed can raise or lower interest rates or buy and sell Treasuries without political interference... But the Fed does not have 'goal independence'. Rather, Congress created the Fed to achieve congressional goals, which have evolved over time."

– Richard Clarida, Former Vice Chair of the Federal Reserve Board of Governors
(Issue 139, May 15)

"Higher rates are the primary reason the fiscal outlook is so concerning. If rates were to fall sharply, I would become far less worried."

– Kenneth Rogoff, Professor, Harvard University
(Issue 140, June 12)

"The GENIUS Act establishes a supervisory system [that] will create a sense of safety in stablecoins, which will drive mass market adoption."

– Brian Brooks, Former US Acting Comptroller of the Currency, Chairman and CEO, Meridian Capital Group
(Issue 141, August 19)

"The Fed *should* bend to political pressure every now and then. We live in a democracy. We do not give technocrats perpetual power to follow their whims."

– John Cochrane, Senior Fellow, Hoover Institution at Stanford University
(Issue 139, May 15)

"Everyone needs to buckle up, because the President is just getting started [on tariffs] and what lies ahead will likely be even more unpredictable than during his first term."

– Jeff Gerrish, Former Deputy US Trade Representative for Asia, Europe, the Middle East, and Industrial Competitiveness
(Issue 136, February 26)

"Generative AI is still essentially autocomplete on steroids."

– Gary Marcus, Professor Emeritus, New York University
(Issue 143, October 22)

"Once the door is open for people to start questioning whether the heads of statistical agencies are motivated by anything other than a desire to measure the truth, it is very difficult to close it, even as leadership shifts."

– Alberto Cavallo, Professor, Harvard Business School
(Issue 142, September 11)

"The US has experimented with private monies—which is essentially what stablecoins are—in the past, often with disastrous consequences for financial stability. I'm concerned that stablecoins could follow a similar path."

– Barry Eichengreen, Professor, University of California, Berkeley
(Issue 141, August 19)

Note: All quotes came from interviews that appeared in GS Top of Mind reports in 2025.
Source: Goldman Sachs GIR.

1

*The year began with **tariff** talk galore,*

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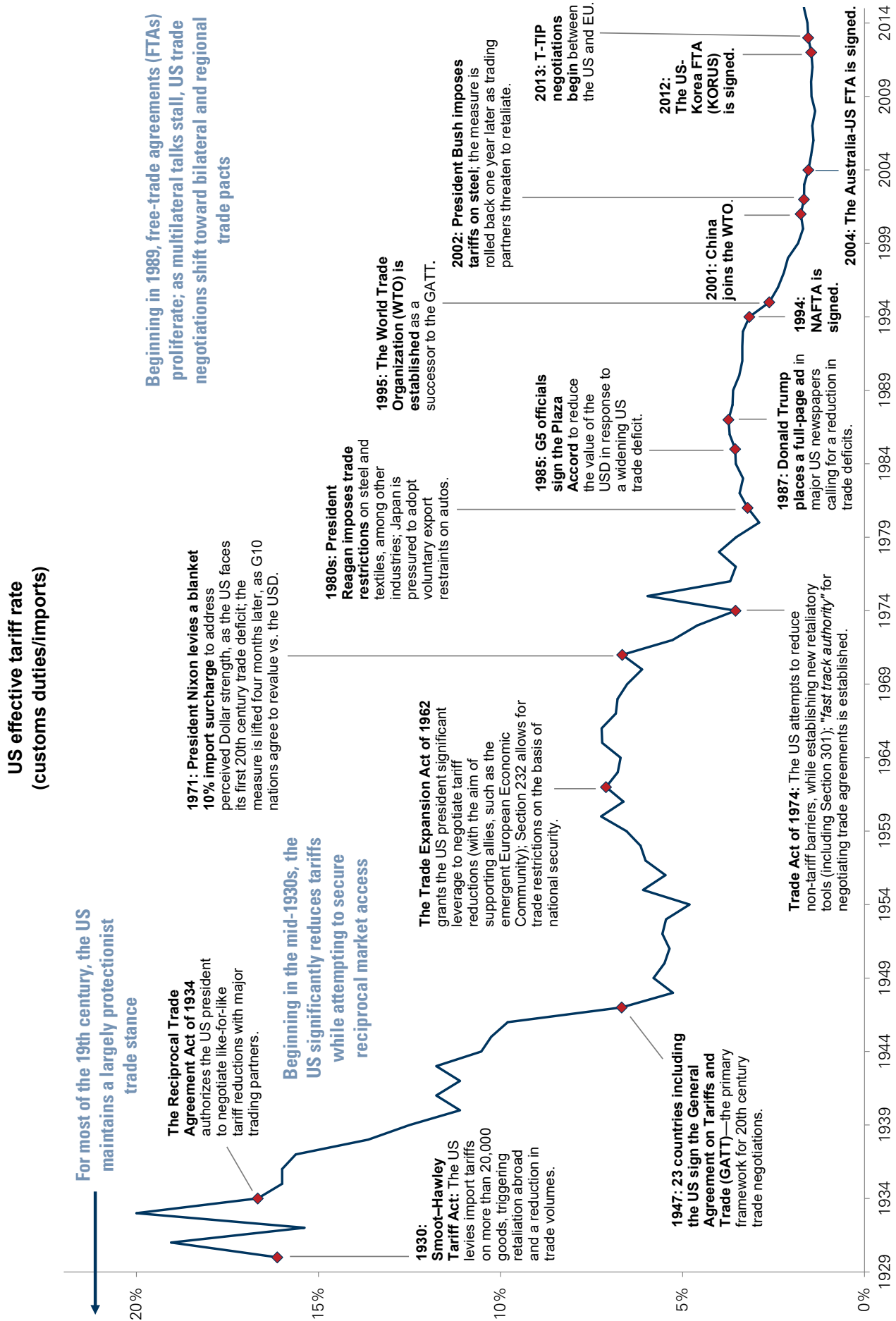
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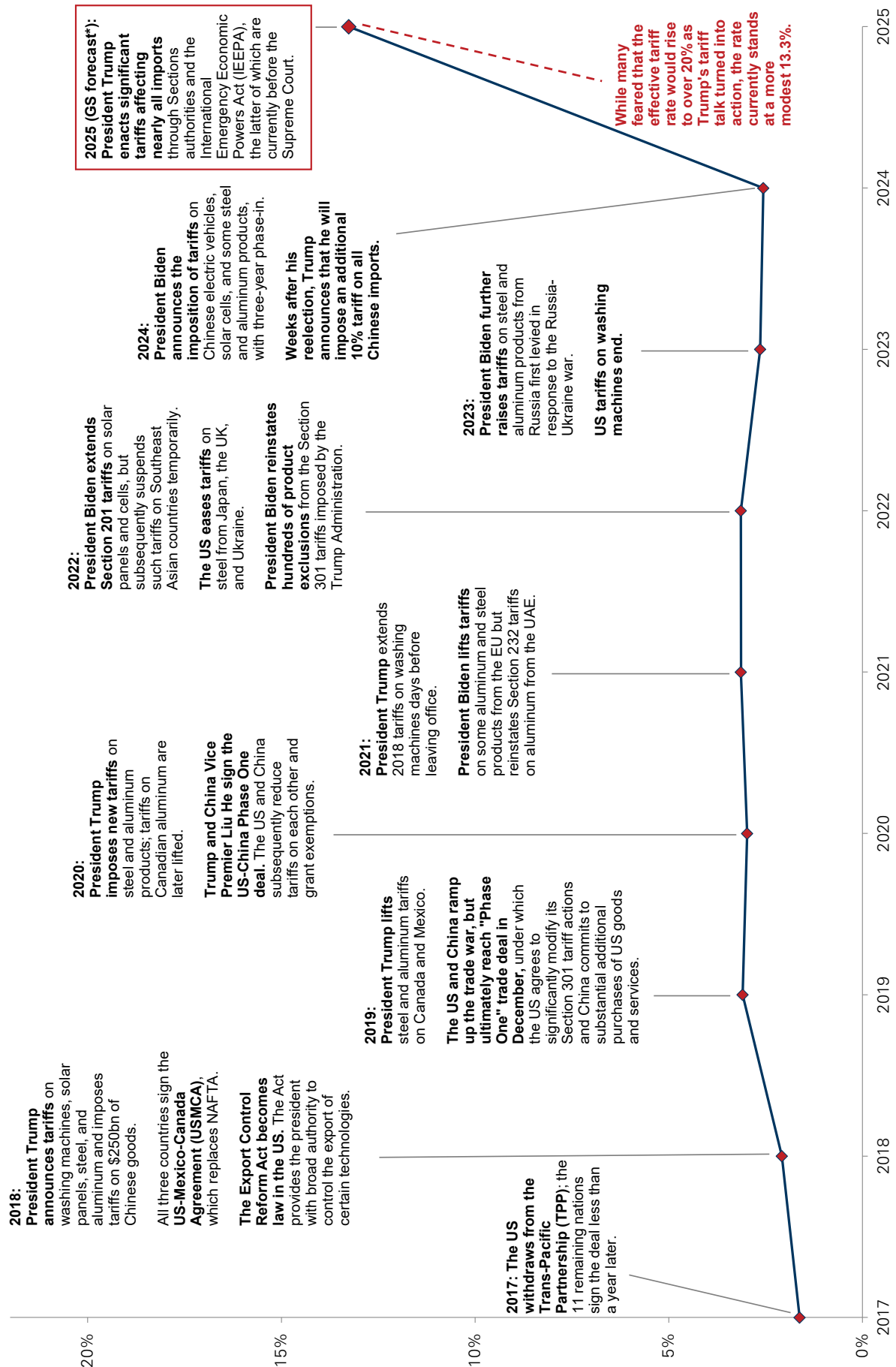
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The US tariff rate has sharply risen...



...but not as much as some had feared...

US effective tariff rate (customs duties/imports)

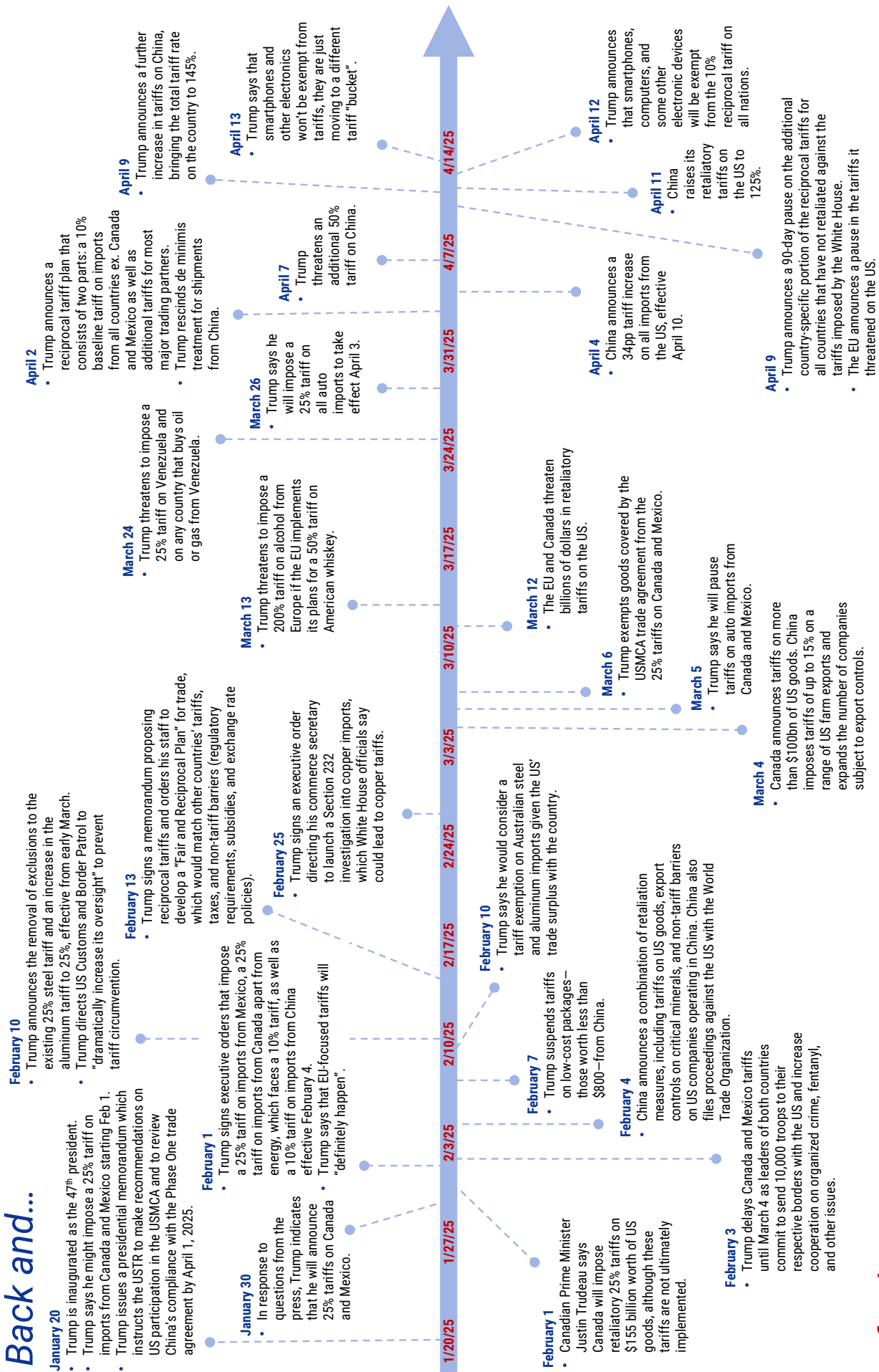


*Special thanks to GS US economist Elsie Peng for 2025 figure (excludes frontloading ahead of expected tariff hikes).

Source: US International Trade Commission, US Department of Commerce, WTO, Irwin, Douglas A., "Clashing over Commerce", Peterson Institute for International Economics, Goldman Sachs GIR.
Charts originally appeared in Issue 136: [Trump tariffs: mostly talk, or big action?](#)

...with lots of back and forth...

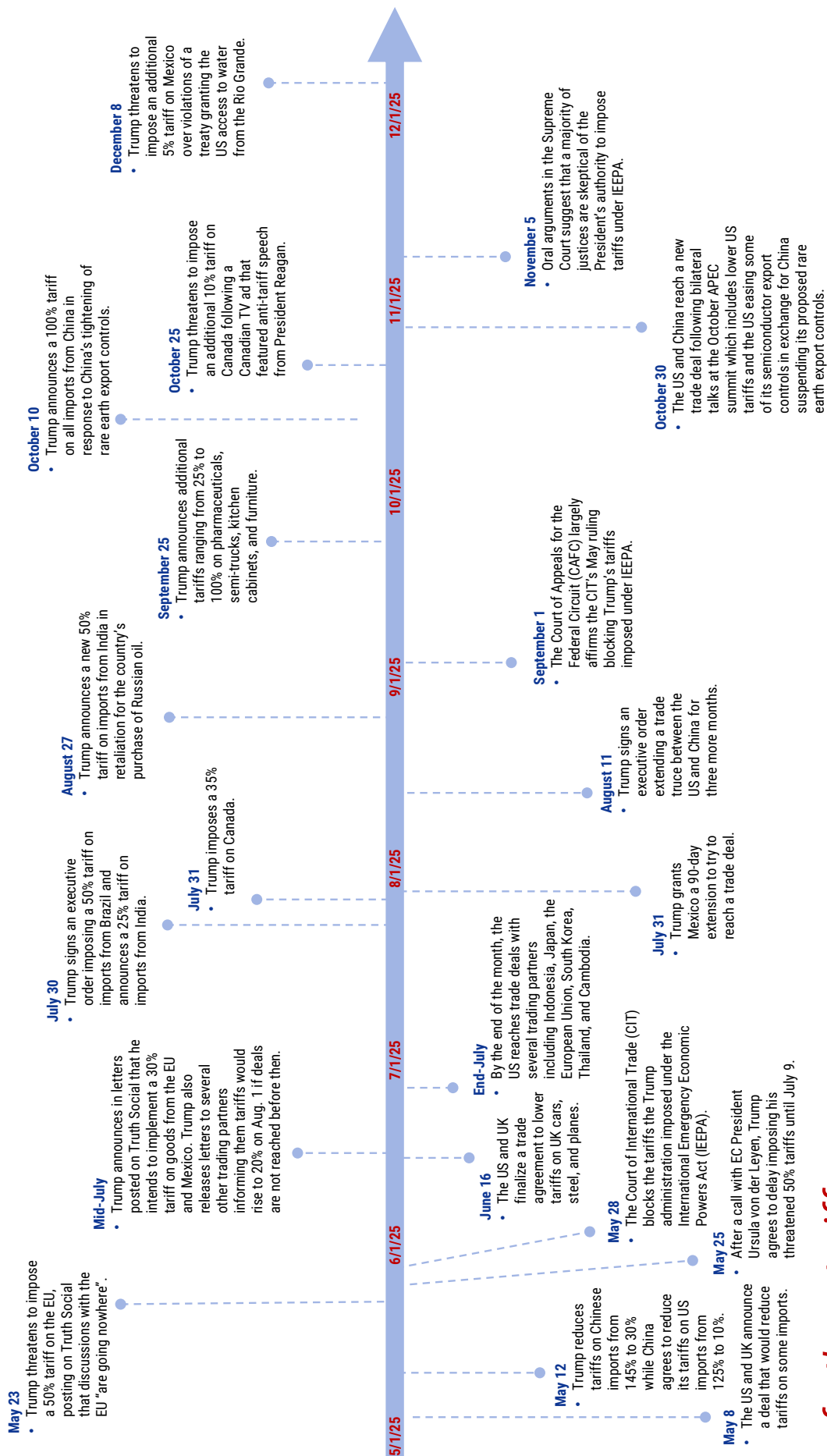
Back and...



...forth...

...resulting in many new trade deals...

...and back and...



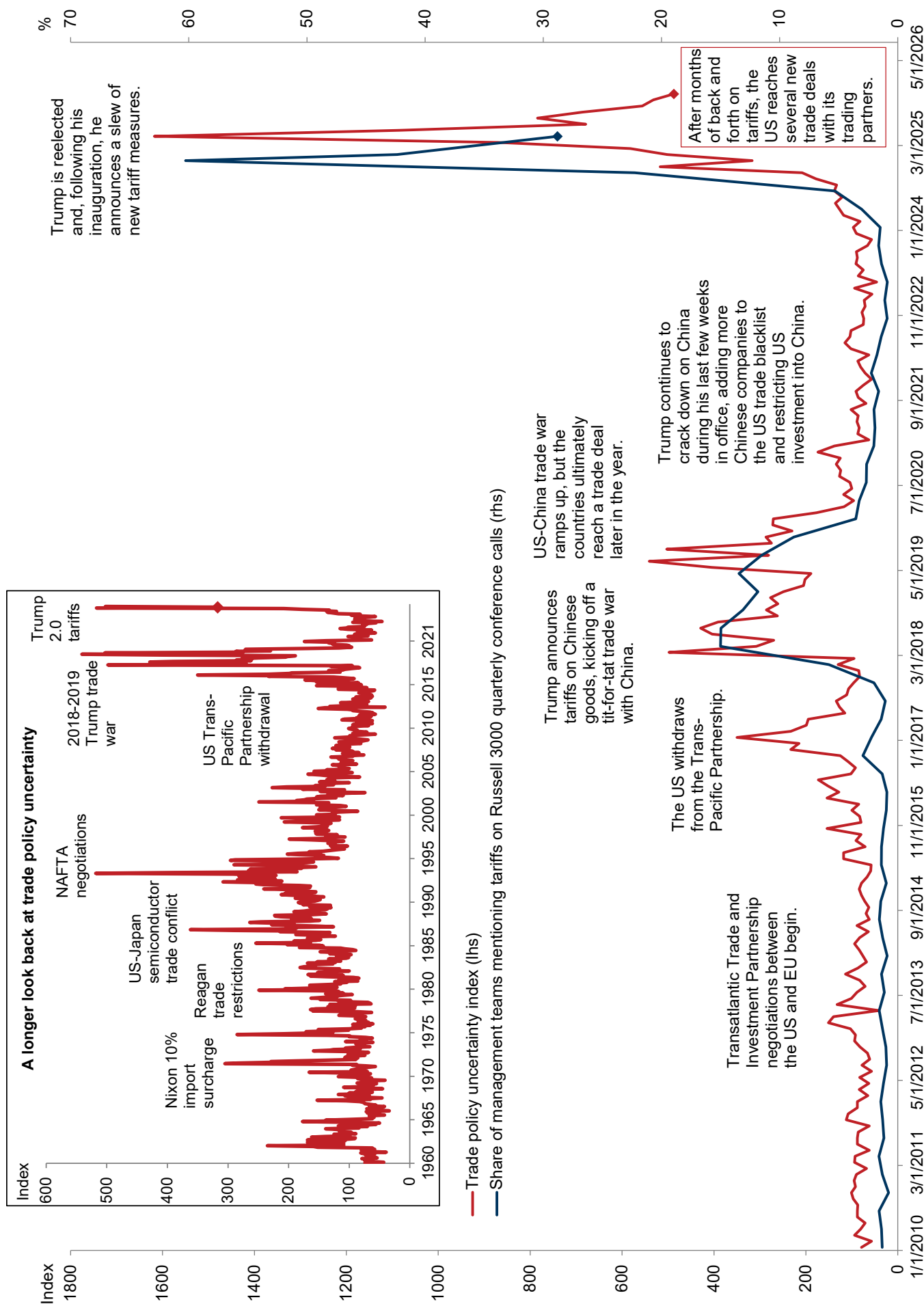
...forth on tariffs

Note: This does not constitute an exhaustive list of all tariff-related developments.

Source: White House, Truth Social, Bloomberg, WSJ, NYT, various news sources, compiled by Goldman Sachs GIR.

Timeline originally appeared in Issue 136: [Trump tariffs: mostly talk, or big action?](#)

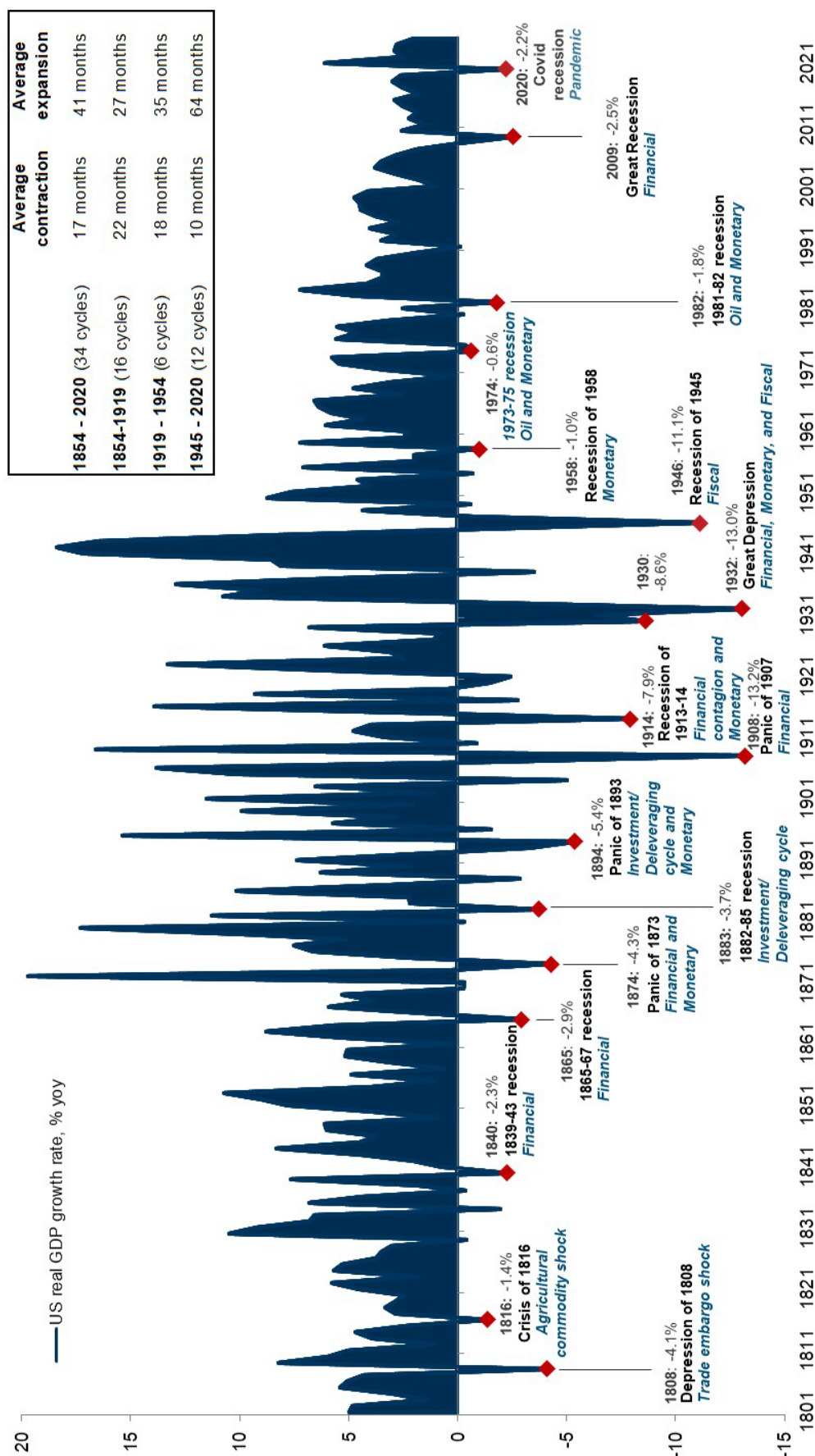
...reducing trade policy uncertainty...



Source: Caldara, Iacoviello, and Molloy, GS Data Works, Goldman Sachs GFR.
 Chart originally appeared in [Issue 136: Trump tariffs: mostly talk, or big action?](#)

...and easing recession concerns

A long history of US recessions



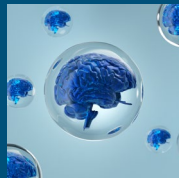
Note: Blue italics denote key contributors to the recession/depression rather than an exhaustive list of drivers, as classified in the June 2017 US Economics Analyst: The Next recession: Lessons from History and the January 2019 US Economics Analyst: Learning from a Century of US Recessions; red markers denote trough in yoy US growth in major recessions; US growth data for 1800-1979 from IMF "Fiscal Prudence and Profligacy".

Source: IMF, NBER, various news sources, Goldman Sachs GIR.

Chart originally appeared in [Issue 138: Tariff-induced recession risk](#).

2

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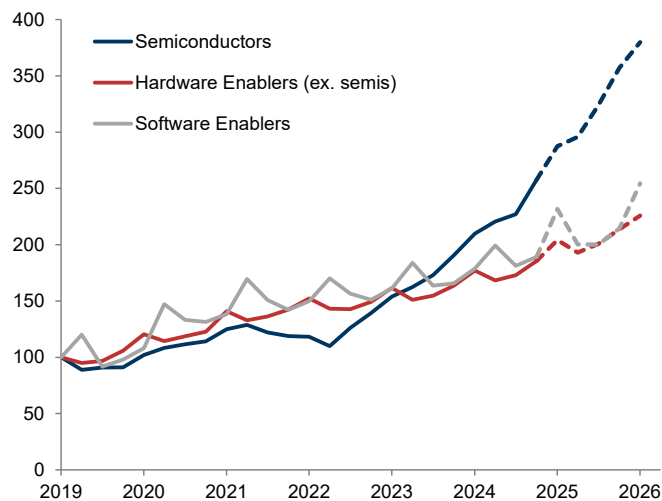


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AI investment growth remains strong...

AI investment growth remains strong...

Actual and forecasted revenues by AI-exposed sectors of Russell 3000, index, 4Q19=100

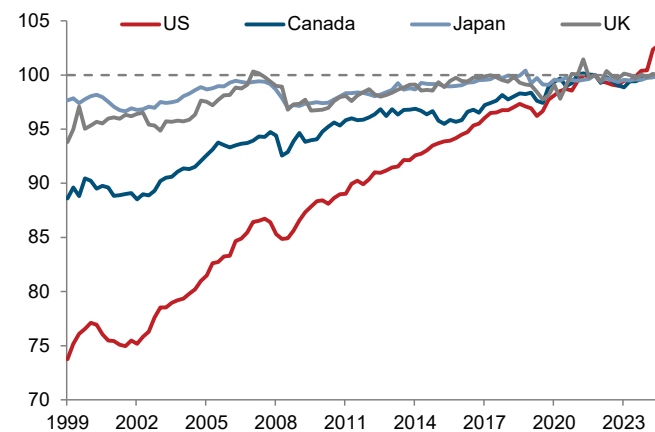


Note: Dashed lines in this chart indicate consensus forecasts.

Source: FactSet, Goldman Sachs GIR.

AI investment growth has been concentrated in the US...

AI-related investment in hardware*: national accounts, log index**, 3Q22=100

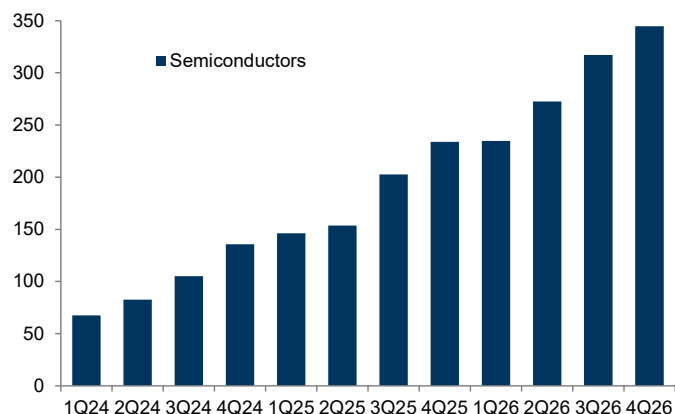


*Categories related to information and communication technologies (ICT) and communications equipment.

Source: Haver Analytics, Goldman Sachs GIR.

The market continues to upgrade its revenue expectations for semiconductor firms...

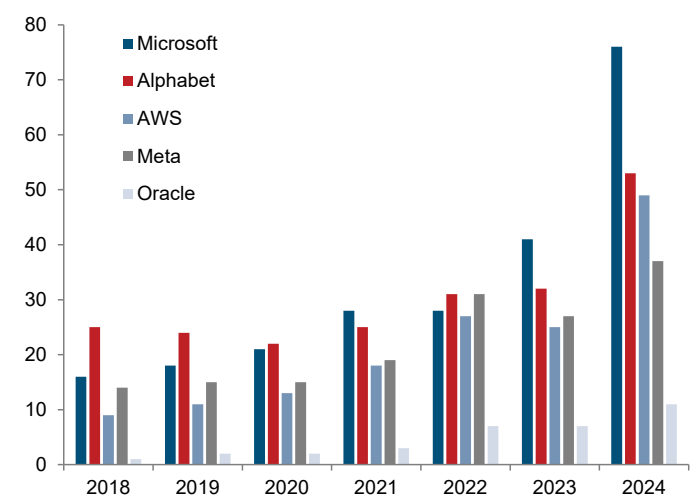
Change in Russell 3000 consensus revenue forecasts since March 2023, \$bn, annualized



Source: FactSet, Goldman Sachs GIR.

...with the hyperscalers having made significant investments in AI infrastructure

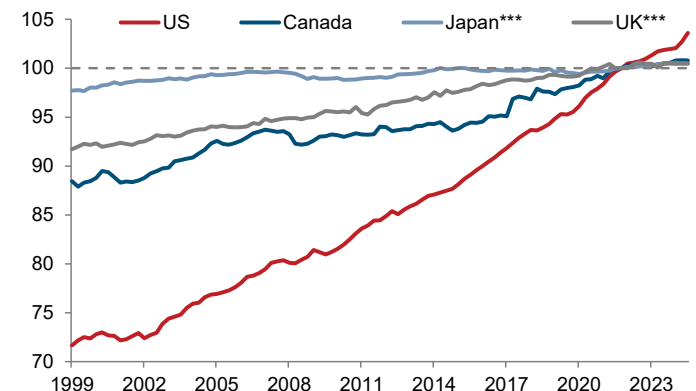
AI infrastructure capex spend by company, \$bn



Source: Company data, Goldman Sachs GIR.

...as US AI-related hardware and software investment has risen

AI-related investment in software: national accounts, log index**, 3Q22=100



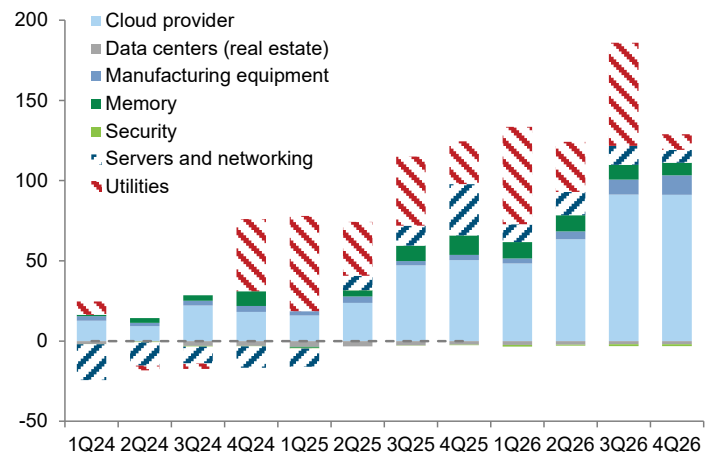
**Shown as log index because software investment grows by different exponential rates across countries. Steady growth in investment would appear as a line with a constant slope, while accelerating growth would appear as a line with an increasing slope.

***Intellectual property products category, which includes software.

Source: Haver Analytics, Goldman Sachs GIR.

...while the broader AI space is still seeing a relatively small upgrade

Change in Russell 3000 consensus revenue forecasts since March 2023, \$bn, annualized

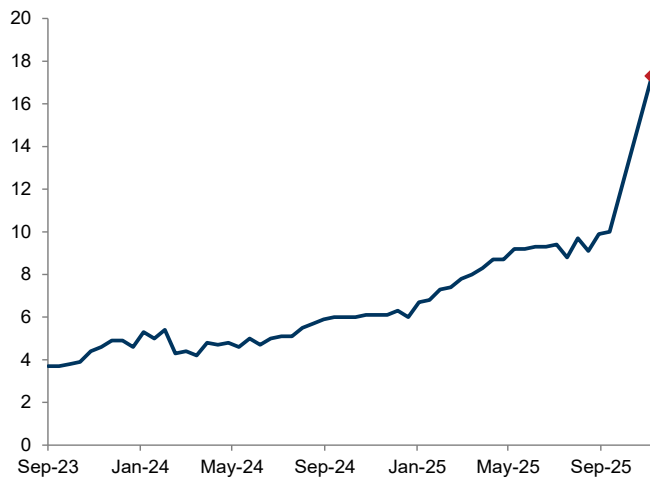


Source: FactSet, Goldman Sachs GIR.

...and AI adoption has risen

The pace of corporate AI adoption rose sharply in the recent quarter, though this most likely reflects a change in the wording of the Census Bureau survey question...

Economy-wide firm AI adoption rate, %



Source: Census Bureau, Goldman Sachs GIR.

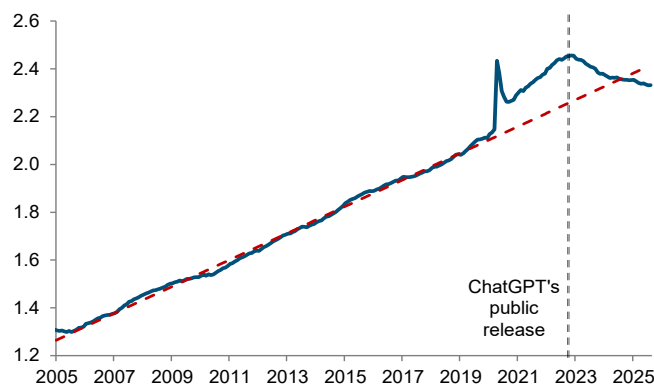
Other surveys also show increased corporate adoption...

Source	Date	Current Adoption
McKinsey	November 2025	88% of surveyed organizations are regularly using AI in at least one business function, up from 78% last year. However, most surveyed organizations are still in the experimentation/piloting phase of AI and nearly two-thirds have not begun scaling AI across the enterprise. 39% reported experimenting with AI agents specifically and the use of AI agents is most commonly reported among TMT and healthcare businesses.
Wharton Human-AI Research and GBK Collective	October 2025	Usage of generative AI among enterprise leaders has risen sharply this year and stands at 46% (+17pp yoy). The most common use cases include data analysis, document summarization, and document editing/writing.
Gartner	June 2025	45% of surveyed firms with high AI maturity reported that their AI initiatives remained in production for 3+ years to ensure sustained impact and value, compared to 20% in low-maturity organizations. 91% of leaders from high-maturity organizations have appointed dedicated AI leaders.
University of Maryland Center for Global Business	June 2025	58% of the business executives and middle managers surveyed reported that their firms had incorporated AI into their business operations in some capacity. 39% reported using generative AI, 30% reported using computer vision, and 27% reported using machine learning.
Paypal and Reimagine Main Street	June 2025	Out of nearly 1000 small businesses surveyed, 25% reported having already integrated AI into their daily operations while another 51% reported experimenting with AI (but not incorporating into regular operations).

Source: Goldman Sachs GIR.

Tech's share of total employment continues to decline relative to its long-run trend...

Tech sector* share of total employment, %



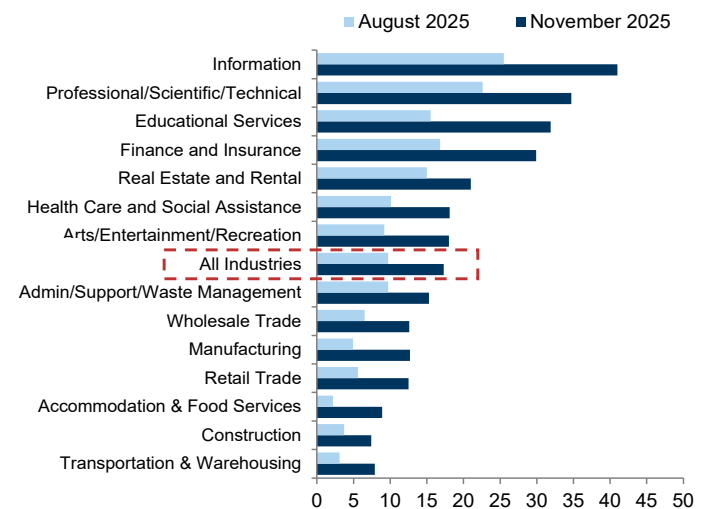
*Tech is the software publishers, data processing and related, web search and related, and computer systems design subsectors. Red line is 2007-2019 trend.
Source: Bureau of Labor Statistics, Haver Analytics, Goldman Sachs GIR.

Special thanks to GS global economists Joseph Briggs and Sarah Dong and equity analyst Jim Schneider and team for charts.

Exhibit originally appeared in [Issue 143: AI: in a bubble?](#)

...with 17.3% of companies across all industries having adopted AI, up significantly from around 10% in Q3

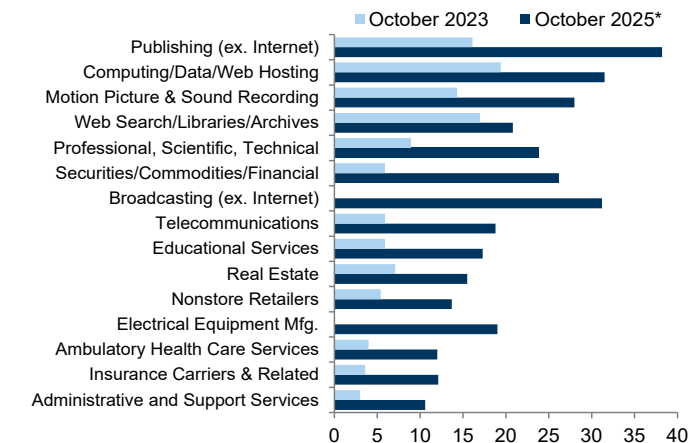
Share of US firms using AI by sector, %



Source: Census Bureau, Goldman Sachs GIR.

...with publishing and computing firms continuing to lead the way

Share of US firms using AI by sector, top 15 subsectors, %

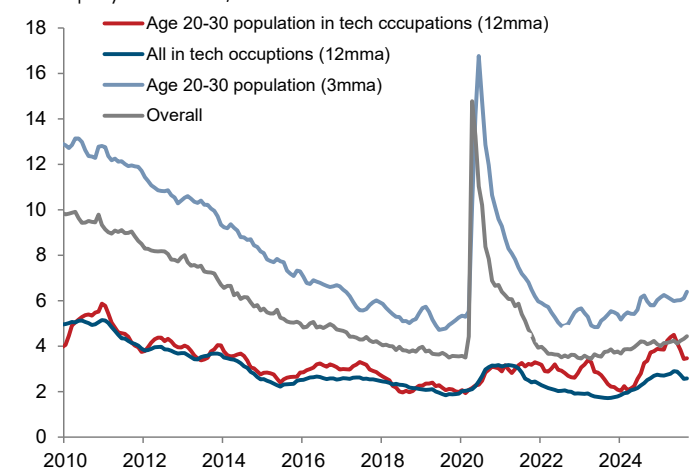


*Latest adoption observation is from early October due to data delays.

Source: Census Bureau, Goldman Sachs GIR.

...and youth unemployment in tech sectors remains elevated

Unemployment rate, %



Source: Bureau of Labor Statistics, iPUMS, Goldman Sachs GIR.

But ongoing developments...

AI's ups...

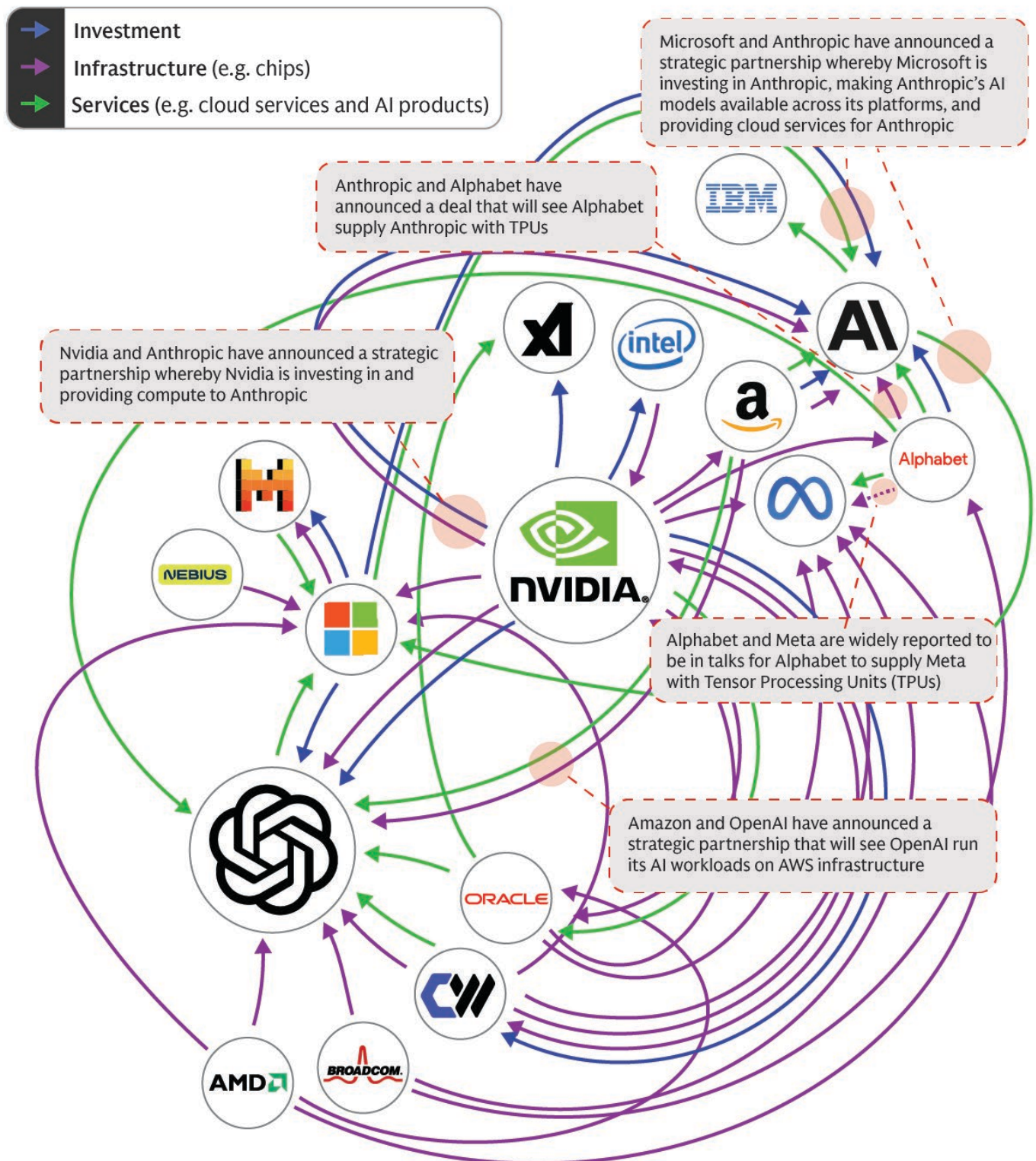


Note: This does not constitute an exhaustive list of all AI-related developments.

Source: BBC, OpenAI, tech.co, Google, WSJ, various news sources, compiled by Goldman Sachs GfR.

Timeline originally appeared in [Issue 143: AI: in a bubble?](#)

...have further increased AI circularity...



*Red dashed text and corresponding arrows represent deals announced since October 22.

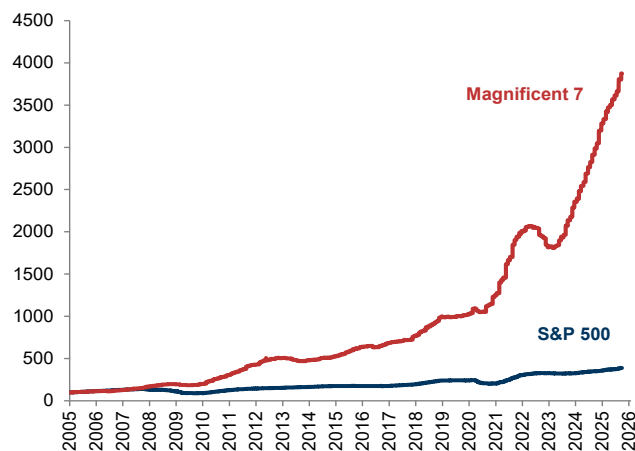
Note: Exhibit does not represent an exhaustive view of the AI ecosystem but rather the most prominent AI model companies, infrastructure providers, and hyperscalers. Source: Company announcements, various news sources, compiled by Goldman Sachs GIR.

Exhibit originally appeared in [Issue 143: AI: in a bubble?](#)

...leaving AI bubble concerns in focus

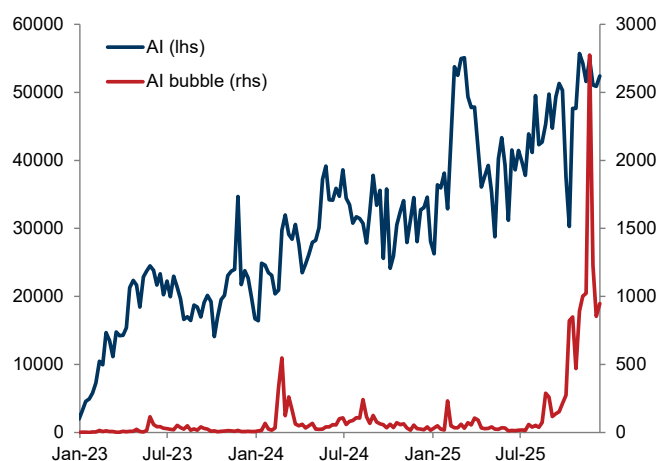
The outsized performance of the Magnificent 7 stocks on the back of AI enthusiasm has raised concerns about an AI bubble

Magnificent 7 and S&P 500, 12m trailing EPS, Jan-2005=100



Source: FactSet, Goldman Sachs GIR.

But news stories about an AI bubble show continued focus...



Source: Bloomberg, Goldman Sachs GIR

AI focus remains broad, with companies continuing to mention AI investment and opportunity...

AI-related mentions in 3Q2025 S&P 500 earnings calls,
scaled by number of mentions



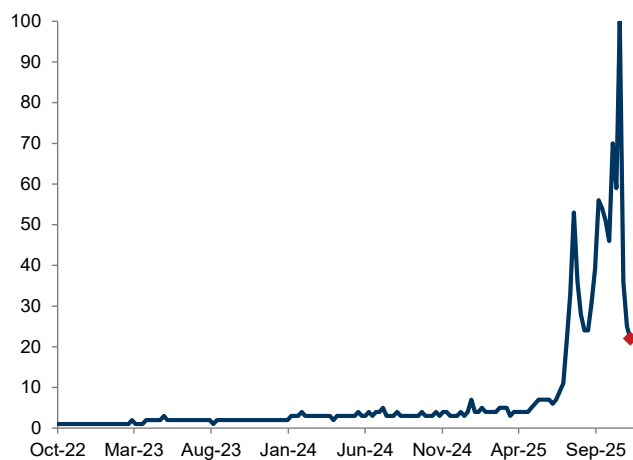
Source: LSEG, Goldman Sachs GIR.

Special thanks to Dan Duggan and GS Data Works for their AI-related analysis.

Exhibit originally appeared in [Issue 143: AI: in a bubble?](#)

Some concern may have abated, with Google searches for “AI bubble” declining from their recent peak

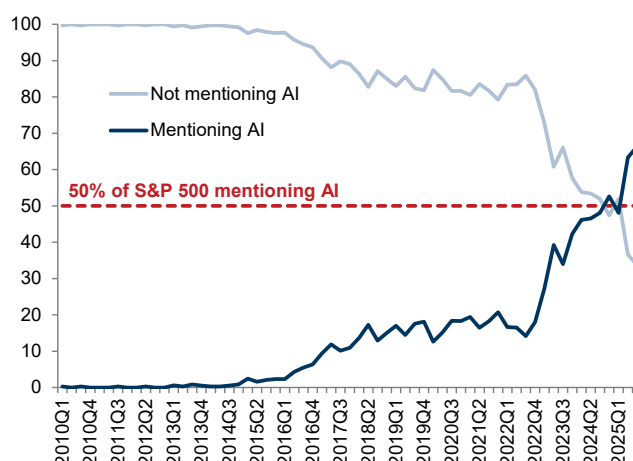
Google mentions of "AI bubble"



Source: Google, Goldman Sachs GIR.

...and AI mentions in earnings calls continue to grow, with over 50% of S&P 500 companies mentioning AI in Q3

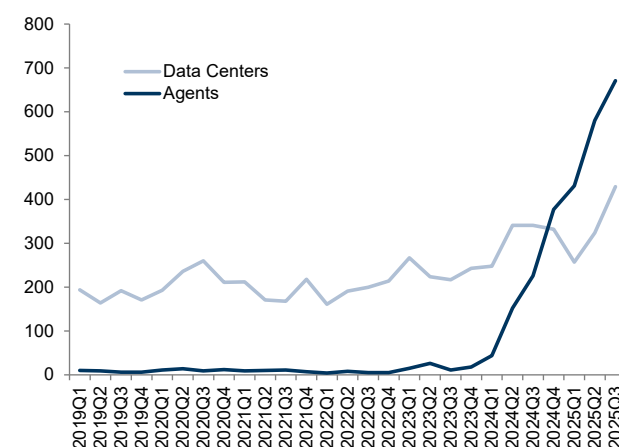
Share of S&P 500 firms mentioning AI in earnings calls, %



Source: LSEG, Goldman Sachs GIR.

...though tech companies have increasingly shifted their AI focus from infrastructure to the application phase, with rising emphasis on AI agents

Mentions in S&P 500 TMT/Telco earnings calls*



*Keyword counts from S&P 500 TMT and Telecommunications companies.

Source: LSEG, Goldman Sachs GIR.

Stablecoins also came to the fore...



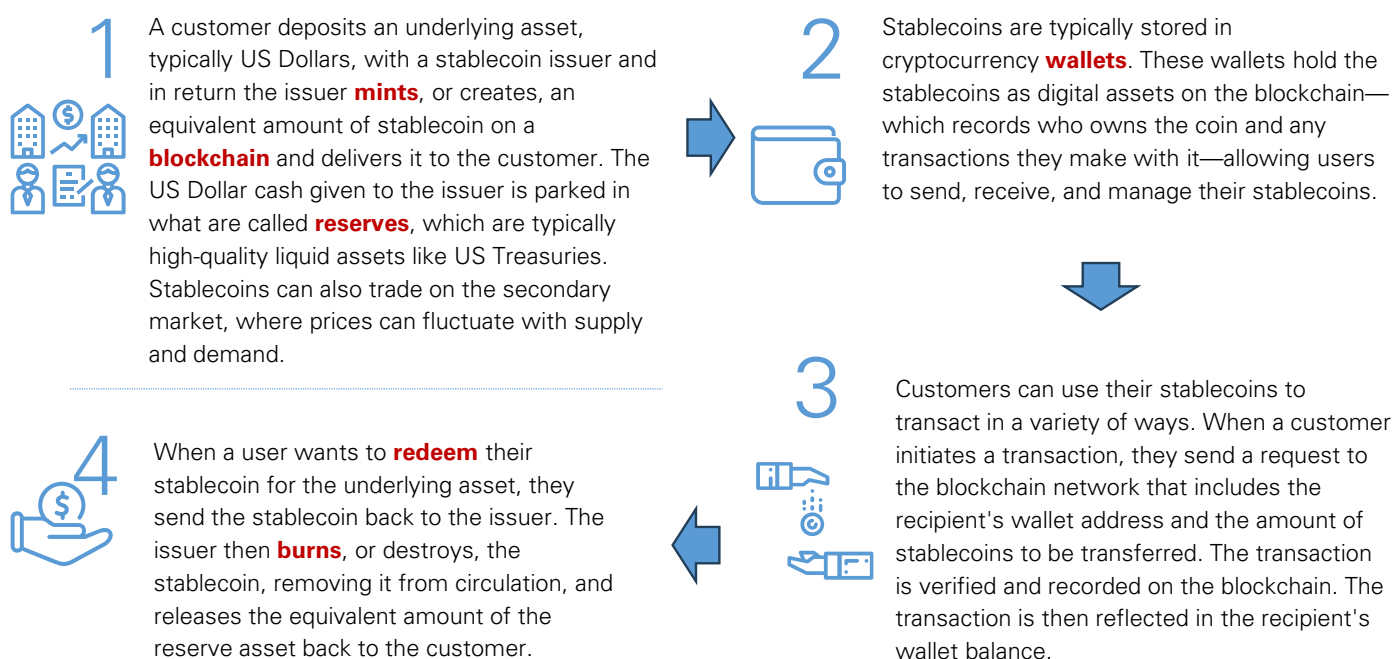
What are stablecoins?

- Stablecoins are a form of tokenized digital money using **blockchain technology** and are designed to maintain a stable value, typically **pegged** one-to-one with conventional fiat currencies, most commonly the US Dollar. This differentiates stablecoins from bitcoin and other cryptocurrencies whose values are determined by supply and demand of the coin. Their primary purpose is to serve as a medium of exchange in digital assets or for payment settlement.
- Stablecoins maintain their fiat currency peg either through market-based mechanisms that adjust supply (**algorithmic stablecoins**) or through explicit asset backing by issuers (**reserve-backed stablecoins**). The latter is significantly larger by market capitalization and more integrated with the existing financial system due to issuers' holdings of financial assets. Issuers typically hold safe, cash-equivalent assets like bank deposits, US Treasuries, repurchase agreements, and commercial paper, though some also hold precious metals and cryptocurrencies. The GENIUS Act requires stablecoins to be fully backed by 'permitted reserves' (primarily safe assets like US Dollars and short-term Treasuries) after a grace period.
- The overall size of the stablecoin market is around \$307bn, with **USDT** (~\$185bn in circulation, issued by Tether) and **USDC** (~\$78bn in circulation, issued by Circle) commanding the vast majority of market share. The remaining stablecoins outstanding are split across a variety of much smaller issuers.

How are stablecoins used?

- Today, stablecoins are primarily used within the **cryptocurrency ecosystem for trading**, providing a stable asset to move in and out of positions without having to convert to fiat currency.
- The other major use case is providing **access to dollars** outside of the US, especially in devaluation-prone regions, as well as for **cross-border payments** and **remittances**.
- Stablecoins can also be used for **consumer payments**, though this usage remains limited so far as consumers continue to rely on traditional payment systems.
- Business-to-business payments (B2B)** could be another use case for stablecoins, although B2B payments have historically been very slow to adopt new payment rails.

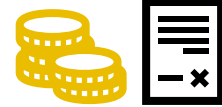
How do stablecoins work?



Source: CoinGecko, DefiLlama, US Treasury, Brookings, Goldman Sachs GIR.

Exhibit originally appeared in [Issue 141: Stablecoin summer](#).

...with the passage of the GENIUS Act...



What is it?

- The GENIUS (Guiding and Establishing National Innovation for U.S. Stablecoins) Act is a piece of US stablecoin legislation that was introduced by Senators Hagerty (R-TN), Scott (R-SC), Gillibrand (D-NY), and Lummis (R-WY).
- The Act provides a regulatory framework for **payment stablecoins**, which it defines as a digital asset designed to be used as a means of payment or settlement.
- The Act was signed into law by President Trump on July 18 after passing the Senate on June 17 on a 68-30 vote and the House of Representatives on July 17 on a 308-122 vote. Two other crypto bills passed in the House alongside it: the Digital Asset Market Clarity Act (also known as the **CLARITY Act**), which would provide a regulatory framework for digital assets more broadly, as well as the **Anti-CBDC Surveillance State Act**, which aims to prevent the Federal Reserve from issuing a central bank digital currency (CBDC) directly to individuals without explicit congressional authorization. Both bills are still before the Senate.

What does it do?

The GENIUS Act:



Defines stablecoin issuers. Stablecoins can be issued by subsidiaries of insured depository institutions, federally-licensed non-bank entities provided they obtain a special license from the Office of the Comptroller of the Currency (OCC), and state-chartered issuers provided their issuance is capped at \$10 billion. So, banks would be permitted to issue their own stablecoins, while non-financial companies would be restricted unless they meet certain standards (which have yet to be established).



Establishes a framework for regulating reserves. Issuers are required to maintain reserves backing stablecoins on an at least 1:1 basis. Reserves must be comprised of US currency, funds held as demand deposits at depository institutions, Treasury bills, notes, or bonds with a maturity of 93 days or less, repo or reverse repo agreements with a maturity of seven days or less, money market funds, or central bank reserve deposits.



Sets out a framework for reserve audits. Stablecoin issuers must disclosure the composition of their reserves monthly on their websites, and issuers with more than \$50 billion in stablecoins outstanding are required to publicly disclose annual financial statements, which must be audited by a registered public accounting firm.



Prohibits interest payments. Stablecoin issuers are prohibited from paying interest to coin holders. However, the bill says nothing about whether third parties or affiliates can pay interest, and issuers can distribute reserve income to affiliates, who can offer rewards to stablecoin holders on their platforms. (Example: Circle can pay reserve income to Coinbase, which can offer rewards to holders of USDC on its platform).



Sets out compliance requirements. Issuers are subject to the Bank Secrecy Act, anti-money laundering compliance, and sanctions requirements. They are also required to conduct transaction monitoring, recordkeeping, and suspicious activity reporting.



Remains neutral on Fed master account access. The Act does not alter who is currently legally eligible for Federal Reserve services or deposit access.

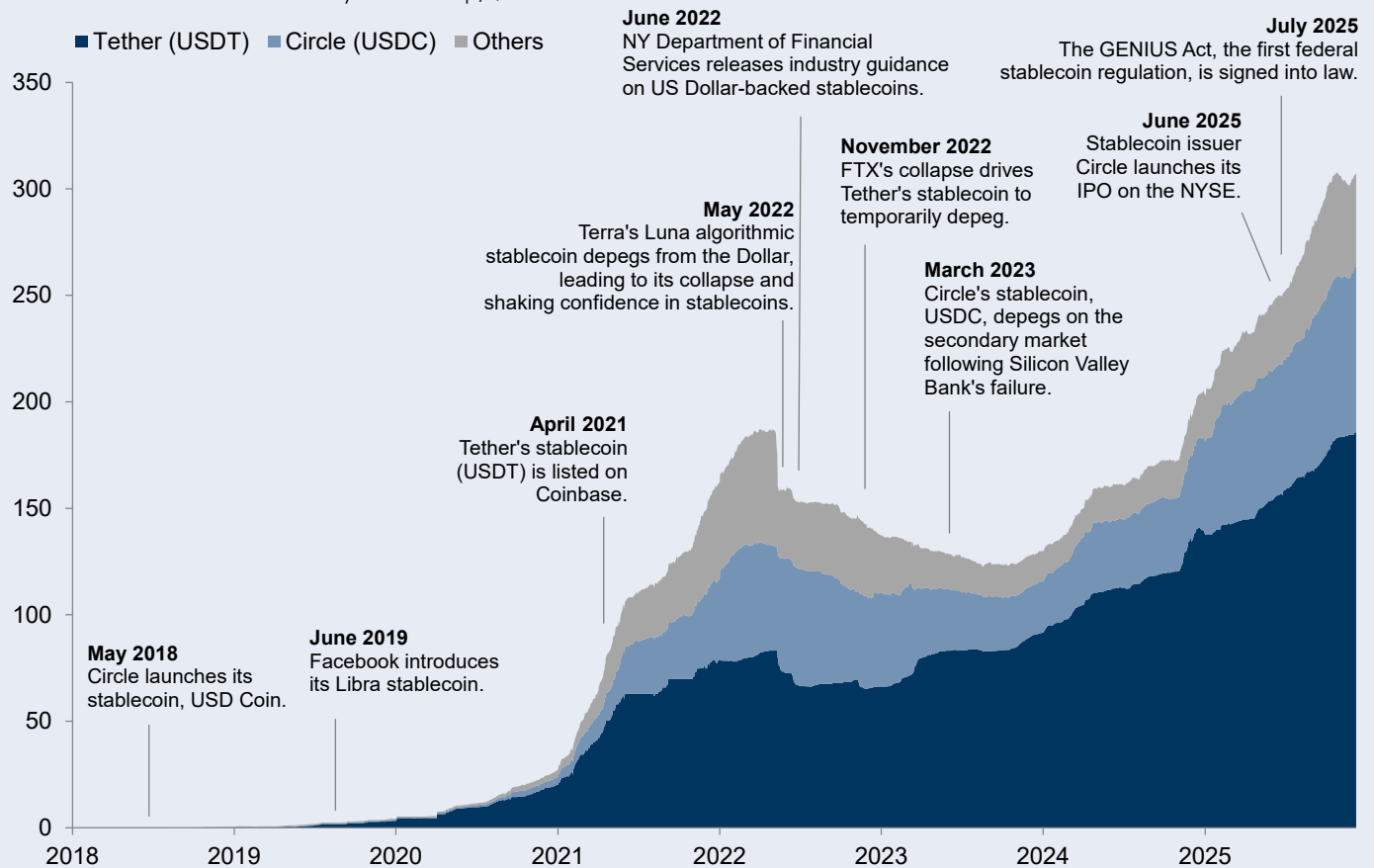
Source: Congress.gov, United States House Committee on Financial Services, United States Senate Committee on Banking, Housing, & Urban Affairs, Goldman Sachs GIR.

Exhibit originally appeared in [Issue 141: Stablecoin summer](#).

...helping drive market cap higher...

The size of the stablecoin market currently sits at ~\$307bn, dominated by two issuers: Tether and Circle

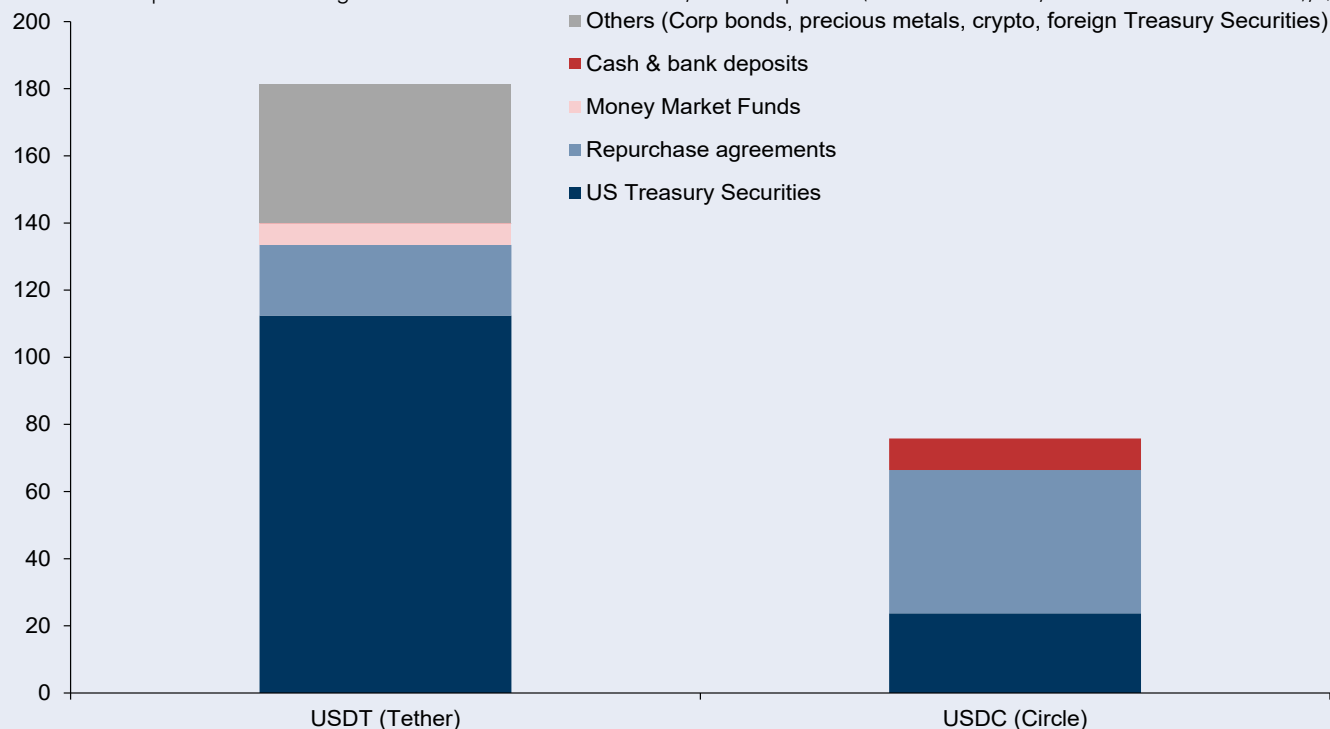
USD stablecoins in circulation by market cap, \$bn



Source: DefiLlama, CoinGecko, Moody's, various news sources, Goldman Sachs GIR.

The vast majority of stablecoin reserves are currently held in USD safe assets, including Treasury securities, repos, money funds, and deposits, though the GENIUS Act now requires payment stablecoins to be **fully backed by safe assets** after a grace period

Reserve composition of the largest two USD stablecoin issuers, latest reported (3Q25 for Tether, October 2025 for Circle), \$bn



Source: Tether, Circle, Goldman Sachs GIR.

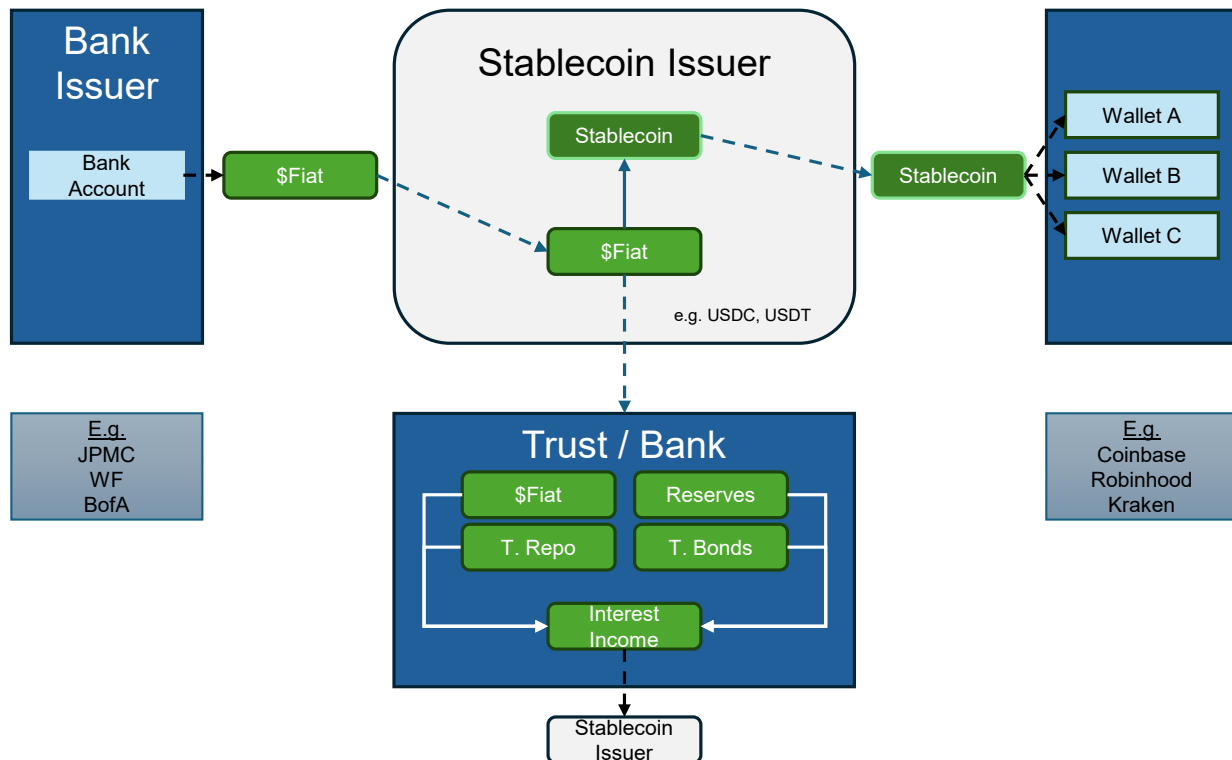
Special thanks to GS Rates research team for charts.

Charts originally appeared in [Issue 141: Stablecoin summer](#).

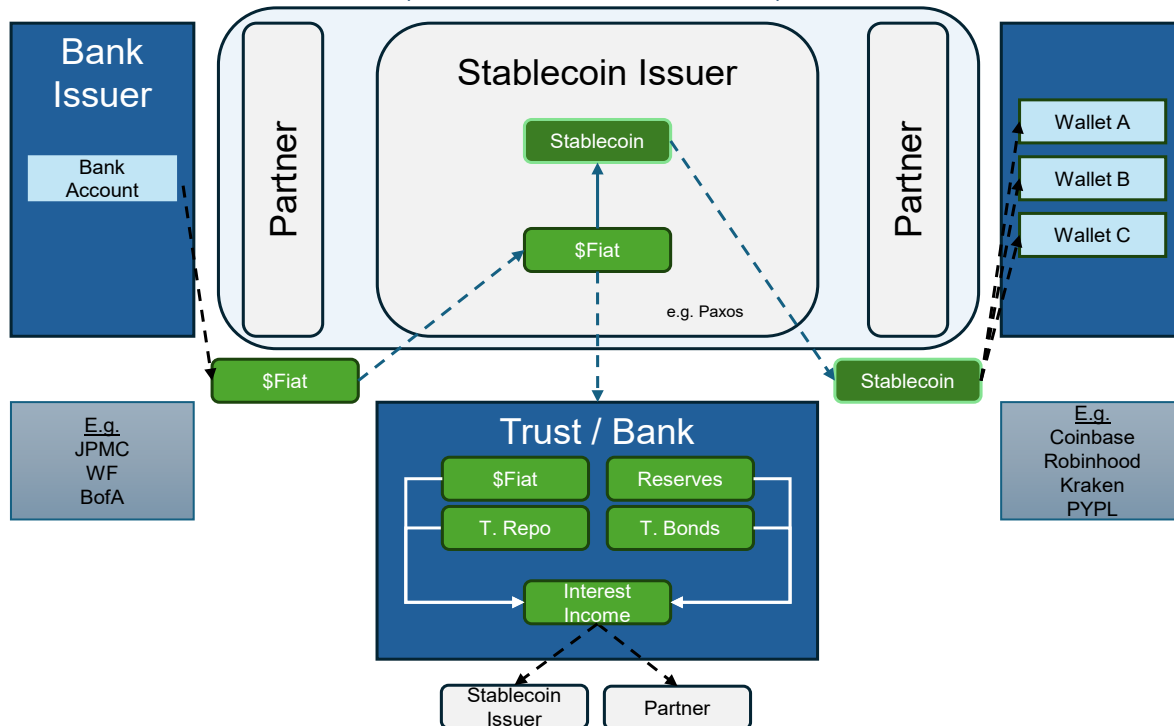
...with issuers focused on monetization

Stablecoin issuers employ one of two primary business models: (1) outright issuance and creation of stablecoins and (2) “white label” stablecoins. In the **outright issuance model**, an issuer “mints” their own stablecoin. In the **white label stablecoin issuance model**, an issuer mints and redeems stablecoins on behalf of a partner, who delivers its customers’ dollars to the issuer for minting and delivers the newly-minted stablecoins to customers. While business models differ, stablecoin issuers generate revenue primarily by earning interest on their reserve assets.

In a **direct issuer model**, an issuer mints and redeems stablecoins on their own behalf...



...while in a **white label issuer model**, an issuer performs these tasks on behalf of a partner



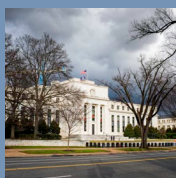
Source: Goldman Sachs GIR.

Special thanks to GS equity analyst James Yaro for exhibits and analysis.

Exhibit originally appeared in [Issue 141: Stablecoin summer](#).

3

Amid all that, *institutional credibility* became an obsession,



With fiscal sustainability, data reliability, and Fed independence all questioned.



These led to unease that higher bond yields may not fade,

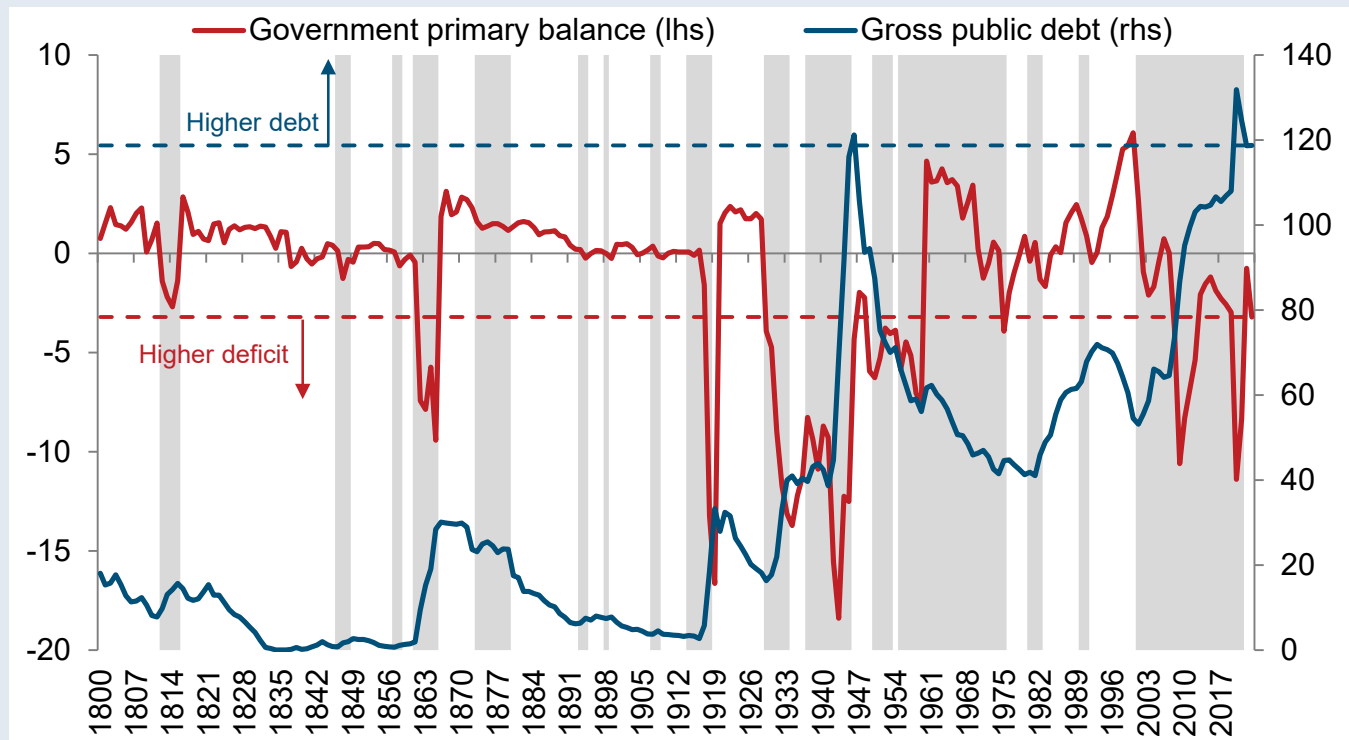
And created a tailwind for the debasement trade.

Pgs. 23-33

The concerning US fiscal situation...

The US deficit and debt are at their highest levels outside of crisis periods...

US government primary balance (lhs) and gross public debt (rhs) as % of GDP (data through 2023)

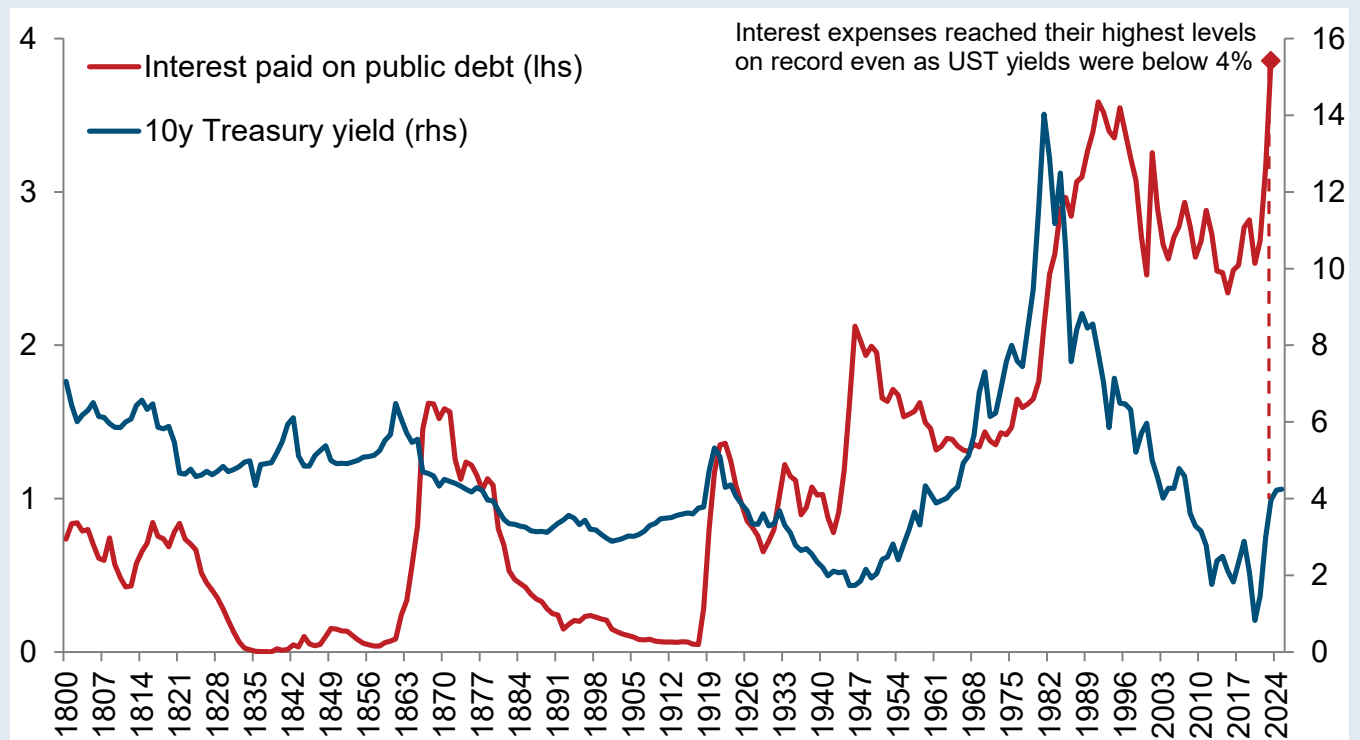


Note: Grey shaded areas represent times of major recessions, crises, or wars in US history.

Source: IMF Public Finances in Modern History database, NBER, Federal Depository Library Program (dates of major wars), compiled by Goldman Sachs GIR.

...while interest expenses were at record highs even before the most recent rise in interest rates

Interest paid on public debt as % of GDP (lhs) (data through 2023), annual average 10y Treasury yield (% , rhs) (current data)



Source: IMF Public Finances in Modern History database, Global Financial Data, Inc., Federal Reserve Board, Haver Analytics, compiled by Goldman Sachs GIR.

Charts originally appeared in [Issue 140: US fiscal worries: is this time different?](#)

...raised doubts about US credit quality

A look at US credit ratings



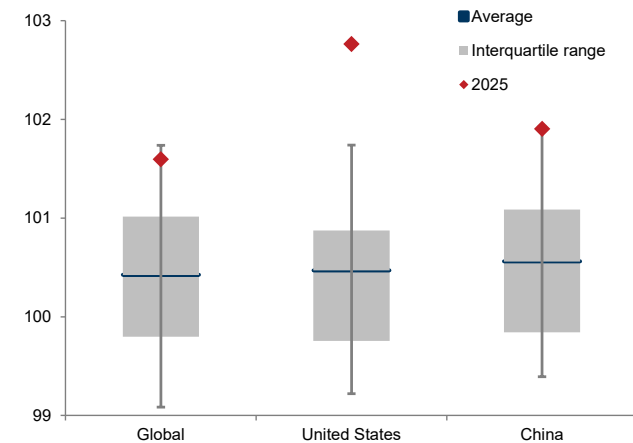
Source: Moody's Ratings, Fitch Ratings, S&P Global Ratings, compiled by Goldman Sachs GIR.

Exhibit originally appeared in [Issue 140: US fiscal worries: is this time different?](#)

(though fiscal worries are global)

Fiscal policy uncertainty around the world remains high...

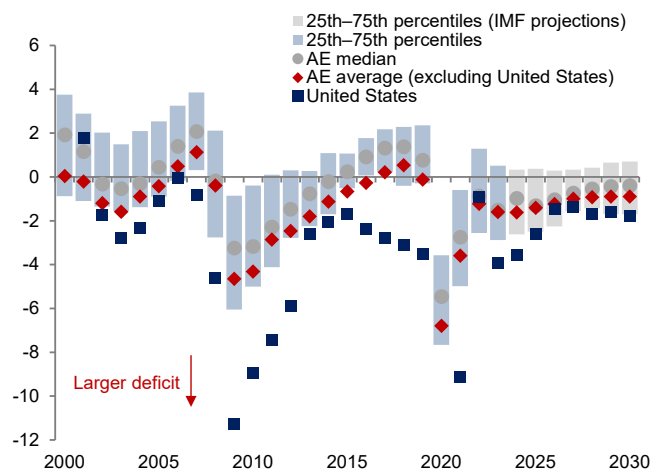
Fiscal policy uncertainty index, 2005-2025



Source: IMF, Goldman Sachs GIR.

...and deficits across advanced economies...

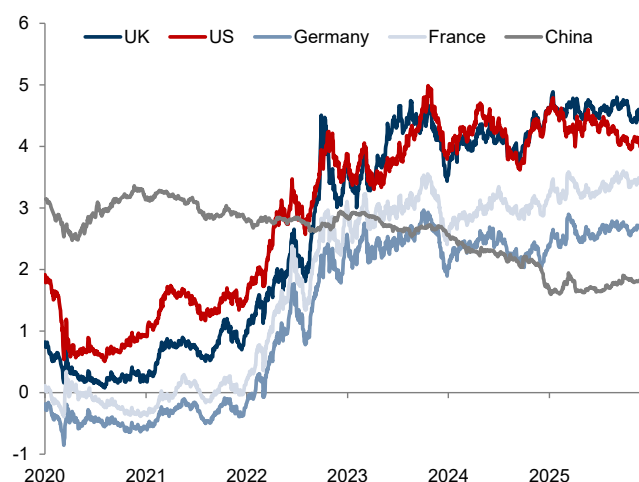
Primary balances in advanced economies (AE), % of GDP



Source: IMF, Goldman Sachs GIR.

At the same time, global interest rates have generally risen recently, although Chinese yields have declined somewhat...

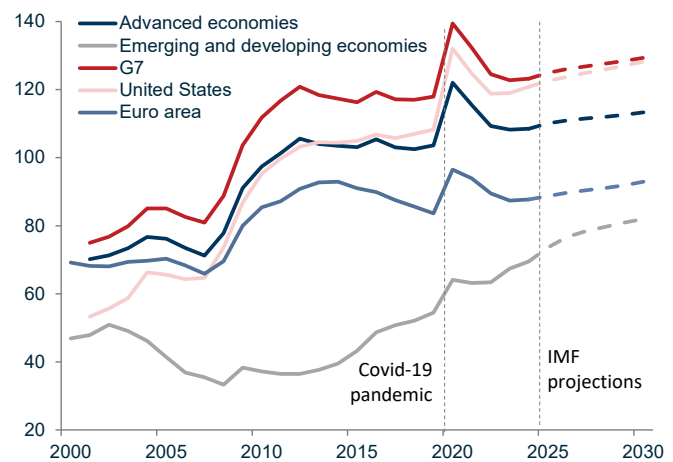
10-year sovereign bond yields, %



Source: IMF, Goldman Sachs GIR.

...at a time when global debt levels are elevated, with debt-to-GDP ratios across regions expected to rise further...

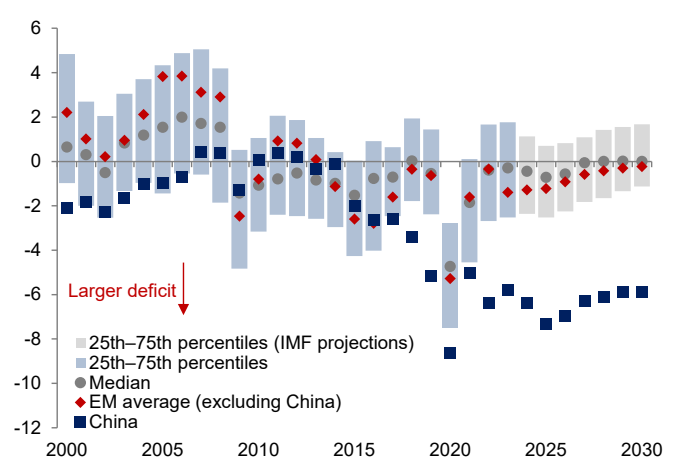
General government debt-to-GDP by country grouping, %



Source: IMF, Goldman Sachs GIR.

...as well as emerging markets are particularly elevated relative to history

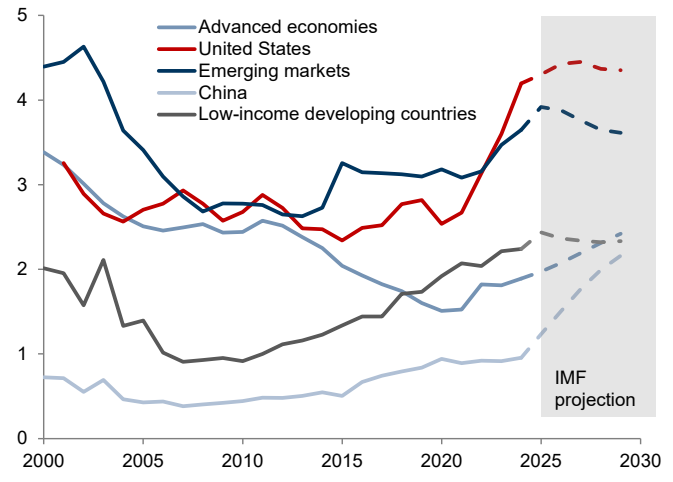
Primary balances in emerging markets (EM), % of GDP



Source: IMF, Goldman Sachs GIR.

...contributing to rising government interest expenses, which are set to increase even further in most regions

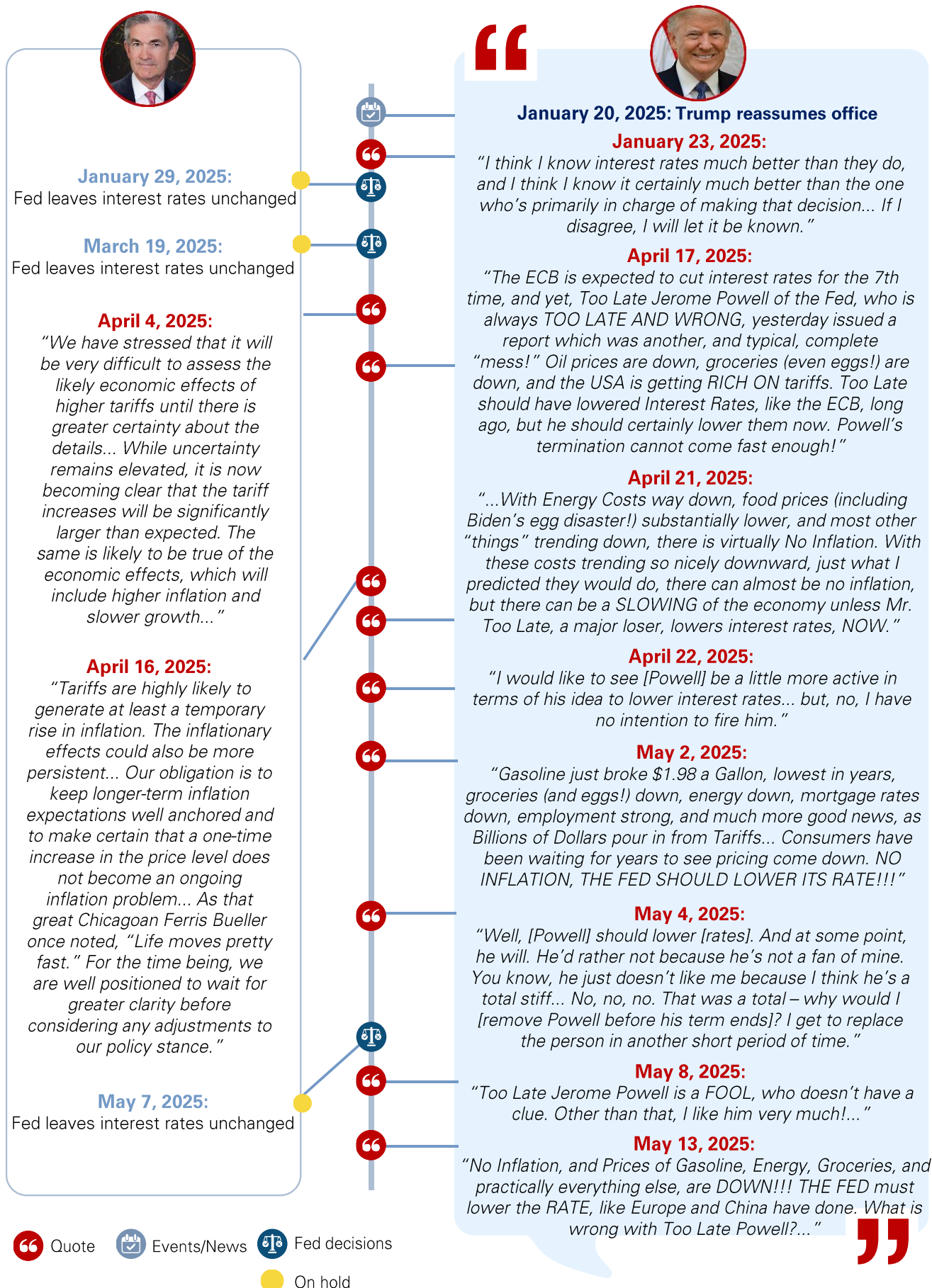
General government interest expense, % of GDP



Source: IMF, World Economic Outlook database, Goldman Sachs GIR.

Charts originally appeared in [Issue 140: US fiscal worries: is this time different?](#)

Plus ongoing concerns about Fed...



Source: Truth Social, various interviews and speeches, Federal Reserve, compiled by Goldman Sachs GIR.

...independence...



June 18, 2025

Fed leaves interest rates unchanged

June 24, 2025:

"For the time being, we are well-positioned to wait to learn more about the likely course of the economy before considering any adjustments to our policy stance."

July 30, 2025:

Fed leaves interest rates unchanged, with two dissents

August 22, 2025:

"With policy in restrictive territory, the baseline outlook and the shifting balance of risks may warrant adjusting our policy stance."

September 17, 2025:

Fed lowers interest rates, with one dissent

September 23, 2025

"Near-term risks to inflation are tilted to the upside and risks to employment to the downside—a challenging situation. Two-sided risks mean that there is no risk-free path. If we ease too aggressively, we could leave the inflation job unfinished and need to reverse course later to fully restore 2 percent inflation. If we maintain restrictive policy too long, the labor market could soften unnecessarily."

October 29, 2025:

Fed lowers interest rates, with two dissents

October 29, 2025

"In the Committee's discussions at this meeting, there were strongly differing views about how to proceed in December. A further reduction in the policy rate at the December meeting is not a foregone conclusion—far from it."

December 10, 2025:

Fed lowers interest rates, with three dissents



June 12, 2025:

"We're going to spend \$600 billion a year, \$600 billion because of one numbskull that sits here [and says] I don't see enough reason to cut the rates now."

June 18, 2025:

"So we have a stupid person. Frankly, you probably won't cut today... Europe had 10 cuts, and we had none. And I guess he's a political guy, I don't know. He's a political guy who's not a smart person, but he's costing the country a fortune."

June 19, 2025:

"Too Late Jerome Powell is costing our Country Hundreds of Billions of Dollars. He is truly one of the dumbest, and most destructive, people in Government, and the Fed Board is complicit. Europe has had 10 cuts, we have had none. We should be 2.5 Points lower, and save \$BILLIONS on all of Biden's Short Term Debt. We have LOW inflation! TOO LATE's an American Disgrace!"

July 15, 2025

"I think he's terrible... But the one thing, I didn't see him as a guy that needed a palace to live in... The one thing I would have never guessed is that he would be spending \$2.5 billion to build a little extension onto the Fed."

August 1, 2025:

"Jerome 'Too Late' Powell, a stubborn MORON, must substantially lower interest rates, NOW. IF HE CONTINUES TO REFUSE, THE BOARD SHOULD ASSUME CONTROL, AND DO WHAT EVERYONE KNOWS HAS TO BE DONE!"

August 19, 2025:

"Could somebody please inform Jerome 'Too Late' Powell that he is hurting the Housing Industry, very badly? People can't get a Mortgage because of him. There is no Inflation, and every sign is pointing to a major Rate Cut. 'Too Late' is a disaster!"

September 10, 2025:

"Just out: No Inflation!!! 'Too Late' must lower the RATE, BIG, right now. Powell is a total disaster, who doesn't have a clue!!!"

November 19, 2025:

*"Scott [Bessent], you've gotta work on this guy... He's got some real mental problems. There's something wrong with him. It's just ridiculous. I'll be honest, I'd love to fire his a**. He should be fired."*



Quote



Events/News



Fed decisions



On hold



Rate cut



Source: Truth Social, various interviews and speeches, Federal Reserve, compiled by Goldman Sachs GIR.

Exhibit originally appeared in [Issue 139: Fed independence: how concerning?](#)

...as the courts continued to weigh in...

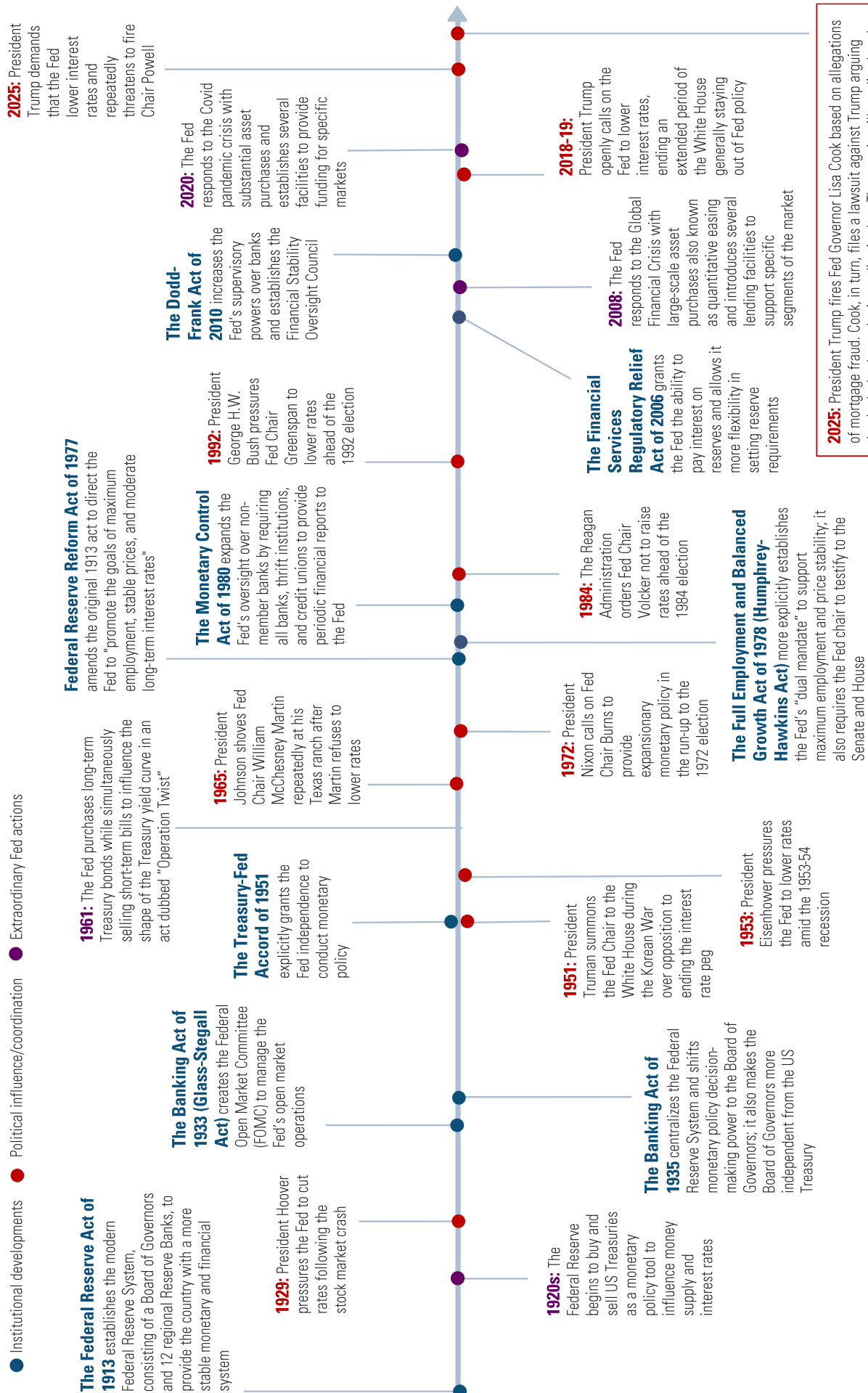
US courts have historically upheld for-cause firing protections to independent boards like the Federal Reserve Board of Governors, but recent decisions suggest support for more removal power

Case	Year	Description	Decision
Humphrey's Executor v. United States	1935	In 1933 President Roosevelt fired William Humphrey from the Federal Trade Commission (FTC) without cause, despite the FTC Act permitting dismissal only for "inefficiency, neglect of duty, or malfeasance in office."	The Court ruled the dismissal invalid, distinguishing FTC commissioners as quasi-legislative or quasi-judicial officers (not executive officers) that may be removed only with procedures consistent with statutory conditions enacted by Congress.
Wiener v. United States	1958	President Eisenhower removed Abraham Wiener, a member of the War Claims Commission, established by Congress to adjudicate claims from parties affected by World War II.	The Court ruled that Wiener's dismissal was invalid, as the Appointments Clause does not guarantee the president's ability to remove an officer of the United States who exercises quasi-judicial authority at will.
Morrison v. Olson	1988	The Chairman of the House Judiciary Committee forwarded a report to the US AG asking that an independent counsel investigate whether Theodore Olson, Assistant AG for the Office of Legal Counsel, had given false testimony regarding EPA enforcement. Olson argued that the Independent Counsel Act was unconstitutional since it created a "hybrid branch" of the government that took executive powers away from the president.	The Court upheld the Independent Counsel Act, affirming that Congress could create independent agencies with limited removal protections for inferior officers that have limited duties and no policymaking role.
Free Enterprise Fund v. Public Company Accounting Oversight Board (PCAOB)	2010	The PCAOB was created by the Sarbanes-Oxley Act of 2002 as a non-profit organization to oversee the audits of US-listed public companies. PCAOB officers could be removed only "for good cause" by SEC officers who could only be removed by the president for "inefficiency, neglect of duty, or malfeasance in office." The Free Enterprise Fund sued in 2006, challenging that the PCAOB violated the appointments clause of the Constitution and vesting clause establishing the separation of powers.	The Court ruled that the appointment provisions of the Act was constitutional, but struck down the for-cause removal provision for PCAOB members. Chief Justice Roberts noted that "the executive power included a power to oversee executive officers through removal" in striking down for-cause protections, but distinguished the case from Humphrey's Executor by noting that it only dealt with principal officers and from Morrison by noting this precedent didn't "address the consequences of more than one level of good-cause tenure".
Seila Law v. Consumer Financial Protection Bureau (CFPB)	2020	The CFPB was established by the 2010 Dodd-Frank Act as an independent agency with a single director selected by the president and confirmed by the Senate for a five-year term. The Act stated that the CFPB could only be removed for "inefficiency, neglect of duty, or malfeasance in office." Seila Law LLC (Seila Law), a law firm that provided debt relief services, challenged the CFPB's constitutionality.	The Court ruled that the CFPB structure was unconstitutional as it violated the separation of powers by establishing a sole director that could only be terminated for cause. While Chief Justice Roberts acknowledged 1) Humphrey's Executor established constraints on the president's removal power for officers of an agency that exercised "no part of the executive power" and performed "specified duties as a legislative or as a judicial aid" and 2) Morrison established similar constraints for "inferior officers with limited duties and no policymaking" role, the CFPB didn't fall under these exemptions.
Collins v. Yellen	2021	The FHFA was established by the Housing and Economic Recovery Act of '08, with the legislation specifying it was to be overseen by a single director that can only be removed "for cause". Shareholders of Fannie Mae and Freddie Mac argued that the government's 2012 decision to allocate nearly all of Fannie and Freddie's profits to the Treasury was invalid since the FHFA is structured unconstitutionally as an agency with a single director who holds power that isn't subject to appropriate checks by the president.	The Supreme Court ruled that the "for cause" restriction on removal of the FHFA director by the president was unconstitutional given the recent Seila Law decision.
Wilcox v. Trump	2025	Trump dismissed Gwynne Wilcox from the National Labor Relations Board (NLRB) in January 2025 even though her term was scheduled to last until August 2028. Wilcox sued, arguing that Humphrey's Executor protects against without cause removal of "principal officers".	On April 7, the DC Court of Appeals blocked the termination, but the Supreme Court issued a stay on this decision.
Dellinger v. Bessent	2025	United States Office of Special Counsel Head Hampton Dellinger was fired by President Trump in February 2025. Dellinger challenged his dismissal, arguing that the Office of Special Counsel is an independent agency where by law the president can only dismiss its head for "inefficiency, neglect of duty, or malfeasance in office."	The DC District Court stayed Dellinger's dismissal in Feb, and the Supreme Court chose to hold the Trump Administration's appeal to vacate the District's court order in abeyance until expiration, effectively allowing the decision to stand. The DC District Court then issued an injunction against Dellinger's firing on Mar 1 which was stayed on Mar 5 by the DC Appeals Court—effectively removing Dellinger from his position—before Dellinger dropped his suit and accepted his dismissal.
Slaughter v. Trump	2025	In March 2025, Trump attempted to fire two FTC Commissioners, Rebecca Slaughter and Alvaro Bedoya, without cause, in violation of the FTC's governing statute. Slaughter and Bedoya subsequently sued in DC District Court arguing their dismissal was illegal under precedent established in Humphrey's Executor.	The case is ongoing with the Supreme Court recently hearing initial oral arguments. The Court has granted a stay, allowing Trump's removal of Slaughter to stand while the case proceeds.
Trump v. Cook	2025	In August 2025, after being fired by President Trump, Fed Governor Lisa Cook sues Trump, the Board of Governors, and Jerome Powell, arguing that Trump lacks the authority to dismiss her.	Litigation is ongoing, but the Supreme Court has ruled that Cook can stay on as governor while the case proceeds. The Court will hear oral arguments in Jan. to deliberate on whether Trump has a legal case for firing Cook.

Source: Justia, Supreme Court of the United States, Goldman Sachs GIR.

Special thanks to GS global economist Joseph Briggs for table, which originally appeared in [Issue 139: Fed independence: how concerning?](#)

...potentially overturning lots of history



Note: Not an exhaustive list of all Federal Reserve-related developments and actions.
Source: Federal Reserve, compiled by Goldman Sachs GIR.

Timeline originally appeared in [Issue 139: Fed independence: how concerning?](#)

Plus US data reliability concerns...

“

“In my opinion, today's Jobs Numbers were RIGGED in order to make the Republicans, and ME, look bad... We need accurate Jobs Numbers. I have directed my Team to fire this Biden Political Appointee, IMMEDIATELY. She will be replaced with someone much more competent and qualified. Important numbers like this must be fair and accurate.”

“Last weeks Job's Report was RIGGED, just like the numbers prior to the Presidential Election were Rigged. That's why, in both cases, there was massive, record setting revisions, in favor of the Radical Left Democrats. Those big adjustments were made to cover up, and level out, the FAKE political numbers that were CONCOCTED in order to make a great Republican Success look less stellar!!! I will pick an exceptional replacement.”

- President Trump, August 2025

“How on earth are businesses supposed to plan—or how is the Fed supposed to conduct monetary policy—when they don't know how many jobs are being added or lost in our economy? It's a serious problem that needs to be fixed immediately... Until it is corrected, the BLS should suspend issuing the monthly job reports.”

“The problems in the BLS data have been evident for three years now, and they still haven't been fixed... The fact that you consistently have large downward revisions means that there are other things wrong with your models and methodologies. Statistical assumptions that may have worked fine before COVID no longer work in today's economy and therefore need to be revised.”

- E.J. Antoni, Former Trump nominee for BLS Commissioner, August 2025

“BLS has tackled the problem of declining response rates from many angles... All data collection activities incorporate quality assurance assessments, including reinterviews with data providers, reviews by senior staff, computer-based data checks, and more.”

- BLS statement, March 2023

Source: BLS, Fox News, CBS News, Truth Social, various news sources, compiled by Goldman Sachs GIR.

“I wouldn't say that I'm concerned about the data today, although there has been a very mild degradation of the scope of the surveys. But I would say the direction of travel is something I'm concerned about. It's really important not just for the Fed, but for Congress and for businesses, frankly, to know what really is going on in the economy. I don't like to see the kind of stories I'm reading and the idea being that the data is going to become more volatile and less reliable. That'll make it more difficult for the private sector and for you and for us.”

- Jerome Powell, Fed Chair, July 2025

“There's a heck of a lot of noisiness in the labor market data right now, which is bad for markets, it's bad for transparency and it's something that I think a fresh set of eyes should really look at and try to fix because the jobs numbers... need to be independently produced but they need to be reliable and not subject to massive revisions.”

“If the data aren't that good, then it's a real problem for the US. And right now the data... have become very unreliable with these massive revisions over the last few years.”

- Kevin Hassett, Director of the National Economic Council, August 2025

“Post-pandemic shifts mean our statistical systems are still catching up. Surveys can lag reality, and we have to adjust our readings accordingly.”

- Austan Goolsbee, Chicago Fed President, April 2025

“I think the most important thing here is that we get back to the integrity of the numbers, because it just became OK, just like so many things in our government, for it to get sloppy... what we want is good data, because you can't make good decisions without good data... It's very likely that the Fed would have been doing something else in June, in July, if they had had this data.”

- Scott Bessent, Treasury Secretary, August 2025

“We've been surprised a few times this year by big revisions. That's just a reminder that the first number you see is not the final story.”

- Raphael Bostic, Atlanta Fed President, August 2025

”

Exhibit originally appeared in [Issue 142: Data \(un\)reliability](#).

...after some data quality problems...

Indicator	MAP	Units	Score	Data Type	Rank: Change in Quality (lower # = greater decline quality)							Response Rate (%)		Standard Error		Standard Deviation		Forecast Surprises^	
					Revision	Sampling	Std. Err.	Noise	Average	Latest*	Historical**	Latest*	Historical**	Latest*	Historical**	Latest*	Historical**		
JOLTS	-	k		Survey	13	1	1	6	5.3	33	63	214.4	127.6	882.4	741.8	371.7	226.5		
NFP	5	Chg. k		Survey	3	3	6	13	6.3	43	60	80	68	79.4	71.3	84.6	63.4		
ADP	3	Chg. k		Admin.	4	16	13	2	8.8	-	-	-	-	98.4	50.4	81.2	43.7		
Retail Sales	4	Chg. %		Survey	9	7	9	11	9.0	57	67	0.3	0.28	0.6	0.5	0.4	0.4		
Core PCE	-	Chg. %		Mixed	8	16	5	7	9.0	-	-	-	-	0.1	0.1	0.1	0.1		
Advance Goods Trade	-	\$bn		Admin.	7	16	13	1	9.3	-	-	-	-	20.3	5.9	14.1	3.1		
Empire Mfg.	2	Index		Survey	2	16	13	8	9.8	50	50	-	-	14.6	12.0	16.6	7.8		
Philly Fed Mfg.	4	Index		Survey	19	10	13	5	11.8	-	45	-	-	16.0	11.7	14.4	8.2		
Personal Spending	1	Chg. %		Mixed	12	11	12	12	11.8	-	-	-	-	0.3	0.3	0.2	0.1		
IP	2	Chg. %		Mixed	15	9	7	21	13.0	-	-	-	-	0.5	0.5	0.3	0.3		
Pending Home Sales	2	Chg. %		Mixed	21	16	13	3	13.3	-	-	-	-	4.3	2.4	4.1	2.0		
Import Prices	-	Chg. %		Mixed	24	4	1	26	13.8	45	55	0.15	0.07	0.4	0.6	0.3	0.3		
SS&P Global Mfg.	-	Index		Survey	11	14	13	19	14.3	75	80	-	-	1.7	1.7	-	-		
Housing Starts	1	Chg. %		Mixed	17	15	10	16	14.5	-	-	7.5	7.1	9.5	9.0	8.0	7.5		
Initial Claims	2	k		Admin.	1	16	13	29	14.8	-	-	-	-	13	26	10.9	11.6		
CPI	-	Chg. %		Mixed	33	8	3	20	16.0	54	65	0.05	0.03	0.2	0.2	0.1	0.1		
ECI	-	Chg. %		Survey	6	2	33	24	16.3	46	66	0.10	0.12	0.08	0.12	0.11	0.12		
NAHB HMI	1	Index		Survey	28	16	13	9	16.5	13	13	-	-	6.1	5.5	3.6	2.7		
New Home Sales	2	Chg. %		Survey	22	16	11	18	16.8	-	-	9.6	9.3	8.3	7.9	7.8	7.7		
Continuing Claims	1	k		Admin.	5	16	13	34	17.0	-	-	-	-	81	237	28.0	31.7		
Existing Home Sales	2	Chg. %		Mixed	26	16	13	15	17.5	-	-	-	-	4.4	4.1	3.4	3.0		
Personal Income	-	Chg. %		Mixed	29	30	7	4	17.5	-	-	-	-	0.3	0.2	0.3	0.2		
Case-Shiller HPI	-	Chg. %		Admin.	25	16	13	17	17.8	-	-	-	-	0.4	0.3	0.2	0.2		
Unemp. Rate	5	%		Survey	31	5	4	32	18.0	70	85	0.13	0.10	0.3	0.6	0.1	0.1		
SS&P Global Serv.	-	Index		Survey	16	29	13	14	18.0	75	85	-	-	2.5	2	-	-		
UMich Sentiment	1	Index		Survey	18	33	13	10	18.5	-	-	-	-	7.3	3.8	-	-		
Building Permits	-	Chg. %		Survey	27	12	13	22	18.5	66	70	-	-	4.4	5.3	3.9	5.2		
Dallas Fed Mfg.	-	Index		Survey	20	13	13	28	18.5	71	-	-	-	10.0	18.3	8.0	7.5		
ISM Serv.	3	Index		Survey	30	16	13	23	20.5	-	-	-	-	1.8	2.2	2.1	1.9		
Conference Board	3	Index		Survey	10	32	13	30	21.3	-	-	-	-	6.5	15.0	5.2	5.0		
KC Fed Mfg.	-	Index		Survey	14	31	13	31	22.3	-	-	-	-	3.9	10.8	4.9	5.3		
ISM Mfg.	5	Index		Survey	32	16	13	33	23.5	-	-	-	-	1.3	4.0	1.1	1.5		
Richmond Mfg.	1	Index		Survey	23	34	13	25	23.8	40	40	-	-	6.9	11.0	6.7	8.9		
Construction Spending	1	Chg. %		Mixed	34	6	34	27	25.3	23	41	0.5	0.8	0.5	0.9	0.4	0.8		

* 2023-present, when available. ** 2015-2019, when available. ^ Reports standard deviation.

Notes: For each ranked category, we use a combination of percent changes and changes in the relevant statistics relative to typical forecast surprises. To rank indicators that mix administrative and survey data, we use the values of their respective source data divided by two. We do not use response rates for UMICH or Conference Board, both of which switched to online surveys in the last five years. Standard errors are not applicable to surveys like the building permit survey that do not use probability samples. For construction spending, we report the response rate for private nonresidential construction. For retail sales, we are reporting the response rate for ARTS (the source data). CPI response rate is a weighted average of the C&S and Housing survey response rates. The sample size of price indexes reflects the number of intended price quotes instead of the number of establishments or households.

Source: Bloomberg, Department of Labor, Department of Commerce, Federal Reserve, Institute for Supply Management, S&P Global Market Intelligence, National Association of Realtors (NAR), National Association of Homebuilders (NAHB), University of Michigan, The Conference Board, Goldman Sachs GIR.

Special thanks to GS US economist Ronnie Walker for table, which originally appeared in [Issue 142: Data \(un\)reliability](#).

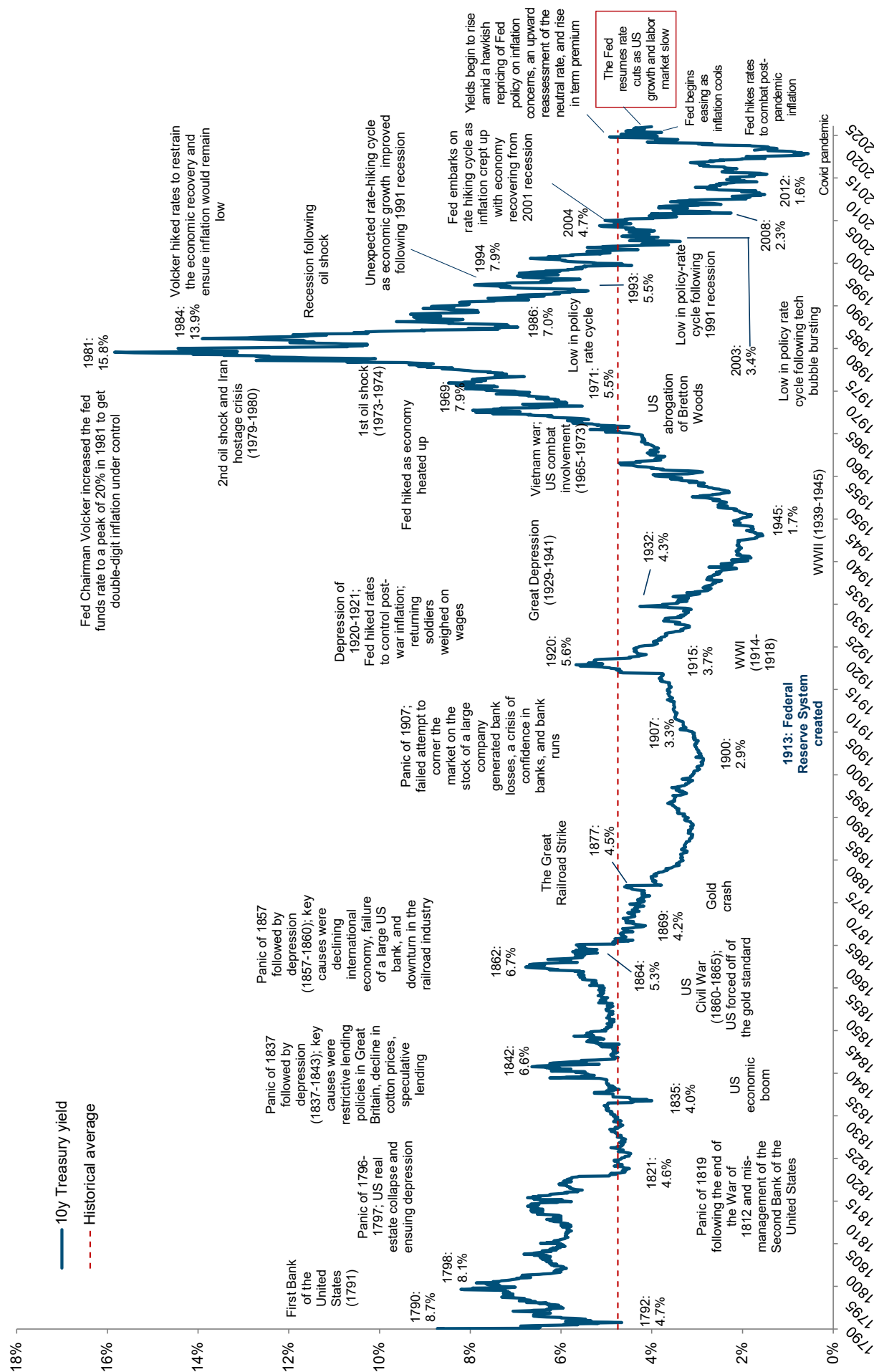
(which, BTW, aren't a new or just US issue)

Country	Agency	Data Release	Issue	Official Quote/Description
US	Bureau of Labor Statistics	Consumer Price Index	Coverage reduction	"To align survey workload with resource levels, BLS suspended data collection for portions of the Consumer Price Index (CPI) sample in select areas across the country starting in April. In April, BLS suspended CPI data collection entirely in Lincoln, NE and Provo, UT. In June, BLS suspended collection entirely in Buffalo, NY. Roughly 15 percent of the sample in the other 72 areas also was suspended from collection, on average. Collection suspension affects both the Commodity and Services Pricing survey and the Housing survey. As a result, the number of collected prices and the number of collected rents used to calculate the CPI has temporarily been reduced."
US	Bureau of Labor Statistics	Current Population Survey Database	Production error	"BLS identified an error in the rounding of median hourly earnings data and associated standard errors in the Current Population Survey database. A total of 162 series were affected. The estimates were rounded to the nearest integer, instead of 2 decimal places (or to the nearest dollar, instead of to the nearest cent). This problem only occurred with the annual averages for 2021."
US	Bureau of Labor Statistics	Current Employment Statistics	Major revisions	"Revisions for May and June were larger than normal. The change in total nonfarm payroll employment for May was revised down by 125,000, from +144,000 to +19,000, and the change for June was revised down by 133,000, from +147,000 to +14,000." (from July Employment Report) "The change in total nonfarm payroll employment for June was revised down by 27,000, from +14,000 to -13,000, and the change for July was revised up by 6,000, from +73,000 to +79,000. With these revisions, employment in June and July combined is 21,000 lower than previously reported." (from August Employment report) "The preliminary estimate of the Current Employment Statistics (CES) national benchmark revision to total nonfarm employment for March 2025 is -911,000 (-0.6 percent)... The annual benchmark revisions over the last 10 years have an absolute average of 0.2 percent of total nonfarm employment." (from Preliminary Benchmark Summary)
US	Census Bureau and Bureau of Labor Statistics	Current Population Survey	Coverage reduction	"The Bureau of Labor Statistics (BLS) and the Census Bureau have paused plans to reduce the Current Population Survey (CPS) sample in January 2025, due to a provision in the recently passed continuing resolution that allows BLS to spend CPS funds at a faster rate. We will continue to monitor the budget situation and will keep the public updated about potential impacts to the CPS sample."
US	University of Michigan	Consumer Survey	Production error	"There was a minor technical error in the generation of the weights used to compute aggregate estimates for January 2025. We have since corrected the weights. Overall, the differences between the estimates originally released and the revised estimates are almost negligible."
UK	Office for National Statistics	Producer Prices Index (PPI) and Construction Price Deflators	Production error	"During work to improve the systems used to create the Producer Price Index (PPI) and the Services Producer Price Indices (SPPI), our quality assurance identified a problem with the chain-linking methods used to calculate these indices... This problem affects the years from 2008 onwards. However, the main impact on annual producer price inflation rates is seen in 2022 and 2023. As a consequence, we are pausing the publication of PPI and SPPI data, which was next due on 26 March, while we rectify this issue. As construction price deflators are partly derived from PPIs and SPPIs, these indices are also affected and publication of them will also be paused. We are planning to recommence publication in the summer and will keep users informed of progress."
UK	Office for National Statistics	UK Labor Survey	Suspended data series	"As we have previously highlighted, falling response rates for household surveys are affecting national statistics institutes internationally. In October 2023, the Office for National Statistics (ONS) suspended publication of the Labour Force Survey (LFS) when its response rates led to increased data uncertainty."
UK	Office for National Statistics	UK Trade Statistics	Production error	"Due to the identification of a further error in UK trade statistics, relating to international trade in services data for 2023, the Office for National Statistics has made the decision to delay the UK Trade: January 2025 release, which was scheduled for Friday March 14." (from the FT)
UK	Office for National Statistics	Consumer Price Inflation	Production error	"An error in car tax data provided by the British government caused the consumer price inflation rate to be overstated by 0.1 percentage points for the year to April [2025]." (from Reuters)
Germany	Destatis	Consumer Spending	Suspended data series	Since May the Federal Statistical Office (Destatis) has not updated time-series data for retail and wholesale sales, as well as revenue from the services sector, hospitality, car dealers, and garages.
Sweden	Statistics Sweden	Unemployment Rate	Production error	In November 2019, Statistics Sweden, which has blamed a subcontractor for the issues, was forced to drastically cut its estimate for unemployment in September. The jobless rate for that month was revised to 6.6% from 7.4%, and the agency has restated nearly all its data since July 2018.
China	National Bureau of Statistics	Broad	Suspended data series	"Land sales measures, foreign investment data and unemployment indicators [including youth unemployment] have gone dark in recent years. Data on cremations and a business confidence index have been cut off. Even official soy sauce production reports are gone. In all, Chinese officials have stopped publishing hundreds of data points once used by researchers and investors." (from WSJ)

Source: BLS, Reuters, WSJ, ONS, FT, UMich, Goldman Sachs GIR.

Special thanks to GS Global Economics research team for table, which originally appeared in [Issue 142: Data \(un\)reliability](#).

...have led to rising US bond yields



Note: Chart shows monthly averages.

Source: Global Financial Data, Inc., Federal Reserve Board, Haver Analytics, Goldman Sachs GfR.

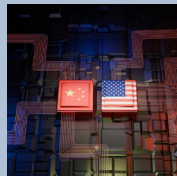
Chart originally appeared in [Issue 135: High bond yields: here to stay?](#)

4

Finally, important **geopolitical developments** continued to take shape,

As President Trump shook up the post-Cold War security landscape,

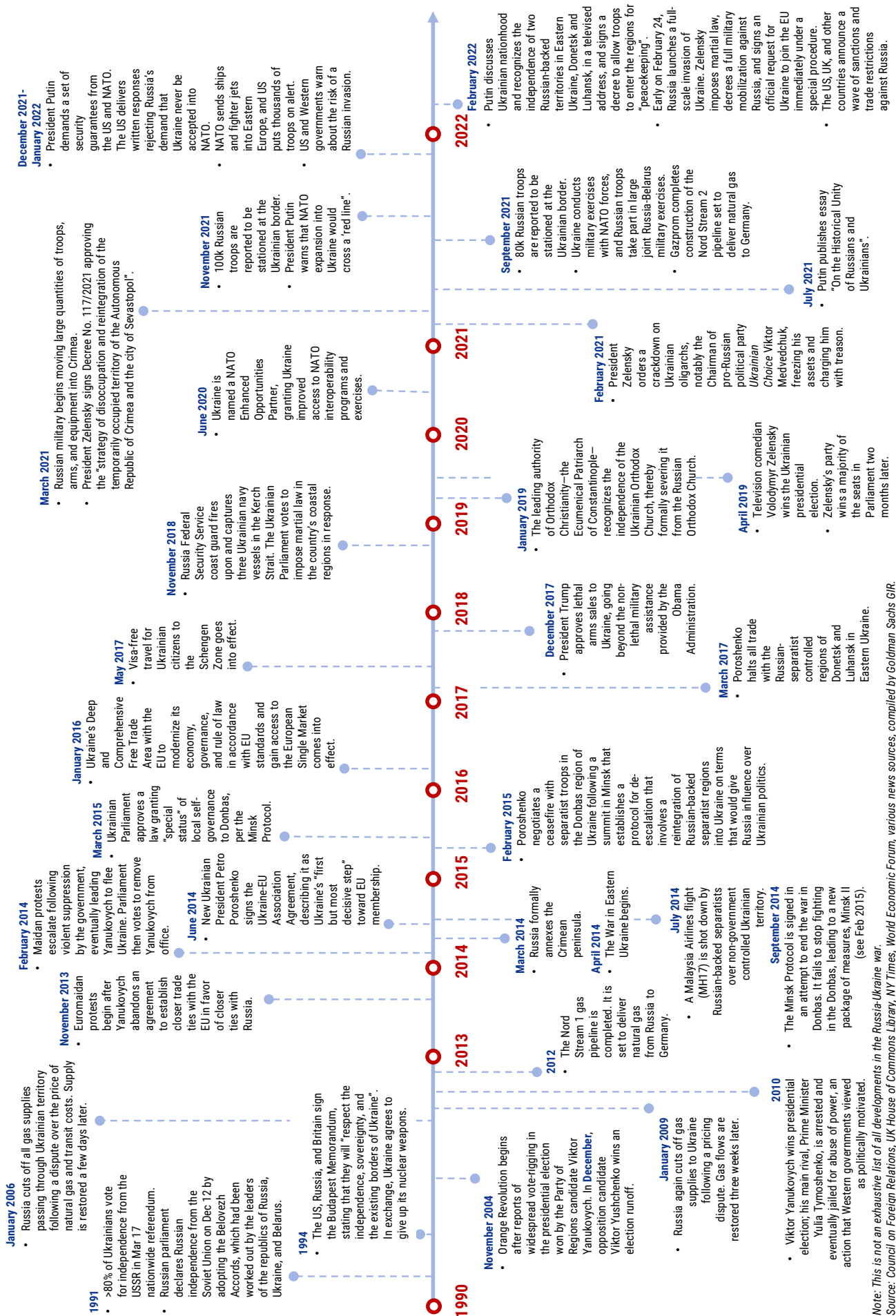
With significant implications for the Russia-Ukraine war and European defense,



While the race between the US and China for tech superiority grew even more intense.

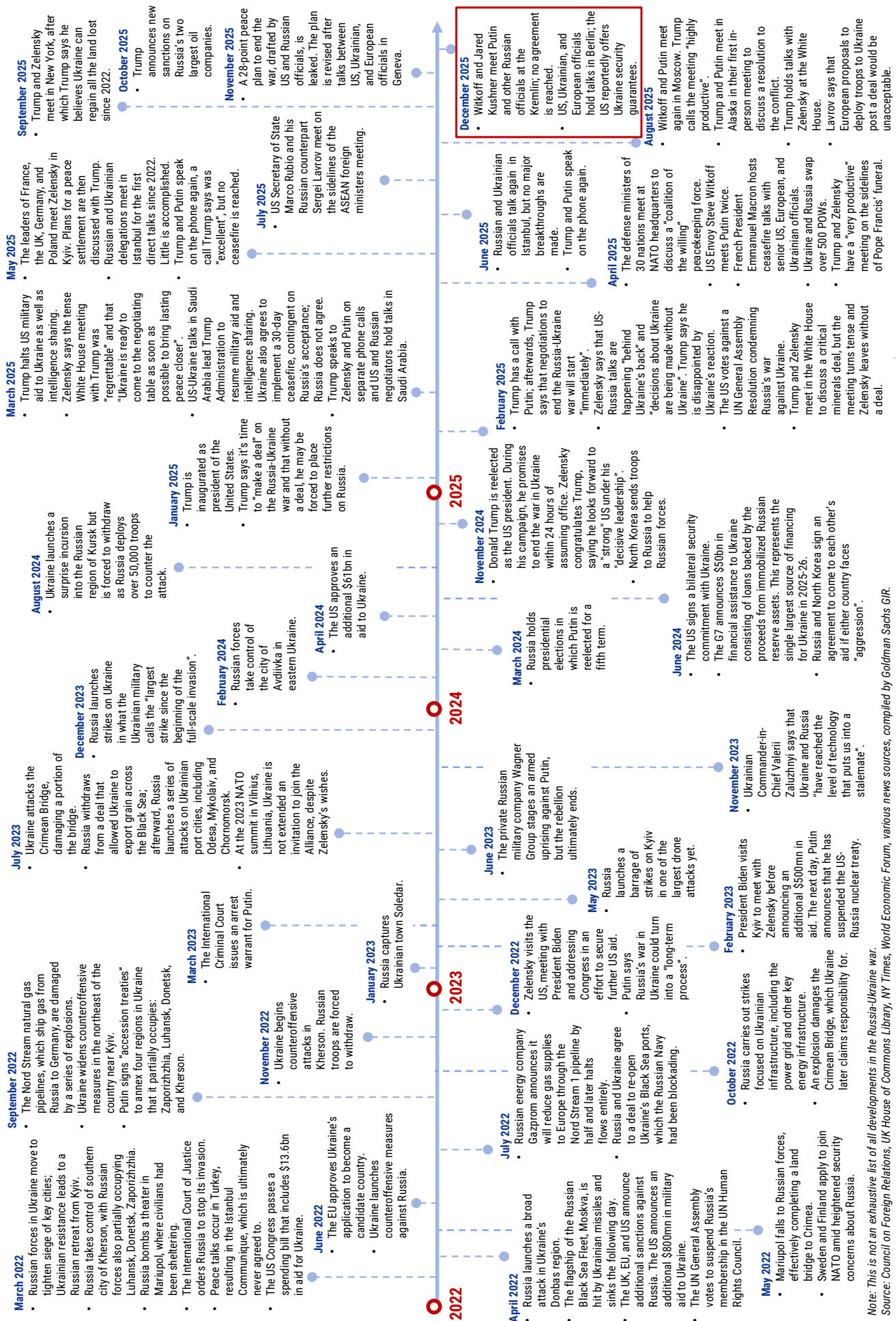
Pgs. 35-43

The Russia-Ukraine war...



Note: This is not an exhaustive list of all developments in the Russia-Ukraine war.
Source: Council on Foreign Relations, UK House of Commons Library, NY Times, World Economic Forum, various news sources, compiled by Goldman Sachs GIR.

...entered its 4th year...



...with little progress in sight...

“

“The so-called civilized world, of which our Western colleagues have self-appointed themselves the only representatives, prefers not to notice [the situation in the Donbas] as if there isn’t a genocide through which nearly four million people are being put through, all simply because these people did not agree to the Western coup of Ukraine in 2014.”

- Address by President Putin, February 2022

“When some propose alternatives, half-hearted settlement plans—so-called sets of principles—it not only ignores the interests and suffering of Ukrainians, who are affected by the war the most, it not only ignores reality, but also gives Putin the political space to continue the war and pressure the world to bring more nations under control. Any parallel or alternative attempts to seek peace are, in fact, efforts to achieve a lull instead of an end to the war, as a global initiative—the Peace Formula—has already existed for two years. And maybe somebody wants a Nobel Prize for their political biography for frozen truce instead of real peace, but the only prizes Putin will give you in return are more suffering and disasters.”

- President Zelensky UN General Assembly Speech, September 2024

“I just had a lengthy and highly productive phone call with President Vladimir Putin of Russia... as we both agreed, we want to stop the millions of deaths taking place in the War with Russia/Ukraine... We agreed to work together, very closely, including visiting each other’s Nations. We have also agreed to have our respective teams start negotiations immediately, and we will begin by calling President Zelenskyy, of Ukraine, to inform him of the conversation...”

- President Trump Truth Social post, February 2025

“What happened at the White House on Friday [February 28], of course, demonstrated how difficult it will be to reach a settlement trajectory around Ukraine. The Kyiv regime and Zelensky do not want peace. They want the war to continue.”

- Remarks by Kremlin spokesman Dmitry Peskov, March 2025

“We are ready to work fast to end the war, and the first stages could be the release of prisoners and truce in the sky—ban on missiles, long-ranged drones, bombs on energy and other civilian infrastructure—and truce in the sea immediately, if Russia will do the same. Then we want to move very fast through all next stages and to work with the US to agree a strong final deal.”

- President Zelensky social media post and letter to President Trump, March 2025

“I appreciate that [Zelensky] sent this letter, just got it a little while ago. Simultaneously, we’ve had serious discussions with Russia and have received strong signals that they are ready for peace.”

- President Trump’s speech to Congress, March 2025

“We will categorically not tolerate such actions. Again, I want to emphasize that this would not be a so-called hybrid involvement but direct, official and undisguised participation of NATO countries in a war against the Russian Federation...”

- Remarks by Russian Foreign Minister Sergei Lavrov regarding the deployment of European troops in Ukraine, March 2025

“Today we made an offer which the Ukrainians have accepted, which is to enter into a ceasefire and into immediate negotiations to end this conflict in a way that’s enduring and sustainable and accounts for their interests, their security, their ability to prosper as a nation... We will take this offer now to the Russians and we hope that they’ll say ‘yes,’ that they’ll say ‘yes’ to peace. The ball is now in their court.”

- Remarks by Secretary of State Marco Rubio following US-Ukraine talks in Saudi Arabia, March 2025

”

Source: Truth Social, X, various speeches, interviews, and texts, compiled by Goldman Sachs GIR.

Exhibit originally appeared in [Issue 137: Europe’s shifting security landscape](#).

...as leaders remain far apart

“

“I am not happy with the Russian strikes on KYIV. Not necessary, and very bad timing. Vladimir, STOP! 5000 soldiers a week are dying. Lets get the Peace Deal DONE!”

- *President Trump Truth Social post, April 2025*

“I have said many times that I consider the Russian and Ukrainian peoples to be one people. In this sense, all of Ukraine is ours... We have an old rule. Wherever a Russian soldier sets foot is ours.”

- *Remarks by President Putin during St. Petersburg International Economic Forum, June 2025*

“There’s no deal until there’s a deal.”

- *Remarks by President Trump following the Alaska summit, August 2025*

“After getting to know and fully understand the Ukraine/Russia Military and Economic situation and, after seeing the Economic trouble it is causing Russia, I think Ukraine, with the support of the European Union, is in a position to fight and WIN all of Ukraine back in its original form.”

- *President Trump Truth Social post, September 2025*

“This is one of the most difficult moments in our history... Currently, the pressure on Ukraine is one of the hardest. Ukraine may now face a very difficult choice, either losing its dignity or the risk of losing a key partner.”

- *Remarks by President Zelensky regarding the 28-point US peace plan proposal, November 2025*

“We firmly believe security decisions about Ukraine must include Ukraine—that was our position in Geneva and always... We all believe that people should decide their own future, and this is, in particular, what sets us apart from regimes like the one in Moscow.”

- *Address by President Zelensky to participants of the Coalition of the Willing meeting, November 2025*

“We are clear on the principle that borders must not be changed by force. We are also concerned by the proposed limitations on Ukraine’s armed forces, which would leave Ukraine vulnerable to future attack. We reiterate that the implementation of elements relating to the European Union and relating to NATO would need the consent of EU and NATO members respectively.”

- *Statement on Ukraine adopted by several European leaders and the PMs of Canada, Japan, and the UK, November 2025*

“Ukraine is prepared for any possible developments, and of course we will work as constructively as possible with all our partners to ensure that peace is achieved—and that it is, after all, a dignified peace. Only a dignified peace provides real security, and we fully understand that this requires—and will continue to require—the support of our partners.”

- *President Zelensky speech, December 2025*

“We want peace. Why aren’t we immediately signing a peace deal?”

- *Remarks by Kremlin spokesman Dmitry Peskov, December 2025*

“It all comes down to this. Either we liberate these territories by force of arms, or Ukrainian troops leave these territories and stop fighting there.”

- *Remarks by President Putin in an interview with India Today, December 2025*

“[Zelensky] is gonna have to get on the ball and start accepting things. You know, when you’re losing, cause he’s losing... you certainly wouldn’t say it’s a victory. I could show you a chart. It’s a lot of land they lost.”

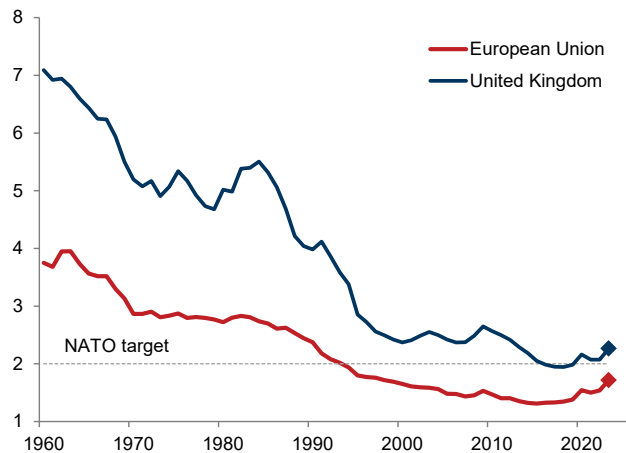
- *Remarks by President Trump in an interview with Politico, December 2025*

”

Source: Truth Social, X, various speeches, interviews, and texts, compiled by Goldman Sachs GIR.

So, Europe turns to building defense...

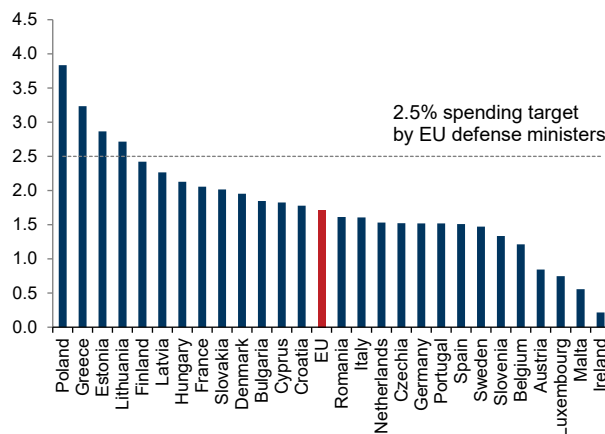
Defense spending in Europe has fallen as a share of GDP since the 1960s, reaching the lowest level in the last decade, but it began to rise in 2022 following Russia's invasion of Ukraine
Military expenditure, % of GDP



Source: World Bank, Goldman Sachs GIR.

Only a few EU countries exceed the 2.5% of GDP defense spending target focused on by EU defense ministers

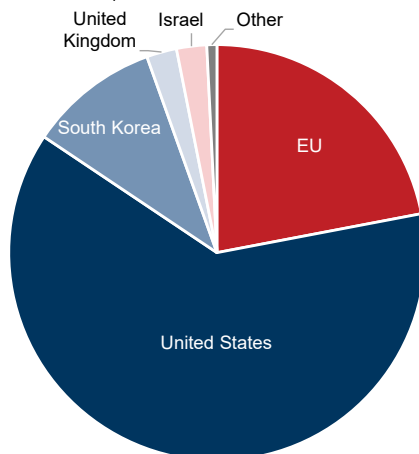
Defense spending by EU countries in 2023, % of GDP



Source: World Bank, Goldman Sachs GIR.

...and Europe imports a large share of the military equipment it spends money on

Import share of equipment sourced in the EU since the start of Russia-Ukraine war, %

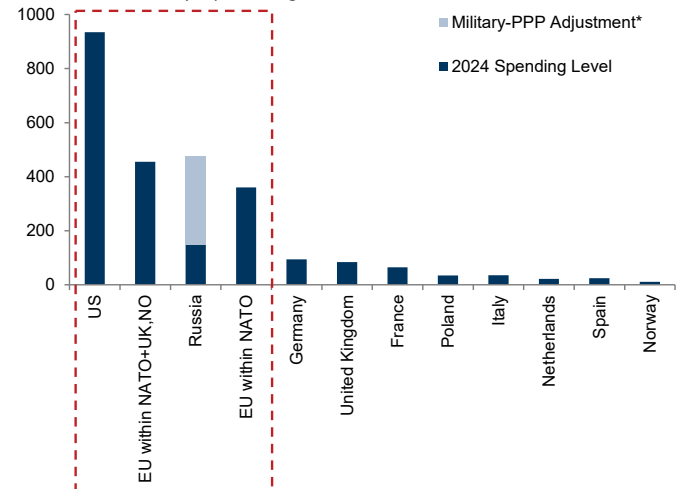


Note: Based on value of defense acquisitions announced by EU countries from the start of the Russia-Ukraine war to June 2023.

Source: Institut de Relations Internationales, Goldman Sachs GIR.

Special thanks to the GS European Economics research team for charts, which originally appeared in [Issue 137: Europe's shifting security landscape](#).

The US significantly outspends Europe on defense, though Europe spends on par with Russia in PPP-adjusted terms
Absolute military spending levels, \$bn

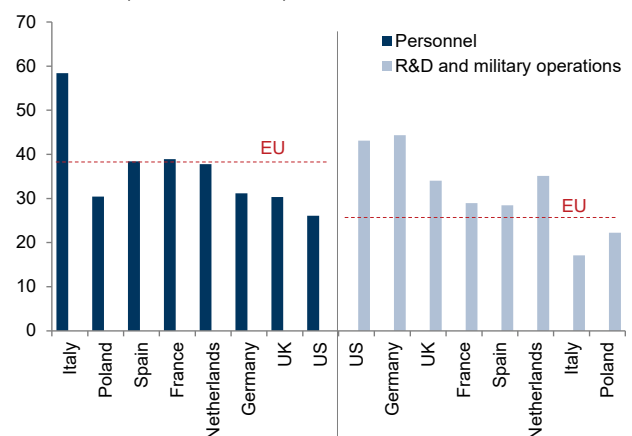


*Adjusted for price differential vs. the US (Robertson, 2021 and SIPRI).

Source: SIPRI, NATO, Robertson (2021), Goldman Sachs GIR.

European defense expenditure remains structurally and technologically outdated, overfunding personnel instead of investing in research and operations...

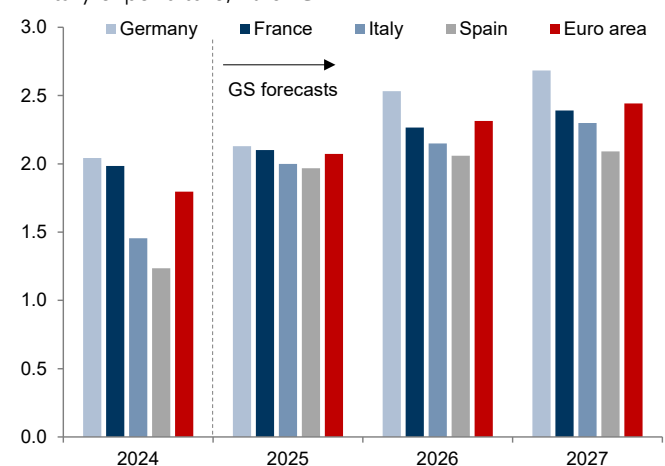
Defense expenditure composition, %



Source: Haver Analytics, Goldman Sachs GIR.

However, increased urgency for greater European defense expenditure suggests more spending ahead; we forecast EU military spending of around 2.5% of GDP by 2027

Military expenditure, % of GDP



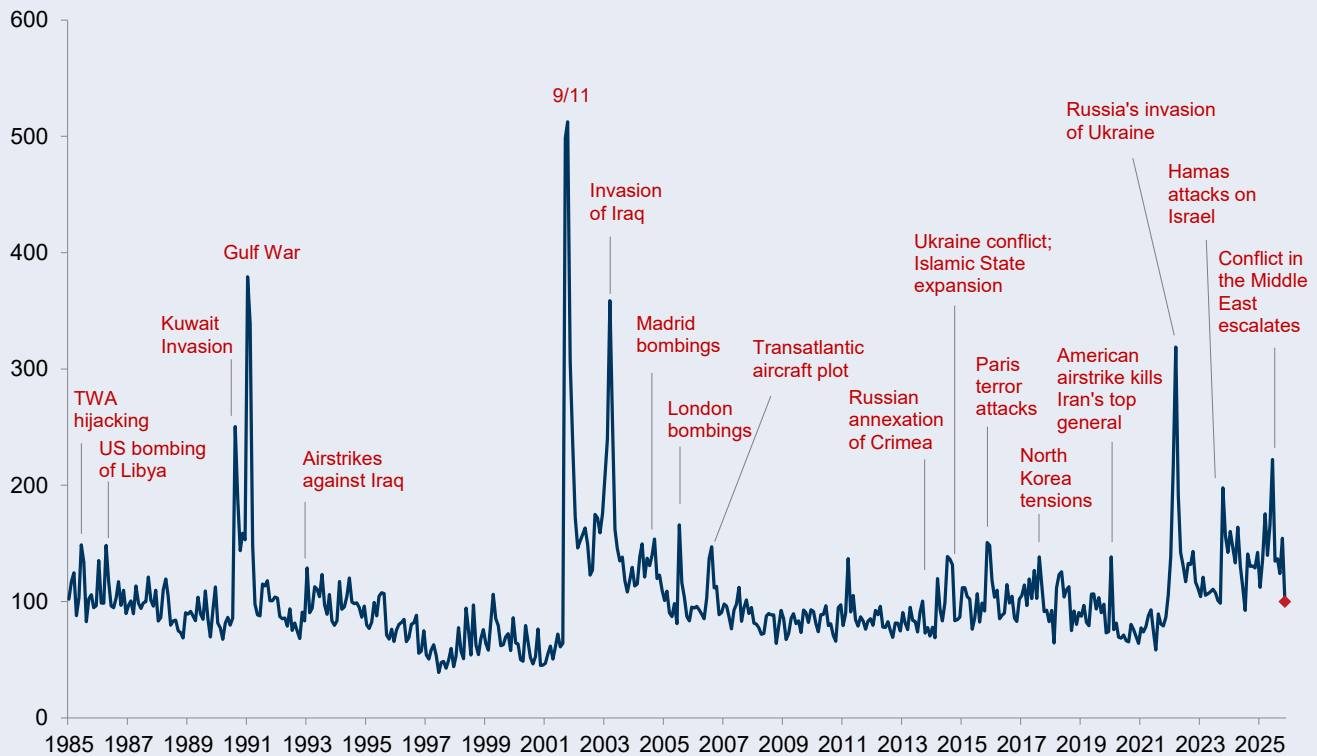
Source: NATO, Goldman Sachs GIR.

...amid still-elevated geopolitical risk

Geopolitical tensions, which take many different forms, are difficult to measure. One proxy for assessing the geopolitical environment is the news-based Geopolitical Risk Index developed by economists from the Federal Reserve Board.

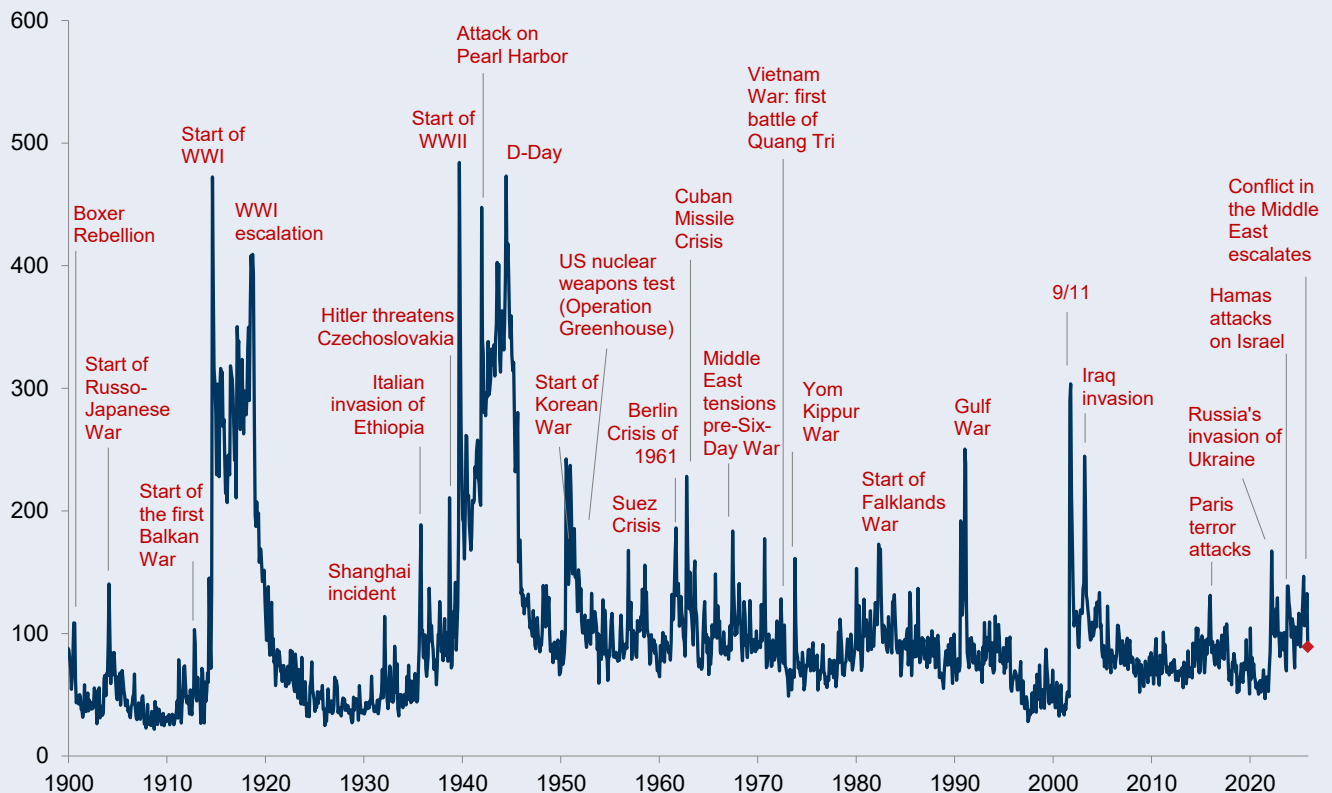
1985-Present

Geopolitical Risk Index, 1900-2019=100



1900-Present

Geopolitical Risk Index, 1985-2019=100



Note: The index from 1985 on counts the number of articles in 11 US, UK, and Canadian newspapers mentioning phrases related to geopolitical tensions. The index from 1900 on performs the same analysis using the archives of three newspapers: the New York Times, the Wall Street Journal, and the Financial Times. The choice of newspapers for both indices implies a measure of geopolitical risk as covered by the Anglo-Saxon press. See [here](#) for more information.

Source: [Dario Caldara and Matteo Iacoviello](#), Federal Reserve Board, Goldman Sachs GIR.

Charts originally appeared in [Issue 137: Europe's shifting security landscape](#).

And the US-China tech race intensified...



March 2018 – The US claims that China has conducted unfair trade practices related to technology, intellectual property, and innovation. The US applies 25% tariffs on ~\$50bn of Chinese goods, including semiconductors and electronics.

April 2018 – The US bans the sale of components to Chinese tech company ZTE.

August 2018 – The US extends tariffs to an additional \$200bn worth of Chinese goods, including a wider range of electronic components and goods.

May 2019 – The US bans the sale of components to Chinese tech company Huawei, though with some exemptions.

November 2020 – The US makes a \$9.6mn contribution to MP Materials efforts to build separation capacity for light rare earths to its California extraction operations site—the US' only rare earth extraction operation.

February 2021 – The US awards Lynas \$30.4mn to develop domestic separation capabilities for light rare earths.

June 2021 – Biden expands Trump-era ban on US investment into Chinese firms with ties to defense or surveillance technology sectors.

February 2022 – The DoD grants \$35mn to MP Materials to separate and process heavy rare earths, creating a full value chain operation, from extraction to manufacturing magnets at a single facility.

August 2022 – The US passes the CHIPS Act, which blocks certain chip exports to China.

October 2022 – The US announces restrictions on exports of advanced computing chips and related equipment to China.

December 2022 – The US blacklists 36 Chinese companies, including advanced chip producers, restricting their use of US semiconductor manufacturing tech.

October 2023 – The US tightens restrictions on the sale of chips that Nvidia designed for the Chinese market.

March 2024 – The DoD establishes a goal to develop a complete 'mine-to-magnet' supply chain for rare earth materials that can meet all US defense needs by 2027.

January 2025 – The Biden Administration establishes the AI Diffusion Rule which limits the export and use of US chip technologies based on a multi-tiered system.

May 2025 – The Trump Administration rescinds the AI Diffusion Rule but warns against the use of Chinese tech.

July 2025 – The DoD announces a deal with MP Materials that includes an offtake commitment and a price floor.

October 2025 – The US announces a 100% tariff on China and additional export controls in response to China's rare earth export controls, but later agrees to loosen restriction after Trump-Xi talks.

December 2025 – Trump says he will allow Nvidia to sell H200 chips to China with 25% of the revenue going to the US government.

2018

2019

2020

2021

2022

2023

2024

2025

June 2019 – China launches a science and technology innovation board to facilitate the financing of high-tech manufacturing companies.

October 2019 – China launches its "Big Fund" Phase II, a government-led investment fund aimed at bolstering the country's semiconductor industry.

January 2020 – As part of the Phase One trade deal, China commits to increase purchases of US goods and services, limit technology transfers, and improve IP enforcement.

May 2020 – The Chinese government grants semiconductor fab SMIC \$2.25bn to build new chip capacity.

August 2020 – The Chinese government vows to increase the self-sufficiency rate of semiconductors to 70% in 2025 from 30% in 2019.

March 2021 – China releases its 14th Five-Year Plan which prioritizes technological self-sufficiency, with a focus on key industries including the semiconductor industry.

December 2022 – China announces a support package for its domestic semiconductor industry to counter heightened US chip restrictions.

March 2023 – China launches cybersecurity review of imports from US memory-chip maker Micron and later bans sales of Micron chips.

July 2023 – China announces new restrictions on gallium and germanium exports.

October 2023 – China imposes export restrictions on certain types of natural graphite.

December 2023 – China announces a ban on rare earth extraction and separation technologies.

August 2024 – China announces new export controls on critical raw materials including antimony and superhard materials.

December 2024 – China bans exports of gallium, germanium, antimony, and superhard materials to the US.

April 2025 – China implements export controls on several rare earth elements and magnets used in the defense, energy, and auto sectors in response to US tariffs.

October 2025 – China expands rare earths export controls, which are later suspended as part of a broader US-China deal. China's 15th Five-Year Plan proposal includes a focus on developing "chokehold technologies" including chips.

Note: not an exhaustive list of all US and Chinese policy developments.

Source: Center for Strategic and International Studies, White House, USTR, Peterson Institute for International Economics, various news sources, Goldman Sachs GIR. Timeline originally appeared in [Issue 144: The US-China tech race](#).

...amid a push for semiconductor and...

What are semiconductors, and who makes them?

Semiconductors are also known as **microchips or chips**. A microchip is a set of electronic circuits layered on a thin wafer of semiconductor material, typically silicon. Transistors located on the chip act as miniature electrical switches that can turn a current on or off. The more transistors that are located on a chip, the more the chip can do. The size of a microchip and the number of transistors on it varies; a microchip the size of a human fingernail can contain billions of transistors.

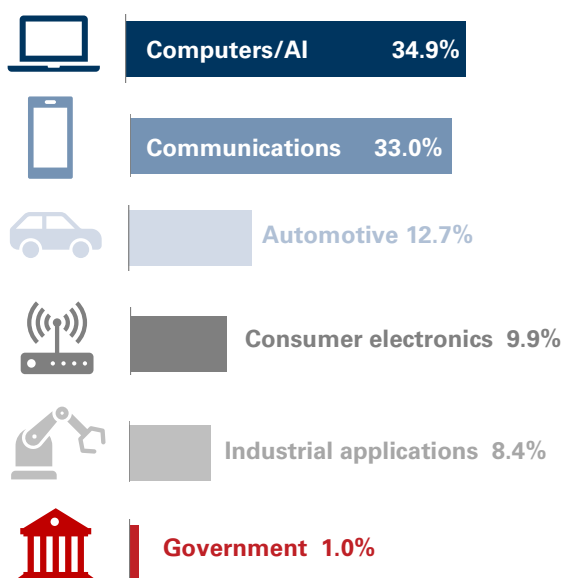
There are three main types of microchip companies: **Integrated Device Manufacturers (IDMs)** (Intel, Samsung), who design and manufacture chips in-house, **Fabless companies** (Nvidia, AMD), who design chips in-house but outsource manufacturing, and **Foundries** (TSMC, GlobalFoundries), who manufacture chips for fabless companies as well as for IDMs who don't have sufficient in-house capacity.

What do semiconductors do?

Semiconductors are the building blocks of technology, and are central inputs in many devices, including cars, computers, smartphones, and medical devices. There are three main types of chips: logic chips, memory chips, and Discrete, Analog, and Other (DAO) chips. Logic chips are the 'brains' of electronics. They process information in order to complete tasks. Central processing units (CPUs) are built for general functionality, graphics processing units (GPUs) are optimized for visual displays, and neural processing units (NPUs) are designed for machine learning applications. Memory chips store information. DRAM chips save data when a device is turned on, while NAND chips save data after a device is turned off. DAO chips transmit, receive, and transform information dealing with continuous parameters, like temperature.

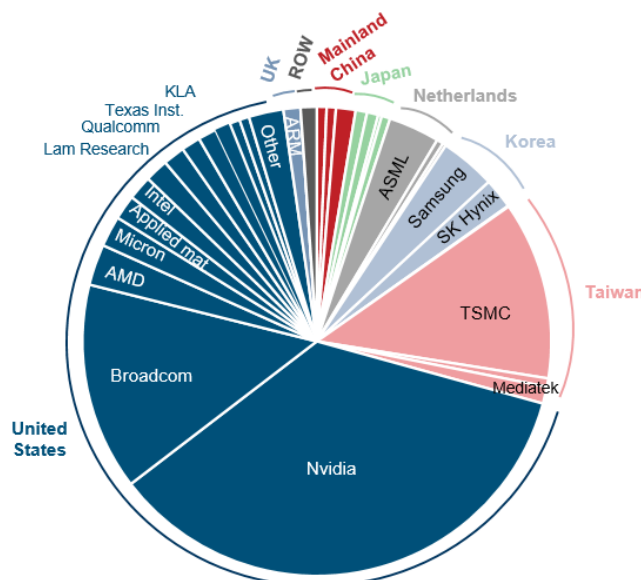
How are semiconductors used?

Global demand share by end use, 2024



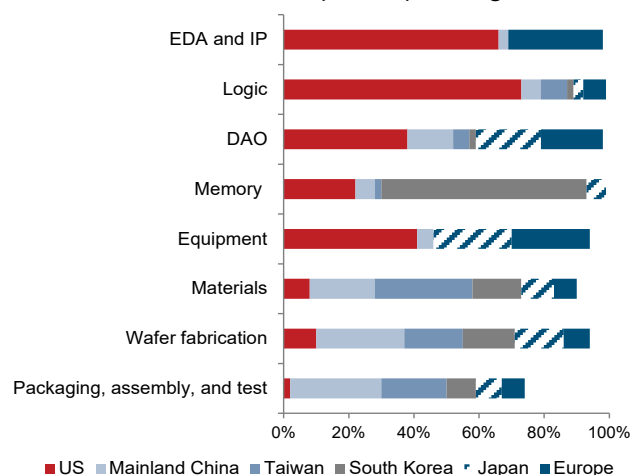
Who are key competitors in the chip industry?

Global semiconductor companies by market cap



What in the supply chain does each region specialize in?

Semiconductor value added by activity and region, 2024, %



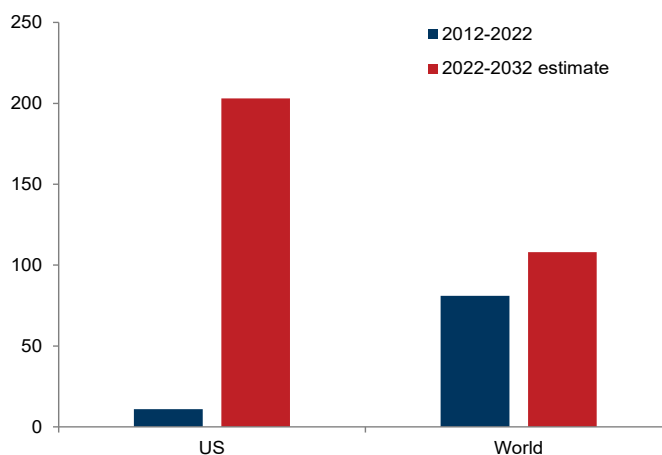
Note: DAO is discrete, analog, other; EDA is electronic design automation.

Source: [Semiconductor Industry Association](#), BCG, government websites, various news sources, companiesmarketcap.com, Goldman Sachs GIR.

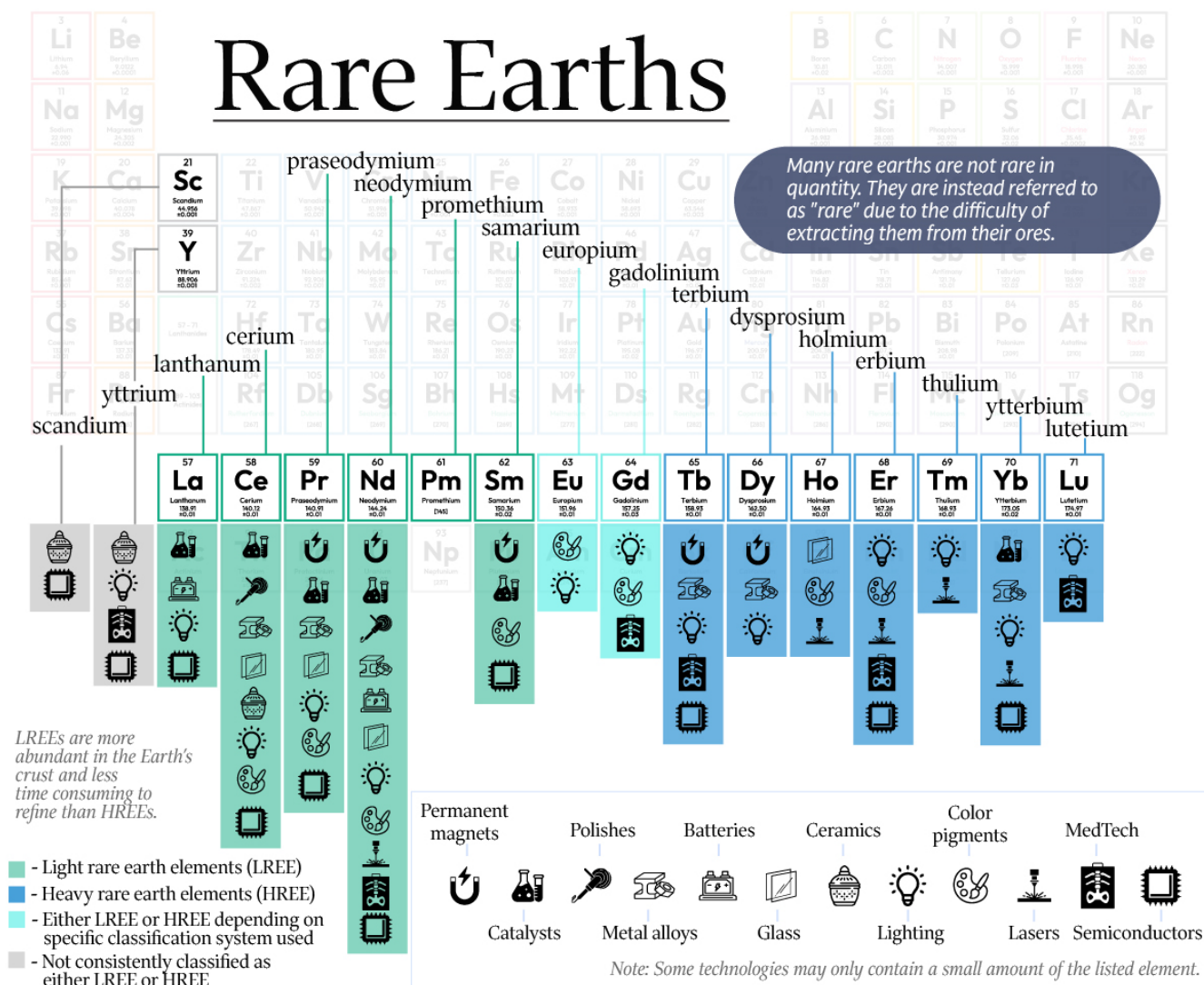
Charts originally appeared in [Issue 144: The US-China tech race](#).

Where is chip-making investment concentrated and how could it evolve?

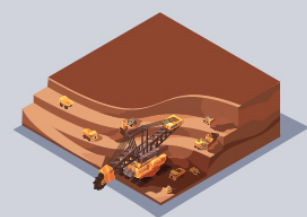
Projected increase in US fab capacity vs. world avg., % chg.



...rare earth self-sufficiency



The rare earth supply chain, at a glance



Mining

Rare earth elements are mined from the ground. These elements are found in various ore deposits. The mining method depends on the type of deposit. Methods include open-pit mining, underground mining, and in-situ leaching.



Processing and Refining

Following mining, the extracted ore undergoes several stages of processing. These include crushing and grinding (the raw ore is crushed and ground into small particles), beneficiation (which separates the rare earth minerals from other waste minerals), and extraction and separation (chemical treatments are used to extract the individual rare earths from concentrate).



Manufacturing of products

The separated oxides are converted into rare earth metals and, in turn, alloys. These alloys are used to create powerful magnets known as permanent magnets, which are a key component of various technologies.



China has the largest share of global rare earth reserves (49%). This includes both light and heavy rare earths, with known HREE reserves extremely scarce outside of China and Myanmar. China also dominates global rare earth mining (69%). Other countries with significant rare earth reserves/mining are:

- Brazil (23% and 0%)*
- India (8% and 1%)
- Australia (6% and 3%)
- US (2% and 12%)



China also dominates global rare earths refining (92%), particularly in heavy rare earths, with Chinese refineries producing the vast majority of the world's supply of dysprosium and terbium metals. Lynas, an Australian company with operations in Malaysia, is one of the only other major players at this stage of the supply chain, accounting for 4% of global refined production. The US accounts for only 1%.



China is the world's largest supplier of permanent magnets, which are critical in many technologies. China accounts for 98% of magnet manufacturing.

** Figures in parentheses represent the share of reserves and mining, respectively.*

Source: IEA, Rareearths.com, Federal Register, various news sources, Goldman Sachs GIR.

Exhibit originally appeared in [Issue 144: The US-China tech race](#).

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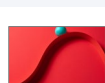
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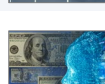
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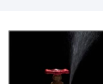
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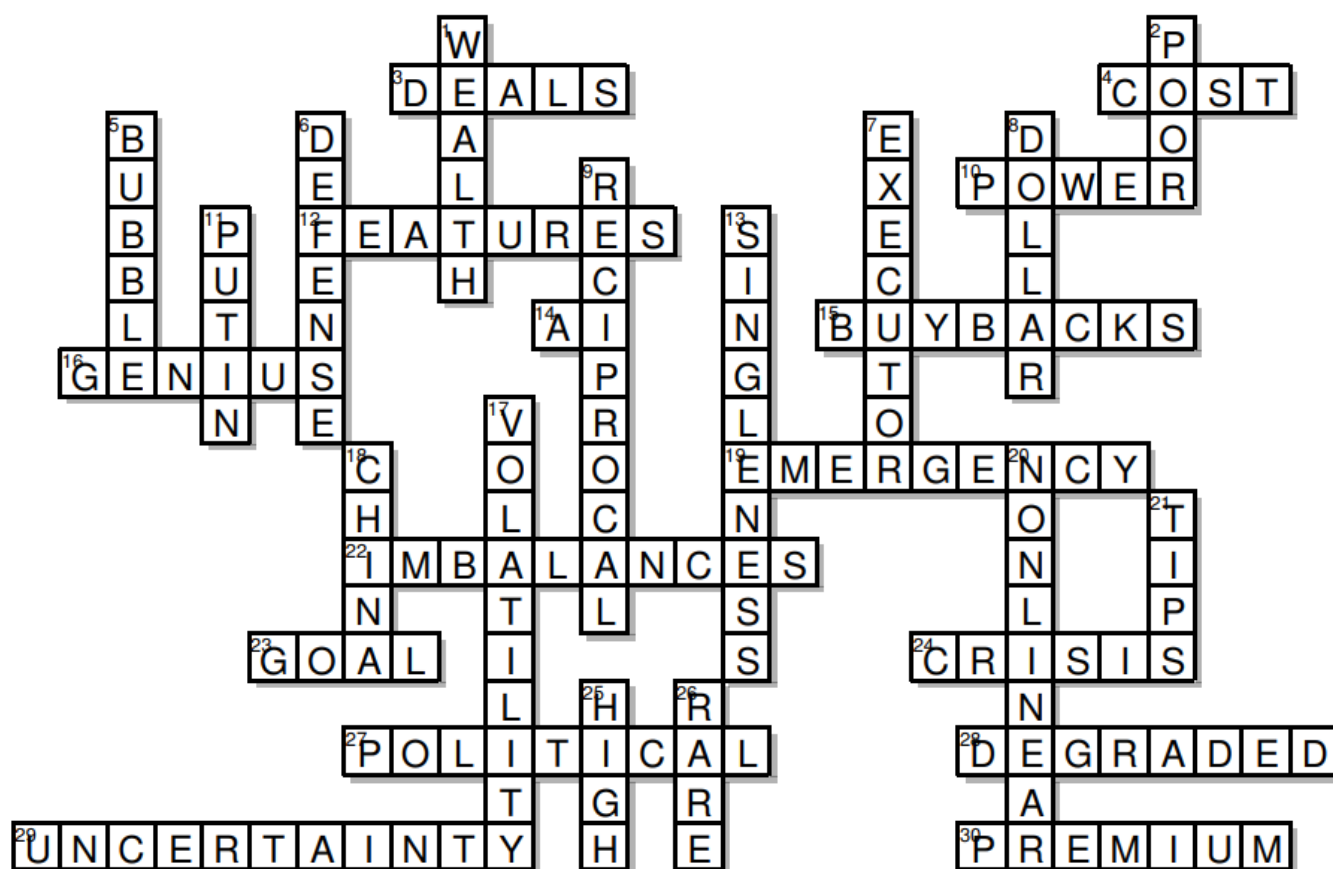


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Source of photos: www.gettyimages.com.

Puzzle solutions

Crossword from pg. 3 of this report:



Glossary of GS proprietary indices

Current Activity Indicator (CAI)

GS CAIs measure the growth signal in a broad range of weekly and monthly indicators, offering an alternative to Gross Domestic Product (GDP). GDP is an imperfect guide to current activity: In most countries, it is only available quarterly and is released with a substantial delay, and its initial estimates are often heavily revised. GDP also ignores important measures of real activity, such as employment and the purchasing managers' indexes (PMIs). All of these problems reduce the effectiveness of GDP for investment and policy decisions. Our CAIs aim to address GDP's shortcomings and provide a timelier read on the pace of growth.

For more, see our CAI page and Global Economics Comment: Technical Updates to Our Global CAIs.

Dynamic Equilibrium Exchange Rates (DEER)

The GSDEER framework establishes an equilibrium (or "fair") value of the real exchange rate based on relative productivity and terms-of-trade differentials.

For more, see our GSDEER page, Global Economics Paper No. 227: Finding Fair Value in EM FX, 26 January 2016, and Global Markets Analyst: A Look at Valuation Across G10 FX, 29 June 2017.

Financial Conditions Index (FCI)

GS FCIs gauge the "looseness" or "tightness" of financial conditions across the world's major economies, incorporating variables that directly affect spending on domestically produced goods and services. FCIs can provide valuable information about the economic growth outlook and the direct and indirect effects of monetary policy on real economic activity.

FCIs for the G10 economies are calculated as a weighted average of a policy rate, a long-term risk-free bond yield, a corporate credit spread, an equity price variable, and a trade-weighted exchange rate; the Euro area FCI also includes a sovereign credit spread. The weights mirror the effects of the financial variables on real GDP growth in our models over a one-year horizon. FCIs for emerging markets are calculated as a weighted average of a short-term interest rate, a long-term swap rate, a CDS spread, an equity price variable, a trade-weighted exchange rate, and—in economies with large foreign-currency-denominated debt stocks—a debt-weighted exchange rate index.

For more, see our FCI page, Global Economics Analyst: Our New G10 Financial Conditions Indices, 20 April 2017, and Global Economics Analyst: Tracking EM Financial Conditions – Our New FCIs, 6 October 2017.

Goldman Sachs Analyst Index (GSAI)

The US GSAI is based on a monthly survey of GS equity analysts to obtain their assessments of business conditions in the industries they follow. The results provide timely "bottom-up" information about US economic activity to supplement and cross-check our analysis of "top-down" data. Based on analysts' responses, we create a diffusion index for economic activity comparable to the ISM's indexes for activity in the manufacturing and nonmanufacturing sectors.

Macro-Data Assessment Platform (MAP)

GS MAP scores facilitate rapid interpretation of new data releases for economic indicators worldwide. MAP summarizes the importance of a specific data release (i.e., its historical correlation with GDP) and the degree of surprise relative to the consensus forecast. The sign on the degree of surprise characterizes underperformance with a negative number and outperformance with a positive number. Each of these two components is ranked on a scale from 0 to 5, with the MAP score being the product of the two, i.e., from -25 to +25. For example, a MAP score of +20 (5;+4) would indicate that the data has a very high correlation to GDP (5) and that it came out well above consensus expectations (+4), for a total MAP value of +20.

Disclosure Appendix

Reg AC

We, Allison Nathan, Jenny Grimberg and Ashley Rhodes, hereby certify that all of the views expressed in this report accurately reflect our personal views, which have not been influenced by considerations of the firm's business or client relationships.

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GS DataWorks analysis provided by Dan Duggan, Ph.D.

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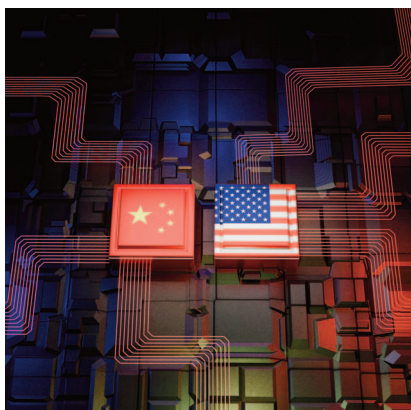
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TOP^{of} MIND

THE US-CHINA TECH RACE



While the US and China have reached a new trade deal, their fierce race to achieve tech superiority remains as intense as ever, with both pursuing policies to develop self-sufficient tech stacks. So, will such policies prove successful, and what investment opportunities do they provide? The Wahba Initiative for Strategic Competition's Mark Kennedy assesses the tech race's current state (the US leads in a key arena but China is catching up or leading in many others) and GS' Alec Phillips and Hui Shan dig into the why, what, and how of US and Chinese policy efforts to win the race. DGA-Albright Stonebridge Group's Paul Triolo and the Critical Minerals Institute's Jack Lifton then explain the intricacies and chokepoints of the two supply chains in focus here—semiconductors and rare earths—and

GS' Daan Struyven digs into power's critical role in determining the winner. Lastly, we explore related investment opportunities, with GS equity analysts and strategists seeing compelling opportunities in rare earths and Asia tech.



It is entirely possible that neither the US nor China emerge as the outright victor in the tech race. I can envision a world in which the US leads in developing the most advanced technologies, while China leads in global installations.

- Mark Kennedy

The desire to reduce our dependence on China [for rare earths] is valid. And there is a viable path to achieving that, even if it won't amount to self-sufficiency. It's just a matter of picking the right companies, technologies, and people.

- Jack Lifton

The US expanding or even maintaining its current lead [in advanced AI] is not a foregone conclusion... much will depend on the Chinese semiconductor industry's ability to overcome the hardware chokepoints.

- Paul Triolo



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Macro news and views

We provide a brief snapshot on the most important economies for the global markets

US

Latest GS proprietary datapoints/major changes in views

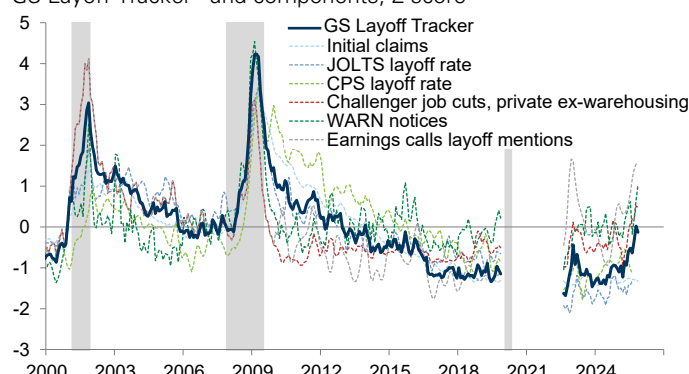
- No major changes in views.

Datapoints/trends we're focused on

- US growth, which we expect to modestly reaccelerate to 2.3% in 2026 (on a Q4/Q4 basis) owing to a reduced tariff drag, tax cuts, and easier financial conditions.
- Fed policy; we expect the Fed to deliver a 25bp rate cut next week followed by two 25bp cuts in 2026 for a terminal rate range of 3-3.25%, but see downside risks to our forecasts given ongoing labor market weakness.
- US productivity outperformance vs. other DMs, which structural factors & AI adoption tailwinds should sustain.

US layoffs: on the rise

GS Layoff Tracker* and components, Z-score



*First principal component of our tracking of WARN notices, initial claims, Challenger layoffs, and earnings calls layoff discussions.

Source: Dept. of Labor, Challenger, Gray & Christmas, Federal Reserve, GS GIR.

Europe

Latest GS proprietary datapoints/major changes in views

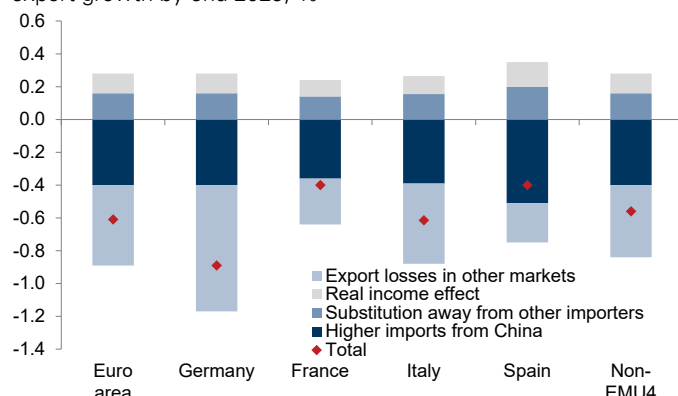
- We lowered our 2026/27 Euro area GDP growth forecasts to 1.2%/1.3% yoy (from 1.3%/1.5%) to reflect increased export competition with China and lower potential growth.
- We raised our 2026 UK GDP growth forecast to 1.1% yoy (from 1.0%) as the Autumn Budget projections suggest a smaller fiscal drag on demand than we originally assumed.

Datapoints/trends we're focused on

- BoE rate cuts, the next one of which we expect this month.
- ECB policy; we expect the ECB to remain on hold, though the risks are tilted toward renewed rate cuts next year.

Europe: an export drag from China's second wave

Estimated country-level real GDP impact of higher Chinese export growth by end-2029, %



Source: Haver Analytics, Goldman Sachs GIR.

Japan

Latest GS proprietary datapoints/major changes in views

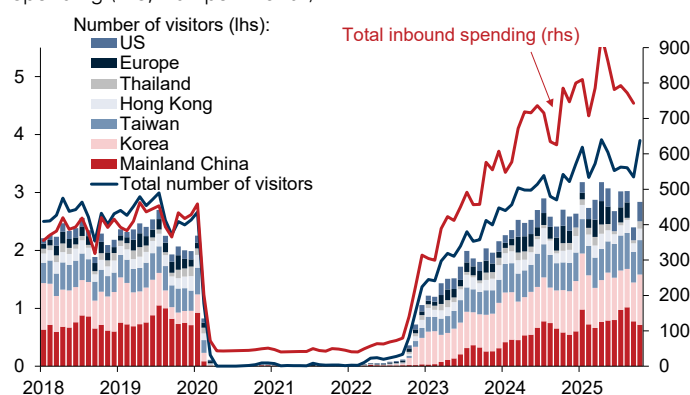
- No major changes in views.

Datapoints/trends we're focused on

- BoJ policy; while Jan 2026 remains our baseline forecast for the next hike, the likelihood of a rate hike this month has risen following a recent hawkish speech by Governor Ueda.
- Heightened Japan-China tensions, which we estimate could result in a 0.2pp hit to Japanese GDP growth from reduced Chinese tourism and lower Japanese exports to China.
- An expected rise in Japanese potential growth over the next several years, driven by labor productivity growth.

Japan: rising China tensions, falling Chinese tourism

Foreign visitors to Japan (lhs, mn per month) and their inbound spending (rhs, ¥bn per month)



Source: JNTO, BoJ, Goldman Sachs GIR.

Emerging Markets (EM)

Latest GS proprietary datapoints/major changes in views

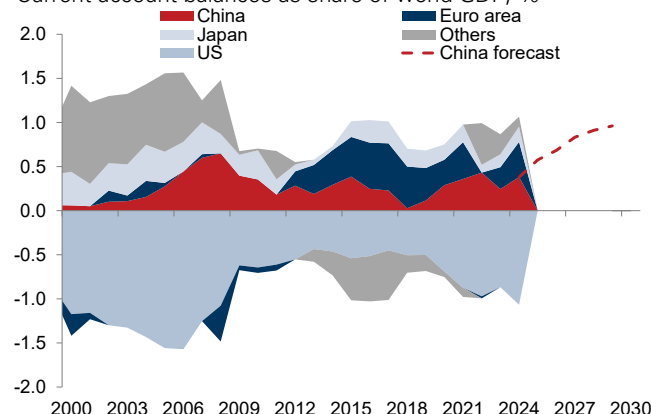
- We recently raised our CY2025/2026 India real GDP growth forecasts to 7.6%/6.6% yoy (from 7.3%/6.5%) following much stronger-than-expected 3Q2025 GDP data.

Datapoints/trends we're focused on

- China shock 2.0; China's renewed focus on export-led growth will likely benefit its economy and raise its current account surplus to 1% of global GDP by 2029, reducing its trading partners' manufacturing output and employment.
- EM growth; we expect growth in China, Brazil, and India to slow in 2026 but pick up in several mid-sized EMs, leaving overall EM growth broadly unchanged.

China shock 2.0

Current account balances as share of world GDP, %



Source: IMF, Haver Analytics, Goldman Sachs GIR.

The US-China tech race

While the US and China reached a new trade deal following bilateral talks at the October APEC summit, their fierce race to achieve technological superiority—driven by a complex interplay of security, economic, and geopolitical interests—remains as intense as ever, with both countries pursuing policies to develop self-sufficient tech stacks. Whether such policies will prove successful, and the investment opportunities they provide, is Top of Mind.

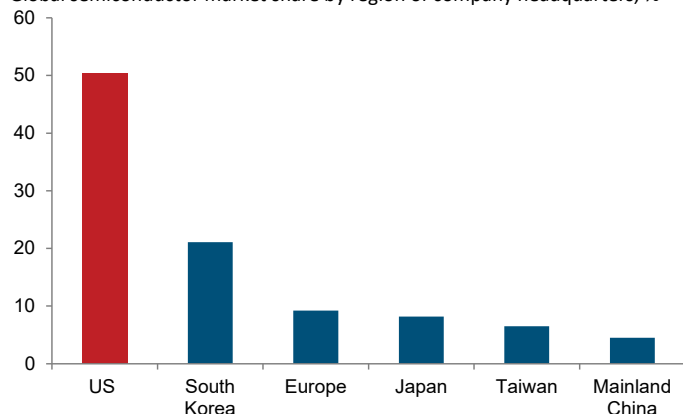
We first ask Mark Kennedy, Founding Director of the Wahba Initiative for Strategic Competition at NYU's Development Research Institute, to assess the current state of the tech race. He explains that while the US leads in many key areas of technological innovation (including semiconductors, AI frameworks, cloud infrastructure, and quantum computing), China is quickly closing the gap or even dominating in three other critical arenas of the race—practical applications like physical AI and robotics, installations of the digital plumbing that underpins technology owing to China's strong presence in the Global South, and technological self-sufficiency as China has worked to reduce its dependence on Western technologies.

GS Chief US Political Economist Alec Phillips and Chief China Economist Hui Shan then dig deeper into the why, what, and how of US and Chinese policy efforts to win the tech race. Phillips explains that US policymakers are combining the playbooks deployed during World War II/the Cold War and the US-Japan economic rivalry of the late 20th century by taking an increased role in some areas (like semiconductors and AI) while relying on financial incentives to boost domestic production and purchases in sensitive sectors (like rare earths). And Shan details Chinese policymakers' "holistic and highly coordinated" approach to supporting technology, which involves systematic planning, significant financial backing, and assisting in other areas from land acquisition to talent development.

We then explore the two supply chains most in focus amid this race: the global semiconductor supply chain—for which the US has a slight edge—and the rare earth supply chain—which China currently controls (see pgs. 10 and 11).

The US accounts for half of the global semiconductor market...

Global semiconductor market share by region of company headquarters, %



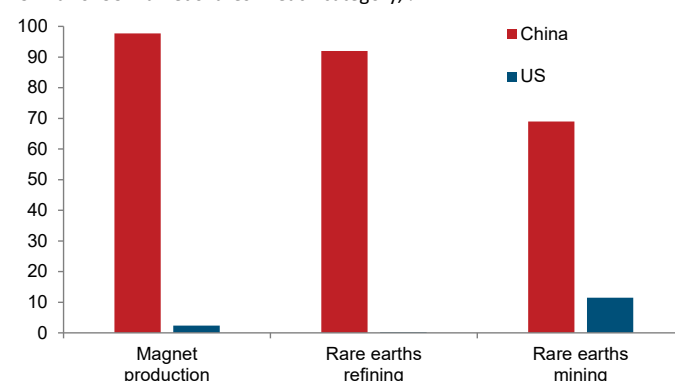
Source: World Semiconductor Trade Statistics (WSTS), Gartner, Omdia, Semiconductor Industry Association, compiled by Goldman Sachs GIR.

We first turn to Paul Triolo, Partner at DGA-Albright Stonebridge Group, for an explanation of the intricacies and key chokepoints of the semiconductor supply chain, which he says makes it difficult for any country to achieve self-sufficiency. But he argues that despite the US' current lead in chip innovation, "if

any country can do it (i.e., achieve self-sufficiency), it's China, owing to its vast resources and pool of engineering talent", though he sees significant technical and financial challenges. But he also believes that recent US policy efforts to strengthen its domestic semiconductor capabilities don't go far enough, leaving open the possibility that China could close the gap with the US in developing the world's most powerful AI models, while also maintaining or even growing its lead in AI adoption given China's embrace of open-source models.

...while China dominates the rare earths space

China vs. US market shares in each category, %



Source: IEA, Woodmac, Goldman Sachs GIR.

We then speak to Jack Lifton, Co-chair of the Critical Minerals Institute, who breaks down China's grip on the supply chain for rare earths, and especially the heavy rare earths used in defense and other critical industries. While he believes China's chokehold can be broken, he argues that current US efforts to do so will not succeed because US policymakers are not investing in "the right companies, technologies, and people." But he also thinks the need for such efforts is overblown given that US rare earth demand is small and acceptable substitutes exist, at least in the military complex.

GS Co-head of Global Commodities Research Daan Struyven nonetheless makes the case that China's dominance in rare earths, together with its abundant power supplies—in sharp contrast to the US' power constraints—give China an edge in the AI and broader tech race. Triolo and Kennedy agree that China's power advantage presents a challenge for the US, which they say sets the stage for the energy-abundant Middle East to play a pivotal role in the US-China tech race.

So, how should investors be positioned as US and Chinese policymakers deploy their playbooks for "winning" this race? GS equity analysts find that the stocks of leading semiconductor manufacturing and equipment companies have already largely priced in US-China tech developments. But they think continued efforts by both countries to develop their own tech stacks could remain a key driver of price action for Western rare earth mining and refining companies. And GS senior Asia Pacific strategists Tim Moe and Kinger Lau see significant opportunities for Asia tech companies—which have outperformed this year—amid these ongoing policy efforts.

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Interview with Mark Kennedy

Mark Kennedy is Founding Director of the Wahba Initiative for Strategic Competition at New York University's Development Research Institute. Below, he lays out the current state of the US-China tech race, arguing that the US leads in a key arena, but China is quickly catching up or even leading in many others.

The views stated herein are those of the interviewee and do not necessarily reflect those of Goldman Sachs or of NYU.



Ashley Rhodes: Technology seems to be at the center of the US-China strategic rivalry. Is that an accurate characterization, or has the role of technology been overplayed?

Mark Kennedy: The role of technology is not being overplayed. Technology isn't just at the center of the US-China rivalry—it's the central switchboard.

Whoever controls how technology, data, and computing power are routed will impact every domain—from military might, to economic influence, to the flow of information. So, the significance of technology in the US-China rivalry is anything but overstated. The profound impact that this technological contest could have on global systems and the resulting geopolitical implications are, if anything, underappreciated.

Ashley Rhodes: Who is currently "winning" the tech race?

Mark Kennedy: It's important to understand that there are four key arenas in this race: technological innovation, practical application of the technology, installation of the digital plumbing or infrastructure underpinning the technology, and technological self-sufficiency. The US is currently leading in most advanced technologies, including semiconductors, AI frameworks, cloud infrastructure, and quantum computing, as well as in attracting global talent. However, China is ahead in areas such as quantum communications, hypersonics, and batteries.

China is also making rapid strides to catch up to and, in some cases, overtake the US in technological application. For example, China deploys robotics in manufacturing on a scale twelve times greater than the US when adjusted for differences in employee income. And while US regulations often limit applications like drone deliveries to your door, China is proactively testing and deploying advanced physical AI and robotics like uncrewed taxis and vertical takeoff vehicles, accelerating their practical adoption.

China is also dominating on the global installations front. It has established a strong presence in the Global South, surpassing the US and other Western nations in building essential digital networks there. And China has made significant strides toward achieving technological self-sufficiency through its dual circulation strategy aimed at reducing its reliance on the West while increasing Western dependence on China. Recent Chinese government measures, such as restricting domestic purchases of Western chips and offering incentives for using domestic alternatives, underscore this push for technological independence. At the same time, China's vast overproduction capacity in batteries and critical minerals has further increased Western dependence on China's supply chains. The US has been ambivalent at best as it relates to this aspect of the tech race and remains reliant on China in many ways. So, on net,

while the US leads in the development of the technology itself, China is rapidly closing the gap—or even leading—in application, infrastructure installations, and tech self-sufficiency.

Ashley Rhodes: Have US export controls on advanced chips materially impeded China's technological progress?

Mark Kennedy: The impact has not been especially significant. China had already planned to reduce its reliance on American semiconductor technology, and the introduction of US export controls has merely accelerated that transition. China is finding ways to adapt by using less advanced chips and still maintains some access to certain US chip technologies. So, while these measures are slowing China's technological progress, they are not significantly impeding it and are unlikely to be the silver bullet that allows the US to maintain its tech lead indefinitely.

Ashley Rhodes: Conversely, will China's grip on rare earths impede the US' ability to maintain tech leadership?

Mark Kennedy: Rare earth elements are a critical input in the tech race, making China's export restrictions a real threat to America's long-term prospects for developing an independent tech ecosystem. While China has granted the US a one-year reprieve on these restrictions, the recent deal between the two countries is fragile and the US will remain vulnerable unless it can rapidly expand its access to rare earths. The US has been aware of this vulnerability for a long time but has admired the problem more than it has taken deliberate steps to address it. America is just now waking up to the seriousness of this threat and beginning to take steps in the right direction, including US government efforts to invest in rare earth companies and take ownership stakes in key firms. But the US is still a long way from achieving self-sufficiency, and true resilience will depend on working closely with allies like Australia and Canada to establish secure, diversified supply chains. Regrettably, US relationships with its key allies have become increasingly fractured, which isn't helpful.

Ashley Rhodes: What else will it take for the US to win?

Mark Kennedy: Beyond diversifying supply chains for critical minerals and other essential technological inputs, the US must more broadly move beyond its historically laissez-faire approach toward technology and lean into winning the tech competition across all arenas. This means significantly increasing investment in research and development, reversing the recent trend of stagnant or declining university funding, and easing restrictions that hinder the entry of global talent.

The US must also expand its energy infrastructure to support the growing power demands of data and computing. While China has gone far in securing energy resources—particularly in renewables such as wind, solar, and nuclear—the US approach has been inconsistent. The current shift toward nuclear energy is promising, but achieving the necessary scale will take time.

And the US must strive to match China's push to dominate installations in the Global South or risk losing influence in these regions. To date, restrictive policies—such as the Biden Administration's AI Diffusion Rule and other restrictions—have limited the reach of US tech. Actions like cutting off Ukraine's satellite access have raised concerns that the US may wield technological access as a form of geopolitical leverage. And where the US withholds its tech, China readily steps in. While recent AI policy discussions have recognized the importance of deploying the domestic tech stack globally, the real challenge is getting multiple government agencies and private companies to act with the same unity and speed as China does.

Ashley Rhodes: The US does seem to be increasing public-private partnerships to make progress toward these goals. How important are such partnerships?

Mark Kennedy: Government involvement in the critical minerals sector will play a crucial role given that China's dominance allows it to lower prices and squeeze out competitors. So, US companies need to have some guarantee that prices will be at a level that will at least sustain their production. These guarantees will be essential to generating the sources of supply at the mineral level that will allow the US to apply its magic at the digital level. And government involvement will be critical to expanding the reach of US technology to other parts of the world, like the Global South.

However, the benefits of other types of government involvement are less clear. For example, it's unclear whether the government taking an ownership stake in Intel will prove helpful beyond the near-term boost in capital and confidence it provided. Such a move could actually be harmful if measures like "golden shares" prevent the company from making the tough decisions that will be required for it to remain globally competitive. Intel's role as the sole domestic foundry of scale is vital, especially as TSMC's investment in the US remains a shadow of the company's efforts to maintain its silicon shield in Taiwan. The trick is providing support without politicized interference. Past US ownership in established commercial enterprises has seldom delivered long-term benefits.

Ashley Rhodes: Conversely, what additional steps must China take to win the tech race?

Mark Kennedy: China is already doing many things right. Its most effective strategy has been sustained, targeted investment in key technologies that will determine the future. And China has long understood the importance of securing and leveraging essential resources like rare earths as well as areas like ports. As we've discussed, China has also been ahead in establishing new markets in key regions like the Global South.

But China recognizes that it needs to accelerate the advancement of its domestic semiconductor industry. To that end, the Chinese government has begun channelling investments into local chip manufacturing, prioritizing long-term development over short-term performance gains.

Another critical priority should be building trust. The Chinese model is characterized by centralized control; their systems are not easily interoperable or replaceable, leading to a degree of lock-in. And China's cybersecurity laws require companies to

provide any requested data to government authorities. All this undermines international trust in Chinese tech. The tech competition is in many ways a battle for trust. The US is currently leading in this area given its history of transparency and reliability, while China has more work to do in ensuring that its technology is not only affordable and reliable but also trustworthy.

Ashley Rhodes: How important are other countries in determining who wins the tech race?

Mark Kennedy: The technological alignment of other countries will be pivotal in determining the outcome of the tech race. Again, China has been aggressively targeting the Global South, offering comprehensive digitalization packages—often as service contracts—which enable it to build, maintain, and exert substantial control over networks and the data they carry. This access to vast amounts of data, in turn, enables China to train AI models that provide insights into how to tailor products or services specifically for those markets, reinforcing its influence and making it increasingly difficult for competitors to match the value China offers.

By contrast, the US has not matched China's government-backed efforts in these markets. Once-prominent programs like USAID, whose grants supporting host country capabilities paired with financing support helped the US prevail in many international infrastructure competitions, have largely been diminished. This has left the US at risk of ceding technological influence while China positions itself to digitally connect up to 70–80% of the world's landmass, aligning it away from US interests. The US really needs to consider whether it should be doing a better job of matching China's thumb on the scale in determining through whose networks the world's data flow.

The Gulf countries will also play a pivotal role in the tech race given their abundant energy resources. Whoever succeeds in establishing influence within these countries will gain privileged access to vast energy resources they can use to power their technological ambitions and capabilities. That said, with both China and the US likely to be increasingly engaged in the Gulf for this reason, establishing security protocols to ensure that tech provided to the Gulf does not drift to China will be essential. Geopolitically, the US must embrace the Gulf, or China will fill the void.

Ashley Rhodes: Ultimately, how do you envision the US-China tech race playing out?

Mark Kennedy: It is entirely possible that neither the US nor China emerge as the outright victor in the tech race. I can envision a world in which the US leads in developing the most advanced technologies, while China leads in global installations, particularly outside of America and its closest allies. China could also prevail in applications as its vast quantity of electric engineering graduates and PhDs could take on a quality of its own. And China could win the "we're independent of you, but you're still dependent on us" race. So, a world could exist in which the US owns the blueprints—leading in software and code—but China owns the buildings—commanding the hardware and circuitry—which would have the potential to significantly reshape economies and geopolitics as we know it.

A US policy roadmap for the tech race

Alec Phillips explores the US policy roadmap for achieving technological self-sufficiency

US policymakers have drawn parallels between the tech rivalry with China and the urgent US industrial and scientific efforts during World War II as well as the US-Soviet tech competition during the Cold War. During both periods, intense scientific and technological rivalries led the US government to organize and fund research and development in several key areas of competition (e.g., nuclear fission and aerospace).

However, many of the technologies central to the US-China tech rivalry today are primarily commercial. While most have benefitted from publicly funded research and have military or strategic uses, the markets for these technologies are much broader and, in many cases, the technology has already been commercialized. Indeed, private sector funding for AI or semiconductors dwarfs government programs. This presents a challenge that goes beyond scaling up public (i.e., military) spending, with arguably greater parallels to the US-Japan economic rivalry in several sectors during the 1970s and 1980s.

US policy appears to be following approaches along both lines, with a larger government role in some areas while primarily relying on financial incentives for production or purchase of domestically-produced products in sensitive sectors.

Focusing investment in strategic sectors

While no formal definition of the sectors of greatest strategic importance exists, the Trump Administration appears to prioritize AI, quantum computing, nuclear technology, and biotechnology, along with related inputs like rare earths/critical minerals and semiconductors. Robotics and drones, shipbuilding, aerospace, and power technology also appear to be priorities, though not as central a focus.

Technologies targeted by several government initiatives

	Genesis Mission (Nov. 2025)	White House memo on R&D priorities (Sep. 2025)	Trump memo to OSTP Dir. (Mar. 2025)	DoD "Critical Technology Areas" (Nov. 2025)	Sec. 232 focused investment (Nov. 2024)	China-focused investment rules (Nov. 2024)	Outbound investment rules (Nov. 2024)	Subsidy program (CHIPS, IRA, OBBBA)	Loan or equity stake (2025)	Trade partner investment goals
AI	•	•	•	•						
Quantum	•	•	•	•						
Nuclear	•		•						•	•
Semis	•				•	•	•	•	•	•
Biotech	•	•		•						
Critical Minerals	•				•			•	•	•
Space	•	•								
Shipbuilding						•		•		•
Aerospace					•			•		•
Steel					•					
Robotics					•				•	
Energy trans.										
/storage								•		•

Source: White House, Department of Defense, USTR, Goldman Sachs GIR.

The Trump Administration has several tools to invest in strategic sectors:

- **CHIPS Act grants:** While around \$43bn of the \$50bn in direct funding in the CHIPS Act (2022) has been allocated, the Commerce Department continues to deploy funding from the Act, including for financing equity stakes in key firms.
- **Advanced manufacturing incentives:** The One Big Beautiful Bill Act (OBBBA) renewed and expanded the Advanced Manufacturing Investment Credit (48D) from 25% to 35%, which subsidizes domestic investment in semiconductor manufacturing capacity and related tech. While 48D was originally enacted in the CHIPS Act, funding is open-ended and separate from the \$50bn grant authority

noted earlier. The OBBBA also preserved the Inflation Reduction Act's (IRA) Advanced Manufacturing Production Credit (45X), which subsidizes domestic production of batteries as well as wind and solar components, but the new legislation reduces the credit for costs attributable to products from Chinese companies. The OBBBA also included broader manufacturing incentives, including immediate expensing of manufacturing structures and equipment.

- **Government financial stakes:** The Trump Administration has announced stakes in firms in the critical minerals, nuclear energy, semiconductor, and steel sectors totaling \$10bn (Intel accounts for nearly \$9bn of this). While some of these financial commitments involve equity, others consist primarily of loans. OBBBA expanded the scale and reach of these programs via a \$1.5bn appropriation allowing up to \$200bn in lending through the Dept. of Defense Office of Strategic Capital, which will likely lead to lower-cost, longer-term loans than these firms would otherwise receive. OBBBA also gives \$2bn to the Defense Innovation Unit, which focuses on scaling commercial tech for military use, and \$10bn to the Industrial Base Fund, for investments in critical minerals supply chains and to bolster the US critical minerals stockpile.
- **Trade partner investment pledges:** The Trump Administration announced agreements with the European Union (\$600bn), Japan (\$550bn), South Korea (\$350bn), and Switzerland (\$200bn) to invest in the US. The structure of these pledges is not entirely clear but the most defined of them, the Japanese investment pledge, suggests that these pledges might function primarily as low-cost loans with a small equity component, targeting a combination of strategic sectors (e.g., semiconductors, energy storage, and data centers) and traditional investments (e.g., natural gas and electric utility infrastructure, fertilizer production, etc.).
- **Lighter domestic regulation:** The Trump Administration has pledged to ease regulation in general, targeting permitting new projects and energy generation, with Trump signing an executive [order](#) in July to accelerate permitting of data center infrastructure. Congress is debating permitting reform, with proponents making another push for enactment in the next few months. If enacted, this could facilitate a buildout of the power grid, where undercapacity threatens to become an obstacle to AI infrastructure investment.

Creating a demand signal for certain industries

While demand is clear in some segments like AI and semiconductors, it is less reliable in others. In those areas, US policy provides assurances/guarantees for certain technologies produced in the US or pursuant to public-private agreements:

- **Offtake agreements, floor prices, and other market interventions:** Recent agreements that the Trump Administration has reached with companies in targeted sectors include offtake agreements and floor prices to limit vulnerability to market and non-market forces (see pg. 17). This is particularly relevant in areas like rare earths, where the periodic lifting of restrictions could lead to substantial price volatility and make domestic production uneconomic, facilitating continued Chinese dominance in the sector.
- **Government procurement and demand subsidies:** Defense spending has been an important contributor to US

innovation and will likely play a role in some sectors. OBBBA provides ~\$30bn for domestic shipbuilding, for example. And the IRA continues to subsidize renewable energy investment and production and electric vehicle purchases, though the OBBBA both tightened and phased these down to place more emphasis on domestically-produced technology.

Reducing demand for Chinese technology

In addition to boosting investment in and demand for US tech, the US has in place or is contemplating several policies that aim to reduce demand for competing Chinese products:

- **Tariffs on Chinese products:** The Trump Administration reduced the 30% tariff imposed on imports from China earlier this year to 20% as part of the October deal, which also paused US export controls in return for China's easing of rare earth export restrictions. The remaining tariff consists of 10% on all imports plus an additional 10% "reciprocal" tariff that exempts several categories, discussed below. These tariffs apply on top of the ~10% weighted average tariff applied on Chinese imports during Trump's first term. We expect tariffs on China to remain at the current level for the foreseeable future, with risks toward lower tariffs if the Administration does not fully replace them should the Supreme Court block them in its upcoming ruling.
- **The threat of tariffs on key sectors:** The Trump Administration launched national security investigations under Sec. 232 into imports of semiconductors, pharmaceuticals, critical minerals, drones, polysilicon, wind turbines, robotics and industrial machinery, as well as medical consumables and devices. While we have assumed that the Administration will follow through with threatened sectoral tariffs, we no longer assume the imposition of any pharmaceutical tariffs in light of the political challenge this could create and the Administration's signaling that most major firms in the space would be exempt. Similarly, tariffs on semiconductors/electronics could be controversial ahead of the midterm elections and would raise the cost of consumer electronics. That said, the Administration will likely continue to use tariffs to protect sensitive sectors in the US, particularly from competition from lower-cost Chinese firms.
- **Regulation of Chinese components in US systems:** During the final days of the Biden Administration, the Office of Information and Communications Technology and Services (OICTS) and the Bureau of Industry and Security (BIS) at the Commerce Department issued rules limiting the use of tech from China or other "foreign adversaries" in connected and self-driving vehicles. Some lawmakers have [urged](#) the Trump Administration to take similar actions in other areas, including AI and cloud infrastructure, robotics, energy generation and storage, network hardware, and chip manufacturing equipment, among other sectors. Separately, the Commerce Department has issued [guidance](#) that the use of advanced Chinese chips would likely violate US export control policies.
- **BIOSECURE Act:** In October, the Senate passed an amendment to the National Defense Authorization Act (NDAA) that would bar federal agencies from contracting with "companies of concern", which could limit US firms dealing with the federal government from using inputs from Chinese entities. While this provision looks unlikely to be included in the final Act, it reflects US policymakers' clear intent to limit reliance on the Chinese health supply chain.

Limiting diffusion of certain US technology

US policy has also focused on limiting the supply of technology to China, particularly in the areas of AI and semiconductors:

- **Export controls:** The US continues to impose export controls on sensitive technologies, particularly advanced semiconductors and chip manufacturing technology. The US "entity list" has grown this year, and the Commerce Dept. expanded the definition of an affiliated entity under a September 2025 rule that would have had a big impact on the ability of Chinese companies to procure US technology. However, implementation of this rule was delayed as part of the October deal and it appears unlikely that the US will impose further export controls on China for the next year.
- **Outbound investment rules:** Restrictions on US investment in Chinese entities that operate in AI, chips, and quantum technologies, which the Biden Administration [finalized](#) in late 2024, are now in effect. Congress is also finalizing its NDAA legislation for 2025 and is considering codifying similar but potentially broader rules, though outbound restrictions look unlikely to be included in the final legislation. While investment in Chinese biotech, aerospace, and other key sectors are not covered under current rules, the White House has [signaled](#) an intent to consider these transactions.
- **GAIN Act and the Diffusion Rule:** The Biden Administration finalized rules at the start of 2025 that would have required US chip companies to retain a sufficient portion of AI-relevant chips for domestic users and strictly limited exports to several countries, including China. The Trump Administration rescinded that rule in May, but the Senate passed an amendment to the NDAA in October that would resurrect some aspects of it, giving US customers priority before allowing export. The final legislation looks unlikely to include these restrictions, but the outcome is uncertain.

A tailored policy roadmap

US policy approaches to the tech race will vary by sector. The AI sector, for example, has little need for financial incentives, but could benefit from regulatory easing and public-private coordination, particularly related to the mobilization of federal research programs and diffusion across the economy (note the Administration's recent "Genesis Mission" to coordinate research and make federal scientific datasets broadly available).

By contrast, small amounts of funding can have a much bigger impact in other sectors, where financial incentives are likely to play a larger role. The total annual market for rare earths, for example, is smaller than several of the individual manufacturing incentives in the IRA, CHIPS Act, or OBBBA. Quantum computing investment is similarly small scale, and federal financial resources seem likely to play a larger role there.

In the near term, US efforts in the tech race with China will likely focus more on domestic incentives rather than on restricting growth of China tech. Following the recent trade deal, it seems unlikely that the White House will take steps that could destabilize the relationship in coming months, but frictions in the relationship are bound to return eventually.

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The why, what, & how of China tech policy

Hui Shan assesses the why, what, and how of China's policy support for technology

The Chinese government has long emphasized the importance of science and technology, with this focus intensifying dramatically in recent years as the US-China tech race has heated up. Indeed, China has implemented several policies to advance technological innovation, including establishing a new Central Science and Technology Commission in the "Party and State Institution Reform" in 2023 to coordinate the construction of the national innovation system, as well as restructuring the Ministry of Science and Technology to optimize the management of the science and technology innovation system. With the US-China tech rivalry showing no signs of slowing, China's drive for technological self-sufficiency will likely remain central to China's policy agenda.

The why

Driving this increased emphasis on science and technology has likely been the Chinese leadership's view that the US and China are locked in long-term strategic competition, making technological self-reliance vital for China's development and national security. Notably, while the 13th Five-Year Plan (FYP) in 2015 contained no mention of the "international situation" (国际形势), the 2018-19 US-China trade war prompted its inclusion in the 14th FYP along with the phrase "a great change unseen in a century" (百年未有之大变局). By 2025, in the wake of the Covid pandemic, Russia's invasion of Ukraine and subsequent Western sanctions on Russia, stricter US semiconductor export controls, and sharply higher US tariffs on China, the phrase "international situation" was cited three times in the 15th FYP proposal, with the language "high winds, rough seas, even raging storms" (风高浪急甚至惊涛骇浪) used to describe the seriousness of the geopolitical challenges that China faces. Against this backdrop, China is determined to strengthen its industrial base, maintain its manufacturing competitiveness, and enhance its domestic innovation capability.

The what

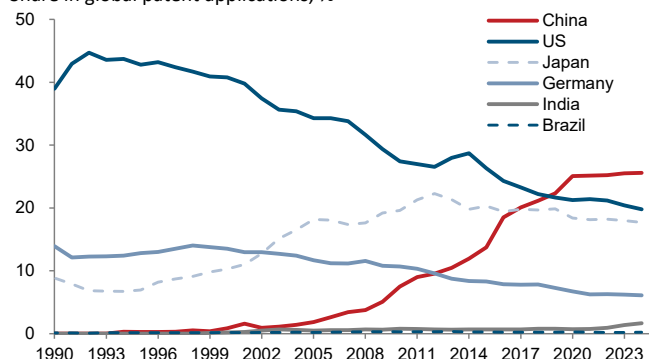
The Chinese government's approach to technological progress demonstrates both continuity and evolution. In 2015, policymakers introduced the "Made in China 2025" plan, which included primary development priorities of information technology, robotics, green energy and electric vehicles (EVs), aerospace equipment, ocean engineering and high-tech ships, railway equipment, power equipment, new materials, pharmaceuticals and medical devices, and agricultural machinery. Since then, China has made remarkable progress in these sectors, securing a dominant position in the global markets for EVs, renewable energy, and shipbuilding.

Looking ahead, the 15th FYP proposal outlines three key areas for future development. The first centers on "chokehold technologies," including integrated circuits, industrial machine tools, high-end instruments, basic software, advanced materials, and biomanufacturing. In these sectors, policymakers

will likely spare no effort or expense to achieve breakthroughs and attain self-reliance given their importance to China's national security and resilience against external pressure.

China has the top number of patent applications in the world

Share in global patent applications, %



Source: World Bank, compiled by Goldman Sachs GIR.

The second area focuses on "emerging industries," including new energy technologies, advanced materials, aerospace, and the low-altitude economy (i.e., drones and urban air mobility). These sectors will likely receive substantial investment as the government aims to make them important drivers of China's economic growth.

The third area targets "future industries," including quantum technology, biomanufacturing, hydrogen energy and nuclear fusion, brain-computer interfaces, embodied intelligence (integrated AI and robotics), and sixth-generation (6G) mobile communications. While these sectors are largely in the early stages of development, policymakers view them as potential game changers that could redefine global technological leadership in the coming decades. The government's ongoing support ensures that China remains at the forefront of research and innovation in these cutting-edge fields, even if widespread commercialization may still be years away.

The how

China's government support for science and technology—and their adoption in industrial sectors—is holistic and highly coordinated. First, it involves systematic planning, including through the FYPs as well as detailed catalogs published by the National Development and Reform Commission (NDRC) identifying industries to encourage, restrict, or eliminate. For example, the latest catalog (released in 2024) encourages certain specialty chemicals, while targeting some small-scale, oil- and gas-based, and polluting chemical productions for restriction or elimination.

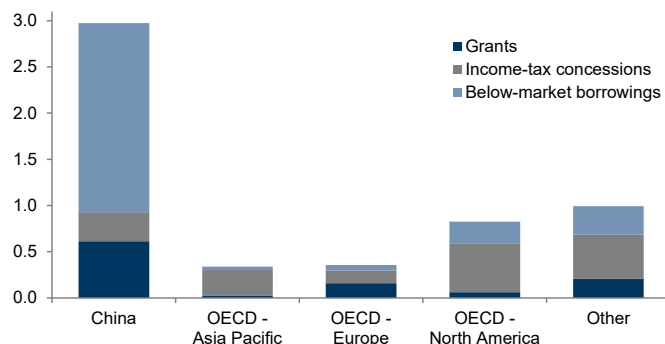
Second, this support includes significant financial backing from various levels of government, including through government-guided funds for tech investments. As of 2024, China had established over 2,000 of these funds, with a targeted total of RMB12.8tn¹. And at the March 2025 Two Sessions, the NDRC announced a RMB1tn government-guided fund for early-stage tech investments in areas like AI, quantum technology, and hydrogen storage. Beyond direct investments, grants, and subsidies, the Ministry of Finance offers tax benefits to tech companies and the PBOC runs a relending program dedicated to technology and innovation. Consequently, Chinese industrial

¹ According to data compiled by the Qingke Research Center.

sectors receive far more financial subsidies than their global counterparts².

China splashes out on industrial subsidies

Industrial subsidies for 14 key industrial sectors, averages over 2005-22, % of annual firm revenue



Source: OECD, compiled by Goldman Sachs GIR.

Third, China's government support extends far beyond traditional tax, credit, and financing assistance. In strategic priority areas, the government often offers cheap land, expedited permits and approvals, and abundant energy supplies while facilitating connections between upstream and downstream industries to foster a complete supply chain. It also helps source and develop needed talent, and direct government procurement toward domestic innovations. For example, local governments often sell land to raise revenues, with industrial land priced at only a tenth of residential land. This allows the government to collect implicit taxes from households and then use those funds to subsidize industrial production. And in 2025, the Ministry of Education added 25 undergraduate majors in higher education, including AI, decarbonization, and digital governance. Such a coordinated and systematic approach is key to China's success in achieving significant advancements in technology and high-tech manufacturing over relatively short periods of time.

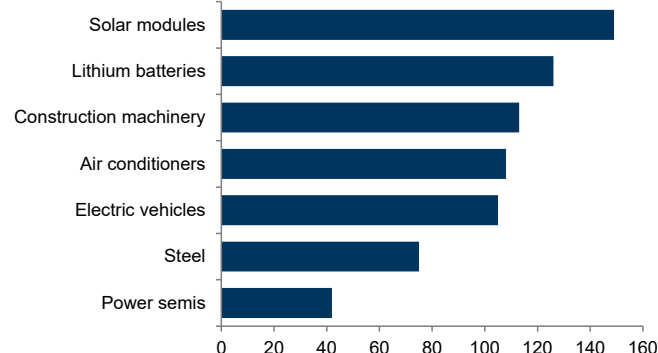
China's system for supporting tech innovation and emerging industries allows it to mobilize resources from all directions and rapidly scale up production following breakthroughs. However, this system also tends to generate wasteful investment and overcapacity as Chinese local governments respond to the same signals from the central government. In recent years, top leadership has become increasingly concerned about these drawbacks, with President Xi recently stating that not all local governments should develop AI, build data centers, and make EVs. Official outlets now emphasize the importance of building a "unified national market" (全国统一大市场) and growing "new-quality productive forces in alignment with local conditions" (因地制宜发展新质生产力). And, since July, the government has launched "anti-involution" initiatives across many industries to curb investment and prevent excessive price cutting.

However, China's repetitive investment and industrial overcapacity issues are deeply rooted in its political system and cannot be easily solved. Local officials, eager to demonstrate political loyalty and competence, often focus policy efforts on sectors prioritized by the central government. Unless the

incentive structure changes—enabling local officials to pursue independent development strategies and achieve promotion without pursuing central directives—accelerated investment in industries explicitly mentioned in the 15th FYP proposal will likely continue. As a result, overcapacity will likely remain a feature of China's economy, even as it shifts from steel and cement in the past to solar panels and EVs today, and potentially to semiconductors and AI in the future.

China's capacity exceeds global demand for several key products

China's capacity as a share of global demand in 2024, %



Source: Goldman Sachs GIR.

Leveraging strengths in the US-China tech competition

China has increasingly relied on two key advantages in its tech competition with the US: its dominance in global supply chains and strong economic ties with the Global South. A notable example of supply chain leverage is China's recent tightening of export controls on critical minerals—including those announced on October 9 covering rare earth minerals, related products like alloys and magnets, as well as refining and processing technology and equipment. Given the importance of rare earths in high-tech manufacturing and China's dominance in refining and processing, such restrictions could significantly disrupt global manufacturing supply chains, prompting the US to ease some of its semiconductor export controls in exchange for China suspending its proposed rare earth export controls, demonstrating the effectiveness of this leverage.

China is also working to establish its technology standards abroad through trade and investment in the Global South. For example, Huawei operates in over 170 countries, encouraging these countries to adopt Chinese telecom technology standards. As more countries follow suit, the resulting network effect increases the attractiveness of China's standards for others. And in 2024, officials called for deeper participation in international standards organizations—including the International Organization for Standardization (ISO), the International Electrotechnical Commission (IEC), and the International Telecommunication Union (ITU)—and emphasized strengthening standard connectivity with Belt and Road Initiative participants. With China now a larger trading partner than the US for over 140 countries, the Chinese government seeks to use these ties to expand markets where Chinese technology prevails.

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² According to OECD calculations.

Semiconductors, at a glance

What are semiconductors, and who makes them?

Semiconductors are also known as **microchips or chips**. A microchip is a set of electronic circuits layered on a thin wafer of semiconductor material, typically silicon. Transistors located on the chip act as miniature electrical switches that can turn a current on or off. The more transistors that are located on a chip, the more the chip can do. The size of a microchip and the number of transistors on it varies; a microchip the size of a human fingernail can contain billions of transistors.

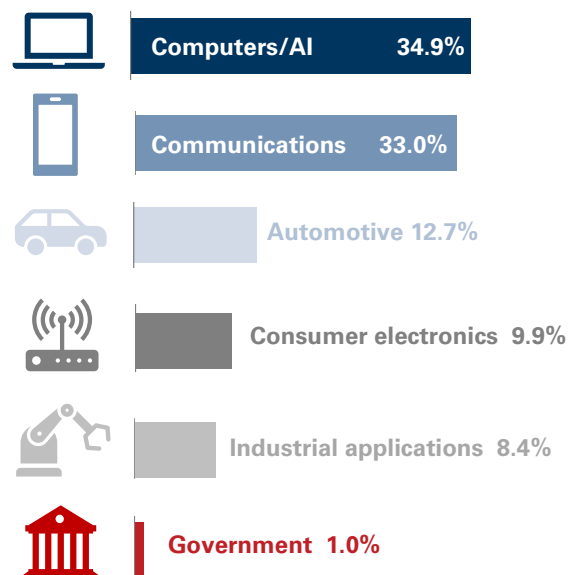
There are three main types of microchip companies: **Integrated Device Manufacturers (IDMs)** (Intel, Samsung), who design and manufacture chips in-house, **Fabless companies** (Nvidia, AMD), who design chips in-house but outsource manufacturing, and **Foundries** (TSMC, GlobalFoundries), who manufacture chips for fabless companies as well as for IDMs who don't have sufficient in-house capacity.

What do semiconductors do?

Semiconductors are the building blocks of technology, and are central inputs in many devices, including cars, computers, smartphones, and medical devices. There are three main types of chips: logic chips, memory chips, and Discrete, Analog, and Other (DAO) chips. Logic chips are the 'brains' of electronics. They process information in order to complete tasks. Central processing units (CPUs) are built for general functionality, graphics processing units (GPUs) are optimized for visual displays, and neural processing units (NPUs) are designed for machine learning applications. Memory chips store information. DRAM chips save data when a device is turned on, while NAND chips save data after a device is turned off. DAO chips transmit, receive, and transform information dealing with continuous parameters, like temperature.

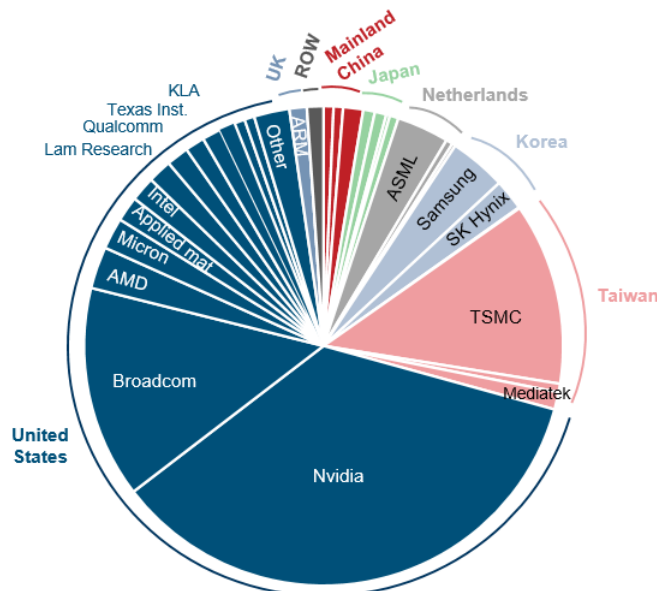
How are semiconductors used?

Global demand share by end use, 2024



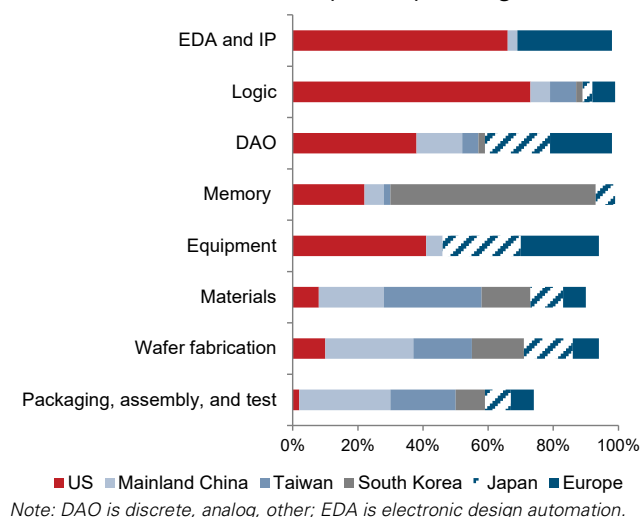
Who are key competitors in the chip industry?

Global semiconductor companies by market cap



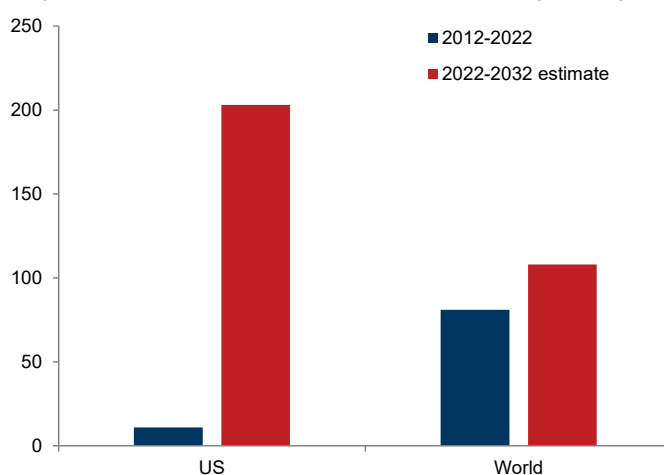
What in the supply chain does each region specialize in?

Semiconductor value added by activity and region, 2024, %



Where is chip-making investment concentrated and how could it evolve?

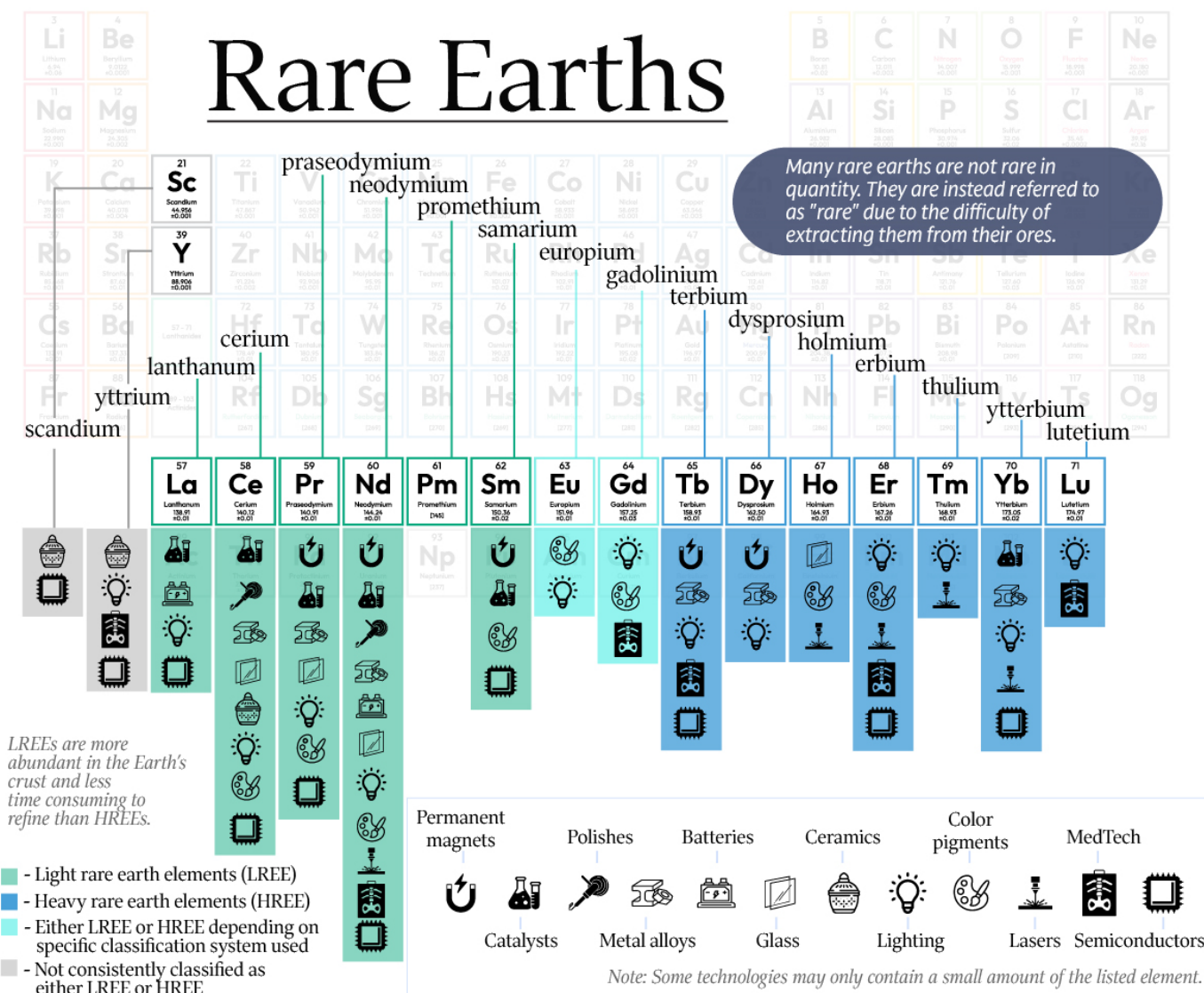
Projected increase in US fab capacity vs. world avg., % chg.



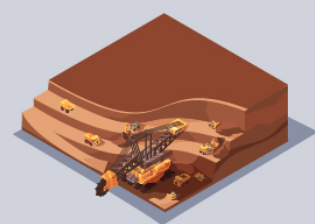
Source: [Semiconductor Industry Association](#), BCG, government websites, various news sources, companiesmarketcap.com, Goldman Sachs GIR.

Rare earths, at a glance

Rare Earths



The rare earth supply chain, at a glance



Mining

Rare earth elements are mined from the ground. These elements are found in various ore deposits. The mining method depends on the type of deposit. Methods include open-pit mining, underground mining, and in-situ leaching.



Processing and Refining

Following mining, the extracted ore undergoes several stages of processing. These include crushing and grinding (the raw ore is crushed and ground into small particles), beneficiation (which separates the rare earths minerals from other waste minerals), and extraction and separation (chemical treatments are used to extract the individual rare earths from concentrate).



Manufacturing of products

The separated oxides are converted into rare earth metals and, in turn, alloys. These alloys are used to create powerful magnets known as permanent magnets, which are a key component of various technologies.



China has the largest share of global rare earth reserves (49%). This includes both light and heavy rare earths, with known HREE reserves extremely scarce outside of China and Myanmar. China also dominates global rare earth mining (69%). Other countries with significant rare earth reserves/mining are:

- Brazil (23% and 0%)*
- India (8% and 1%)
- Australia (6% and 3%)
- US (2% and 12%)



China also dominates global rare earths refining (92%), particularly in heavy rare earths, with Chinese refineries producing the vast majority of the world's supply of dysprosium and terbium metals. Lynas, an Australian company with operations in Malaysia, is one of the only other major players at this stage of the supply chain, accounting for 4% of global refined production. The US accounts for only 1%.



China is the world's largest supplier of permanent magnets, which are critical in many technologies. China accounts for 98% of magnet manufacturing.

* Figures in parentheses represent the share of reserves and mining, respectively.

Source: IEA, Rareearths.com, Federal Register, various news sources, Goldman Sachs GIR.

Interview with Paul Triolo

Paul Triolo is Partner at DGA-Albright Stonebridge Group. Below, he argues that it will be hard for any country to achieve self-sufficiency in semiconductors, but if any can, it's China.

The views stated herein are those of the interviewee and do not necessarily reflect those of Goldman Sachs.



Allison Nathan: Where are the critical chokepoints across the global semiconductor supply chain?

Paul Triolo: Let me first clarify that there is not one semiconductor supply chain but rather multiple interlocking ones. And it's not a one-way supply chain—a semiconductor product can crisscross the globe several dozen

times before being inserted into a final product. So, the term "supply chain" is not representative of the reality of the global semiconductor industry. A more accurate characterization would be "the most complicated set of technologies on Earth."

Within that are three key chokepoints. The first is advanced lithography, the tool that prints the intricate patterns of semiconductor circuits onto silicon wafers at feature sizes far smaller than the width of a single hair. Feature sizes have shrunk rapidly over the past several decades, which has been the driving force behind Moore's Law. The Dutch company ASML is currently the only firm capable of manufacturing the Extreme Ultraviolet (EUV) lithography machines critical to produce advanced chips—those with feature sizes below 5 nanometers (nm)—as it has successfully integrated the cutting-edge light sources, precision optics, and complex computational software for wafer alignment necessary for these machines.

The second chokepoint is etching, the process that exposes the underlying circuit after lithography. Sophisticated etching equipment is needed for chips produced with EUV lithography, and only a handful of toolmakers—like the US' Lam Research and Applied Materials and Japan's TEL—produce it. The third chokepoint is advanced chip manufacturing. Taiwan's TSMC dominates this space, producing over 90% of the world's most advanced chips, because its business model of focusing solely on manufacturing products designed by its clients rather than competing with them, strong relationships with toolmakers, and ability to deliver quality and quantity at scale have made it the trusted partner for integrated device manufacturers like Intel and Samsung and fabless companies like Nvidia.

Allison Nathan: Is it possible for any country to achieve self-sufficiency in semiconductors given these hurdles?

Paul Triolo: If any country can do it, it's China, owing to its vast resources and pool of engineering talent. And it would be wrong to think that just because China isn't self-sufficient today it won't be in the future. Despite the common argument that China always intended to be self-sufficient in this space, the reality is that as long as Chinese companies had access to the most advanced tools, they had little incentive to develop or use local alternatives. China now has the incentive because the US and its allies have largely cut it off from access to EUV lithography and other tooling, including advanced etch, deposition, and metrology. But such technology is difficult to replicate because it requires integrating several complex

systems at scale, including advanced optics, specialized light sources, intricate software, and materials like photoresists, all of which must work seamlessly together with other manufacturing tools. And this is only half the battle. As an industry insider once told me, building a prototype is easy, but servicing hundreds of sets of this equipment 24/7 is incredibly challenging owing to the complex nature of the machinery and the need for a very capable workforce to support deployments of these complex tools for customers worldwide that are expecting maximum uptime and few, if any, defects. So, competing with ASML in this space is very difficult. And even if new entrants could develop the know-how, leading-edge semiconductor manufacturers would be reluctant to take a risk on an unproven technology given the enormous costs involved.

Allison Nathan: So, are you saying that US policies have been effective in slowing China's efforts to compete in the semiconductor space and, ultimately, the AI race?

Paul Triolo: Whether US policies have been effective depends on the goal. The US and its Western partners have prevented Chinese companies from purchasing EUV equipment, and the US has banned the sale of advanced GPUs to China through export controls, though this has now been partially reversed. So, US policies have succeeded in restricting China's access to advanced lithography and chips. But these restrictions have spurred China to improve its own capabilities. China is exploring alternative technologies to generate UV light for lithography systems, and Chinese firms are working to improve their GPU design and development capabilities. And US policies have been wholly ineffective in impeding China's ability to develop advanced AI models. All leading open-source and open-weight generative AI models today are the work of Chinese firms like Alibaba, Baidu, Tencent, Moonshot, Zhipu AI, and DeepSeek, in stark contrast to a year ago when Western firms like Meta and Mistral topped the leaderboards. While the US continues to develop the world's leading AI models, the open-source and open-weight nature of Chinese models has allowed them to increasingly dominate real-world adoption due to their lower cost and accessibility. While the issue of model diffusion does not get a lot of airtime, it's vital because winning the AI race will depend not only on model capabilities but also on usage.

Allison Nathan: How have Chinese companies achieved so much without access to advanced lithography and chips?

Paul Triolo: China's progress owes to successes at both the company and national levels. Chinese firms have become incredibly well-versed in how hardware and software interact with AI models and how to train these models using less compute. DeepSeek, for example, successfully used low-level hardware programming, which provides high performance but requires more advanced programming skills, to maximize the use of the limited hardware it had, including the A100 Nvidia GPU the company obtained before the US imposed export controls in October 2022 and the downgraded version of the H100 GPU Nvidia was permitted to sell. Simultaneously, the

Chinese government has supported the domestic AI industry and made it easier for domestic companies to gain access to compute by building data centers nationwide and ensuring reliable access to power by investing heavily in the grid and expanding renewable energy capacity. While major companies like Alibaba, Tencent, and Baidu have their own cloud-based infrastructure and so don't need state support, these government efforts are a lifeline for smaller companies across China. But again, Chinese companies lack access to the most advanced Western GPUs, so despite these favorable state policies, overcoming the compute problem remains difficult.

Allison Nathan: So, ultimately, can China close the gap with the US in developing the most powerful AI models?

Paul Triolo: The idea that the US can maintain its lead in advanced AI is premised on scaling—the belief that increasing computational resources will lead to better models. But the jury is still out on that, and a new technology could come along that obviates the need for training large models on millions of GPUs. Chinese firms, particularly Alibaba and DeepSeek, have also proven capable of developing advanced models with much less compute than US leaders. So, the US expanding or even maintaining its current lead is not a foregone conclusion.

Of course, much will depend on the Chinese semiconductor industry's ability to overcome the hardware chokepoints we've discussed. Making semiconductors is one of the most complex things humans do. The sheer knowledge required is immense and takes decades to acquire. And China started decades behind the US. But US controls have ironically turned all Chinese fabs restricted from accessing advanced US tools into critical tech incubators for China's domestic toolmakers, accelerating their move up the value chain and speeding China's ability to get to more advanced semiconductor production for AI.

China's lack of deep capital markets also poses a challenge. Unlike the US, where abundant patient capital has supported long-term investments in the semiconductor space and companies can readily access deep and liquid financial markets to fund large data center buildouts, China's capital markets are underdeveloped, leading companies to rely more on the government for financing. While that may change over time, the US currently enjoys a significant advantage in this regard.

Again, if any country can overcome these challenges, it's China. But with no historical precedent for an undertaking of this scale, predicting China's chances of success and the timing of key breakthroughs such as advanced lithography is difficult.

Allison Nathan: What challenges does the US face as it tries to strengthen its leadership in semiconductors and AI?

Paul Triolo: While the US is the global leader in chip design and tooling, manufacturing is largely dominated by Asian firms—particularly TSMC and, to a lesser extent, Samsung and some Japanese firms. So, the US is in a strange position—it pioneered the semiconductor industry, but manufacturing has largely gone to Asia as Intel lost its spot as the global leader in manufacturing by missing out on the mobile and AI revolutions and entering the foundry business very late. Intel remains a

major CPU producer and is still a huge player in the space, but the market has moved away from its historic core business.

The US has had some success in reshoring advanced manufacturing, with TSMC building a facility in Arizona that now produces 4nm GPUs. But the Arizona facility will likely only reduce the US' reliance on Taiwan for advanced manufacturing by 10% by 2030—far short of the 20% goal of end-to-end share of global capacity laid out as the goal of the CHIPS Act. The major obstacle to significant reshoring is the cost and complexity. Building a fab costs ~\$30-40bn, and the US lacks enough engineers and tradespeople as well as the institutional knowledge to pull this off owing to three decades of offshoring. So, just like Mainland China, the US faces sizable challenges.

Allison Nathan: Does China's dominance in rare earths and extensive buildout of power generation capacity provide it with significant advantages over the US in the AI race?

Paul Triolo: I don't view the rare earth issue as particularly salient in this context. While some rare earth elements are used in semiconductor manufacturing, and the industry utilizes rare earth permanent magnets, the numbers are tiny, and enough alternatives exist across the supply chain. So, while rare earths will likely be a constant irritation of ongoing US-China tensions, they won't affect the buildout of large-scale AI infrastructure. China's power advantage is the more critical issue for the US, because energy will be a key gating factor in the AI race. The US has a massive supply chain problem when it comes to supplying the hardware needed to generate enough power. It also lacks a unified grid, which makes it hard to move power around. The Middle East, with its significant access to cheap power, will have an important role to play here, with the US already starting to leverage partners in the region to offset China's power advantage and US weaknesses in this area.

Allison Nathan: The US government has invested and taken stakes in companies to strengthen domestic semiconductor and AI capabilities. Will such efforts be enough to address all these challenges?

Paul Triolo: They are a step in the right direction but likely not enough. While the government's investment in Intel via the CHIPS Act has helped put it on the road to regaining its footing, Intel's long-term viability hinges on securing a large client base over which to amortize the high costs of chip manufacturing. The government has tried to help with that by encouraging Nvidia and Apple to shift production from TSMC to Intel, and may do more of that now that it has taken an equity stake. But to achieve meaningful progress, the Commerce Department must become an industrial policy juggernaut. The CHIPS Program Office, which the Biden Administration established to implement the semiconductor incentives of the CHIPS Act, was a good start. But the Trump Administration has downsized it and is instead trying to reshore semiconductor production via tariffs, which are unlikely to persuade firms to build US facilities given the costs and complexities. What's needed is a credible, long-term commitment to industrial policy. The governments of Taiwan, Japan, and South Korea have long pursued industrial policy, which is a key reason why these places are home to successful chip industries. So, US policymakers need to step up their game for America to prevail in this tech race.

Interview with Jack Lifton

Jack Lifton is Co-chair of the Critical Minerals Institute. Below, he breaks down the global rare earth supply chain and China's role as a chokepoint, but he argues that current US attempts to achieve rare earth self-sufficiency are misguided and the need for them is overblown.

The views stated herein are those of the interviewee and do not necessarily reflect those of Goldman Sachs.



Jenny Grimberg: What does the rare earth supply chain look like?

Jack Lifton: The rare earth supply chain is mostly a “mine to magnet” supply chain. The process begins with the extraction of rare earth elements, which are found, economically, in deposits of the minerals bastnaesite and monazite, and in ionic adsorption

clays. Two primary rare earth mines—those mined only for rare earths—exist in the world: Mountain Pass in California, operated by MP Materials and containing one of the world's largest deposits of bastnaesite, and Mount Weld in Australia, operated by Lynas Rare Earths and containing monazite. But the largest rare earth mine in the world is located in Inner Mongolia, where rare earth elements are produced as companion minerals of iron ores.

After mining, the ore goes through an initial processing stage to concentrate the rare earth minerals followed by a separation stage, during which the rare earth elements are individually separated and purified into oxides. Selected separated oxides are then converted into rare earth metals, which are alloyed, or combined, with other elements to form the precursor alloys for magnets. Lastly, these alloys are processed into “permanent magnets” that produce their own persistent magnetic field.

Jenny Grimberg: As many investors are now attempting to get up to speed on the rare earth supply chain given the role of rare earths in critical technologies, what would you caution are the biggest misconceptions around it?

Jack Lifton: Many people seem to believe that all the money is made in the mine, but very little of it is. If money could be made in the mine, the large mining companies, like Australia's BHP and Rio Tinto, the UK's Anglo American, and the US' Newmont and Kennecott, would simply buy rare earth deposits and develop them. But these mining giants have largely stayed out of primary rare earth mining.

The money is instead in the magnet, with permanent magnets accounting for around 70% of global rare earth product revenues. It's also widely underappreciated that permanent magnets come in different forms. Electric vehicle drive motors and military equipment like aircraft, tanks, and weapons utilize specialty rare earth permanent magnets, which are designed to maintain their magnetic strength throughout high temperature cycles that could reasonably be encountered during warfare, especially nuclear warfare. Car accessory motors like power steering, by contrast, utilize ordinary rare earth permanent magnets, which don't perform well in high temperatures, but don't need to given their uses.

Jenny Grimberg: China is widely considered the critical chokepoint in the rare earth supply chain. Is that true?

Jack Lifton: Yes. Despite their moniker, many rare earths are not that rare. For example, neodymium, a light rare earth element used in permanent magnets, is more abundant in the Earth's crust than lead. And many countries, including the US, have significant rare earth reserves. But China has the largest reserves of heavy rare earth elements like dysprosium and terbium, which are critical for temperature cycling degradation resistant specialty neodymium-iron-boron magnets—the most powerful permanent magnets. Even more crucially, China has a dramatic advantage over the rest of the world in the extraction, separation, and refining of these elements because it has done this work nonstop for several decades and developed an immense internal infrastructure for it.

China long ago figured out how to extract dysprosium and terbium from the ionic adsorption clays they principally come from. Today, it uses this method in Myanmar, where the majority of the heavy rare earths that China uses are now produced, while processing remains in China. As Myanmar sites are degrading in quality, China is increasingly looking to other countries in Southeast Asia with similar deposits—Laos, Thailand, Vietnam, Indonesia, and particularly Malaysia—and offering large sums of money for the right to build a rare earth industry in these countries. The catch is that China would bring in its own technicians and equipment, ensuring that it maintains complete control of the global production of heavy rare earths.

Jenny Grimberg: Can't the US break China's chokehold on the heavy rare earth supply chain with the significant investments the US government is making in domestic rare earth companies?

Jack Lifton: No. While the US does have significant rare earth reserves, these are mainly light rare earth deposits; there are no known heavy rare earth deposits in the US. And just having deposits isn't enough. The material has to be of a high enough concentration in a location where currently available technology can recover it economically. I suspect that New York's Central Park has as many rare earths as Ramaco Resources and American Rare Earths recently discovered in Wyoming. But that doesn't make it worthwhile to mine Central Park. Again, this is an important reason why large mining companies haven't developed US deposits—it's just not economic to do so.

The US also lacks the rest of the supply chain. Rare earths are so similar chemically that ordinary chemical operations can't separate them; specialized separation facilities are necessary. Mountain Pass is home to a light rare earths separation facility, which dates back to the 1960s and mainly arose out of the need to separate europium to produce the red phosphor for color televisions. I was actually involved in this research. But even if there were accessible, economic deposits from which

heavy rare earths could be extracted, the US has no commercial-scale facility for heavy rare earth separation today.

And while the US was a pioneer in the development of rare earth permanent magnets, with General Motors research and Japan's Sumitomo developing the world's first commercial rare earth permanent magnet in the late 1970s, the US currently has limited capacity to produce such magnets. All the equipment in use today has come from China. I recently attended the opening ceremonies of Canadian company Neo Performance Materials' new rare earth magnet manufacturing plant in Estonia and saw that the equipment is mostly Chinese. But China has since banned the export of technology for extracting, separating, and processing rare earths.

And as a chemical engineer who has worked in metallurgy of rare metals, I can tell you that successfully replicating such technologies is a much more daunting task than many people seem to think. On top of that, the institutional knowledge necessary to make rare earth permanent magnets, especially the specialty magnets, is all but gone. I know all the people left in America with experience making such magnets, and you could fit them into a handful of phone booths.

Overcoming all of these challenges is not just a question of money. Policymakers need to invest in the right companies, technologies, and people, which isn't happening today.

“The institutional knowledge necessary to make rare earth permanent magnets, especially the specialty magnets, is all but gone. I know all the people left in America with experience making such magnets, and you could fit them into a handful of phone booths.”

Jenny Grimberg: So, are you saying current government efforts to achieve US self-sufficiency won't succeed?

Jack Lifton: Yes. Bear in mind that the annual capacity of the largest modern rare earth permanent magnet plant ever built in the Western world—the Estonia plant I recently visited—is 2,000 tons. So, current plans to build a 10,000-ton US magnet plant, funded by the government and private equity investors, seem ridiculous given everything we've discussed. The government's current plan to achieve US self-sufficiency is simply unattainable in the short run and will not succeed.

Jenny Grimberg: How, then, will the US be able to meet its rare earth needs if it can't rely on China?

Jack Lifton: Let me first clarify the other aspect of rare earths that is underappreciated: the actual amount of rare earth minerals that the US needs is small. The US consumes only around 3% of global rare earth mineral supplies, and specialty rare earth permanent magnets are a lesser subset of that. The 16mn total cars sold in the US last year required roughly 10,000 tons of ordinary magnets, but the 1.2mn electric vehicles sold in America last year required only around 3,000 tons of specialty magnets. And while the military's demand for

specialty magnets is classified information, calculations I've performed jointly in the recent past with the Department of Defense suggest the military demand for specialty magnets is probably only around 1,000 tons/year. And a portion of that demand is for samarium-cobalt magnets, which were the first rare earth permanent magnets used in the automotive industry and don't typically use heavy rare earth elements. So, US needs for heavy rare earths is a trivial amount of world demand.

“The government's current plan to achieve US self-sufficiency is simply unattainable in the short run and will not succeed.”

Jenny Grimberg: Even if the actual amount of rare earth minerals the US needs is small, aren't rare earths nonetheless crucial for defense and other technologies?

Jack Lifton: Absolutely not, at least in the case of defense. Almost any magnet application can be addressed by iron-based ferrite magnets, aluminum-nickel-cobalt (Alnico) magnets, or rare earth magnets of either the neodymium-iron-boron or samarium-cobalt variety. The choice ultimately depends on physical size requirements and whether the magnet needs to maintain its magnetic strength under different temperature regimes and be able to withstand the disruptive effects of an electromagnetic pulse. Samarium-cobalt magnets are resistant to electromagnetic pulse exposure, and neodymium-iron-boron magnets modified by dysprosium and terbium are resistant to demagnetization by high temperature exposure. When no such exposures are likely, non-rare earth permanent magnets can be used as substitutes. So, acceptable workarounds for rare earths exist for most applications in the military complex.

I should also point out that the Department of Defense has contracted with Vacuumschmelze, a former German company now owned by a US private equity firm, to build a magnet manufacturing plant in South Carolina with an announced capacity of 2,000 tons/year. If that's successful, it will more than cover the US military's rare earth magnet needs.

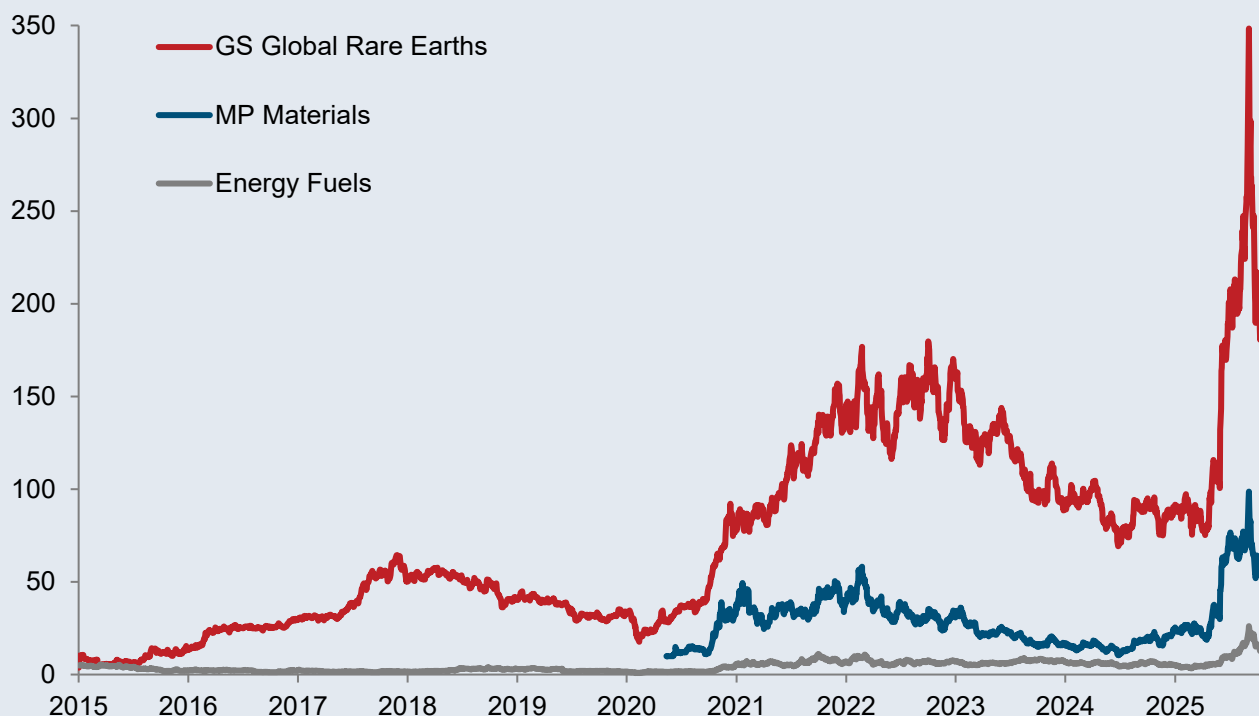
Jenny Grimberg: So, is the apparent urgency for the US to achieve self-sufficiency in rare earths overblown?

Jack Lifton: I believe so. That said, the desire to reduce our dependence on China in this regard is valid. And there is a viable path to achieving that, even if it won't amount to self-sufficiency. It's just a matter of picking the right companies, technologies, and people. The US could source light rare earths—neodymium, praseodymium, and samarium—from Mountain Pass, which, again, is teeming with them, and heavy rare earths from Malaysia, which will likely start producing them by 1Q2026. And it could use domestic capacity that currently exists or is expected to come online in the coming years for the separation work. Lastly, it could leverage overseas partners outside of China to produce the rare earth magnets themselves. Good options are available in Japan and Estonia, and these options would be far less costly than building overly large brand-new facilities in the US, which the US government unfortunately seems intent on financing.

Pricing rare earths

While rare earth stocks have retraced sharply following the US-China trade deal, they are still up significantly year to date amid the ongoing push to diversify rare earth supply chains...

Performance of GS Global Rare Earths basket* and largest components of the basket



*Basket composed of the most exposed and liquid companies in the rare earths space. These are: MP Materials (US, 44% weight in basket), Energy Fuels (US, 32% weight in basket), Lynas Rare Earths (AU, 7.7% weight in basket), USA Rare Earths (US, 7.5% weight in basket), NioCorp Developments (US, 3.1% weight in basket), Iluka Resources (AU, 2.8% weight in basket), JL Mag Rare-Earth Co (HK, 2.4% weight in basket), Neo Performance Materials (CA, 0.4% weight in basket), and Arafura Rare Earths (AU, 0.2% weight in basket).

Source: Goldman Sachs FICC and Equities, Goldman Sachs GIR. Basket developed by the GS Global Banking & Markets division.

...and the spot prices of key rare earth oxides have risen over the past year, though they remain well below their peaks

China prices of select light and heavy rare earth oxides (before VAT is excluded), \$/metric kg



Source: Asian Metal Inc., Bloomberg, Goldman Sachs GIR.

Special thanks to GS Australia metals & mining analyst Chris Bulgin for pricing data.

Tech race company impacts: rare earths

How are US and Chinese efforts to develop their own tech stacks impacting the companies in your coverage universe, and do you expect continued such efforts to be an important driver of price action ahead?

Australia Metals & Mining

Paul Young and Chris Bulgin, GS Equity Research

While China has temporarily lifted the export restrictions it imposed on seven rare earth (RE) elements (scandium, yttrium, samarium, gadolinium, terbium, dysprosium, and lutetium) in April 2025, the country's clear effort to control its tech stack has fostered the emergence of two distinct markets—the China spot market and a new term floor price market.

The emergence of a term floor price market is a strategic effort by Western governments to establish a guaranteed minimum price for RE elements and broader critical minerals to encourage domestic magnet production and reduce reliance on China. Government agencies and companies, including the Department of Defense (DoD) and MP Materials (MP) in the US and Carester and Solvay in France, have responded with significant funding and announced agreements. The DoD and MP Materials agreement includes an "offtake" commitment whereby the DoD has agreed to purchase MP Materials' output for a specified period as well as a \$110/kg price floor for neodymium-praseodymium oxide (NdPrO), whose price is the most referenced in the RE space due to the oxide's key presence in end-use applications (e.g., rare earth permanent magnets).

For companies which possess refining capabilities, these developments are highly impactful. We anticipate that from around mid-2027, these companies will contract the majority of their sales at a floor price exceeding \$100/kg (real\$). This is separate from the China spot price for NdPrO, which we now forecast will rise to \$85/kg (from \$75/kg previously, both real\$) as we expect deficits over the medium term to 2027 and perhaps beyond. The higher floor price, benchmarked by the DoD-MP deal, is crucial. It reflects the strategic importance of refining capacity, which is a significant bottleneck for new Western rare earth permanent magnet supply, and is designed to ensure an implied high return on capital.

Separately, the scarcity of heavy rare earths like terbium (Tb) and dysprosium (Dy) in Europe, driven by export restrictions as well as China's limited production (1,800tpa) and its reliance on Myanmar imports, has led TbDy prices to rise by ~3x according to channel checks with EU metal makers. Consequently, we have lifted our contracted price forecasts for dysprosium and terbium to approximately \$600-700/kg and \$2,000-2,500/kg, respectively.

The continued efforts by both the US and Chinese governments to secure their own tech stacks will undoubtedly remain an important driver of price action. Our upgraded price forecasts for NdPrO and heavy rare earths, along with the expectation of new Western magnet capacity by mid-2027, indicate that these strategic shifts are not currently fully priced in but are actively shaping future market dynamics and contract structures.

US rare earths

Brian Lee and Nick Cash, GS Equity Research

In addition to providing floor pricing for NdPrO and guaranteeing offtakes for MP Materials' magnet expansion, the DoD took an equity stake in MP and provided capital for the company to expand its separation capabilities and accelerate the downstream production of neodymium-iron-boron (NdFeB) magnets as part of their deal.

This will ultimately reduce the US' reliance on China-related rare earth permanent magnet supply chains. We estimate US magnet demand of roughly 25,000mt in 2025, of which defense needs represent roughly 3,000mt-4,000mt according to [estimates cited by Vulcan Elements](#).

That said, the US will still likely rely on China for commercial NdFeB magnets unless additional recycling capabilities are built out (which would reduce the need for new mining/processing) or the US is able to obtain more NdPrO from sources outside of China. As a result, we believe any further efforts by the US and China to control/develop their own tech stacks could result in price swings for NdPr and NdFeB magnets.

Tech race company impacts: semis

How are US and Chinese efforts to develop their own tech stacks impacting the companies in your coverage universe, and do you expect continued such efforts to be an important driver of price action ahead?

US Semiconductors, Data Centers, and IT Services

Jim Schneider, GS Equity Research

The US has implemented several policies to strengthen its control over key technologies and limit the export of such technologies to China. This began in 2012 with the Obama Administration's addition of Chinese organizations to the US Entity List, which prevented these organizations from acquiring US technologies. Restrictions have broadened significantly in subsequent years. Since 2012, the number of US restrictions on exports to many countries, including China, has greatly expanded, especially post 2019. On January 15, 2025, the Biden Administration published the AI Diffusion Rule, which limited the export and use of key semiconductor technologies based on certain performance thresholds, with additional geographical restrictions based on a multi-tiered system of geographies ranging from those requiring no license to complete prohibitions. While the Trump Administration rescinded the rule on May 13 before it went into effect, the Administration also warned against the use of specific AI processor products from Huawei and the use of China-based LLMs as well as other AI technologies. Electronic Design Automation (EDA) software vendors Cadence and Synopsys also acknowledged the receipt of Commerce Department orders restricting the export and sale of certain software tools used in semiconductor design in June, though these restrictions were lifted in July.

While shifting US export controls represent the greatest fundamental risk to the companies in our coverage, these risks have been increasingly reflected in (somewhat) de-risked estimates and priced into stocks. For AI computing companies we believe the risk of further controls is largely priced in at this stage and see minimal further downside risk to estimates.

For AI-related networking and EDA software companies, our base case assumes continued strategic competition without further bans. We continue to see modest incremental risks to semiconductor capital equipment given incremental additions to the Entity List.

And when it comes to companies in "traditional" markets including PCs, smartphone, server, automotive, and industrial/Internet of Things, we see low probability of a complete ban across most traditional segments, consistent with past statements from the Administration.

Greater China Tech

Allen Chang and Xuan Zhang, GS Equity Research

China has implemented a range of policies in an effort to build a comprehensive domestic semiconductor ecosystem. The government has focused on addressing supply chain bottlenecks with government-backed investment, including the China Integrated Circuit Industry Investment Fund ([Phases 1 and 2](#) totaled \$47bn and [Phase 3](#), which is currently ongoing, totals another \$47bn). These investments have supported companies across the local semiconductor ecosystem, including Integrated Circuit (IC) design, equipment, IC manufacturing, advanced packaging, Electronic Design Automation (EDA), and materials. The government has also provided subsidies, including an up to 50% [power subsidy](#) for data centers using local AI chips and subsidies for capacity expansions that [some local governments have offered](#). R&D expenses have also been made tax deductible, with the government implementing a "[China R&D super deduction](#)" that allows companies to claim an additional 100% of eligible R&D expenses as pre-tax deductions.

This comprehensive suite of government policies has helped Chinese semiconductor companies increase their product offerings, develop more high-end technology, and receive more orders from domestic customers that prefer to use China-produced products. The data center power subsidies have also contributed to the rising share prices of domestic AI chip suppliers. However, the long-term drivers of share prices remain order deliveries and product expansions. Indeed, expectations of continuous government support are already priced in, with investors now placing greater emphasis on sustainable growth, healthy competition, and continuous innovation rather than anticipating more government incentives.

We expect leading domestic Semiconductor Production Equipment (SPE) players and high-end semiconductor suppliers with significant technological moats to benefit as domestic supply and demand increase.

Asia Pacific Technology

Bruce Lu and Ryan Huang, GS Equity Research

TSMC is increasingly anchored to the US side of the semiconductor ecosystem. Specifically, TSMC has announced its intention to invest \$165bn in advanced semiconductor manufacturing in the US. This includes plans for six advanced wafer manufacturing fabs in Arizona, two advanced packaging fabs, and a major R&D center to support strong multi-year demand from customers, particularly US Cloud Service Providers (CSPs) and AI GPU/Application Specific Integrated Circuit (ASIC) leaders, who are the primary growth drivers for TSMC's advanced node demand.

European Communications Technology, Semis, and Hardware

Alex Duval and Anant Jakhar, GS Equity Research

Chinese efforts to pursue self-sufficiency in semiconductor production necessarily require lithography tools to print circuits onto chips. ASML has a monopoly in Extreme Ultraviolet (EUV) lithography systems, which are critical for manufacturing the most advanced and powerful chips (those with the smallest node sizes). Since 2019, regulation has precluded ASML from sending EUV tools to China, and indeed no such tools have been exported to the geography.

As such, press reports have suggested that China may try to create its own EUV machines. But such efforts appear unlikely to succeed, at least for the next decade, given that ASML's roadmap for EUV came to fruition nearly two decades later than envisaged despite the company's significant scale, experience, and knowledge in this domain owing to the complexity of generating and leveraging EUV light. Moreover, any new producer of such machines would need to ramp their technology based on feedback from advanced customers. ASML also has an important ecosystem of suppliers—the result of decades of work—that is absent in China, further complicating the country's efforts to produce EUV. As a result, a serious competitor to ASML that could dent its global market share seems unlikely to emerge over the next decade, if ever.

By contrast, ASML has been allowed to sell Deep Ultraviolet (DUV) lithography systems—which are less advanced in their capabilities to produce chips with very small node sizes—to China. However, in 2024, the Dutch government imposed significant restrictions on such sales, especially on the more advanced DUV systems.

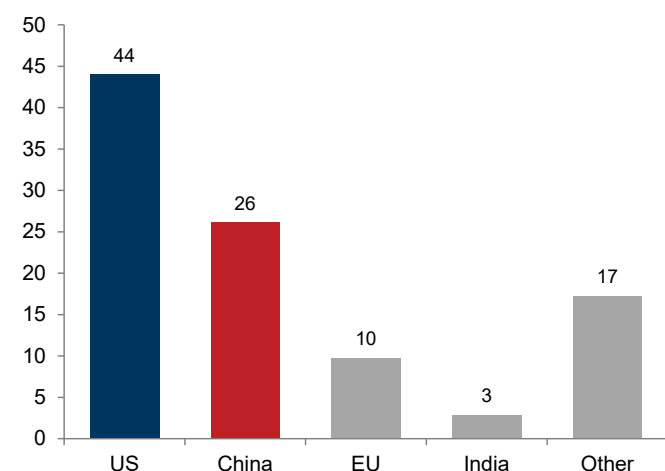
China's physical edge in the tech race

Daan Struyven argues that China's edge in the supply of rare earths and power critical to the tech race will be hard for the US to overcome

The global tech race is heating up, with fierce competition centered on semiconductors, rare earths access, energy supply, talent, and AI adoption. The US is now the frontrunner in the AI race, hosting 44% of global data center capacity vs. 26% for China, and home to leading-edge chips. However, China has a substantial edge in the supply of certain commodities and power critical to the tech race, which will be difficult for the US to overcome.

The US hosts 44% of global data center capacity

Installed data center capacity, as of June 2025, % of global



Source: IEA, Goldman Sachs GIR.

China's critical metals dominance

Chinese policymakers have engineered dominant positions in critical input markets for the buildout of chips, data centers, the associated power grid, and other leading-edge technologies. This dominance is most extreme for rare earth elements (REE), which was foreshadowed by former Chinese leader Deng Xiaoping's 1992 remarks, "the Middle East has oil; China has rare earths."

Dominance in REE, magnets, and critical metals markets reduces China's dependence on other countries in the AI and defense industries and is a powerful source of economic and geopolitical leverage. While the rare earths market is 33 times smaller than the copper market, rare earths, magnets, and other critical minerals are vital for the production of chips, batteries, AI, and defense equipment. So, supply disruptions of these critical elements could result in substantial economic losses.

China is particularly dominant in magnet manufacturing (98% market share) and refining of rare earths (92% market share) but also has a 69% market share in mining rare earths. And while China [has agreed to a one-year suspension](#) of the US rare earth export controls announced on October 9, the April restrictions primarily concerning the geologically scarcer so-called heavy rare earths remain in place.

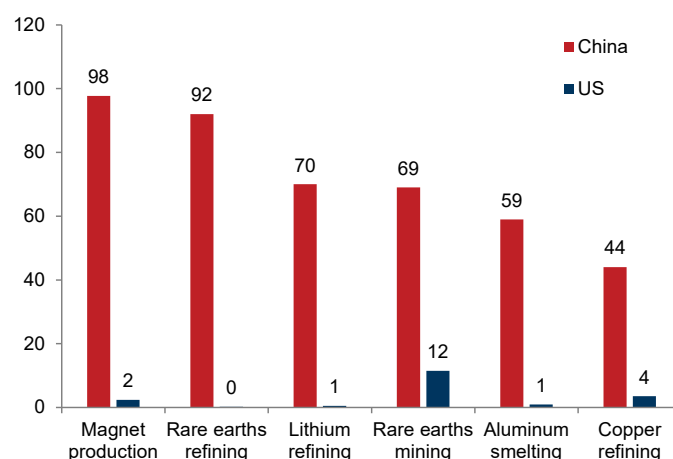
While countries outside of China, especially the US, are accelerating efforts to build independent REE and magnet supply chains, they face two major challenges. First, known

reserves of some rare earths, such as the recently restricted heavy rare earths, are very scarce outside of Myanmar and China, and the process of building a mine typically takes 8-10 years. Second, refining rare earths is technically complex and requires deep expertise. Building and ramping up refineries usually takes around five years.

Looking beyond rare earths, while geology matters when it comes to reserves, decades of industrial policy have enabled China to dominate refining across critical minerals. China's refining market share dramatically exceeds US market share for many minerals, including lithium (critical for storing energy and modernizing the grid), copper (critical for distributing power locally), and aluminum (critical for transmitting power over long distances).

China dominates critical minerals refining and magnet production

China vs. US market shares, %



Source: IEA, Woodmac, Goldman Sachs GIR.

Diverging power capacity

China's ample power market spare capacity is set to grow further on a policy-driven supply boom since a 2021 power crunch, when high energy demand and coal shortages led to power rationing. At the same time, already tight US power markets are tightening further. As a result, power bottlenecks are likely to at least temporarily slow US progress in the AI race by 2030, especially in its regional power markets, where spare power capacity is most scarce.

Specifically, eight out of the thirteen US regional power markets are already at or below critical spare capacity levels based on our estimates of peak summer effective spare power generation capacity—our summary measure of power availability and reliability. This US power market tightness already triggered large spikes in both real-time power prices last summer and power generation capacity prices in the PJM market, which includes Virginia, the world's data center capital.

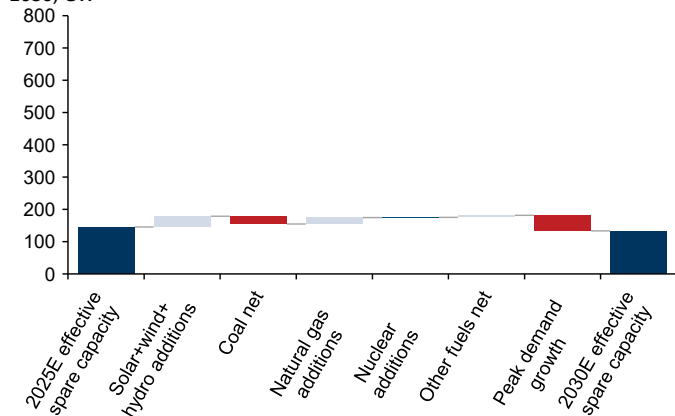
We expect US power spare capacity to decline further as renewable and natural gas power builds are insufficient to offset scheduled coal retirements and meet rapid power demand growth, which is now outpacing GDP growth.

China, in contrast, already has ample spare capacity, which we expect to grow further. By 2030, we project that China's effective power spare capacity will be equivalent to over three times the world's expected data center power demand (~400

GW vs. ~120 GW), positioning China to fuel rapid data center expansion.

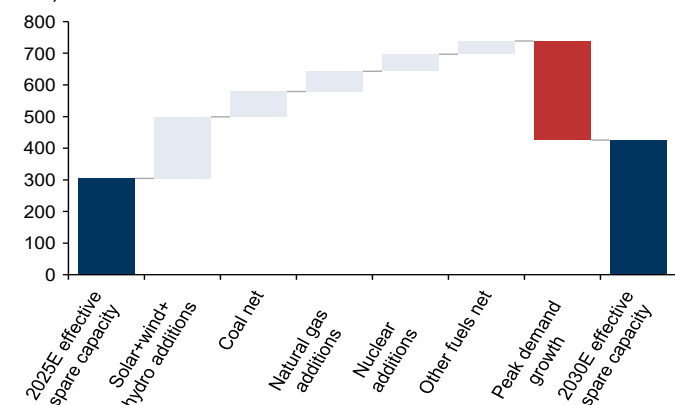
We project power spare capacity to decrease further in the US...

Estimated peak US summer effective power spare capacity from 2025 to 2030, GW



...but to rise in China

Estimated peak China summer effective power spare capacity from 2025 to 2030, GW

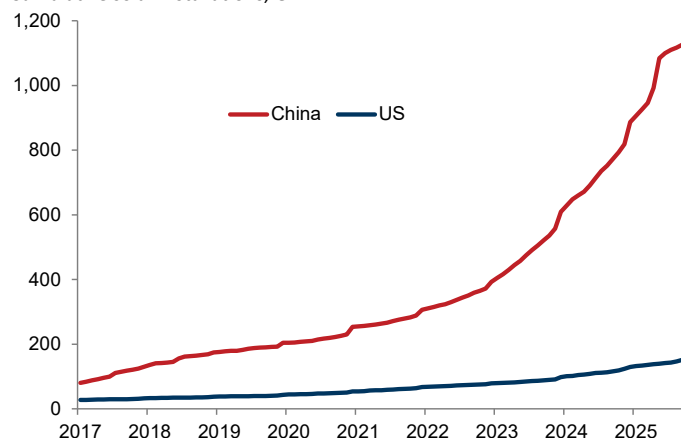


Source: IEA, Wind, EIA, Goldman Sachs GIR.

The most remarkable feature of China's ongoing power capacity boom over the past five years has been the rapid expansion of all power supply sources to ensure energy security. The scale and speed of China capacity additions especially stand out for solar energy—with installed capacity in China now roughly six times higher than in the US—and for nuclear energy. While building nuclear reactors has been a slow and expensive process in the US, learning-by-doing has reduced the real construction cost of nuclear reactors in China.

China has rapidly expanded power capacity for both solar...

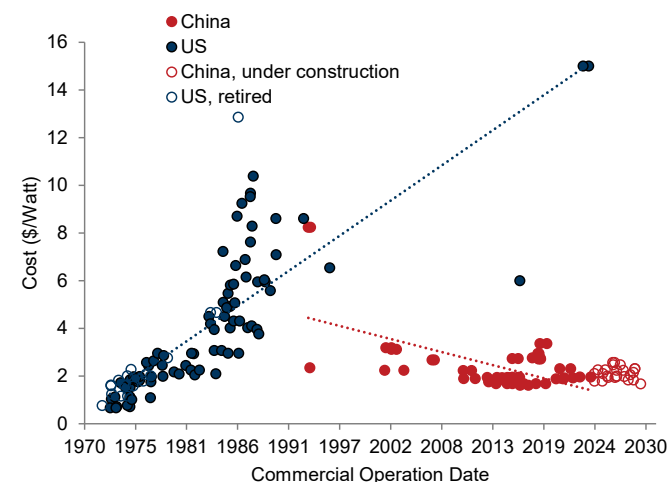
Cumulative solar installations, GW



Source: Ember, Nature, Goldman Sachs GIR.

...and nuclear energy

Construction costs of nuclear reactors



Note: Construction costs don't account for financing costs (such as interest during construction) and exclude operating and fuel costs, as these have remained low and relatively stable over time. All costs are rescaled to their equivalent 2020 value.

Source: Ember, Nature, Goldman Sachs GIR.

Broader implications: more China leverage, higher power and copper prices, and greater insurance value of commodities in portfolios

Beyond the risk that critical commodity and power bottlenecks slow US progress in the tech/AI race, we see four other major implications of China's dominance in these crucial areas:

1. China's tight control over rare earths and other critical minerals will likely **constrain the scope for other countries to impose significant trade barriers against China**. This will likely allow China to maintain some access to advanced chips as well as support rapid growth in China manufacturing exports and its current account surplus.
2. **US power markets are at risk of significantly higher prices and even outages**. This risk is particularly acute in local power markets where data centers are booming, with 72% of all US datacenters sitting in 1% of US counties.
3. As the need to invest in the power grid—a vulnerable link in the energy supply chain—becomes more acute amid the global AI race, rising geopolitical tensions, and the shift to hybrid warfare, **solid copper demand growth along with lagging mine supply should drive a rally in copper prices** from 2028 to \$15,000 by 2035.
4. The growing concentration of the supply of critical metals and commodities in countries competing for AI, broader tech and geopolitical leadership **raises the risk of supply disruptions and underscores the insurance value of commodities in portfolios**.

Daan Struyven, Co-head of Global Commodities Research

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Goldman Sachs & Co. LLC

China's tech drive, Asia's opportunity

Tim Moe and Kinger Lau argue that China's drive to build its own tech stack creates significant opportunities for Asia tech

While Presidents Trump and Xi recently reached an agreement to reduce trade frictions between their two countries, the strategic competition between the US and China in multiple domains—most importantly, technology—continues. As part of this, China has doubled down on its strategic support to key sectors of the economy, with policy initiatives like those in the 15th Five-Year Plan proposal underscoring the Chinese government's commitment to building an indigenous tech stack, particularly in next-gen information technology (e.g., semiconductors), AI, and high-end manufacturing (e.g., robotics and drones) (see pgs. 8-9 for a detailed policy roadmap). These strategic priorities create significant opportunities for technology companies across China and Asia more broadly.

AI is the name of the game...

The technology sector has driven much of the gains in Asian equities this year. Tech hardware & semiconductors and the internet/media/entertainment sector—which includes large Chinese e-commerce companies—have each rallied over 50% year-to-date. The AI theme has powered most of these moves. Indeed, index constituents with tangible revenue connections to AI have accounted for over 90% of the gains in Taiwan equities and 50-60% of the gains in Mainland China, Japan, and Korea equities. As Mainland China continues to pursue technological self-sufficiency and the fundamental picture for the sector remains favorable, we believe the technology sector's leadership can persist and so stay overweight.

Our implementation focus remains on AI and the adjacent theme of power investment spurred by the electricity demands of AI data centers. Specifically, we see opportunities in all the major AI components: (1) hardware, (2) semiconductors, (3) Artificial Intelligence Generated Content (AIGC) applications, a broad category of software that uses AI to produce new content, and (4) physical AI such as robotics, assisted driving, and Artificial Intelligence of Things (AIoT), which combines AI technologies with the Internet of Things infrastructure. That said, AIGC applications and physical AI have lagged hardware and semiconductors, which suggests better entry points for the stocks levered to those components.

Within the power investment theme, we strategically favor nuclear, though its strong performance and the small number of listed direct plays may warrant some caution. Renewables may also benefit tactically from China's anti-involution policies as well as over the longer term as China continues making significant investments in solar, wind, and other green technologies to achieve technological and energy self-sufficiency.

...but broader tech also provides opportunities

China's focus on self-sufficiency in advanced technologies could also benefit the biotech sector. China's biotech industry is already at an inflection point as its innovative drug assets are gaining global recognition. Merck KGaA, for example, recently

executed a worldwide commercialization agreement for Shanghai-based Abbisko Therapeutics' pimicotinib medication. And we think defense tech, quantum computing, and the emerging low-altitude (eVTOL, which refers to electric vertical takeoff and landing aircraft that operate in low altitudes) and space economies are also worth keeping a close eye on and could present interesting investment opportunities.

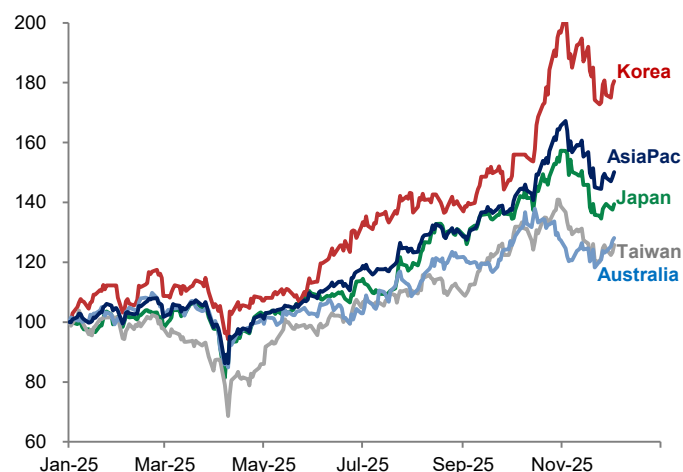
More broadly, our Chinese Prominent 10 (10 existing market leaders that appear well-placed to capitalize on their past successes and further elevate their dominance in the Chinese stock market) and Going Global (25 Buy-rated companies that look well positioned to compete globally) themes provide exposure to the government's continued focus on technological development and manufacturing competitiveness.

US' tech drive is beneficial for Asia too

The parallel US effort to reduce its technological reliance on Mainland China and develop its own tech stack also presents opportunities for Asia-Pacific companies. Taiwan's TSMC is now manufacturing close to 90% of Nvidia's chips (by value), including the [Blackwell wafers](#) it recently began producing at its Phoenix facility. Japan's Mitsubishi Electric is supplying power systems and components for next-gen US data centers while Hitachi and Toshiba are making significant investments in US energy infrastructure. And the US and Australia recently signed [a critical minerals and rare earths agreement](#) to invest at least \$1bn in projects in each country, which should benefit select companies in the Australia mining sector. We see a number of companies in Australia, Japan, Korea, and Taiwan that could benefit from the theme of enhanced US technological self-sufficiency, part of the larger theme of US reindustrialization.

US reindustrialization-related Asian stocks have performed well year to date, and we think this theme has further room to run

Indexed performance of US reindustrialization beneficiaries (in USD)



Note: Cap-weighted backtesting performance with a 4% cap applied to the AsiaPac portfolio at the start of 2025.

Source: FactSet, Goldman Sachs GIR.

Tim Moe, Chief Asia Pacific Equity Strategist

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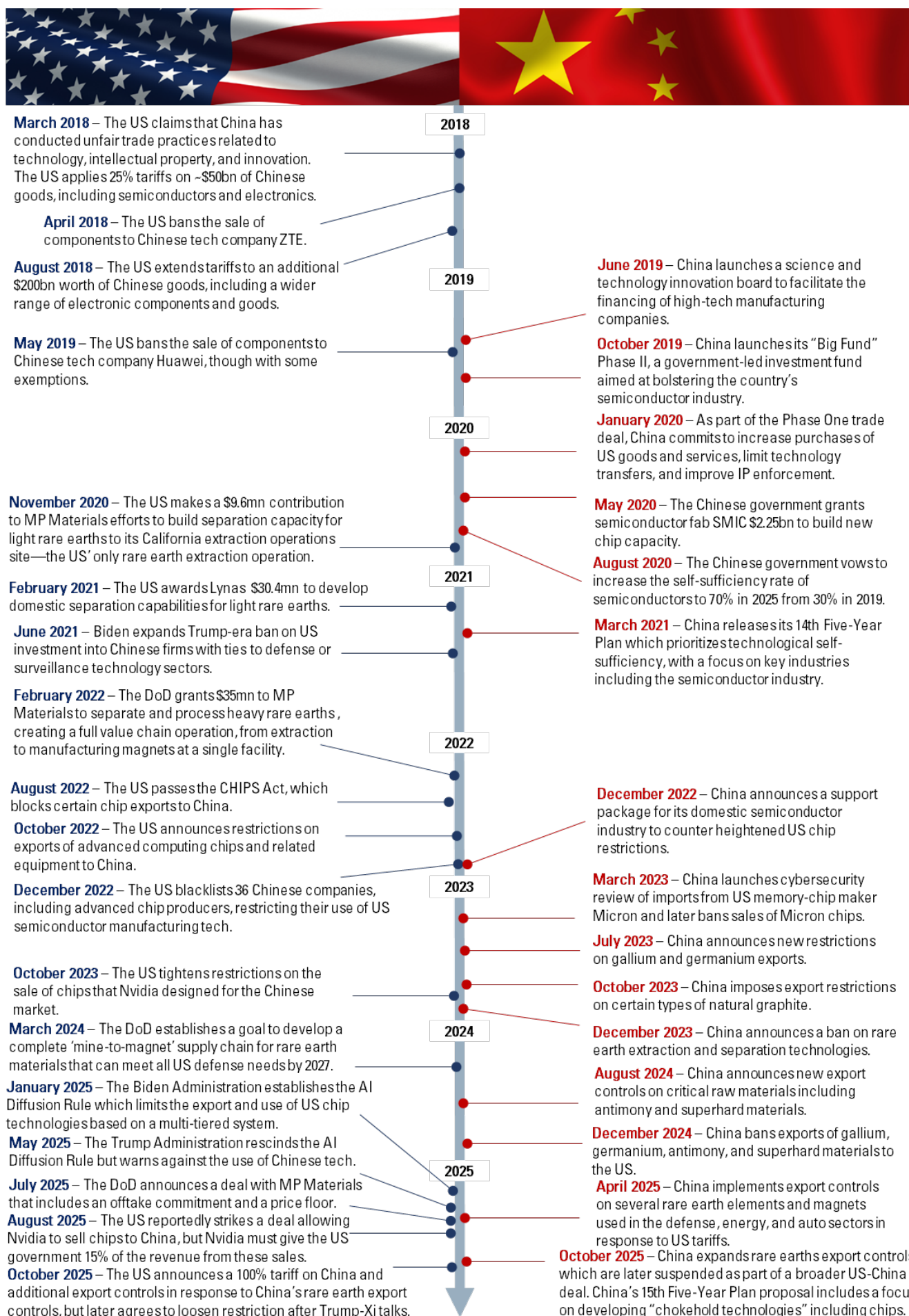
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Goldman Sachs (Asia) LLC

Policies driving the US-China tech race



Note: Exhibit does not constitute an exhaustive list of US and Chinese policy developments.

Source: Center for Strategic and International Studies, White House, USTR, Peterson Institute for International Economics, various news sources, Goldman Sachs GIR.

Summary of our key forecasts

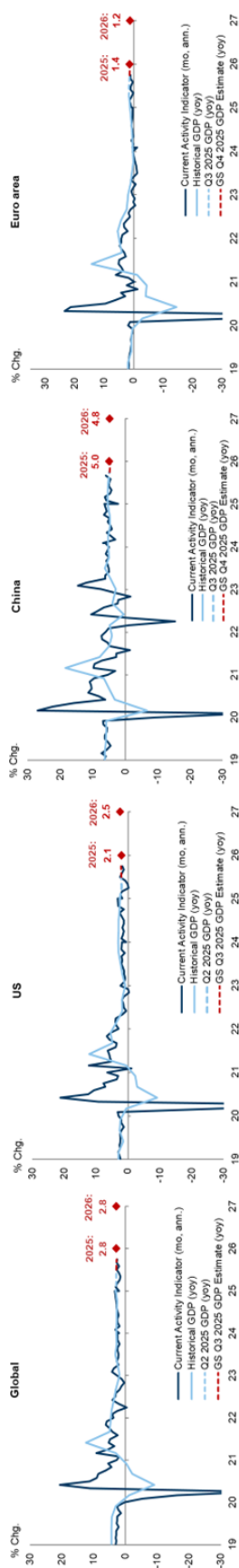
GS GIR: Macro at a glance

Watching

- **Globally**, we expect real GDP growth of 2.8% yoy in 2026 amid fading headwinds from US tariffs and rising real income growth. We expect global core inflation to decline to 2.2% by the end of 2026 as the tariff-driven boost fades and shelter and wage inflation fall further.
- **In the US**, we expect real GDP growth to rise to 2.3% on a Q4/Q4 basis in 2026 as the tariff drag fades, fiscal policy turns more expansionary, and the Fed continues to ease. We expect core PCE inflation to decline to 2.3% yoy by the end of 2026 as tariff pass-through fades, wage growth slows further, and rent inflation remains benign. We expect the unemployment rate to end 2026 at 4.3%.
- **We expect the Fed** to deliver one more 25bp rate cut this year in December followed by two 25bp cuts next year in March and June for a terminal rate range of 3-3.25%.
- **In the Euro area**, we expect real GDP growth of 1.2% yoy in 2026 reflecting receding trade-related growth headwinds, real income gains, and a fading fiscal policy drag but growing headwinds from more intense export competition with China. We expect core inflation to decline to 1.8% by end-2026 amid lower energy prices, a stronger Euro, and the potential for a negative inflation effect from Chinese trade rerouting.
- **We expect the ECB** to remain on hold for the foreseeable future given our and the ECB's forecast of better growth ahead and target-consistent inflation over the medium term.
- **In China**, we expect above-consensus real GDP growth of 4.8% yoy in 2026 given resilient export growth, continued government easing, and a decreasing drag from the ongoing property market downturn. On the inflation front, we believe China's overcapacity problem will take time and effort to solve and expect CPI/PPI inflation of 0.6%/-0.7% next year.
- **WATCH US LABOR MARKET, FISCAL DYNAMICS AND GEOPOLITICAL DEVELOPMENTS.** Uncertainty about the US labor market remains high, and the recent rise in some layoff indicators could mean that the weakness in the labor market is becoming too entrenched to be checked by a modest cyclical growth acceleration. Fiscal pressures—including in the US, UK, France, and Japan—also remain important to watch. And geopolitical developments pose risks worth watching, especially as a potential Russia-Ukraine peace deal remains highly uncertain and US-China relations remain fragile.

Goldman Sachs Global Investment Research.

Growth



Source: Haver Analytics, Goldman Sachs Global Investment Research.
Note: GS CAI is a measure of current growth. For more information on the methodology of the CAI please see "Technical Updates to Our Global CAIs," Global Economics Comment, Sep. 01, 2025.

Forecasts

Economics						Markets										Equities																						
GDP growth (%)			2025		2026		Interest rates 10Yr(%)		Last		E2025		E2026		FX		Last		3m		12m		S&P 500		E2025		E2026		Returns (%)		12m		YTD		E2025 P/E			
			GS (Q4/Q4)	Cons. (CY)	GS (CY)	Cons. (CY)																			GS	Cons.	GS	Cons.										
Global	2.6	--	2.8	2.5	2.8	2.4	US	4.06	4.20	4.20	4.20	EUR/US	1.17	1.20	1.25	Price	6,800	7,600	S&P 500	5.1	16.5	25.8x																
US	2.1	1.7	2.1	2.0	2.5	2.0	Germany	2.75	3.00	3.25	GBP/US	1.33	1.35	1.36	EPS	\$280	\$271	MIX4PJ	14.0	24.0	17.4x																	
China	4.5	4.4	5.0	4.9	4.8	4.4	Japan	1.89	1.70	1.90	JP/JPY	155	150	145	Growth	7%	10%	Topix	5.9	19.7	17.7x																	
Euro area	1.2	1.1	1.4	1.4	1.2	1.1	UK	4.37	4.25	4.00	US/GBP	7.04	6.95	6.85	STOXX 600	3.3	13.5	16.3x																				
Policy rates (%)			2025		2026		Commodities		Last		3m		12m		Credit (bp)		Last		2025		4Q25		Consumer		2025		2026		Wage Tracker 2025 (%)		Q1		Q2		Q3		Q4	
			GS	Mkt.	GS	Mkt.	Crude Oil, Brent (\$/bbl)	63	58	57	IG	79 <td>88</td> <td>85</td> <td>US</td> <td>2.7</td> <td>4.5</td> <td>2.5</td> <td>4.3</td> <td>3.9</td> <td>4.0</td> <td>--</td>	88	85	US	2.7	4.5	2.5	4.3	3.9	4.0	--																
US	3.63	3.66	3.13	2.99	Nat Gas, NYMEX (\$/mmBtu)	5.00	4.50	5.00	USD	IG	79	88	85	US	2.7	4.5	2.5	4.3	3.9	4.0	--																	
Euro area	2.00	1.93	2.00	1.89	Nat Gas, TTF EUR/MMWh	28.19	30	28	HY	269	303	300	Euro area	2.1	6.4	1.8	6.5	--	--	--																		
China	1.40	1.44	1.20	--	Copper (\$/mt)	11,576	10,725	10,610	EUR	IG	90	102	100	China	0.0	--	0.6	--	--	--	--																	
Japan	0.50	0.66	1.00	1.06	Gold (\$/mty oz)	4,210	4,160	4,745	HY	278	310	307																										

Source: Bloomberg, Goldman Sachs Global Investment Research. For important disclosures, see the Disclosure Appendix or go to www.gs.com/research/hedge.html.

Market pricing as of December 3, 2025

Glossary of GS proprietary indices

Current Activity Indicator (CAI)

GS CAIs measure the growth signal in a broad range of weekly and monthly indicators, offering an alternative to Gross Domestic Product (GDP). GDP is an imperfect guide to current activity: In most countries, it is only available quarterly and is released with a substantial delay, and its initial estimates are often heavily revised. GDP also ignores important measures of real activity, such as employment and the purchasing managers' indexes (PMIs). All of these problems reduce the effectiveness of GDP for investment and policy decisions. Our CAIs aim to address GDP's shortcomings and provide a timelier read on the pace of growth.

For more, see our CAI page and Global Economics Comment: Technical Updates to Our Global CAIs.

Dynamic Equilibrium Exchange Rates (DEER)

The GSDEER framework establishes an equilibrium (or "fair") value of the real exchange rate based on relative productivity and terms-of-trade differentials.

For more, see our GSDEER page, Global Economics Paper No. 227: Finding Fair Value in EM FX, 26 January 2016, and Global Markets Analyst: A Look at Valuation Across G10 FX, 29 June 2017.

Financial Conditions Index (FCI)

GS FCIs gauge the "looseness" or "tightness" of financial conditions across the world's major economies, incorporating variables that directly affect spending on domestically produced goods and services. FCIs can provide valuable information about the economic growth outlook and the direct and indirect effects of monetary policy on real economic activity.

FCIs for the G10 economies are calculated as a weighted average of a policy rate, a long-term risk-free bond yield, a corporate credit spread, an equity price variable, and a trade-weighted exchange rate; the Euro area FCI also includes a sovereign credit spread. The weights mirror the effects of the financial variables on real GDP growth in our models over a one-year horizon. FCIs for emerging markets are calculated as a weighted average of a short-term interest rate, a long-term swap rate, a CDS spread, an equity price variable, a trade-weighted exchange rate, and—in economies with large foreign-currency-denominated debt stocks—a debt-weighted exchange rate index.

For more, see our FCI page, Global Economics Analyst: Our New G10 Financial Conditions Indices, 20 April 2017, and Global Economics Analyst: Tracking EM Financial Conditions – Our New FCIs, 6 October 2017.

Goldman Sachs Analyst Index (GSAI)

The US GSAI is based on a monthly survey of GS equity analysts to obtain their assessments of business conditions in the industries they follow. The results provide timely "bottom-up" information about US economic activity to supplement and cross-check our analysis of "top-down" data. Based on analysts' responses, we create a diffusion index for economic activity comparable to the ISM's indexes for activity in the manufacturing and nonmanufacturing sectors.

Macro-Data Assessment Platform (MAP)

GS MAP scores facilitate rapid interpretation of new data releases for economic indicators worldwide. MAP summarizes the importance of a specific data release (i.e., its historical correlation with GDP) and the degree of surprise relative to the consensus forecast. The sign on the degree of surprise characterizes underperformance with a negative number and outperformance with a positive number. Each of these two components is ranked on a scale from 0 to 5, with the MAP score being the product of the two, i.e., from -25 to +25. For example, a MAP score of +20 (5;+4) would indicate that the data has a very high correlation to GDP (5) and that it came out well above consensus expectations (+4), for a total MAP value of +20.

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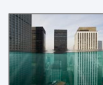
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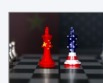
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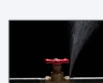
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Disclosure Appendix

Reg AC

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We, Allen Chang, Paul Young, Alexander Duval, Brian Lee, CFA, Bruce Lu, and James Schneider, Ph.D., hereby certify that all of the views expressed in this report accurately reflect our personal views about the subject company or companies and its or their securities. We also certify that no part of our compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

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	Buy	Hold	Sell	Buy	Hold	Sell
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